

120-MHz, 32-bit RX MCU, on-chip FPU, 240 DMIPS, up to 1-MB flash memory, 256-KB SRAM, various communications interfaces including Ethernet MAC, SD host interface (optional), SD slave interface (optional), quad SPI, and CAN, 12-bit A/D converter, RTC, encryption (optional), CMOS camera interface

Features

■ 32-bit RXv2 CPU core

- Max. operating frequency: 120 MHz
Capable of 240 DMIPS in operation at 120 MHz
- Single precision 32-bit IEEE-754 floating point
- Two types of multiply-and-accumulation unit (between memories and between registers)
- 32-bit multiplier (fastest instruction execution takes one CPU clock cycle)
- Divider (fastest instruction execution takes two CPU clock cycles)
- Fast interrupt
- CISC Harvard architecture with 5-stage pipeline
- Variable-length instructions: Ultra-compact code
- Supports the memory protection unit (MPU)
- JTAG and FINE (one-line) debugging interfaces

■ Low-power design and architecture

- Operation from a single 2.7- to 3.6-V supply
- Low power consumption: A product that supports all peripheral functions draws only 0.19 mA/MHz (Typ.).
- RTC is capable of operation from a dedicated power supply.
- Four low-power modes

■ On-chip code flash memory

- Supports versions with up to 1 Mbytes of ROM
- No wait cycles at up to 50 MHz or when the ROM cache is hit, one-wait state at up to 100 MHz, two-wait state at above 100 MHz
- User code is programmable by on-board or off-board programming.

■ On-chip SRAM

- 256 Kbytes of SRAM (no wait cycles)
- 8 Kbytes of standby RAM (backup on deep software standby)

■ Data transfer

- DMACAa: 8 channels
- DTCb: 1 channel
- EXDMAC: 2 channels
- DMAC for the Ethernet controller: 1 channel

■ Reset and supply management

- Power-on reset (POR)
- Low voltage detection (LVD) with voltage settings

■ Clock functions

- External crystal oscillator or internal PLL for operation at 8 to 24 MHz
- Internal 240-kHz LOCO and HOCO selectable from 16, 18, and 20 MHz
- 120-kHz clock for the IWDtA

■ Real-time clock

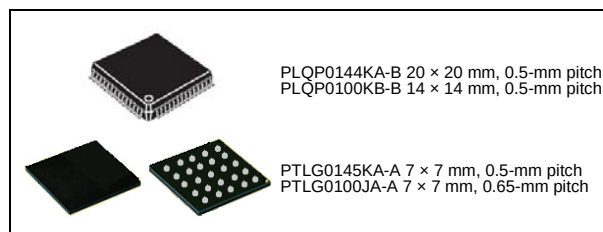
- Adjustment functions (30 seconds, leap year, and error)
- Real-time clock counting and binary counting modes are selectable
- Time capture function
(for capturing times in response to event-signal input)

■ Independent watchdog timer

- 120-kHz (1/2 LOCO frequency) clock operation

■ Useful functions for IEC60730 compliance

- Oscillation-stoppage detection, frequency measurement, CRCA, IWDtA, self-diagnostic function for the A/D converter, etc.
- Register write protection function can protect values in important registers against overwriting.



■ Various communications interfaces

- Ethernet MAC (1 channel)
- PHY layer (1 channel) for host/function or OTG controller (1 channel) with full-speed USB 2.0 transfer
- CAN (compliant with ISO11898-1), incorporating 32 mailboxes (up to 2 channels)
- SCiG and SCiH with multiple functionalities (up to 11 channels)
Choose from among asynchronous mode, clock-synchronous mode, smart-card interface mode, simplified SPI, simplified I²C, and extended serial mode.
- SCiI with 16-byte transmission and reception FIFOs (up to 2 channels)
- I²C bus interface for transfer at up to 1 Mbps (up to 2 channels)
- Four-wire QSPI (1 channel) in addition to RSPiC (3 channels)
- Parallel data capture unit (PDC) for the CMOS camera interface (not in 100-pin products)
- SD host interface (optional: 1 channel) with a 1- or 4-bit SD bus for use with SD memory or SDIO
- SD slave interface (optional: 1 channel) with a 1- or 4-bit SD bus for use with SD host interface
- MMCIF with 1-, 4-, or 8-bit transfer bus width

■ External address space

- Buses for full-speed data transfer (max. operating frequency of 60 MHz)
- 8 CS areas
- 8-, or 16-bit bus space is selectable per area
- Independent SDRAM area (128 Mbytes)

■ Up to 25 extended-function timers

- 16-bit TPUa, and MTU3a
- 8-bit TMRa (4 channels), 16-bit CMT (4 channels), 32-bit CMTW (2 channels)

■ 12-bit A/D converter

- Two 12-bit units (8 channels for unit 0; 21 channels for unit 1)
- Self diagnosis
- Detection of analog input disconnection

■ 12-bit D/A converter: 2 channels

■ Temperature sensor for measuring temperature within the chip

■ Encryption (optional)

- AES (key lengths: 128, 192, and 256 bits)

■ Up to 111 pins for general I/O ports

- 5-V tolerance, open drain, input pull-up, switchable driving ability

■ Operating temp. range

- -40°C to +85°C

1. Overview

1.1 Outline of Specifications

Table 1.1 lists the specifications in outline, and Table 1.2 gives a comparison of the functions of products in different packages.

Table 1.1 shows the outline of maximum specifications, and the number of peripheral module channels differs depending on the pin number on the package. For details, see Table 1.2, Comparison of Functions for Different Packages.

Table 1.1 Outline of Specifications (1/8)

Classification	Module/Function	Description
CPU	CPU	- Maximum operating frequency: 120 MHz 32-bit RX CPU (RXv2) - Minimum instruction execution time: One instruction per state (cycle of the system clock) - Address space: 4-Gbyte linear - Register set of the CPU - General purpose: Sixteen 32-bit registers - Control: Ten 32-bit registers - Accumulator: Two 72-bit registers - Basic instructions: 75 - Floating-point instructions: 11 - DSP instructions: 23 - Addressing modes: 11 - Data arrangement - Instructions: Little endian - Data: Selectable as little endian or big endian - On-chip 32-bit multiplier: $32 \times 32 \rightarrow 64$ bits - On-chip divider: $32 / 32 \rightarrow 32$ bits - Barrel shifter: 32 bits
	FPU	- Single precision (32-bit) floating point - Data types and floating-point exceptions in conformance with the IEEE754 standard
Memory	Code flash memory	- Capacity: 512 Kbytes/768 Kbytes/1 Mbyte $50 \text{ MHz} \leq$ No-wait cycle access $100 \text{ MHz} \leq$ 1-wait cycle access $100 \text{ MHz} \geq$ 2-wait cycle access - Instructions hitting the ROM cache or operand = 120 MHz: No-wait access - On-board programming: Four types - Off-board programming (parallel programmer mode) - The trusted memory (TM) function protects against the reading of programs from blocks 8 and 9.
	RAM	- Capacity: 256 Kbytes 120 MHz, no-wait access
	Standby RAM	- Capacity: 8 Kbytes - Operation synchronized with PCLKB: Up to 60 MHz, two-cycle access
Operating modes		- Operating modes by the mode-setting pins at the time of release from the reset state - Single-chip mode - Boot mode (for the SCI interface) - Boot mode (for the USB interface) - Boot mode (for the FINE interface) - Selection of operating mode by register setting - Single-chip mode - On-chip ROM disabled extended mode - On-chip ROM enabled extended mode - Endian selectable

Table 1.1 Outline of Specifications (2/8)

Classification	Module/Function	Description
Clock	Clock generation circuit	- Main clock oscillator, sub clock oscillator, low-speed/high-speed on-chip oscillator, PLL frequency synthesizer, and IWDI-dedicated on-chip oscillator - The peripheral module clocks can be set to frequencies above that of the system clock. - Main-clock oscillation stoppage detection - Separate frequency-division and multiplication settings for the system clock (ICLK), peripheral module clocks (PCLKA, PCLKB, PCLKC, PCLKD), flash-IF clock (FCLK) and external bus clock (BCLK) The CPU and other bus masters run in synchronization with the system clock (ICLK): Up to 120 MHz Peripheral modules of MTU3, RSPI, SCli, ETHERC, EDMAC, and AES run in synchronization with PCLKA, which operates at up to 120 MHz. Other peripheral modules run in synchronization with PCLKB: Up to 60 MHz ADCLK in the S12AD (unit 0) runs in synchronization with PCLKC: Up to 60 MHz ADCLK in the S12AD (unit 1) runs in synchronization with PCLKD: Up to 60 MHz Flash IF run in synchronization with the flash-IF clock (FCLK): Up to 60 MHz Devices connected to the external bus run in synchronization with the external bus clock (BCLK): Up to 60 MHz - Multiplication is possible with using the high-speed on-chip oscillator (HOCO) as a reference clock of the PLL circuit
Reset		Nine types of reset - RES# pin reset: Generated when the RES# pin is driven low. - Power-on reset: Generated when the RES# pin is driven high and VCC = AVCC0 = AVCC1 rises. - Voltage-monitoring 0 reset: Generated when VCC = AVCC0 = AVCC1 falls. - Voltage-monitoring 1 reset: Generated when VCC = AVCC0 = AVCC1 falls. - Voltage-monitoring 2 reset: Generated when VCC = AVCC0 = AVCC1 falls. - Deep software standby reset: Generated in response to an interrupt to trigger release from deep software standby. - Independent watchdog timer reset: Generated when the independent watchdog timer underflows, or a refresh error occurs. - Watchdog timer reset: Generated when the watchdog timer underflows, or a refresh error occurs. - Software reset: Generated by register setting.
Power-on reset		If the RES# pin is at the high level when power is supplied, an internal reset is generated. After VCC = AVCC0 = AVCC1 has exceeded the voltage detection level and the specified period has elapsed, the reset is cancelled.
Voltage detection circuit (LVDA)		Monitors the voltage being input to the VCC = AVCC0 = AVCC1 pins and generates an internal reset or internal interrupt. - Voltage detection circuit 0 - Capable of generating an internal reset - The option-setting memory can be used to select enabling or disabling of the reset. - Voltage detection level: Selectable from three different levels (2.94 V, 2.87 V, 2.80 V) - Voltage detection circuits 1 and 2 - Voltage detection level: Selectable from three different levels (2.99 V, 2.92 V, 2.85 V) - Digital filtering (1/2, 1/4, 1/8, and 1/16 LOCO frequency) - Capable of generating an internal reset - Two types of timing are selectable for release from reset - An internal interrupt can be requested. - Detection of voltage rising above and falling below thresholds is selectable. - Maskable or non-maskable interrupt is selectable - Voltage detection monitoring - Event linking
Low power consumption	Low power consumption function	- Module stop function - Four low power consumption modes - Sleep mode, all-module clock stop mode, software standby mode, and deep software standby mode
	Battery backup function	When the voltage on the VCC pin drops, battery power from the VBATT pin is supplied to keep the real-time clock (RTC) operating.

Table 1.1 Outline of Specifications (3/8)

Classification	Module/Function	Description
Interrupt	Interrupt controller (ICUB)	- Peripheral function interrupts: 243 sources - External interrupts: 16 (pins IRQ0 to IRQ15) - Software interrupts: 2 sources - Non-maskable interrupts: 7 sources - Sixteen levels specifiable for the order of priority - Method of interrupt source selection: The interrupt vectors consist of 256 vectors (128 sources are fixed. The remaining 128 vectors are selected from among the other 112 sources.)
External bus extension		- The external address space can be divided into eight areas (CS0 to CS7), each with independent control of access settings. Capacity of each area: 16 Mbytes (CS0 to CS7) A chip-select signal (CS0# to CS7#) can be output for each area. Each area is specifiable as an 8-, or 16-bit bus space. The data arrangement in each area is selectable as little or big endian (only for data). - SDRAM interface connectable - Bus format: Separate bus, multiplex bus - Wait control - Write buffer facility
DMA	DMA controller (DMACa)	8 channels - Three transfer modes: Normal transfer, repeat transfer, and block transfer - Activation sources: Software trigger, external interrupts, and interrupt requests from peripheral functions
	EXDMA controller (EXDMACa)	2 channels - Four transfer modes: Normal transfer, repeat transfer, block transfer, and cluster transfer - Single-address transfer enabled with the EDACKn signal - Activation sources: Software trigger, external DMA requests (EDREQn), and interrupt requests from peripheral functions
	Data transfer controller (DTCb)	- Three transfer modes: Normal transfer, repeat transfer, and block transfer - Activation sources: External interrupts and interrupt requests from peripheral functions - Sequence transfer
I/O ports	Programmable I/O ports	- I/O ports for the 145-pin TFLGA and 144-pin LQFP I/O pins: 111 Input pin: 1 Pull-up resistors: 111 Open-drain outputs: 111 5-V tolerance: 18 - I/O ports for the 100-pin TFLGA and 100-pin LQFP I/O pins: 78 Input pin: 1 Pull-up resistors: 78 Open-drain outputs: 78 5-V tolerance: 17
Event link controller (ELC)		- Event signals such as interrupt request signals can be interlinked with the operation of functions such as timer counting, eliminating the need for intervention by the CPU to control the functions. 83 internal event signals can be freely combined for interlinked operation with connected functions. - Event signals from peripheral modules can be used to change the states of output pins (of ports B and E). - Changes in the states of pins (of ports B and E) being used as inputs can be interlinked with the operation of peripheral modules.
Timers	16-bit timer pulse unit (TPUa)	(16 bits × 6 channels) × 1 unit - Maximum of 16 pulse-input/output possible - Select from among seven or eight counter-input clock signals for each channel - Input capture/output compare function - Output of PWM waveforms in up to 15 phases in PWM mode - Support for buffered operation, phase-counting mode (two phase encoder input) and cascade-connected operation (32 bits × 2 channels) depending on the channel. - PPG output trigger can be generated - Capable of generating conversion start triggers for the A/D converters - Digital filtering of signals from the input capture pins - Event linking by the ELC

Table 1.1 Outline of Specifications (4/8)

Classification	Module/Function	Description
Timers	Multifunction timer pulse unit (MTU3a)	• 9 channels (16 bits × 8 channels, 32 bits × 1 channel) • Maximum of 28 pulse-input/output and 3 pulse-input possible • Select from among 14 counter-input clock signals for each channel (PCLKA/1, PCLKA/2, PCLKA/4, PCLKA/8, PCLKA/16, PCLKA/32, PCLKA/64, PCLKA/256, PCLKA/1024, MTCLKA, MTCLKB, MTCLKC, MTCLKD, MTIOC1A) 14 of the signals are available for channel 0, 11 are available for channels 1, 3, 4, 6 to 8, 12 are available for channel 2, and 10 are available for channel 5. • Input capture function • 39 output compare/input capture registers • Counter clear operation (synchronous clearing by compare match/input capture) • Simultaneous writing to multiple timer counters (TCNT) • Simultaneous register input/output by synchronous counter operation • Buffered operation • Support for cascade-connected operation • 43 interrupt sources • Automatic transfer of register data • Pulse output mode Toggle/PWM/complementary PWM/reset-synchronized PWM • Complementary PWM output mode Outputs non-overlapping waveforms for controlling 3-phase inverters Automatic specification of dead times PWM duty cycle: Selectable as any value from 0% to 100% Delay can be applied to requests for A/D conversion. Non-generation of interrupt requests at peak or trough values of counters can be selected. Double buffer configuration • Reset synchronous PWM mode Three phases of positive and negative PWM waveforms can be output with desired duty cycles. • Phase-counting mode: 16-bit mode (channels 1 and 2); 32-bit mode (channels 1 and 2) • Counter functionality for dead-time compensation • Generation of triggers for A/D converter conversion • A/D converter start triggers can be skipped • Digital filter function for signals on the input capture and external counter clock pins • PPG output trigger can be generated • Event linking by the ELC
	Port output enable 3 (POE3a)	• Control of the high-impedance state of the MTU3 waveform output pins • 5 pins for input from signal sources: POE0#, POE4#, POE8#, POE10#, POE11# • Initiation on detection of short-circuited outputs (detection of simultaneous PWM output to the active level) • Initiation by oscillation-stoppage detection or software • Additional programming of output control target pins is enabled
	Programmable pulse generator (PPG)	• (4 bits × 4 groups) × 2 units • Pulse output with the MTU or TPU output as a trigger • Maximum of 32 pulse-output possible
	8-bit timers (TMRb)	• (8 bits × 2 channels) × 2 units • Select from among seven internal clock signals (PCLKB/1, PCLKB/2, PCLKB/8, PCLKB/32, PCLKB/64, PCLKB/1024, PCLKB/8192) and one external clock signal • Capable of output of pulse trains with desired duty cycles or of PWM signals • The 2 channels of each unit can be cascaded to create a 16-bit timer • Generation of triggers for A/D converter conversion • Capable of generating baud-rate clocks for SCI5, SCI6, and SCI12 • Event linking by the ELC
	Compare match timer (CMT)	• (16 bits × 2 channels) × 2 units • Select from among four internal clock signals (PCLKB/8, PCLKB/32, PCLKB/128, PCLKB/512) • Event linking by the ELC
	Compare match timer W (CMTW)	• (32 bits × 1 channel) × 2 units • Compare-match, input-capture input, and output-comparison output are available. • Select from among four internal clock signals (PCLKB/8, PCLKB/32, PCLKB/128, PCLKB/512) • Interrupt requests can be output in response to compare-match, input-capture, and output-comparison events. • Event linking by the ELC

Table 1.1 Outline of Specifications (5/8)

Classification	Module/Function	Description
Timers	Realtime clock (RTCd)	• Clock sources: Main clock, sub clock • Selection of the 32-bit binary count in time count/second unit possible • Clock and calendar functions • Interrupt sources: Alarm interrupt, periodic interrupt, and carry interrupt • Battery backup operation • Time-capture facility for three values • Event linking by the ELC
	Watchdog timer (WDTA)	• 14 bits × 1 channel • Select from among 6 counter-input clock signals (PCLKB/4, PCLKB/64, PCLKB/128, PCLKB/512, PCLKB/2048, PCLKB/8192)
	Independent watchdog timer (IWDtA)	• 14 bits × 1 channel • Counter-input clock: IWDt-dedicated on-chip oscillator • Dedicated clock/1, dedicated clock/16, dedicated clock/32, dedicated clock/64, dedicated clock/128, dedicated clock/256 • Window function: The positions where the window starts and ends are specifiable (the window defines the timing with which refreshing is enabled and disabled). • Event linking by the ELC
Communication function	Ethernet controller (ETHERC)	• Input and output of Ethernet/IEEE 802.3 frames • Transfer at 10 or 100 Mbps • Full- and half-duplex modes • MII (Media Independent Interface) or RMII (Reduced Media Independent Interface) as defined in IEEE 802.3u • Detection of Magic Packets™*1 or output of a “wake-on-LAN” signal (WOL) • Compliance with flow control as defined in IEEE 802.3x standards
	DMA controller for Ethernet controller (EDMACa)	• Alleviation of CPU load by the descriptor control method • Transmission FIFO: 2 Kbytes; Reception FIFO: 2 Kbytes
	USB 2.0 FS host/function module (USBb)	• Includes a UDC (USB Device Controller) and transceiver for USB 2.0 FS • One port • Compliance with the USB 2.0 specification • Transfer rate: Full speed (12 Mbps), low speed (1.5 Mbps) (host only) • Both self-power mode and bus power are supported • OTG (On the Go) operation is possible (low-speed is not supported) • Incorporates 2 Kbytes of RAM as a transfer buffer • External pull-up and pull-down resistors are not required
	Serial communications interfaces (SCIg, SCIlh, SCIl)	• 13 channels (SCIg: 10 channels + SCIlh: 1 channel + SCIl: 2 channels) • SCIg, SCIlh, SCIl - Serial communications modes: Asynchronous, clock synchronous, and smart-card interface - Multi-processor function - On-chip baud rate generator allows selection of the desired bit rate - Choice of LSB-first or MSB-first transfer - Start-bit detection: Level or edge detection is selectable. - Simple I²C - Simple SPI 9-bit transfer mode - Bit rate modulation - Double-speed mode • SCIg, SCIlh - Average transfer rate clock can be input from TMR timers for SCIl5, SCIl6, and SCIl12 - Event linking by the ELC (only on channel 5) • SCIlh - Supports the serial communications protocol, which contains the start frame and information frame - Supports the LIN format • SCIl - Data can be transmitted or received in sequence by the 16-byte FIFO buffers of the transmission and reception unit

Table 1.1 Outline of Specifications (6/8)

Classification	Module/Function	Description
Communication function	I ² C bus interface (RIICa)	2 channels (only channel 0 can be used in fast-mode plus) - Communication formats - I²C bus format/SMBus format - Supports the multi-master - Max. transfer rate: 1 Mbps (channel 0) - Event linking by the ELC
	CAN module (CAN)	2 channels - Compliance with the ISO11898-1 specification (standard frame and extended frame) 32 mailboxes per channel
	Serial peripheral interface (RSPIc)	3 channel - RSPI transfer facility - Using the MOSI (master out, slave in), MISO (master in, slave out), SSL (slave select), and RSPCK (RSPI clock) signals enables serial transfer through SPI operation (four lines) or clock-synchronous operation (three lines) - Capable of handling serial transfer as a master or slave - Data formats - Switching between MSB first and LSB first - The number of bits in each transfer can be changed to any number of bits from 8 to 16, or to 20, 24, or 32 bits. 128-bit buffers for transmission and reception - Up to four frames can be transmitted or received in a single transfer operation (with each frame having up to 32 bits) - Transit/receive data can be swapped in byte units - Buffered structure - Double buffers for both transmission and reception - RSPCK can be stopped with the receive buffer full for master reception. - Event linking by the ELC
	Quad serial peripheral interface (QSPI)	1 channel - Connectable with serial flash memory equipped with multiple input and output lines (i.e. for single, dual, or quad operation) - Programmable bit length and selectable active sense and phase of the clock signal - Sequential execution of transfer - LSB or MSB first is selectable
	SD host interface (SDHI)* ³	1 channel - Transfer speed: Supports high-speed mode (15 MB/s) and default speed mode (11.9 MB/s) - One interface for SD memory and I/O cards (supporting 1- and 4-bit SD buses) - SD specifications - Part 1: Physical Layer Specification Ver.3.01 compliant (DDR not supported) - Part E1: SDIO Specification Ver. 3.00 - Error checking: CRC7 for commands and CRC16 for data - Interrupt requests: Card access interrupt, SDIO access interrupt, card detection interrupt - DMA transfer requests: SD_BUF write and SD_BUF read - Support for card detection and write protection
	SD slave interface (SDSI)* ³	1 channel - Compliant with the SDIO Card Specification Ver.2.00 (CSA is not supported) 1-bit SD/4-bit SD/SPI mode - SDIO Proprietary command is supported - SD/SPI Mandatory command is supported - Interrupt requests: 6
	MMC host interface (MMCIF)	1 channel - Transfer speed: Data transfer mode (30 MB/s), backward compatible mode (25 MB/s) - Compliant with JEDEC STANDARD JESD84-A441 (DDR is not supported) - Interface for Multimedia Cards (MMCs) - Device buses: Support for 1-, 4-, and 8-bit MMC buses - Interrupt requests: Card detection interrupt, error/timeout interrupt, normal operation interrupt - DMA transfer requests: CE_DATA write and CE_DATA read - Support for card detection, boot operation, high priority interrupt (HPI)
	Parallel data capture unit (PDC)	1 channel - Acquisition of synchronization through external 8-bit horizontal and vertical synchronization signals - Setting of the image size when clipping of the output for a one-frame image is required

Table 1.1 Outline of Specifications (7/8)

Classification	Module/Function	Description
12-bit A/D converter (S12ADFa)		12 bits × 2 units (unit 0: 8 channels; unit 1: 21 channels) 12-bit resolution (switchable between 8, 10, and 12 bits) - Conversion time 0.48 μs per channel (for 12-bit conversion) 0.45 μs per channel (for 10-bit conversion) 0.42 μs per channel (for 8-bit conversion) - Operating mode - Scan mode (single scan mode, continuous scan mode, or 3 group scan mode) - Group priority control (only for 3 group scan mode) - Sample-and-hold function - Common sample-and-hold circuit included - In addition, channel-dedicated sample-and-hold function (3 channels: in unit 0 only) included - Sampling variable - Sampling time can be set up for each channel. - Digital comparison - Method: Comparison to detect voltages above or below thresholds and window comparison - Measurement: Comparison of two results of conversion or comparison of a value in the comparison register and a result of conversion - Self-diagnostic function - The self-diagnostic function internally generates three analog input voltages (unit 0: VREFL0, VREFH0 × 1/2, VREFH0; unit 1: AVSS1, AVCC1 × 1/2, AVCC1) - Double trigger mode (A/D conversion data duplicated) - Detection of analog input disconnection - Three ways to start A/D conversion - Software trigger, timer (MTU3, TMR, TPU) trigger, external trigger - Event linking by the ELC
12-bit D/A converter (R12DA)		2 channels 12-bit resolution - Output voltage: 0 V to AVCC1 - Event linking by the ELC
Temperature sensor		1 channel - Relative precision: ±1°C - The voltage of the temperature is converted into a digital value by the 12-bit A/D converter (unit 1).
Safety	Memory protection unit (MPU)	- Protection area: Eight areas (max.) can be specified in the range from 0000 0000h to FFFF FFFFh. - Minimum protection unit: 16 bytes - Reading from, writing to, and enabling the execution access can be specified for each area. - An address exception occurs when the detected access is not in the permitted area.
	Trusted Memory (TM) Function	- Protects against the reading of programs from blocks 8 and 9 of the code flash memory - Instruction fetching by the CPU is the only form of access to these areas when the TM function is enabled.
	Register write protection function	Protects important registers from being overwritten for in case a program runs out of control.
	CRC calculator (CRCA)	- Generation of CRC codes for 8-/32-bit data 8-bit data - Selectable from the following three polynomials $X^8 + X^2 + X + 1$, $X^{16} + X^{15} + X^2 + 1$, $X^{16} + X^{12} + X^5 + 1$ 32-bit data - Selectable from the following two polynomials $X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$, $X^{32} + X^{28} + X^{27} + X^{26} + X^{25} + X^{23} + X^{22} + X^{20} + X^{19} + X^{18} + X^{14} + X^{13} + X^{11} + X^{10} + X^9 + X^8 + X^6 + 1$ - Generation of CRC codes for use with LSB-first or MSB-first communications is selectable
	Main clock oscillation stop function	Main clock oscillation stop detection: Available

Table 1.1 Outline of Specifications (8/8)

Classification	Module/Function	Description
Safety	Clock frequency accuracy measurement circuit (CAC)	Monitors the clock output from the main clock oscillator, sub-clock oscillator, low- and high-speed on-chip oscillators, the PLL frequency synthesizer, IWDG-dedicated on-chip oscillator, and PCLKB, and generates interrupts when the setting range is exceeded.
	Data operation circuit (DOC)	The function to compare, add, or subtract 16-bit data
Encryption function	AESa*2	- Key lengths: 128, 192, and 256 bits - Support for CFB, OFB, and CMAC operating modes - Speed of calculations: 128-bit key length in 22 cycles 192-bit key length in 26 cycles 256-bit key length in 30 cycles - Compliant with FIPS PUB 197
	True random number generator (RNGa)*2	- Length of random numbers: 16 bits - Generation of random-number-generated interrupts after a number is generated - Random number generation time: 1.9 ms (typ)
Operating frequency		Up to 120 MHz
Power supply voltage		VCC = AVCC0 = AVCC1 = VCC_USB = 2.7 to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$, $V_{BATT} = 2.0$ to 3.6 V
Operating temperature		D-version: -40 to +85°C G-version: -40 to +105°C (in planning)
Package		145-pin TFLGA (PTLG0145KA-A) 144-pin LFQFP (PLQP0144KA-B) 100-pin TFLGA (PTLG0100JA-A) 100-pin LFQFP (PLQP0100KB-B)
On-chip debugging system		- E1 emulator (JTAG and FINE interfaces) - E20 emulator (JTAG interface)

Note 1. Magic Packet™ is a registered trademark of Advanced Micro Devices, Inc.

Note 2. The product part number differs according to whether or not it supports encryption.

Note 3. The product part number differs according to whether or not it includes an SDHI (SD host interface) / SD SI (SD slave interface).

Table 1.2 Comparison of Functions for Different Packages

Functions		RX65N Group		RX651 Group	
Package		145 Pins, 144 Pins	100 Pins	145 Pins, 144 Pins	100 Pins
External bus	External bus width	16 bits		16 bits	
	SDRAM area controller	Available	Not Available	Available	Not Available
DMA	DMA controller	Ch. 0 to 7		Ch. 0 to 7	
	Data transfer controller	Available		Available	
	EXDMA controller	Ch. 0 and 1		Ch. 0 and 1	
Timers	16-bit timer pulse unit	Ch. 0 to 5		Ch. 0 to 5	
	Multi-function timer pulse unit 3	Ch. 0 to 8		Ch. 0 to 8	
	General-purpose PWM timer	Ch. 0 to 3		Ch. 0 to 3	
	Port output enable 3	Available		Available	
	Programmable pulse generator	Ch. 0 and 1		Ch. 0 and 1	
	8-bit timers	Ch. 0 to 3		Ch. 0 to 3	
	Compare match timer	Ch. 0 to 3		Ch. 0 to 3	
	Compare match timer W	Ch. 0 and 1		Ch. 0 and 1	
	Realtime clock	Available		Available	
	Watchdog timer	Available		Available	
	Independent watchdog timer	Available		Available	
Communi- cation function	Ethernet controller	Ch. 0		Not available	
	DMAC controller for ethernet	Ch. 0		Not available	
	USB 2.0 FS host/function module	Ch. 0		Ch. 0	
	Serial communications interfaces (SCIg)	Ch. 0 to 9	Ch. 0 to 3, 5, 6, 8, 9	Ch. 0 to 9	Ch. 0 to 3, 5, 6, 8, 9
	Serial communications interfaces (SCIh)	Ch. 12			
	Serial communications interfaces (SCIi)	Ch. 10 to 11			
	I ² C bus interfaces	Ch. 0 and 2			
	Serial peripheral interface	Ch. 0 to 2			
	CAN module	Ch. 0 and 1			
	Quad serial peripheral interface	Ch. 0			
	SD host interface	Ch. 0			
	SD slave interface	Ch. 0			
	MMC host interface	Ch. 0			
	Parallel data capture unit	Available	Not available	Available	Not available
12-bit A/D converter		AN000 to 007 (unit 0: 8 channels) AN100 to 120 (unit 1: 21 channels)	AN000 to 007 (unit 0: 8 channels) AN100 to 113 (unit 1: 14 channels)	AN000 to 007 (unit 0: 8 channels) AN100 to 120 (unit 1: 21 channels)	AN000 to 007 (unit 0: 8 channels) AN100 to 113 (unit 1: 14 channels)
12-bit D/A converter		Ch. 0 and 1	Ch. 1	Ch. 0 and 1	Ch. 1
Temperature sensor		Available			
CRC calculator		Available			
Data operation circuit		Available			
Clock frequency accuracy measurement circuit		Available			
AES		Available			
RNG		Available			
Event link controller		Available			

1.2 List of Products

Table 1.3 is a list of products, and Figure 1.1 shows how to read the product part no.

Table 1.3 List of Products (1/3)

Group	Part No.	Package	Code Flash Memory Capacity	RAM Capacity	Operating Frequency (Max.)	Encryption Module	SDHI/SDSI
RX65N	R5F565N4ADFB	PLQP0144KA-B	512 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N4BDFB	PLQP0144KA-B	512 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F565N4EDFB	PLQP0144KA-B	512 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F565N4FDFB	PLQP0144KA-B	512 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F565N4ADFP	PLQP0100KB-B	512 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N4BDFP	PLQP0100KB-B	512 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F565N4EDFP	PLQP0100KB-B	512 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F565N4FDFP	PLQP0100KB-B	512 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F565N4ADLK	PTLG0145KA-A	512 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N4BDLK	PTLG0145KA-A	512 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F565N4EDLK	PTLG0145KA-A	512 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F565N4FDLK	PTLG0145KA-A	512 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F565N4ADLJ	PTLG0100JA-A	512 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N4BDLJ	PTLG0100JA-A	512 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F565N4EDLJ	PTLG0100JA-A	512 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F565N4FDLJ	PTLG0100JA-A	512 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F565N7ADFB	PLQP0144KA-B	768 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N7BDFB	PLQP0144KA-B	768 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F565N7EDFB	PLQP0144KA-B	768 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F565N7FDFB	PLQP0144KA-B	768 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F565N7ADFP	PLQP0100KB-B	768 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N7BDFP	PLQP0100KB-B	768 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F565N7EDFP	PLQP0100KB-B	768 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F565N7FDFP	PLQP0100KB-B	768 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F565N7ADLK	PTLG0145KA-A	768 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N7BDLK	PTLG0145KA-A	768 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F565N7EDLK	PTLG0145KA-A	768 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F565N7FDLK	PTLG0145KA-A	768 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F565N7ADLJ	PTLG0100JA-A	768 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N7BDLJ	PTLG0100JA-A	768 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F565N7EDLJ	PTLG0100JA-A	768 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F565N7FDLJ	PTLG0100JA-A	768 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F565N9ADFB	PLQP0144KA-B	1 Mbyte	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N9BDFB	PLQP0144KA-B	1 Mbyte	256 Kbytes	120 MHz	Not supported	Available
	R5F565N9EDFB	PLQP0144KA-B	1 Mbyte	256 Kbytes	120 MHz	Available	Not supported
	R5F565N9FDFB	PLQP0144KA-B	1 Mbyte	256 Kbytes	120 MHz	Available	Available
	R5F565N9ADFP	PLQP0100KB-B	1 Mbyte	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N9BDFP	PLQP0100KB-B	1 Mbyte	256 Kbytes	120 MHz	Not supported	Available
	R5F565N9EDFP	PLQP0100KB-B	1 Mbyte	256 Kbytes	120 MHz	Available	Not supported
	R5F565N9FDFP	PLQP0100KB-B	1 Mbyte	256 Kbytes	120 MHz	Available	Available

Table 1.3 List of Products (2/3)

Group	Part No.	Package	Code Flash Memory Capacity	RAM Capacity	Operating Frequency (Max.)	Encryption Module	SDHI/SDSI
RX65N	R5F565N9ADLK	PTLG0145KA-A	1 Mbyte	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N9BDLK	PTLG0145KA-A	1 Mbyte	256 Kbytes	120 MHz	Not supported	Available
	R5F565N9EDLK	PTLG0145KA-A	1 Mbyte	256 Kbytes	120 MHz	Available	Not supported
	R5F565N9FDLK	PTLG0145KA-A	1 Mbyte	256 Kbytes	120 MHz	Available	Available
	R5F565N9ADLJ	PTLG0100JA-A	1 Mbyte	256 Kbytes	120 MHz	Not supported	Not supported
	R5F565N9BDLJ	PTLG0100JA-A	1 Mbyte	256 Kbytes	120 MHz	Not supported	Available
	R5F565N9EDLJ	PTLG0100JA-A	1 Mbyte	256 Kbytes	120 MHz	Available	Not supported
	R5F565N9FDLJ	PTLG0100JA-A	1 Mbyte	256 Kbytes	120 MHz	Available	Available
RX651	R5F56514ADFB	PLQP0144KA-B	512 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56514BDFB	PLQP0144KA-B	512 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F56514EDFB	PLQP0144KA-B	512 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F56514FDFB	PLQP0144KA-B	512 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F56514ADFP	PLQP0100KB-B	512 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56514BDFP	PLQP0100KB-B	512 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F56514EDFP	PLQP0100KB-B	512 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F56514FDFP	PLQP0100KB-B	512 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F56514ADLK	PTLG0145KA-A	512 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56514BDLK	PTLG0145KA-A	512 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F56514EDLK	PTLG0145KA-A	512 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F56514FDLK	PTLG0145KA-A	512 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F56514ADLJ	PTLG0100JA-A	512 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56514BDLJ	PTLG0100JA-A	512 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F56514EDLJ	PTLG0100JA-A	512 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F56514FDLJ	PTLG0100JA-A	512 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F56517ADFB	PLQP0144KA-B	768 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56517BDFB	PLQP0144KA-B	768 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F56517EDFB	PLQP0144KA-B	768 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F56517FDFB	PLQP0144KA-B	768 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F56517ADFP	PLQP0100KB-B	768 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56517BDFP	PLQP0100KB-B	768 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F56517EDFP	PLQP0100KB-B	768 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F56517FDFP	PLQP0100KB-B	768 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F56517ADLK	PTLG0145KA-A	768 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56517BDLK	PTLG0145KA-A	768 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F56517EDLK	PTLG0145KA-A	768 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F56517FDLK	PTLG0145KA-A	768 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F56517ADLJ	PTLG0100JA-A	768 Kbytes	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56517BDLJ	PTLG0100JA-A	768 Kbytes	256 Kbytes	120 MHz	Not supported	Available
	R5F56517EDLJ	PTLG0100JA-A	768 Kbytes	256 Kbytes	120 MHz	Available	Not supported
	R5F56517FDLJ	PTLG0100JA-A	768 Kbytes	256 Kbytes	120 MHz	Available	Available
	R5F56519ADFB	PLQP0144KA-B	1 Mbyte	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56519BDFB	PLQP0144KA-B	1 Mbyte	256 Kbytes	120 MHz	Not supported	Available
	R5F56519EDFB	PLQP0144KA-B	1 Mbyte	256 Kbytes	120 MHz	Available	Not supported
	R5F56519FDFB	PLQP0144KA-B	1 Mbyte	256 Kbytes	120 MHz	Available	Available
	R5F56519ADFP	PLQP0100KB-B	1 Mbyte	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56519BDFP	PLQP0100KB-B	1 Mbyte	256 Kbytes	120 MHz	Not supported	Available
	R5F56519EDFP	PLQP0100KB-B	1 Mbyte	256 Kbytes	120 MHz	Available	Not supported
	R5F56519FDFP	PLQP0100KB-B	1 Mbyte	256 Kbytes	120 MHz	Available	Available

Table 1.3 List of Products (3/3)

Group	Part No.	Package	Code Flash Memory Capacity	RAM Capacity	Operating Frequency (Max.)	Encryption Module	SDHI/SDSI
RX651	R5F56519ADLK	PTLG0145KA-A	1 Mbyte	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56519BDLK	PTLG0145KA-A	1 Mbyte	256 Kbytes	120 MHz	Not supported	Available
	R5F56519EDLK	PTLG0145KA-A	1 Mbyte	256 Kbytes	120 MHz	Available	Not supported
	R5F56519FDLK	PTLG0145KA-A	1 Mbyte	256 Kbytes	120 MHz	Available	Available
	R5F56519ADLJ	PTLG0100JA-A	1 Mbyte	256 Kbytes	120 MHz	Not supported	Not supported
	R5F56519BDLJ	PTLG0100JA-A	1 Mbyte	256 Kbytes	120 MHz	Not supported	Available
	R5F56519EDLJ	PTLG0100JA-A	1 Mbyte	256 Kbytes	120 MHz	Available	Not supported
	R5F56519FDLJ	PTLG0100JA-A	1 Mbyte	256 Kbytes	120 MHz	Available	Available

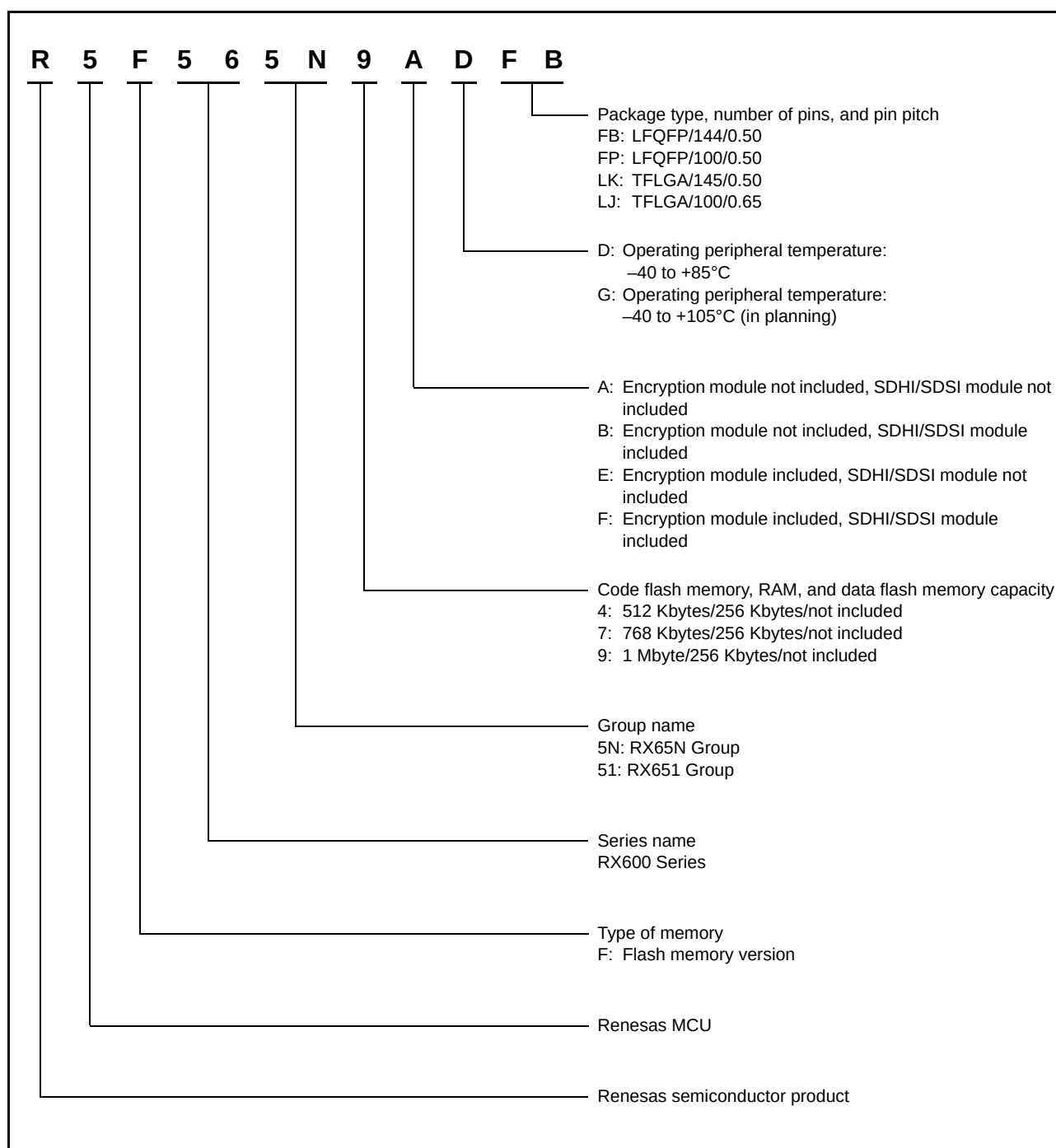


Figure 1.1 How to Read the Product Part Number

1.3 Block Diagram

Figure 1.2 shows a block diagram.

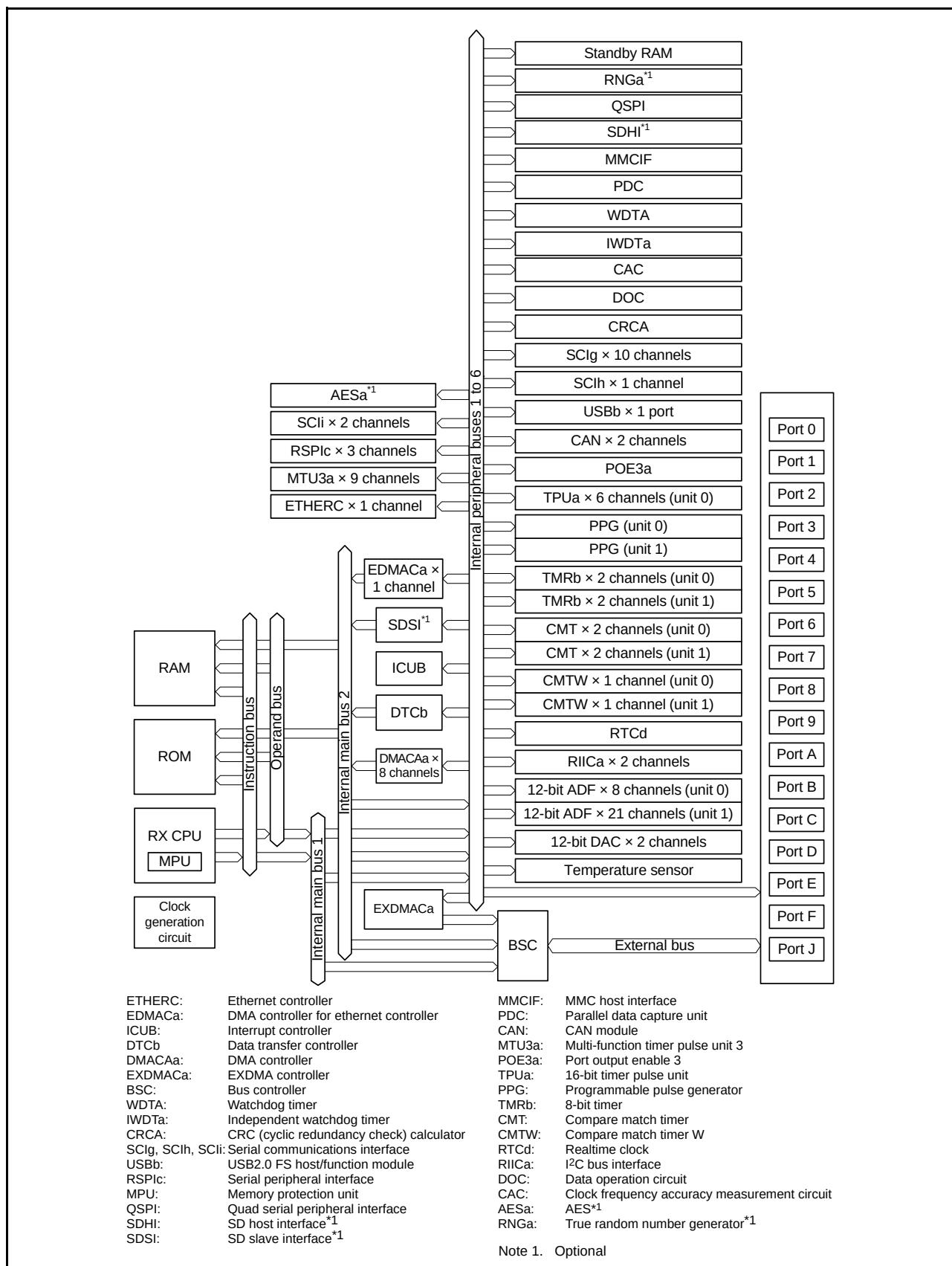


Figure 1.2 Block Diagram

1.4 Pin Functions

Table 1.4 lists the pin functions.

Table 1.4 Pin Functions (1/7)

Classifications	Pin Name	I/O	Description
Digital power supply	VCC	Input	Power supply pin. Connect this pin to the system power supply. Connect the pin to VSS via a 0.1-μF multilayer ceramic capacitor. The capacitor should be placed close to the pin.
	VCL	Input	Connect this pin to VSS via a 0.22-μF multilayer ceramic capacitor. The capacitor should be placed close to the pin.
	VSS	Input	Ground pin. Connect it to the system power supply (0 V).
	VBATT	Input	Backup power pin
Clock	XTAL	Output	Pins for a crystal resonator. An external clock signal can be input through the EXTAL pin.
	EXTAL	Input	
	BCLK	Output	Outputs the external bus clock for external devices.
	SDCLK	Output	Outputs the SDRAM-dedicated clock.
	XCOUT	Output	Input/output pins for the sub clock oscillator. Connect a crystal resonator between XCOUT and XCIN.
	XCIN	Input	
Clock frequency accuracy measurement	CACREF	Input	Reference clock input pin for the clock frequency accuracy measurement circuit
Operating mode control	MD	Input	Pins for setting the operating mode. The signal levels on these pins must not be changed during operation.
	UB	Input	USB boot mode enable pin
	UPSEL	Input	Selects the power supply method in USB boot mode. The low level selects self-power mode and the high level selects bus power mode.
System control	RES#	Input	Reset signal input pin. This LSI enters the reset state when this signal goes low.
	EMLE	Input	Input pin for the on-chip emulator enable signal. When the on-chip emulator is used, this pin should be driven high. When not used, it should be driven low.
	BSCANP	Input	Boundary scan enable pin. Boundary scan is enabled when this pin goes high. When not used, it should be driven low.
On-chip emulator	FINED	I/O	Fine interface pin
	TRST#	Input	On-chip emulator or boundary scan pins. When the EMLE pin is driven high, these pins are dedicated for the on-chip emulator.
	TMS	Input	
	TDI	Input	
	TCK	Input	
	TDO	Output	
	TRCLK	Output	This pin outputs the clock for synchronization with the trace data.
	TRSYNC TRSYNC1	Output	This pin indicates that output from the TRDATA0 to TRDATA7 pins is valid.
	TRDATA0 TRDATA1 TRDATA2 TRDATA3 TRDATA4 TRDATA5 TRDATA6 TRDATA7	Output	These pins output the trace information.
Address bus	A0 to A23	Output	Output pins for the address
Data bus	D0 to D15	I/O	Input and output pins for the bidirectional data bus

Table 1.4 Pin Functions (2/7)

Classifications	Pin Name	I/O	Description
Multiplexed bus	A0/D0 to A15/D15	I/O	Address/data multiplexed bus
Bus control	RD#	Output	Strobe signal which indicates that reading from the external bus interface space is in progress
	WR#	Output	Strobe signal which indicates that writing to the external bus interface space is in progress, in 1-write strobe mode
	WR0# and WR1#	Output	Strobe signals which indicate that either group of data bus pins (D7 to D0, D15 to D8) is valid in writing to the external bus interface space, in byte strobe mode
	BC0# and BC1#	Output	Strobe signals which indicate that either group of data bus pins (D7 to D0, D15 to D8) is valid in access to the external bus interface space, in 1-write strobe mode
	ALE	Output	Address latch signal when address/data multiplexed bus is selected
	WAIT#	Input	Input pin for wait request signals in access to the external space
	CS0# to CS7#	Output	Select signals for CS areas
	CKE	Output	SDRAM clock enable signal
	SDCS#	Output	SDRAM chip select signal
	RAS#	Output	SDRAM row address strobe signal
	CAS#	Output	SDRAM column address strobe signal
	WE#	Output	SDRAM write enable pin
	DQM0 and DQM1	Output	SDRAM I/O data mask enable signals
EXDMA controller	EDREQ0, EDREQ1	Input	External DMA transfer request pins
	EDACK0, EDACK1	Output	Single address transfer acknowledge signals
Interrupt	NMI	Input	Non-maskable interrupt request pin
	IRQ0 to IRQ15, IRQ0-DS to IRQ15-DS	Input	Maskable interrupt request pins
Multi-function timer pulse unit 3	MTIOC0A, MTIOC0B, MTIOC0C, MTIOC0D	I/O	The TGRA0 to TGRD0 input capture input/output compare output/PWM output pins
	MTIOC1A, MTIOC1B	I/O	The TGRA1 and TGRB1 input capture input/output compare output/PWM output pins
	MTIOC2A, MTIOC2B	I/O	The TGRA2 and TGRB2 input capture input/output compare output/PWM output pins
	MTIOC3A, MTIOC3B, MTIOC3C, MTIOC3D	I/O	The TGRA3 to TGRD3 input capture input/output compare output/PWM output pins
	MTIOC4A, MTIOC4B, MTIOC4C, MTIOC4D	I/O	The TGRA4 to TGRD4 input capture input/output compare output/PWM output pins
	MTIC5U, MTIC5V, MTIC5W	Input	The TGRU5, TGRV5, and TGRW5 input capture input/dead time compensation input pins
	MTIOC6A, MTIOC6B, MTIOC6C, MTIOC6D	I/O	The TGRA6 to TGRD6 input capture input/output compare output/PWM output pins
	MTIOC7A, MTIOC7B, MTIOC7C, MTIOC7D	I/O	The TGRA7 to TGRD7 input capture input/output compare output/PWM output pins
	MTIOC8A, MTIOC8B, MTIOC8C, MTIOC8D	I/O	The TGRA8 to TGRD8 input capture input/output compare output/PWM output pins
	MTCLKA, MTCLKB, MTCLKC, MTCLKD	Input	Input pins for external clock signals or for phase counting mode clock signals
Port output enable 3	POE0#, POE4#, POE8#, POE10#, POE11#	Input	Input pins for request signals to place the MTU in the high impedance state

Table 1.4 Pin Functions (3/7)

Classifications	Pin Name	I/O	Description
16-bit timer pulse unit	TIOCA0, TIOCB0, TIOCC0, TIOCD0	I/O	The TGRA0 to TGRD0 input capture input/output compare output/PWM output pins
	TIOCA1, TIOCB1	I/O	The TGRA1 and TGRB1 input capture input/output compare output/PWM output pins
	TIOCA2, TIOCB2	I/O	The TGRA2 and TGRB2 input capture input/output compare output/PWM output pins
	TIOCA3, TIOCB3, TIOCC3, TIOCD3	I/O	The TGRA3 to TGRD3 input capture input/output compare output/PWM output pins
	TIOCA4, TIOCB4	I/O	The TGRA4 and TGRB4 input capture input/output compare output/PWM output pins
	TIOCA5, TIOCB5	I/O	The TGRA5 and TGRB5 input capture input/output compare output/PWM output pins
	TCLKA, TCLKB, TCLKC, TCLKD	Input	Input pins for external clock signals or for phase counting mode clock signals
Programmable pulse generator	PO0 to PO31	Output	Output pins for the pulse signals
8-bit timer	TMO0 to TMO3	Output	Compare match output pins
	TMCI0 to TMCI3	Input	Input pins for external clocks to be input to the counter
	TMRI0 to TMRI3	Input	Input pins for the counter reset
Compare match timer W	TIC0 to TIC3	Input	Input pins for CMTW
	TOC0 to TOC3	Output	Output pins for CMTW
Serial communications interface (SCIg)	• Asynchronous mode/clock synchronous mode		
	SCK0 to SCK9	I/O	Input/output pins for the clock
	RXD0 to RXD9	Input	Input pins for received data
	TXD0 to TXD9	Output	Output pins for transmitted data
	CTS0# to CTS9#	Input	Input pins for controlling the start of transmission and reception
	RTS0# to RTS9#	Output	Output pins for controlling the start of transmission and reception
	• Simple I ² C mode		
	SSCL0 to SSCL9	I/O	Input/output pins for the I ² C clock
	SSDA0 to SSDA9	I/O	Input/output pins for the I ² C data
	• Simple SPI mode		
	SCK0 to SCK9	I/O	Input/output pins for the clock
	SMISO0 to SMISO9	I/O	Input/output pins for slave transmission of data
	SMOSI0 to SMOSI9	I/O	Input/output pins for master transmission of data
	SS0# to SS9#	Input	Chip-select input pins

Table 1.4 Pin Functions (4/7)

Classifications	Pin Name	I/O	Description
Serial communications interface (SCIh)	• Asynchronous mode/clock synchronous mode		
	SCK12	I/O	Input/output pin for the clock
	RXD12	Input	Input pin for received data
	TXD12	Output	Output pin for transmitted data
	CTS12#	Input	Input pin for controlling the start of transmission and reception
	RTS12#	Output	Output pin for controlling the start of transmission and reception
	• Simple I ² C mode		
	SSCL12	I/O	Input/output pin for the I ² C clock
	SSDA12	I/O	Input/output pin for the I ² C data
	• Simple SPI mode		
	SCK12	I/O	Input/output pin for the clock
	SMISO12	I/O	Input/output pin for slave transmission of data
	SMOSI12	I/O	Input/output pin for master transmission of data
	SS12#	Input	Chip-select input pin
	• Extended serial mode		
	RXDX12	Input	Input pin for received data
	TXDX12	Output	Output pin for transmitted data
	SIOX12	I/O	Input/output pin for received or transmitted data
Serial communications interface (SCIi)	• Asynchronous mode/clock synchronous mode		
	SCK10 and SCK11	I/O	Input/output pin for the clock
	RXD10 and RXD11	Input	Input pin for received data
	TXD10 and TXD11	Output	Output pin for transmitted data
	CTS10# and CTS11#	Input	Input pin for controlling the start of transmission and reception
	RTS10# and RTS11#	Output	Output pin for controlling the start of transmission and reception
	• Simple I ² C mode		
	SSCL10 and SSCL11	I/O	Input/output pin for the I ² C clock
	SSDA10 and SSDA11	I/O	Input/output pin for the I ² C data
	• Simple SPI mode		
	SCK10 and SCK11	I/O	Input/output pin for the clock
	SMISO10 and SMISO11	I/O	Input/output pin for slave transmission of data
	SMOSI10 and SMOSI11	I/O	Input/output pin for master transmission of data
	SS10# and SS11#	Input	Chip-select input pin
I ² C bus interface	SCL0[FM+], SCL2, SCL2-DS	I/O	Input/output pins for clocks. Bus can be directly driven by the N-channel open drain
	SDA0[FM+], SDA2, SDA2-DS	I/O	Input/output pins for data. Bus can be directly driven by the N-channel open drain

Table 1.4 Pin Functions (5/7)

Classifications	Pin Name	I/O	Description
Ethernet controller	REF50CK0	Input	50-MHz reference clocks. These pins input reference signals for transmission/reception timings in RMII mode.
	RMII0_CRS_DV	Input	Indicate that there are carrier detection signals and valid receive data on RMII0_RXD1 and RMII0_RXD0 in RMII mode.
	RMII0_TXD0, RMII0_TXD1	Output	2-bit transmit data in RMII mode
	RMII0_RXD0, RMII0_RXD1	Input	2-bit receive data in RMII mode
	RMII0_TXD_EN	Output	Output pins for data transmit enable signals in RMII mode
	RMII0_RX_ER	Input	Indicate an error has occurred during reception of data in RMII mode.
	ET0_CRS	Input	Carrier detection/data reception enable pins
	ET0_RX_DV	Input	Indicate that there are valid receive data on ET0_ERXD3 to ET0_ERXD0.
	ET0_EXOUT	Output	General-purpose external output pins
	ET0_LINKSTA	Input	Input link status from the PHY-LSI.
	ET0_ETXD0 to ET0_ETXD3	Output	4 bits of MII transmit data
	ET0_ERXD0 to ET0_ERXD3	Input	4 bits of MII receive data
	ET0_TX_EN	Output	Transmit enable pins. Function as signals indicating that transmit data is ready on ET0_ETXD3 to ET0_ETXD0.
	ET0_TX_ER	Output	Transmit error pins. Function as signals notifying the PHY-LSI of an error during transmission.
	ET0_RX_ER	Input	Receive error pins. Function as signals to recognize an error during reception.
	ET0_TX_CLK	Input	Transmit clock pins. These pins input reference signals for output timings from ET0_TX_EN, ET0_ETXD3 to ET0_ETXD0, and ET0_TX_ER.
	ET0_RX_CLK	Input	Receive clock pins. These pins input reference signals for input timings to ET0_RX_DV, ET0_ERXD3 to ET0_ERXD0, and ET0_RX_ER.
	ET0_COL	Input	Input collision detection signals.
	ET0_WOL	Output	Receive Magic packets.
	ET0_MDC	Output	Output reference clock signals for information transfer via ET0_MDIO.
	ET0_MDIO	I/O	Input or output bidirectional signals for exchange of management information between this MCU and the PHY-LSI.
USB 2.0 host/function module	VCC_USB	Input	Power supply pins
	VSS_USB	Input	Ground pins
	USB0_DP	I/O	Input or output USB transceiver D+ data.
	USB0_DM	I/O	Input or output USB transceiver D- data.
	USB0_EXICEN	Output	Connect to the OTG power IC.
	USB0_ID	Input	Connect to the OTG power IC.
	USB0_VBUSEN	Output	USB VBUS power enable pins
	USB0_OVRCURA/ USB0_OVRCURB	Input	USB overcurrent pins
CAN module	USB0_VBUS	Input	USB cable connection/disconnection detection input pins
	CRX0, CRX1, CRX1-DS	Input	Input pins
	CTX0, CTX1	Output	Output pins

Table 1.4 Pin Functions (6/7)

Classifications	Pin Name	I/O	Description
Serial peripheral interface	RSPCKA-A/RSPCKA-B/ RSPCKB-A/RSPCKB-B/ RSPCKC	I/O	Clock input/output pin
	MOSIA-A/MOSIA-B/ MOSIB-A/MOSIB-B/ MOSIC	I/O	Inputs or outputs data output from the master
	MISOA-A/MISOA-B/ MISOB-A/MISOB-B/ MISOC	I/O	Inputs or outputs data output from the slave
	SSLA0-A/SSLA0-B/ SSLB0-A/SSLB0-B/ SSLC0	I/O	Input or output pin for slave selection
	SSLA1-A/SSLA1-B/ SSLB1-A/SSLB1-B/ SSLC1, SSLA2-A/SSLA2-B/ SSLB2-A/SSLB2-B/ SSLC2, SSLA3-A/SSLA3-B/ SSLB3-A/SSLB3-B/ SSLC3	Output	Output pin for slave selection
Quad serial peripheral interface	QSPCLK-A/-B	Output	QSPI clock output pin
	QSSL-A/-B	Output	QSPI slave output pin
	QMO-A/-B, QIO0-A/-B	I/O	Master transmit data/data 0
	QMI-A/-B, QIO1-A/-B	I/O	Master input data/data 1
	QIO2-A/-B, QIO3-A/-B	I/O	Data 2, data 3
MMC host interface	MMC_CLK-A/ MMC_CLK-B	Output	MMC clock pin
	MMC_CMD-A/ MMC_CMD-B	I/O	Command/response pin
	MMC_D7-A/MMC_D7-B to MMC_D0-A/MMC_D0-B	I/O	Transmit data/receive data
	MMC_CD-A/MMC_CD-B	Input	Card detection pin
	MMC_RES#-A/MMC_RES#-B	Output	MMC reset output pin
SD host interface	SDHI_CLK-A/SDHI_CLK-B	Output	SD clock output pin
	SDHI_CMD-A/SDHI_CMD-B	I/O	SD command output, response input signal pin
	SDHI_D3-A/SDHI_D3-B to SDHI_D0-A/SDHI_D0-B	I/O	SD data bus pins
	SDHI_CD-A/SDHI_CD-B	Input	SD card detection pin
	SDHI_WP-A/SDHI_WP-B	Input	SD write-protect signal
SD slave interface	SDSI_CLK-A/SDSI_CLK-B	Input	SD clock input pin
	SDSI_CMD-A/SDSI_CMD-B	I/O	SD command input, response output signal pin
	SDSI_D3-A/SDSI_D3-B, SDSI_D2-A/SDSI_D2-B, SDSI_D1-A/SDSI_D1-B, SDSI_D0-A/SDSI_D0-B	I/O	SD data bus pins
Parallel data capture unit	PIXCLK	Input	Image transfer clock pin
	VSYNC	Input	Vertical synchronization signal pin
	HSYNC	Input	Horizontal synchronization signal pin
	PIXD0 to PIXD7	Input	8-bit image data pins
	PCKO	Output	Output pin for dot clock
Realtime clock	RTCOUT	Output	Output pin for 1-Hz/64-Hz clock
	RTCIC0 to RTCIC2	Input	Time capture event input pins

Table 1.4 Pin Functions (7/7)

Classifications	Pin Name	I/O	Description
12-bit A/D converter	AN000 to AN007, AN100 to AN120	Input	Input pins for the analog signals to be processed by the A/D converter
	ADTRG0#, ADTRG1#	Input	Input pins for the external trigger signals that start the A/D conversion
	ANEX0	Output	Extended analog output pin
	ANEX1	Input	Extended analog input pin
12-bit D/A converter	DA0, DA1	Output	Output pins for the analog signals to be processed by the D/A converter
Analog power supply	AVCC0	Input	Analog voltage supply pin for the 12-bit A/D converter (unit 0). Connect this pin to a branch from the VCC power supply. Connect the pin to AVSS0 via a 0.1-μF multilayer ceramic capacitor. The capacitor should be placed close to the pin.
	AVSS0	Input	Analog ground pin for the 12-bit A/D converter (unit 0). Connect this pin to a branch from the VSS ground power supply. Connect the pin to AVCC0 via a 0.1-μF multilayer ceramic capacitor. The capacitor should be placed close to the pin.
	VREFH0	Input	Analog reference voltage supply pin for the 12-bit A/D converter (unit 0). Connect this pin to VCC if the 12-bit A/D converter is not to be used.
	VREFL0	Input	Analog reference ground pin for the 12-bit A/D converter (unit 0). Connect this pin to VSS if the 12-bit A/D converter is not to be used.
	AVCC1	Input	Analog voltage supply and reference voltage supply pin for the 12-bit A/D converter (unit 1) and D/A converter. This pin also supplies the analog voltage to the temperature sensor. Connect this pin to a branch from the VCC power supply. Connect the pin to AVSS1 via a 0.1-μF multilayer ceramic capacitor. The capacitor should be placed close to the pin.
	AVSS1	Input	Analog voltage supply and reference voltage supply pin for the 12-bit A/D converter (unit 1) and D/A converter. This pin also supplies the analog ground voltage to the temperature sensor. Connect this pin to a branch from the VSS ground power supply. Connect the pin to AVCC1 via a 0.1-μF multilayer ceramic capacitor. The capacitor should be placed close to the pin.
I/O ports	P00 to P03, P05, P07	I/O	6-bit input/output pins
	P12 to P17	I/O	6-bit input/output pins
	P20 to P27	I/O	8-bit input/output pins
	P30 to P37	I/O	8-bit input/output pins (P35: input pin)
	P40 to P47	I/O	8-bit input/output pins
	P50 to P56	I/O	7-bit input/output pins
	P60 to P67	I/O	8-bit input/output pins
	P70 to P77	I/O	8-bit input/output pins
	P80 to P83, P86, P87	I/O	6-bit input/output pins
	P90 to P93	I/O	4-bit input/output pins
	PA0 to PA7	I/O	8-bit input/output pins
	PB0 to PB7	I/O	8-bit input/output pins
	PC0 to PC7	I/O	8-bit input/output pins
	PD0 to PD7	I/O	8-bit input/output pins
	PE0 to PE7	I/O	8-bit input/output pins
	PF5	I/O	1-bit input/output pin
	PJ3, PJ5	I/O	2-bit input/output pins

Note: Note the following regarding pin names. For details, see section 1.5, Pin Assignments.

- We recommend using pins that have a letter (“-A”, “-B”, etc.) to indicate group membership appended to their names as groups.
For the RSPI, QSPI, SDHI, SDSI, and MMC interfaces, the AC portion of the electrical characteristics is measured for each group.
- Pins that have “-DS” appended to their names can be used as triggers for release from deep software standby.
- IIC pin functions that have [FM+] appended to their names support fast-mode plus.

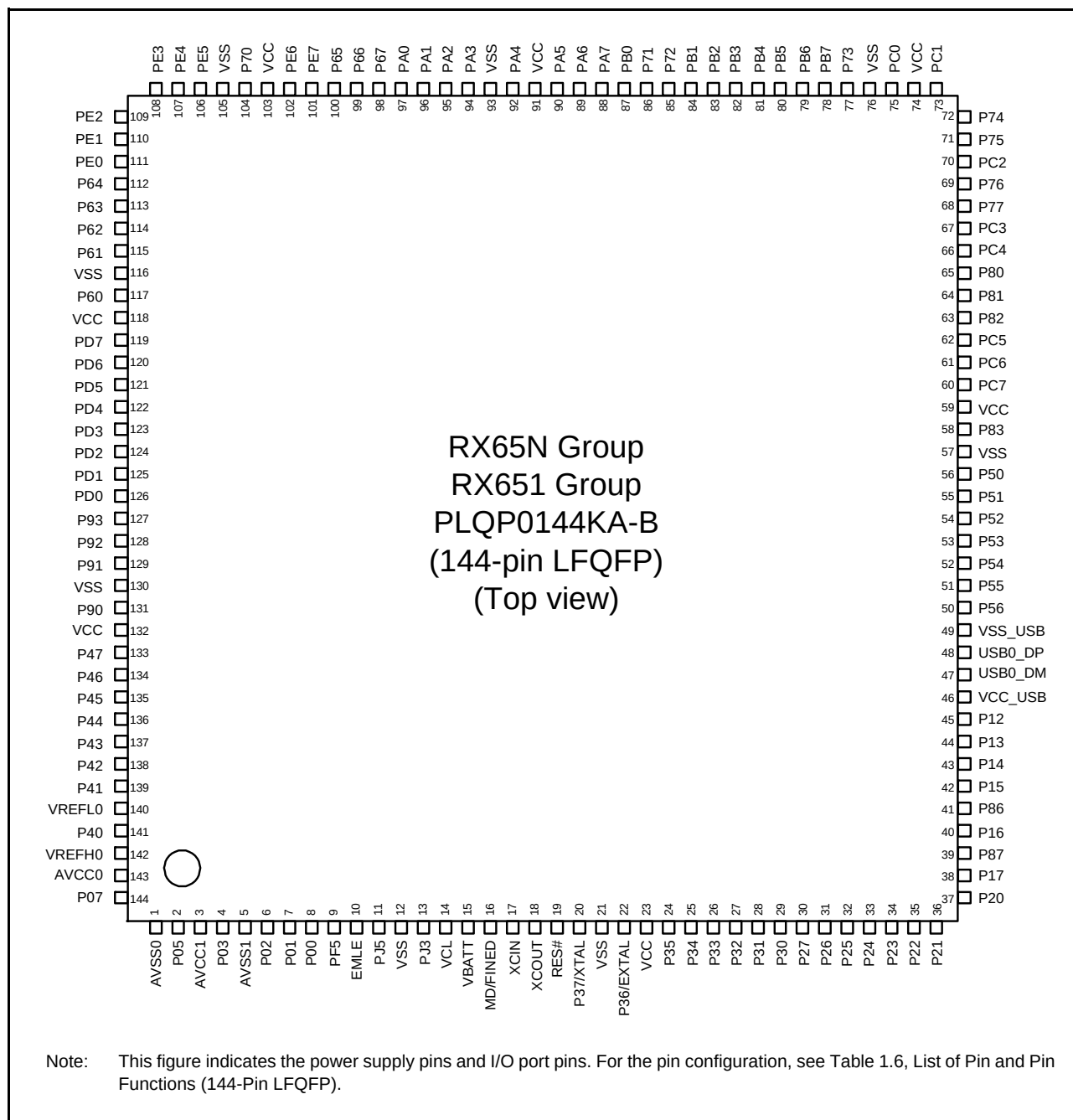
1.5 Pin Assignments

Figure 1.3 to Figure 1.6 show the pin assignments. Table 1.5 to Table 1.8 show the lists of pins and pin functions.

	A	B	C	D	E	F	G	H	J	K	L	M	N	
13	PE3	PE4	VSS	PE6	P67	PA2	PA4	PA7	PB1	PB5	VSS	VCC	P74	13
12	PE1	PE2	P70	PE5	P65	PA1	VCC	PB0	PB2	PB6	P73	PC1	P75	12
11	P62	P61	PE0	VCC	P66	VSS	PA6	P71	PB4	PB7	PC2	PC0	PC3	11
10	VSS	VCC	P63	PE7	PA0	PA3	PA5	P72	PB3	P76	PC4	P77	P82	10
9	PD6	PD4	PD7	P64	RX65N Group, RX651 Group PTLG0145KA-A (145-Pin TFLGA) (Upper Perspective View)					P80	PC5	P81	PC7	9
8	PD2	PD0	PD3	P60						VCC	P83	PC6	VSS	8
7	P92	P91	PD1	PD5						P51	P52	P50	P55	7
6	P90	P47	VSS	P93						P53	P56	VSS_ USB	USB0_ DP	6
5	P45	P43	P46	VCC	P44					P54	P13	VCC_ USB	USB0_ DM	5
4	P42	VREFL0	P41	P01	EMLE	VBATT	BSCANP	P35	P30	P15	P24	P12	P14	4
3	P40	P05	VREFH0	P03	PJ5	PJ3	MD/ FINED	VSS	P32	P31	P16	P86	P87	3
2	P07	AVCC0	P02	PF5	VCL	XCOUT	RES#	VCC	P33	P26	P23	P17	P20	2
1	AVSS0	AVCC1	AVSS1	P00	VSS	XCIN	XTAL	EXTAL	P34	P27	P25	P22	P21	1
	A	B	C	D	E	F	G	H	J	K	L	M	N	

Note: This figure indicates the power supply pins and I/O port pins. For the pin configuration, see Table 1.5, List of Pin and Pin Functions (145-Pin TFLGA).

Figure 1.3 Pin Assignment (145-Pin TFLGA)

**Figure 1.4 Pin Assignment (144-Pin LFQFP)**

RX65N Group, RX651 Group
PTLG0100JA-A (100-Pin TFLGA)
(Upper Perspective View)

	A	B	C	D	E	F	G	H	J	K	
10	PE2	PE3	PE4	PA0	PA3	VSS	VCC	PB7	PC1	PC2	10
9	PE1	PD7	PE5	PA1	PA5	PA7	PB1	PB6	PC0	PC3	9
8	PE0	PD6	PD5	PE7	PA4	PB0	PB4	PC6	PC4	PC5	8
7	PD4	PD3	PD2	PE6	PA6	PB2	PB5	PC7	P50	P51	7
6	PD0	PD1	P47	P46	PA2	PB3	P52	P54	VCC_ USB	USB0_ DP	6
5	P43	P44	P42	P45	P41	P12	P53	P55	VSS_ USB	USB0_ DM	5
4	VREFL0	P40	VREFH0	VBATT	P34	P32	P27	P15	P13	P14	4
3	P07	AVCC0	PJ3	MD/ FINED	RES#	P35	P30	P16	P17	P20	3
2	AVCC1	AVSS0	AVSS1	XCOUT	VSS	VCC	P31	P25	P21	P22	2
1	P05	EMLE	VCL	XCIN	XTAL	EXTAL	P33	P26	P24	P23	1
	A	B	C	D	E	F	G	H	J	K	

Note: This figure indicates the power supply pins and I/O port pins. For the pin configuration, see Table 1.7, List of Pin and Pin Functions (100-Pin TFLGA).

Figure 1.5 Pin Assignment (100-Pin TFLGA)

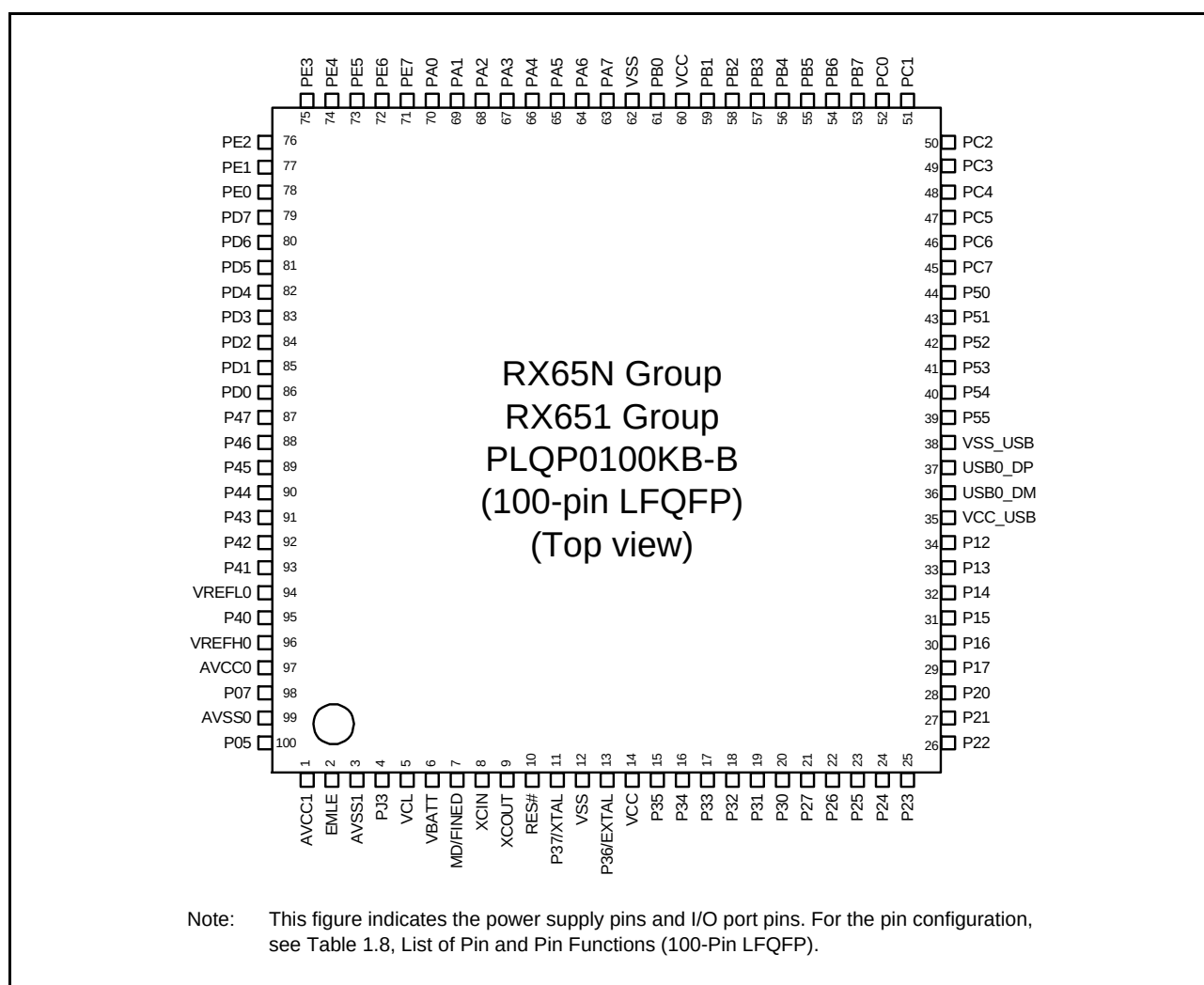


Figure 1.6 Pin Assignment (100-Pin LFQFP)

Table 1.5 List of Pin and Pin Functions (145-Pin TFLGA) (1/6)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCIh, SCii, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
A1	AVSS0							
A2		P07					IRQ15	ADTRG0#
A3		P40					IRQ8-DS	AN000
A4		P42					IRQ10-DS	AN002
A5		P45					IRQ13-DS	AN005
A6		P90	A16		TXD7/SMOSI7/ SSDA7			AN114
A7		P92	A18	POE4#	RXD7/SMISO7/ SSCL7			AN116
A8		PD2	D2[A2/D2]	MTIOC4D/TIC2	CRX0/MISOC	MMC_D2-B/ SDHI_D2-B/ QIO2-B	IRQ2	AN110
A9		PD6	D6[A6/D6]	MTIC5V/MTIOC8A/ POE4#	SSLC2	MMC_D0-B/ SDHI_D0-B/ QIO0-B/ QMO-B	IRQ6	AN106
A10	VSS							
A11		P62	CS2#/RAS#					
A12		PE1	D9[A9/D9]	MTIOC4C/ MTIOC3B/PO18	TXD12/SMOSI12/ SSDA12/TXDX12/ SIOX12/SSLB2-B	MMC_D5-B		ANEX1
A13		PE3	D11[A11/D11]	MTIOC4B/PO26/ POE8#/TOC3	CTS12#/RTS12#/ SS12#/ET0_ERXD3/	MMC_D7-B		AN101
B1	AVCC1							
B2	AVCC0							
B3		P05					IRQ13	DA1
B4	VREFL0							
B5		P43					IRQ11-DS	AN003
B6		P47					IRQ15-DS	AN007
B7		P91	A17		SCK7			AN115
B8		PD0	D0[A0/D0]	POE4#			IRQ0	AN108
B9		PD4	D4[A4/D4]	MTIOC8B/POE11#	SSLC0	MMC_CMD-B/ SDHI_CMD-B/ QSSL-B	IRQ4	AN112
B10	VCC							
B11		P61	CS1#/SDCS#					
B12		PE2	D10[A10/D10]	MTIOC4A/PO23/ TIC3	RXD12/SMISO12/ SSCL12/RXDX12/ SSLB3-B	MMC_D6-B	IRQ7-DS	AN100
B13		PE4	D12[A12/D12]	MTIOC4D/ MTIOC1A/PO28	ET0_ERXD2/ SSLB0-B			AN102
C1	AVSS1							
C2		P02		TMC11	SCK6		IRQ10	AN120
C3	VREFH0							
C4		P41					IRQ9-DS	AN001
C5		P46					IRQ14-DS	AN006
C6	VSS							
C7		PD1	D1[A1/D1]	MTIOC4B/POE0#	CTX0/MOSIC		IRQ1	AN109
C8		PD3	D3[A3/D3]	MTIOC8D/POE8#/ TOC2	RSPCKC	MMC_D3-B/ SDHI_D3-B/ QIO3-B	IRQ3	AN111
C9		PD7	D7[A7/D7]	MTIC5U/POE0#	SSLC3	MMC_D1-B/ SDHI_D1-B/ QIO1-B/QMI-B	IRQ7	AN107

Table 1.5 List of Pin and Pin Functions (145-Pin TFLGA) (2/6)

Pin Number				Timer	Communication	Memory Interface Camera Interface		
145-Pin TFLGA	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	(MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	(ETHERC, SCiG, SCiH, SCiI, RSPI, RIIC, CAN, USB)	(QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
C10		P63	CS3#/CAS#					
C11		PE0	D8[A8/D8]	MTIOC3D	SCK12/SSLB1-B	MMC_D4-B		ANEX0
C12		P70	SDCLK					
C13	VSS							
D1		P00		TMRI0	TXD6/SMOSI6/SSDA6		IRQ8	AN118
D2		PF5					IRQ4	
D3		P03					IRQ11	DA0
D4		P01		TMCi0	RXD6/SMISO6/SSCL6		IRQ9	AN119
D5	VCC							
D6		P93	A19	POE0#	CTS7#/RTS7#/SS7#			AN117
D7		PD5	D5[A5/D5]	MTIC5W/MTIOC8C/ POE10#	SSLC1	MMC_CLK-B/ SDHI_CLK-B/ QSPCLK-B	IRQ5	AN113
D8		P60	CS0#					
D9		P64	CS4#/WE#					
D10		PE7	D15[A15/D15]	MTIOC6A/TOC1	MISOB-B	MMC_RES#-B/ SDHI_WP-B	IRQ7	AN105
D11	VCC							
D12		PE5	D13[A13/D13]	MTIOC4C/MTIOC2B	ET0_RX_CLK/ REF50CK0/ RSPCKB-B		IRQ5	AN103
D13		PE6	D14[A14/D14]	MTIOC6C/ TIC1	MOSIB-B	MMC_CD-B/ SDHI_CD-B	IRQ6	AN104
E1	VSS							
E2	VCL							
E3		PJ5		POE8#	CTS2#/RTS2#/SS2#			
E4	EMLE							
E5		P44					IRQ12-DS	AN004
E10		PA0	A0/BC0#	MTIOC4A/ MTIOC6D/TIOCA0/ CACREF/PO16	SSLA1-B/ ET0_TX_EN/ RMII0_TXD_EN			
E11		P66	CS6#/DQM0	MTIOC7D				
E12		P65	CS5#/CKE					
E13		P67	CS7#/DQM1	MTIOC7C			IRQ15	
F1	XCIN							
F2	XCOUT							
F3		PJ3	EDACK1	MTIOC3C	ET0_EXOUT/ CTS6#/RTS6#/ CTS0#/RTS0#/ SS6#/SS0#			
F4	VBATT							
F10		PA3	A3	MTIOC0D/MTCLKD/ TIOC0D/TCLKB/ PO19	RXD5/SMISO5/ SSCL5/ET0_MDIO		IRQ6-DS	
F11	VSS							
F12		PA1	A1	MTIOC0B/MTCLKC/ MTIOC7B/TIOCB0/ PO17	SCK5/SSLA2-B/ ET0_WOL		IRQ11	
F13		PA2	A2	MTIOC7A/PO18	RXD5/SMISO5/ SSCL5/SSLA3-B			
G1	XTAL	P37						
G2	RES#							
G3	MD/FINED							

Table 1.5 List of Pin and Pin Functions (145-Pin TFLGA) (3/6)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCiH, SCiI, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
G4	BSCANP							
G10		PA5	A5	MTIOC6B/TIOCB1/ PO21	RSPCKA-B/ ET0_LINKSTA			
G11		PA6	A6	MTIC5V/MTCLKB/ TIOCA2/TMC13/ PO22/POE10#	CTS5#/RTS5#/ SS5#/MOSIA-B/ ET0_EXOUT			
G12	VCC							
G13		PA4	A4	MTIC5U/MTCLKA/ TIOCA1/TMRI0/ PO20	TXD5/SMOSI5/ SSDA5/SSLA0-B/ ET0_MDC		IRQ5-DS	
H1	EXTAL	P36						
H2	VCC							
H3	VSS							
H4	UPSEL	P35					NMI	
H10		P72	A19/CS2#		ET0_MDC			
H11		P71	A18/CS1#		ET0_MDIO			
H12		PB0	A8	MTIC5W/TIOCA3/ PO24	RXD4/RXD6/ SMISO4/SMISO6/ SSCL4/SSCL6/ ET0_ERXD1/ RMII0_RXD1		IRQ12	
H13		PA7	A7	TIOCB2/PO23	MISOA-B/ET0_WOL			
J1	TRST#	P34		MTIOC0A/TMC13/ PO12/POE10#	SCK6/SCK0/ ET0_LINKSTA		IRQ4	
J2		P33	EDREQ1	MTIOC0D/TIOCD0/ TMRI3/PO11/ POE4#/POE11#	RXD6/RXD0/ SMISO6/SMISO0/ SSCL6/SSCL0/ CRX0	PCKO	IRQ3-DS	
J3		P32		MTIOC0C/TIOCC0/ TMO3/PO10/ RTICOUT/RTICIC2/ POE0#/POE10#	TXD6/TXD0/ SMOSI6/SMOSI0/ SSDA6/SSDA0/ CTX0/ USB0_VBUSEN	VSYNC	IRQ2-DS	
J4	TDI	P30		MTIOC4B/TMRI3/ PO8/RTICIC0/ POE8#	RXD1/SMISO1/ SSCL1/MISOB-A		IRQ0-DS	
J10		PB3	A11	MTIOC0A/ MTIOC4A/TIOCD3/ TCLKD/TMO0/ PO27/POE11#	SCK4/SCK6/ ET0_RX_ER/ RMII0_RX_ER	SDSI_D3-B		
J11		PB4	A12	TIOCA4/PO28	CTS9#/RTS9#/ SS9#/ET0_TX_EN/ RMII0_TXD_EN/ CTS11#/RTS11#/ SS11	SDSI_CMD-B		
J12		PB2	A10	TIOCC3/TCLKC/ PO26	CTS4#/RTS4#/ CTS6#/RTS6#/ SS4#/SS6#/ ET0_RX_CLK/ REF50CK0	SDSI_D2-B		
J13		PB1	A9	MTIOC0C/ MTIOC4C/TIOCB3/ TMC10/PO25	TXD4/TXD6/ SMOSI4/SMOSI6/ SSDA4/SSDA6/ ET0_ERXD0/ RMII0_RXD0		IRQ4-DS	
K1	TCK	P27	CS7#	MTIOC2B/TMC13/PO7	SCK1/RSPCKB-A			
K2	TDO	P26	CS6#	MTIOC2A/TMO1/PO6	TXD1/CTS3#/ RTS3#/SMOSI1/ SS3#/SSDA1/ MOSIB-A			
K3	TMS	P31		MTIOC4D/TMC12/ PO9/RTIC1	CTS1#/RTS1#/ SS1#/SSLB0-A		IRQ1-DS	

Table 1.5 List of Pin and Pin Functions (145-Pin TFLGA) (4/6)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCIh, SCiI, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
K4		P15		MTIOC0B/MTCLKB/ TIOC0B/TCLKB/ TMC12/PO13	RXD1/SCK3/ SMISO1/SSCL1/ CRX1-DS	PIXD0	IRQ5	
K5	TRDATA2	P54	ALE/EDACK0	MTIOC4B/TMC11	CTS2#/RTS2#/ SS2#/CTX1/ ET0_LINKSTA			
K6		P53	BCLK					
K7		P51	WR1#/BC1#/ WAIT#		SCK2/SSLB2-A			
K8	VCC							
K9	TRDATA0	P80	EDREQ0	MTIOC3B/PO26	SCK10/RTS10#/ ET0_TX_EN/ RMII0_TXD_EN	MMC_D2-A/ SDHI_WP-A/ QIO2-A		
K10	TRDATA6	P76	CS6#	PO22	RXD11/SMISO11/ SSCL11/ ET0_RX_CLK/ REF50CK0	MMC_CMD-A/ SDHI_CMD-A/ SDSI_CMD-A/ QSSL-A		
K11		PB7	A15	MTIOC3B/TIOCB5/ PO31	TXD9/SMOSI9/ SSDA9/ET0_CRS/ RMII0_CRS_DV/ TXD11/SMOSI11/ SSDA11	SDSI_D1-B		
K12		PB6	A14	MTIOC3D/TIOCA5/ PO30	RXD9/SMISO9/ SSCL9/ET0_ETXD1/ RMII0_TXD1/ RXD11/SMISO11/ SSCL11	SDSI_D0-B		
K13		PB5	A13	MTIOC2A/ MTIOC1B/TIOCB4/ TMRI1/PO29/ POE4#	SCK9/ET0_ETXD0/ RMII0_TXD0/SCK11	SDSI_CLK-B		
L1		P25	CS5#/ EDACK1	MTIOC4C/MTCLKB/ TIOCA4/PO5	RXD3/SMISO3/ SSCL3	HSYNC		ADTRG0#
L2		P23	EDACK0	MTIOC3D/MTCLKD/ TIOC03/PO3	TXD3/CTS0#/ RTS0#/SMOSI3/ SS0#/SSDA3	PIXD7		
L3		P16		MTIOC3C/ MTIOC3D/TIOCB1/ TCLKC/TMO2/ PO14/RTCOUT	TXD1/RXD3/ SMOSI1/SMISO3/ SSDA1/SSCL3/ SCL2-DS/ USB0_VBUS/ USB0_VBUSEN/ USB0_OVRCURB		IRQ6	ADTRG0#
L4		P24	CS4#/ EDREQ1	MTIOC4A/MTCLKA/ TIOCB4/TMRI1/PO4	SCK3/ USB0_VBUSEN	PIXCLK		
L5		P13		MTIOC0B/TIOCA5/ TMO3/PO13	TXD2/SMOSI2/ SSDA2/SDA0[FM+]		IRQ3	ADTRG1#
L6		P56	EDACK1	MTIOC3C/TIOCA1				
L7		P52	RD#		RXD2/SMISO2/ SSCL2/SSLB3-A			
L8	TRCLK	P83	EDACK1	MTIOC4C	CTS10#/SS10#/ ET0_CRS/ RMII0_CRS_DV/ SCK10			
L9		PC5	A21/CS2#/ WAIT#	MTIOC3B/MTCLKD/ TMRI2/PO29	SCK8/RSPCKA-A/ ET0_ETXD2/SCK10	MMC_D5-A		
L10		PC4	A20/CS3#	MTIOC3D/MTCLKC/ TMC11/PO25/ POE0#	SCK5/CTS8#/ RTS8#/SS8#SSLA0- A/ET0_TX_CLK/ CTS10#/RTS10#/ SS10#	MMC_D1-A/ SDHI_D1-A/ SDSI_D1-A/ QIO1-A/ QMI-A		
L11		PC2	A18	MTIOC4B/TCLKA/ PO21	RXD5/SMISO5/ SSCL5/SSLA3-A/ ET0_RX_DV	MMC_CD-A/ SDHI_D3-A/ SDSI_D3-A		

Table 1.5 List of Pin and Pin Functions (145-Pin TFLGA) (5/6)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCIh, SCiI, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
L12	TRDATA4	P73	CS3#	PO16	ET0_WOL			
L13	VSS							
M1		P22	EDREQ0	MTIOC3B/MTCLKC/ TIOCC3/TMO0/PO2	SCK0/ USB0_OVRCURB	PIXD6		
M2		P17		MTIOC3A/ MTIOC3B/ MTIOC4B/TIOCB0/ TCLKD/TMO1/ PO15/POE8#	SCK1/TXD3/ SMOSI3/SSDA3/ SDA2-DS	PIXD3	IRQ7	ADTRG1#
M3		P86		MTIOC4D/TIOCA0	RXD10/SMISO10/ SSCL10	PIXD1		
M4		P12		TMC11	RXD2/SMISO2/ SSCL2/SCL0[FM+]		IRQ2	
M5	VCC_USB							
M6	VSS_USB							
M7		P50	WR0#/WR#		TXD2/SMOSI2/ SSDA2/SSLB1-A			
M8		PC6	A22/CS1#	MTIOC3C/MTCLKA/ TMC12/TIC0/PO30	RXD8/SMISO8/ SSCL8/MOSIA-A/ ET0_ETXD3/RXD10/ SMISO10/SSCL10	MMC_D6-A	IRQ13	
M9	TRDATA1	P81	EDACK0	MTIOC3D/PO27	RXD10/SMISO10/ SSCL10/ ET0_ETXD0/ RMII0_TXD0	MMC_D3-A/ SDHI_CD-A/ QIO3-A		
M10	TRDATA7	P77	CS7#	PO23	TXD11/SMOSI11/ SSDA11/ ET0_RX_ER/ RMII0_RX_ER	MMC_CLK-A/ SDHI_CLK-A/ SDSI_CLK-A/ QSPCLK-A		
M11		PC0	A16	MTIOC3C/TCLKC/ PO17	CTS5#/RTS5#/ SS5#/SSLA1-A/ ET0_ERXD3		IRQ14	
M12		PC1	A17	MTIOC3A/TCLKD/ PO18	SCK5/SSLA2-A/ ET0_ERXD2		IRQ12	
M13	VCC							
N1		P21		MTIOC1B/ MTIOC4A/TIOCA3/ TMC10/PO1	RXD0/SMISO0/ SSCL0/ USB0_EXICEN/	PIXD5	IRQ9	
N2		P20		MTIOC1A/TIOCB3/ TMR10/PO0	TXD0/SMOSI0/ SSDA0/USB0_ID	PIXD4	IRQ8	
N3		P87		MTIOC4C/TIOCA2	TXD10 /SMOSI10/SSDA10	PIXD2		
N4		P14		MTIOC3A/MTCLKA/ TIOCB5/TCLKA/ TMR12/PO15	CTS1#/RTS1#/ SS1#/CTX1/ USB0_OVRCURA		IRQ4	
N5					USB0_DM			
N6					USB0_DP			
N7	TRDATA3	P55	WAIT#/ EDREQ0	MTIOC4D/TMO3	CRX1/ET0_EXOUT		IRQ10	
N8	VSS							
N9	UB	PC7	A23/CS0#	MTIOC3A/MTCLKB/ TMO2/TOC0/PO31/ CACREF	TXD8/SMOSI8/ SSDA8/MISOA-A/ ET0_COL/TXD10/ SMOSI10/SSDA10	MMC_D7-A	IRQ14	
N10	TRSYNC	P82	EDREQ1	MTIOC4A/PO28	TXD10/SMOSI10/ SSDA10/ ET0_ETXD1/ RMII0_TXD1	MMC_D4-A		

Table 1.5 List of Pin and Pin Functions (145-Pin TFLGA) (6/6)

Pin Number				Timer	Communication	Memory Interface Camera Interface		
145-Pin TFLGA	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	(MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	(ETHERC, SC1g, SCIh, SC1i, RSPI, R1IC, CAN, USB)	(QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
N11		PC3	A19	MTIOC4D/TCLKB/ PO24	TXD5/SMOSI5/ SSDA5/ET0_TX_ER	MMC_D0-A/ SDHI_D0-A/ SDSI_D0-A/ QIO0-A/ QMO-A		
N12	TRSYNC1	P75	CS5#	PO20	SCK11/RTS11#/ ET0_ERXD0/ RMII0_RXD0	MMC_RES#- A/SDHI_D2-A/ SDSI_D2-A		
N13	TRDATA5	P74	A20/CS4#	PO19	CTS11#/SS11#/ ET0_ERXD1/ RMII0_RXD1			

Table 1.6 List of Pin and Pin Functions (144-Pin LFQFP) (1/6)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCIh, SCii, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
1	AVSS0							
2		P05					IRQ13	DA1
3	AVCC1							
4		P03					IRQ11	DA0
5	AVSS1							
6		P02		TMC11	SCK6		IRQ10	AN120
7		P01		TMC10	RXD6/SMISO6/SSCL6		IRQ9	AN119
8		P00		TMR10	TXD6/SMOSI6/SSDA6		IRQ8	AN118
9		PF5					IRQ4	
10	EMLE							
11		PJ5		POE8#	CTS2#/RTS2#/SS2#			
12	VSS							
13		PJ3	EDACK1	MTIOC3C	ET0_EXOUT/ CTS6#/RTS6#/ CTS0#/RTS0#/ SS6#/SS0#			
14	VCL							
15	VBATT							
16	MD/FINED							
17	XCIN							
18	XCOUT							
19	RES#							
20	XTAL	P37						
21	VSS							
22	EXTAL	P36						
23	VCC							
24	UPSEL	P35					NMI	
25	TRST#	P34		MTIOC0A/TMC13/ PO12/POE10#	SCK6/SCK0/ ET0_LINKSTA		IRQ4	
26		P33	EDREQ1	MTIOC0D/TIOC0D/ TMR13/PO11/ POE4#/POE11	RXD6/RXD0/ SMISO6/SMISO0/ SSCL6/SSCL0/ CRX0	PCKO	IRQ3-DS	
27		P32		MTIOC0C/TIOCC0/ TMO3/PO10/ RTCOU/RTCI2/ POE0#/POE10#	TXD6/TXD0/ SMOSI6/SMOSI0/ SSDA6/SSDA0/ CTX0/ USB0_VBUSEN	VSYNC	IRQ2-DS	
28	TMS	P31		MTIOC4D/TMC12/ PO9/RTIC1	CTS1#/RTS1#/ SS1#/SSLB0-A		IRQ1-DS	
29	TDI	P30		MTIOC4B/TMR13/ PO8/RTIC0/ POE8#	RXD1/SMISO1/ SSCL1/MISOB-A		IRQ0-DS	
30	TCK	P27	CS7#	MTIOC2B/TMC13/PO7	SCK1/RSPCKB-A			
31	TDO	P26	CS6#	MTIOC2A/TMO1/ PO6	TXD1/CTS3#/ RTS3#/SMOSI1/ SS3#/SSDA1/ MOSIB-A			
32		P25	CS5#/ EDACK1	MTIOC4C/MTCLKB/ TIOCA4/PO5	RXD3/SMISO3/ SSCL3	HSYNC		ADTRG0#
33		P24	CS4#/ EDREQ1	MTIOC4A/MTCLKA/ TIOCB4/TMRI1/PO4	SCK3/ USB0_VBUSEN	PIXCLK		
34		P23	EDACK0	MTIOC3D/MTCLKD/ TIOC3D/PO3	TXD3/CTS0#/ RTS0#/SMOSI3/ SS0#/SSDA3	PIXD7		

Table 1.6 List of Pin and Pin Functions (144-Pin LFQFP) (2/6)

Pin Number				Timer	Communication	Memory Interface		
144-Pin LFQFP	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	(MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	(ETHERC, SCiG, SCiH, SCiI, RSPI, RIIC, CAN, USB)	(QSPI, SDHI, SDSi, MMCIF, PDC)	Interrupt	S12AD, R12DA
35		P22	EDREQ0	MTIOC3B/MTCLKC/ TIOCC3/TMO0/PO2	SCK0/ USB0_OVRCURB	PIXD6		
36		P21		MTIOC1B/ MTIOC4A/TIOCA3/ TMCIO/PO1	RXD0/SMISO0/ SSCL0/ USB0_EXICEN	PIXD5	IRQ9	
37		P20		MTIOC1A/TIOCB3/ TMRIO/PO0	TXD0/SMOSI0/ SSDA0/USB0_ID	PIXD4	IRQ8	
38		P17		MTIOC3A/ MTIOC3B/ MTIOC4B/TIOCB0/ TCLKD/TMO1/ PO15/POE8#	SCK1/TXD3/ SMOSI3/SSDA3/ SDA2-DS	PIXD3	IRQ7	ADTRG1#
39		P87		MTIOC4C/TIOCA2	TXD10/SMOSI10/ SSDA10	PIXD2		
40		P16		MTIOC3C/ MTIOC3D/TIOCB1/ TCLKC/TMO2/ PO14/RTCOUNT	TXD1/RXD3/ SMOSI1/SMISO3/ SSDA1/SSCL3/ SCL2-DS/ USB0_VBUS/ USB0_VBUSEN/ USB0_OVRCURB		IRQ6	ADTRG0#
41		P86		MTIOC4D/TIOCA0	RXD10/SMISO10/ SSCL10	PIXD1		
42		P15		MTIOC0B/MTCLKB/ TIOCB2/TCLKB/ TMC12/PO13	RXD1/SCK3/ SMISO1/SSCL1/ CRX1-DS	PIXD0	IRQ5	
43		P14		MTIOC3A/MTCLKA/ TIOCB5/TCLKA/ TMR12/PO15	CTS1#/RTS1#/ SS1#/CTX1/ USB0_OVRCURA		IRQ4	
44		P13		MTIOC0B/TIOCA5/ TMO3/PO13	TXD2/SMOSI2/ SSDA2/SDA0[FM+]		IRQ3	ADTRG1#
45		P12		TMC11	RXD2/SMISO2/ SSCL2/SCL0[FM+]		IRQ2	
46	VCC_USB							
47					USB0_DM			
48					USB0_DP			
49	VSS_USB							
50		P56	EDACK1	MTIOC3C/TIOCA1				
51	TRDATA3	P55	WAIT#/ EDREQ0	MTIOC4D/TMO3	CRX1/ET0_EXOUT		IRQ10	
52	TRDATA2	P54	ALE/EDACK0	MTIOC4B/TMC11	CTS2#/RTS2#/ SS2#/CTX1/ ET0_LINKSTA			
53		P53	BCLK					
54		P52	RD#		RXD2/SMISO2/ SSCL2/SSLB3-A			
55		P51	WR1#/BC1#/ WAIT#		SCK2/SSLB2-A			
56		P50	WR0#/WR#		TXD2/SMOSI2/ SSDA2/SSLB1-A			
57	VSS							
58	TRCLK	P83	EDACK1	MTIOC4C	CTS10#/SS10#/ ET0_CRS/ RMII0_CRS_DV/ SCK10			
59	VCC							
60	UB	PC7	A23/CS0#	MTIOC3A/MTCLKB/ TMO2/TOC0/PO31/ CACREF	TXD8/SMOSI8/ SSDA8/MISOA-A/ ET0_COL/TXD10/ SMOSI10/SSDA10	MMC_D7-A	IRQ14	

Table 1.6 List of Pin and Pin Functions (144-Pin LFQFP) (3/6)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCIh, SCiI, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
61		PC6	A22/CS1#	MTIOC3C/MTCLKA/ TMCI2/TIC0/PO30	RXD8/SMISO8/ SSCL8/MOSIA-A/ ET0_ETXD3/RXD10/ SMISO10/SSCL10	MMC_D6-A	IRQ13	
62		PC5	A21/CS2#/ WAIT#	MTIOC3B/MTCLKD/ TMRI2/PO29	SCK8/RSPCKA-A/ ET0_ETXD2/SCK10	MMC_D5-A		
63	TRSYNC	P82	EDREQ1	MTIOC4A/PO28	TXD10/SMOSI10/ SSDA10/ ET0_ETXD1/ RMII0_TXD1	MMC_D4-A		
64	TRDATA1	P81	EDACK0	MTIOC3D/PO27	RXD10/SMISO10/ SSCL10/ ET0_ETXD0/ RMII0_TXD0	MMC_D3-A/ SDHI_CD-A/ QIO3-A		
65	TRDATA0	P80	EDREQ0	MTIOC3B/PO26	SCK10/RTS10#/ ET0_TX_EN/ RMII0_TXD_EN	MMC_D2-A/ SDHI_WP-A/ QIO2-A		
66		PC4	A20/CS3#	MTIOC3D/MTCLKC/ TMCI1/PO25/ POE0#	SCK5/CTS8#/ RTS8#/SS8#/ SSLA0-A/ ET0_TX_CLK/ CTS10#/RTS10#/ SS10#	MMC_D1-A/ SDHI_D1-A/ SDSI_D1-A/ QIO1-A/QMI-A		
67		PC3	A19	MTIOC4D/TCLKB/ PO24	TXD5/SMOSI5/ SSDA5/ET0_TX_ER	MMC_D0-A/ SDHI_D0-A/ SDSI_D0-A/ QIO0-A/ QMO-A		
68	TRDATA7	P77	CS7#	PO23	TXD11/SMOSI11/ SSDA11/ ET0_RX_ER/ RMII0_RX_ER	MMC_CLK-A/ SDHI_CLK-A/ SDSI_CLK-A/ QSPCLK-A		
69	TRDATA6	P76	CS6#	PO22	RXD11/SMISO11/ SSCL11/ ET0_RX_CLK/ REF50CK0	MMC_CMD-A/ SDHI_CMD-A/ SDSI_CMD-A/ QSSL-A		
70		PC2	A18	MTIOC4B/TCLKA/ PO21	RXD5/SMISO5/ SSCL5/SSLA3-A/ ET0_RX_DV	MMC_CD-A/ SDHI_D3-A/ SDSI_D3-A		
71	TRSYNC1	P75	CS5#	PO20	SCK11/RTS11#/ ET0_ERXD0/ RMII0_RXD0	MMC_RES#- A/SDHI_D2-A/ SDSI_D2-A		
72	TRDATA5	P74	A20/CS4#	PO19	CTS11#/SS11#/ ET0_ERXD1/ RMII0_RXD1			
73		PC1	A17	MTIOC3A/TCLKD/ PO18	SCK5/SSLA2-A/ ET0_ERXD2		IRQ12	
74	VCC							
75		PC0	A16	MTIOC3C/TCLKC/ PO17	CTS5#/RTS5#/ SS5#/SSLA1-A/ ET0_ERXD3		IRQ14	
76	VSS							
77	TRDATA4	P73	CS3#	PO16	ET0_WOL			
78		PB7	A15	MTIOC3B/TIOCB5/ PO31	TXD9/SMOSI9/ SSDA9/ET0_CRS/ RMII0_CRS_DV/ TXD11/SMOSI11/ SSDA11	SDSI_D1-B		
79		PB6	A14	MTIOC3D/TIOCA5/ PO30	RXD9/SMISO9/ SSCL9/ET0_ETXD1/ RMII0_TXD1/ RXD11/SMISO11/ SSCL11	SDSI_D0-B		

Table 1.6 List of Pin and Pin Functions (144-Pin LFQFP) (4/6)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCIh, SCii, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
80		PB5	A13	MTIOC2A/ MTIOC1B/TIOCB4/ TMRI1/PO29/ POE4#	SCK9/ET0_ETXD0/ RMII0_TXD0/SCK11	SDSI_CLK-B		
81		PB4	A12	TIOCA4/PO28	CTS9#/RTS9#/ SS9#/ET0_TX_EN/ RMII0_TXD_EN/ CTS11#/RTS11#/ SS11#	SDSI_CMD-B		
82		PB3	A11	MTIOC0A/ MTIOC4A/TIOCD3/ TCLKD/TMO0/ PO27/POE11#	SCK4/SCK6/ ET0_RX_ER/ RMII0_RX_ER	SDSI_D3-B		
83		PB2	A10	TIOCC3/TCLKC/ PO26	CTS4#/RTS4#/ CTS6#/RTS6#/ SS4#/SS6#/ ET0_RX_CLK/ REF50CK0	SDSI_D2-B		
84		PB1	A9	MTIOC0C/ MTIOC4C/TIOCB3/ TMC10/PO25	TXD4/TXD6/ SMOSI4/SMOSI6/ SSDA4/SSDA6/ ET0_ERXD0/ RMII0_RXD0		IRQ4-DS	
85		P72	A19/CS2#		ET0_MDC			
86		P71	A18/CS1#		ET0_MDIO			
87		PB0	A8	MTIC5W/TIOCA3/ PO24	RXD4/RXD6/ SMISO4/SMISO6/ SSCL4/SSCL6/ ET0_ERXD1/ RMII0_RXD1		IRQ12	
88		PA7	A7	TIOCB2/PO23	MISOA-B/ET0_WOL			
89		PA6	A6	MTIC5V/MTCLKB/ TIOCA2/TMC13/ PO22/POE10#	CTS5#/RTS5#/SS5#/ MOSIA-B/ ET0_EXOUT			
90		PA5	A5	MTIOC6B/TIOCB1/ PO21	RSPCKA-B/ ET0_LINKSTA			
91	VCC							
92		PA4	A4	MTIC5U/MTCLKA/ TIOCA1/TMRI0/ PO20	TXD5/SMOSI5/ SSDA5/SSLA0-B/ ET0_MDC		IRQ5-DS	
93	VSS							
94		PA3	A3	MTIOC0D/MTCLKD/ TIOC0D/TCLKB/ PO19	RXD5/SMISO5/ SSCL5/ET0_MDIO		IRQ6-DS	
95		PA2	A2	MTIOC7A/PO18	RXD5/SMISO5/ SSCL5/SSLA3-B			
96		PA1	A1	MTIOC0B/MTCLKC/ MTIOC7B/TIOCB0/ PO17	SCK5/SSLA2-B/ ET0_WOL		IRQ11	
97		PA0	A0/BC0#	MTIOC4A/ MTIOC6D/TIOCA0/ CACREF/PO16	SSLA1-B/ ET0_TX_EN/ RMII0_TXD_EN			
98		P67	CS7#/DQM1	MTIOC7C			IRQ15	
99		P66	CS6#/DQM0	MTIOC7D				
100		P65	CS5#/CKE					
101		PE7	D15[A15/D15]	MTIOC6A/TOC1	MISOB-B	MMC_RES#-B/ SDHI_WP-B	IRQ7	AN105
102		PE6	D14[A14/D14]	MTIOC6C/TIC1	MOSIB-B	MMC_CD-B/ SDHI_CD-B	IRQ6	AN104
103	VCC							
104		P70	SDCLK					

Table 1.6 List of Pin and Pin Functions (144-Pin LQFP) (5/6)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCIh, SCii, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
105	VSS							
106		PE5	D13[A13/D13]	MTIOC4C/MTIOC2B	ET0_RX_CLK/ REF50CK0/ RSPCKB-B		IRQ5	AN103
107		PE4	D12[A12/D12]	MTIOC4D/ MTIOC1A/PO28	ET0_ERXD2/ SSLB0-B			AN102
108		PE3	D11[A11/D11]	MTIOC4B/PO26/ POE8#/TOC3	CTS12#/RTS12#/ SS12#/ET0_ERXD3/	MMC_D7-B		AN101
109		PE2	D10[A10/D10]	MTIOC4A/PO23/ TIC3	RXD12/SMISO12/ SSCL12/RXDX12/ SSLB3-B	MMC_D6-B	IRQ7-DS	AN100
110		PE1	D9[A9/D9]	MTIOC4C/ MTIOC3B/PO18	TXD12/SMOSI12/ SSDA12/TXDX12/ SIOX12/SSLB2-B	MMC_D5-B		ANEX1
111		PE0	D8[A8/D8]	MTIOC3D	SCK12/SSLB1-B	MMC_D4-B		ANEX0
112		P64	CS4#/WE#					
113		P63	CS3#/CAS#					
114		P62	CS2#/RAS#					
115		P61	CS1#/SDCS#					
116	VSS							
117		P60	CS0#					
118	VCC							
119		PD7	D7[A7/D7]	MTIC5U/POE0#	SSLC3	MMC_D1-B/ SDHI_D1-B/ QIO1-B/QMI-B	IRQ7	AN107
120		PD6	D6[A6/D6]	MTIC5V/MTIOC8A/ POE4#	SSLC2	MMC_D0-B/ SDHI_D0-B/ QIO0-B/ QMO-B	IRQ6	AN106
121		PD5	D5[A5/D5]	MTIC5W/MTIOC8C/ POE10#	SSLC1	MMC_CLK-B/ SDHI_CLK-B/ QSPCLK-B	IRQ5	AN113
122		PD4	D4[A4/D4]	MTIOC8B/POE11#	SSLC0	MMC_CMD-B/ SDHI_CMD-B/ QSSL-B	IRQ4	AN112
123		PD3	D3[A3/D3]	MTIOC8D/POE8#/ TOC2	RSPCKC	MMC_D3-B/ SDHI_D3-B/ QIO3-B	IRQ3	AN111
124		PD2	D2[A2/D2]	MTIOC4D/TIC2	CRX0/MISOC	MMC_D2-B/ SDHI_D2-B/ QIO2-B	IRQ2	AN110
125		PD1	D1[A1/D1]	MTIOC4B/POE0#	CTX0/MOSIC		IRQ1	AN109
126		PD0	D0[A0/D0]	POE4#			IRQ0	AN108
127		P93	A19	POE0#	CTS7#/RTS7#/SS7#			AN117
128		P92	A18	POE4#	RXD7/SMISO7/SSCL7			AN116
129		P91	A17		SCK7			AN115
130	VSS							
131		P90	A16		TXD7/SMOSI7/SSDA7			AN114
132	VCC							
133		P47					IRQ15-DS	AN007
134		P46					IRQ14-DS	AN006
135		P45					IRQ13-DS	AN005
136		P44					IRQ12-DS	AN004
137		P43					IRQ11-DS	AN003
138		P42					IRQ10-DS	AN002

Table 1.6 List of Pin and Pin Functions (144-Pin LFQFP) (6/6)

Pin Number				Timer	Communication	Memory Interface Camera Interface		
144-Pin LFQFP	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	(MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	(ETHERC, SC1g, SCIh, SCli, RSPI, RIIC, CAN, USB)	(QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
139		P41					IRQ9-DS	AN001
140	VREFL0							
141		P40					IRQ8-DS	AN000
142	VREFH0							
143	AVCC0							
144		P07					IRQ15	ADTRG0#

Table 1.7 List of Pin and Pin Functions (100-Pin TFLGA) (1/4)

Pin Number								
100-Pin TFLGA	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SC1g, SC1h, SC1i, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
A1		P05					IRQ13	DA1
A2	AVCC1							
A3		P07					IRQ15	ADTRG0#
A4	VREFL0							
A5		P43					IRQ11-DS	AN003
A6		PD0	D0[A0/D0]	POE4#			IRQ0	AN108
A7		PD4	D4[A4/D4]	MTIOC8B/POE11#	SSLC0	MMC_CMD-B/ SDHI_CMD-B/ QSSL-B	IRQ4	AN112
A8		PE0	D8[A8/D8]	MTIOC3D	SCK12/SSLB1-B	MMC_D4-B		ANEX0
A9		PE1	D9[A9/D9]	MTIOC4C/ MTIOC3B/PO18	TXD12/SMOS12/ SSDA12/TXD12/ SIOX12/SSLB2-B	MMC_D5-B		ANEX1
A10		PE2	D10[A10/ D10]	MTIOC4A/PO23/ TIC3	RXD12/SMISO12/ SSCL12/RXD12/ SSLB3-B	MMC_D6-B	IRQ7-DS	AN100
B1	EMLE							
B2	AVSS0							
B3	AVCC0							
B4		P40					IRQ8-DS	AN000
B5		P44					IRQ12-DS	AN004
B6		PD1	D1[A1/D1]	MTIOC4B/POE0#	CTX0/MOSIC		IRQ1	AN109
B7		PD3	D3[A3/D3]	MTIOC8D/POE8#/ TOC2	RSPCKC	MMC_D3-B/ SDHI_D3-B/ QIO3-B	IRQ3	AN111
B8		PD6	D6[A6/D6]	MTIC5V/MTIOC8A/ POE4#	SSLC2	MMC_D0-B/ SDHI_D0-B/ QIO0-B/ QMO-B	IRQ6	AN106
B9		PD7	D7[A7/D7]	MTIC5U/POE0#	SSLC3	MMC_D1-B/ SDHI_D1-B/ QIO1/QMI-B	IRQ7	AN107
B10		PE3	D11[A11/ D11]	MTIOC4B/PO26/ POE8#/TOC3	CTS12#/RTS12#/ SS12#/ET0_ERXD3	MMC_D7-B		AN101
C1	VCL							
C2	AVSS1							
C3		PJ3	EDACK1	MTIOC3C	ET0_EXOUT/ CTS6#/RTS6#/ CTS0#/RTS0#/SS6#/ SS0#			
C4	VREFH0							
C5		P42					IRQ10-DS	AN002
C6		P47					IRQ15-DS	AN007
C7		PD2	D2[A2/D2]	MTIOC4D/TIC2	CRX0/MISOC	MMC_D2-B/ SDHI_D2-B/ QIO2-B	IRQ2	AN110
C8		PD5	D5[A5/D5]	MTIC5W/MTIOC8C/ POE10#	SSLC1	MMC_CLK-B/ SDHI_CLK-B/ QSPCLK-B	IRQ5	AN113
C9		PE5	D13[A13/ D13]	MTIOC4C/MTIOC2B	ET0_RX_CLK/ REF50CK0/ RSPCKB-B		IRQ5	AN103
C10		PE4	D12[A12/ D12]	MTIOC4D/ MTIOC1A/PO28	ET0_ERXD2/ SSLB0-B			AN102
D1	XCIN							
D2	XCOUT							

Table 1.7 List of Pin and Pin Functions (100-Pin TFLGA) (2/4)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SC1g, SC1h, SC1i, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
D3	MD/FINED							
D4	VBATT							
D5		P45					IRQ13-DS	AN005
D6		P46					IRQ14-DS	AN006
D7		PE6	D14[A14/ D14]	MTIOC6C/TIC1	MOSIB-B	MMC_CD-B/ SDHI_CD-B	IRQ6	AN104
D8		PE7	D15[A15/ D15]	MTIOC6A/TOC1	MISOB-B	MMC_RES#-B/ SDHI_WP-B	IRQ7	AN105
D9		PA1	A1	MTIOC0B/MTCLKC/ MTIOC7B/TIOCB0/ PO17	SCK5/SSLA2-B/ ET0_WOL		IRQ11	
D10		PA0	A0/BC0#	MTIOC4A/ MTIOC6D/TIOCA0/ CACREF/PO16	SSLA1-B/ ET0_TX_EN/ RMII0_TXD_EN			
E1	XTAL	P37						
E2	VSS							
E3	RES#							
E4	TRST#	P34		MTIOC0A/TMC13/ PO12/POE10#	SCK6/SCK0/ ET0_LINKSTA		IRQ4	
E5		P41					IRQ9-DS	AN001
E6		PA2	A2	MTIOC7A/PO18	RXD5/SMISO5/ SSCL5/SSLA3-B			
E7		PA6	A6	MTIC5V/MTCLKB/ TIOCA2/TMC13/ PO22/POE10#	CTS5#/RTS5#/SS5#/ MOSIA-B/ ET0_EXOUT			
E8		PA4	A4	MTIC5U/MTCLKA/ TIOCA1/TMRI0/ PO20	TXD5/SMOSI5/ SSDA5/SSLA0-B/ ET0_MDC		IRQ5-DS	
E9		PA5	A5	MTIOC6B/TIOCB1/ PO21	RSPCKA-B/ ET0_LINKSTA			
E10		PA3	A3	MTIOC0D/MTCLKD/ TIOC0D/TCLKB/ PO19	RXD5/SMISO5/ SSCL5/ET0_MDIO		IRQ6-DS	
F1	EXTAL	P36						
F2	VCC							
F3	UPSEL	P35					NMI	
F4		P32		MTIOC0C/TIOCC0/ TMO3/PO10/ RTCOU/RTIC2/ POE0#/POE10#	TXD6/TXD0/ SMOSI6/SMOSI0/ SSDA6/SSDA0/ CTX0/ USB0_VBUSEN		IRQ2-DS	
F5		P12		TMC11	RXD2/SMISO2/ SSCL2/SCL0[FM+]		IRQ2	
F6		PB3	A11	MTIOC0A/ MTIOC4A/TIOC0D3/ TCLKD/TMO0/ PO27/POE11#	SCK6/ET0_RX_ER/ RMII0_RX_ER	SDSI_D3-B		
F7		PB2	A10	TIOCC3/TCLKC/ PO26	CTS6#/RTS6#SS6#/ ET0_RX_CLK/ REF50CK0	SDSI_D2-B		
F8		PB0	A8	MTIC5W/TIOCA3/ PO24	RXD6/SMISO6/ SSCL6/ET0_ERXD1/ RMII0_RXD1		IRQ12	
F9		PA7	A7	TIOCB2/PO23	MISOA-B/ET0_WOL			
F10	VSS							

Table 1.7 List of Pin and Pin Functions (100-Pin TFLGA) (3/4)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCiG, SCiH, SCiI, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
G1		P33	EDREQ1	MTIOC0D/TIOC0D/ TMRI3/PO11/ POE4#/POE11#	RXD6/RXD0/ SMISO6/SMISO0/ SSCL6/SSCL0/ CRX0		IRQ3-DS	
G2	TMS	P31		MTIOC4D/TMC12/ PO9/RTCIC1	CTS1#/RTS1#/SS1#/ SSLB0-A		IRQ1-DS	
G3	TDI	P30		MTIOC4B/TMRI3/ PO8/RTCIC0/POE8#	RXD1/SMISO1/ SSCL1/MISOB-A		IRQ0-DS	
G4	TCK	P27	CS7#	MTIOC2B/TMC13/PO7	SCK1/RSPCKB-A			
G5		P53	BCLK					
G6		P52	RD#		RXD2/SMISO2/ SSCL2/SSLB3-A			
G7		PB5	A13	MTIOC2A/ MTIOC1B/TIOCB4/ TMRI1/PO29/POE4#	SCK9/ET0_ETXD0/ RMII0_TXD0/SCK11	SDSI_CLK-B		
G8		PB4	A12	TIOCA4/PO28	CTS9#/RTS9#/SS9#/ ET0_TX_EN/ RMII0_TXD_EN/ CTS11#/RTS11#/ SS11#	SDSI_CMD-B		
G9		PB1	A9	MTIOC0C/MTIOC4C/ TIOCB3/TMC10/PO25	TXD6/SMOSI6/ SSDA6/ ET0_ERXD0/ RMII0_RXD0		IRQ4-DS	
G10	VCC							
H1	TDO	P26	CS6#	MTIOC2A/TMO1/PO6	TXD1/CTS3#/ RTS3#/SMOSI1/ SS3#/SSDA1/ MOSIB-A			
H2		P25	CS5#/ EDACK1	MTIOC4C/MTCLKB/ TIOCA4/PO5	RXD3/SMISO3/ SSCL3			ADTRG0#
H3		P16		MTIOC3C/MTIOC3D/ TIOCB1/TCLKC/ TMO2/PO14/ RTCOUT	TXD1/RXD3/ SMOSI1/SMISO3/ SSDA1/SSCL3/ SCL2-DS/ USB0_VBUS/ USB0_VBUSEN/ USB0_OVRCURB		IRQ6	ADTRG0#
H4		P15		MTIOC0B/MTCLKB/ TIOCB2/TCLKB/ TMC12/PO13	RXD1/SCK3/ SMISO1/SSCL1/ CRX1-DS		IRQ5	
H5		P55	WAIT#/ EDREQ0	MTIOC4D/TMO3	CRX1/ET0_EXOUT		IRQ10	
H6		P54	ALE/ EDACK0	MTIOC4B/TMC11	CTS2#/RTS2#/SS2#/ CTX1/ET0_LINKSTA			
H7	UB	PC7	A23/CS0#	MTIOC3A/MTCLKB/ TMO2/TOC0/PO31/ CACREF	TXD8/SMOSI8/ SSDA8/MISOA-A/ ET0_COL/TXD10/ SMOSI10/SSDA10		IRQ14	
H8		PC6	A22/CS1#	MTIOC3C/MTCLKA/ TMC12/TIC0/PO30	RXD8/SMISO8/ SSCL8/MOSIA-A/ ET0_ETXD3/RXD10/ SMISO10/SSCL10		IRQ13	
H9		PB6	A14	MTIOC3D/TIOCA5/ PO30	RXD9/SMISO9/ SSCL9/ET0_ETXD1/ RMII0_TXD1/RXD11/ SMISO11/SSCL11	SDSI_D0-B		
H10		PB7	A15	MTIOC3B/TIOCB5/ PO31	TXD9/SMOSI9/ SSDA9/ET0_CRS/ RMII0_CRS_DV/ TXD11/SMOSI11/ SSDA11	SDSI_D1-B		

Table 1.7 List of Pin and Pin Functions (100-Pin TFLGA) (4/4)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC SDRAMC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SC1g, SC1h, SC1i, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
J1		P24	CS4#/EDREQ1	MTIOC4A/MTCLKA/ TIOCB4/TMRI1/PO4	SCK3/ USB0_VBUSEN			
J2		P21		MTIOC1B/ MTIOC4A/TIOCA3/ TMCI0/PO1	RXD0/SMISO0/ SSCL0/ USB0_EXICEN		IRQ9	
J3		P17		MTIOC3A/ MTIOC3B/ MTIOC4B/TIOCB0/ TCLKD/TMO1/ PO15/POE8#	SCK1/TXD3/ SMOSI3/SSDA3/ SDA2-DS		IRQ7	ADTRG1#
J4		P13		MTIOC0B/TIOCA5/ TMO3/PO13	TXD2/SMOSI2/ SSDA2/SDA0[FM+]		IRQ3	ADTRG1#
J5	VSS_USB							
J6	VCC_USB							
J7		P50	WR0#/WR#		TXD2/SMOSI2/ SSDA2/SSLB1-A			
J8		PC4	A20/CS3#	MTIOC3D/MTCLKC/ TMCI1/PO25/POE0#	SCK5/CTS8#/ RTS8#/SS8#/ SSLA0-A/ ET0_TX_CLK/ CTS10#/RTS10#/ SS10#			
J9		PC0	A16	MTIOC3C/TCLKC/ PO17	CTS5#/RTS5#/SS5#/ SSLA1-A/ ET0_ERXD3		IRQ14	
J10		PC1	A17	MTIOC3A/TCLKD/ PO18	SCK5/SSLA2-A/ ET0_ERXD2		IRQ12	
K1		P23	EDACK0	MTIOC3D/MTCLKD/ TIOCD3/PO3	TXD3/CTS0#/ RTS0#/SMOSI3/ SS0#/SSDA3			
K2		P22	EDREQ0	MTIOC3B/MTCLKC/ TIOCC3/TMO0/PO2	SCK0/ USB0_OVRCURB			
K3		P20		MTIOC1A/TIOCB3/ TMRI0/PO0	TXD0/SMOSI0/ SSDA0/USB0_ID		IRQ8	
K4		P14		MTIOC3A/MTCLKA/ TIOCB5/TCLKA/ TMRI2/PO15	CTS1#/RTS1#/SS1#/ CTX1/ USB0_OVRCURA		IRQ4	
K5					USB0_DM			
K6					USB0_DP			
K7		P51	WR1#/ BC1#/ WAIT#		SCK2/SSLB2-A			
K8		PC5	A21/CS2#/ WAIT#	MTIOC3B/MTCLKD/ TMRI2/PO29	SCK8/RSPCKA-A/ ET0_ETXD2/SCK10			
K9		PC3	A19	MTIOC4D/TCLKB/ PO24	TXD5/SMOSI5/ SSDA5/ET0_TX_ER			
K10		PC2	A18	MTIOC4B/TCLKA/ PO21	RXD5/SMISO5/ SSCL5/SSLA3-A/ ET0_RX_DV			

Table 1.8 List of Pin and Pin Functions (100-Pin LFQFP) (1/4)

Pin Number 100-Pin LFQFP	Power Supply Clock System Control	I/O Port	Bus EXDMAC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SC1g, SC1h, SC1i, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
1	AVCC1							
2	EMLE							
3	AVSS1							
4		PJ3	EDACK1	MTIOC3C	ET0_EXOUT CTS6#/RTS6#/CTS0#/ RTS0#/SS6#/SS0#			
5	VCL							
6	VBATT							
7	MD/FINED							
8	XCIN							
9	XCOUT							
10	RES#							
11	XTAL	P37						
12	VSS							
13	EXTAL	P36						
14	VCC							
15	UPSEL	P35					NMI	
16	TRST#	P34		MTIOC0A/TMCI3/ PO12/POE10#	SCK6/SCK0/ ET0_LINKSTA		IRQ4	
17		P33	EDREQ1	MTIOC0D/TIOC0D/ TMRI3/PO11/POE4#/ POE11#	RXD6/RXD0/SMISO6/ SMISO0/SSCL6/ SSCL0/CRX0		IRQ3-DS	
18		P32		MTIOC0C/TIOCC0/ TMO3/PO10/ RTCOUT/RTCIC2/ POE0#/POE10#	TXD6/TXD0/SMOSI6/ SMOSI0/SSDA6/ SSDA0/CTX0/ USB_VBUSEN		IRQ2-DS	
19	TMS	P31		MTIOC4D/TMCI2/ PO9/RTCIC1	CTS1#/RTS1#/ SS1#/SSLB0-A		IRQ1-DS	
20	TDI	P30		MTIOC4B/TMRI3/ PO8/RTCIC0/POE8#	RXD1/SMISO1/ SSCL1/MISOB-A		IRQ0-DS	
21	TCK	P27	CS7#	MTIOC2B/TMCI3/PO7	SCK1/RSPCKB-A			
22	TDO	P26	CS6#	MTIOC2A/TMO1/PO6	TXD1/CTS3#/ RTS3#/SMOSI1/ SS3#/SSDA1/ MOSIB-A			
23		P25	CS5#/ EDACK1	MTIOC4C/MTCLKB/ TIOCA4/PO5	RXD3/SMISO3/ SSCL3			ADTRG0#
24		P24	CS4#/ EDREQ1	MTIOC4A/MTCLKA/ TIOCB4/TMRI1/PO4	SCK3/ USB_VBUSEN			
25		P23	EDACK0	MTIOC3D/MTCLKD/ TIOC3D/PO3	TXD3/CTS0#/ RTS0#/SMOSI3/ SS0#/SSDA3			
26		P22	EDREQ0	MTIOC3B/MTCLKC/ TIOCC3/TMO0/PO2	SCK0/ USB_OVRCURB			
27		P21		MTIOC1B/ MTIOC4A/TIOCA3/ TMCI0/PO1	RXD0/SMISO0/ SSCL0/ USB_EXICEN		IRQ9	
28		P20		MTIOC1A/TIOCB3/ TMRI0/PO0	TXD0/SMOSI0/ SSDA0/USB_ID		IRQ8	
29		P17		MTIOC3A/ MTIOC3B/ MTIOC4B/TIOCB0/ TCLKD/TMO1/ PO15/POE8#	SCK1/TXD3/ SMOSI3/SSDA3/ SDA2-DS		IRQ7	ADTRG1#

Table 1.8 List of Pin and Pin Functions (100-Pin LFQFP) (2/4)

Pin Number	Power Supply Clock System Contro	I/O Port	Bus EXDMAC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SC1g, SCIh, SC1i, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
30		P16		MTIOC3C/MTIOC3D/ TIOCBI/TCLKC/ TMO2/PO14/ RTCOUT	TXD1/RXD3/SMOSI1/ SMISO3/SSDA1/ SSCL3/SCL2-DS/ USB0_VBUS/ USB0_VBUSEN/ USB0_OVRCURB		IRQ6	ADTRG0#
31		P15		MTIOC0B/MTCLKB/ TIOCBI/TCLKB/ TMC12/PO13	RXD1/SCK3/ SMISO1/SSCL1/ CRX1-DS		IRQ5	
32		P14		MTIOC3A/MTCLKA/ TIOCB5/TCLKA/ TMR12/PO15	CTS1#/RTS1#/SS1#/ CTX1/ USB0_OVRCURA		IRQ4	
33		P13		MTIOC0B/TIOCA5/ TMO3/PO13	TXD2/SMOSI2/ SSDA2/SDA0[FM+]		IRQ3	ADTRG1#
34		P12		TMC11	RXD2/SMISO2/ SSCL2/SCL0[FM+]		IRQ2	
35	VCC_USB							
36					USB0_DM			
37					USB0_DP			
38	VSS_USB							
39		P55	WAIT#/ EDREQ0	MTIOC4D/TMO3	CRX1/ET0_EXOUT		IRQ10	
40		P54	ALE/EDACK0	MTIOC4B/TMC11	CTS2#/RTS2#/SS2#/ CTX1/ET0_LINKSTA			
41		P53	BCLK					
42		P52	RD#		RXD2/SMISO2/ SSCL2/SSLB3-A			
43		P51	WR1#/BC1#/ WAIT#		SCK2/SSLB2-A			
44		P50	WR0#/WR#		TXD2/SMOSI2/ SSDA2/SSLB1-A			
45	UB	PC7	A23/CS0#	MTIOC3A/MTCLKB/ TMO2/TOC0/PO31/ CACREF	TXD8/SMOSI8/ SSDA8/MISOA-A/ ET0_COL/TXD10/ SMOSI10/SSDA10		IRQ14	
46		PC6	A22/CS1#	MTIOC3C/MTCLKA/ TMC12/TIC0/PO30	RXD8/SMISO8/ SSCL8/MOSIA-A/ ET0_ETXD3/RXD10/ SMISO10/SSCL10		IRQ13	
47		PC5	A21/CS2#/ WAIT#	MTIOC3B/MTCLKD/ TMR12/PO29	SCK8/RSPCKA-A/ ET0_ETXD2/SCK10			
48		PC4	A20/CS3#	MTIOC3D/MTCLKC/ TMC11/PO25/ POE0#	SCK5/CTS8#/ RTS8#/SS8#/ SSLA0-A/ ET0_TX_CLK/ CTS10#/RTS10#/ SS10#			
49		PC3	A19	MTIOC4D/TCLKB/ PO24	TXD5/SMOSI5/ SSDA5/ET0_TX_ER			
50		PC2	A18	MTIOC4B/TCLKA/ PO21	RXD5/SMISO5/ SSCL5/SSLA3-A/ ET0_RX_DV			
51		PC1	A17	MTIOC3A/TCLKD/ PO18	SCK5/SSLA2-A/ ET0_ERXD2		IRQ12	
52		PC0	A16	MTIOC3C/TCLKC/ PO17	CTS5#/RTS5#/SS5#/ SSLA1-A/ET0_ERXD3		IRQ14	
53		PB7	A15	MTIOC3B/TIOCB5/ PO31	TXD9/SMOSI9/ SSDA9/ET0_CRS/ RMII0_CRS_DV/ TXD11/SMOSI11/ SSDA11	SDSI_D1-B		

Table 1.8 List of Pin and Pin Functions (100-Pin LFQFP) (3/4)

Pin Number	Power Supply Clock System Control	I/O Port	Bus EXDMAC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SCIg, SCIh, SCII, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
54		PB6	A14	MTIOC3D/TIOCA5/ PO30	RXD9/SMISO9/ SSCL9/ET0_ETXD1/ RMII0_TXD1/ RXD11/SMISO11/ SSCL11	SDSI_D0-B		
55		PB5	A13	MTIOC2A/MTIOC1B/ TIOCB4/TMRI1/PO29/ POE4#	SCK9/ET0_ETXD0/ RMII0_TXD0/SCK11	SDSI_CLK-B		
56		PB4	A12	TIOCA4/PO28	CTS9#/RTS9#/ SS9#/ET0_TX_EN/ RMII0_TXD_EN/ CTS11#/RTS11#/ SS11#	SDSI_CMD-B		
57		PB3	A11	MTIOC0A/MTIOC4A/ TIOCD3/TCLKD/ TMO0/PO27/POE11#	SCK6/ET0_RX_ER/ RMII0_RX_ER	SDSI_D3-B		
58		PB2	A10	TIOCC3/TCLKC/ PO26	CTS6#/RTS6#SS6#/ ET0_RX_CLK/ REF50CK0	SDSI_D2-B		
59		PB1	A9	MTIOC0C/MTIOC4C/ TIOCB3/TMCI0/PO25	TXD6/SMOSI6/ SSDA6/ET0_ERXD0/ RMII0_RXD0		IRQ4-DS	
60	VCC							
61		PB0	A8	MTIC5W/TIOCA3/ PO24	RXD6/SMISO6/ SSCL6/ET0_ERXD1/ RMII0_RXD1		IRQ12	
62	VSS							
63		PA7	A7	TIOCB2/PO23	MISOA-B/ET0_WOL			
64		PA6	A6	MTIC5V/MTCLKB/ TIOCA2/TMCI3/ PO22/POE10#	CTS5#/RTS5#SS5#/ MOSIA-B/ ET0_EXOUT			
65		PA5	A5	MTIOC6B/TIOCB1/ PO21	RSPCKA-B/ ET0_LINKSTA			
66		PA4	A4	MTIC5U/MTCLKA/ TIOCA1/TMRI0/ PO20	TXD5/SMOSI5/ SSDA5/SSLA0-B/ ET0_MDC		IRQ5-DS	
67		PA3	A3	MTIOC0D/MTCLKD/ TIOCD0/TCLKB/PO19	RXD5/SMISO5/ SSCL5/ET0_MDIO		IRQ6-DS	
68		PA2	A2	MTIOC7A/PO18	RXD5/SMISO5/ SSCL5/SSLA3-B			
69		PA1	A1	MTIOC0B/MTCLKC/ MTIOC7B/TIOCB0/ PO17	SCK5/SSLA2-B/ ET0_WOL		IRQ11	
70		PA0	A0/BC0#	MTIOC4A/ MTIOC6D/TIOCA0/ CACREF/PO16	SSLA1-B/ ET0_TX_EN/ RMII0_TXD_EN			
71		PE7	D15[A15/D15]	MTIOC6A/TOC1	MISOB-B	MMC_RES#-B/ SDHI_WP-B	IRQ7	AN105
72		PE6	D14[A14/D14]	MTIOC6C/ TIC1	MOSIB-B	MMC_CD-B/ SDHI_CD-B	IRQ6	AN104
73		PE5	D13[A13/D13]	MTIOC4C/ MTIOC2B	ET0_RX_CLK/ REF50CK0/ RSPCKB-B		IRQ5	AN103
74		PE4	D12[A12/D12]	MTIOC4D/MTIOC1A/ PO28	ET0_ERXD2/SSLB0-B			AN102
75		PE3	D11[A11/D11]	MTIOC4B/PO26/ POE8#/TOC3	CTS12#/RTS12#/ SS12#/ET0_ERXD3	MMC_D7-B		AN101
76		PE2	D10[A10/D10]	MTIOC4A/PO23/ TIC3	RXD12/SMISO12/ SSCL12/RXD12/ SSLB3-B	MMC_D6-B	IRQ7-DS	AN100

Table 1.8 List of Pin and Pin Functions (100-Pin LQFP) (4/4)

Pin Number	Power Supply Clock System Contro	I/O Port	Bus EXDMAC	Timer (MTU, TPU, TMR, PPG, RTC, CMTW, POE, CAC)	Communication (ETHERC, SC1g, SC1h, SC1i, RSPI, RIIC, CAN, USB)	Memory Interface Camera Interface (QSPI, SDHI, SDSI, MMCIF, PDC)	Interrupt	S12AD, R12DA
77		PE1	D9[A9/D9]	MTIOC4C/ MTIOC3B/PO18	TXD12/SMOS12/ SSDA12/TXDX12/ SIOX12/SSLB2-B	MMC_D5-B		ANEX1
78		PE0	D8[A8/D8]	MTIOC3D	SCK12/SSLB1-B	MMC_D4-B		ANEX0
79		PD7	D7[A7/D7]	MTIC5U/POE0#	SSLC3	MMC_D1-B/ SDHI_D1-B/ QIO1-B/ QMI-B	IRQ7	AN107
80		PD6	D6[A6/D6]	MTIC5V/MTIOC8A/ POE4#	SSLC2	MMC_D0-B/ SDHI_D0-B/ QIO0-B/ QMO-B	IRQ6	AN106
81		PD5	D5[A5/D5]	MTIC5W/MTIOC8C/ POE10#	SSLC1	MMC_CLK-B/ SDHI_CLK-B/ QSPCLK-B	IRQ5	AN113
82		PD4	D4[A4/D4]	MTIOC8B/POE11#	SSLC0	MMC_CMD-B/ SDHI_CMD-B/ QSSL-B	IRQ4	AN112
83		PD3	D3[A3/D3]	MTIOC8D/POE8#/ TOC2	RSPCKC	MMC_D3-B/ SDHI_D3-B/ QIO3-B	IRQ3	AN111
84		PD2	D2[A2/D2]	MTIOC4D/TIC2	CRX0/MISOC	MMC_D2-B/ SDHI_D2-B/ QIO2-B	IRQ2	AN110
85		PD1	D1[A1/D1]	MTIOC4B/POE0#	CTX0/MOSIC		IRQ1	AN109
86		PD0	D0[A0/D0]	POE4#			IRQ0	AN108
87		P47					IRQ15-DS	AN007
88		P46					IRQ14-DS	AN006
89		P45					IRQ13-DS	AN005
90		P44					IRQ12-DS	AN004
91		P43					IRQ11-DS	AN003
92		P42					IRQ10-DS	AN002
93		P41					IRQ9-DS	AN001
94	VREFL0							
95		P40					IRQ8-DS	AN000
96	VREFH0							
97	AVCC0							
98		P07					IRQ15	ADTRG0#
99	AVSS0							
100		P05					IRQ13	DA1

2. CPU

Figure 2.1 shows register set of the CPU.

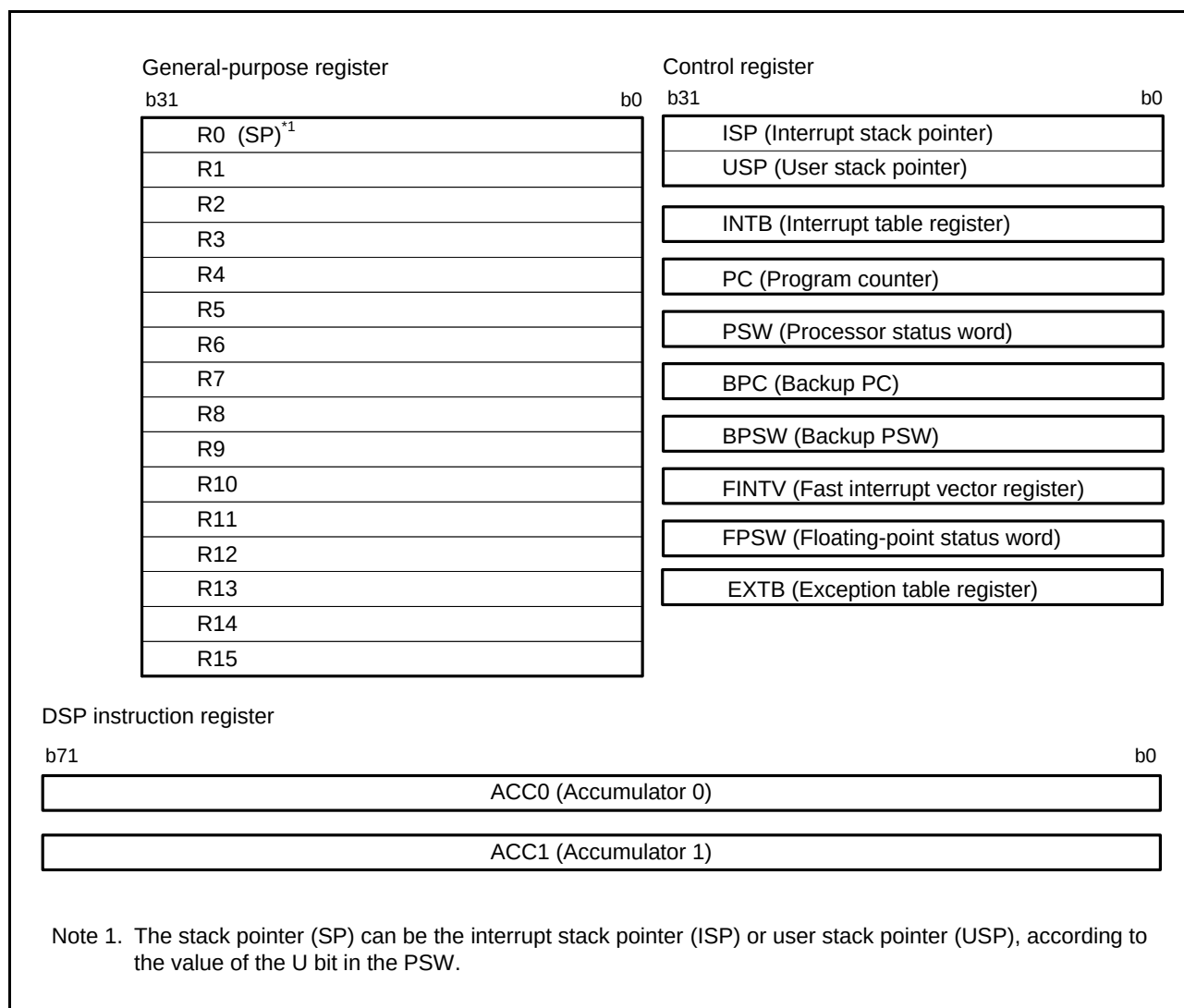


Figure 2.1 Register Set of the CPU

2.1 General-Purpose Registers (R0 to R15)

This CPU has sixteen 32-bit general-purpose registers (R0 to R15). R0 to R15 can be used as data registers or address registers.

R0, a general-purpose register, also functions as the stack pointer (SP).

The stack pointer is switched to operate as the interrupt stack pointer (ISP) or user stack pointer (USP) by the value of the stack pointer select bit (U) in the processor status word (PSW).

2.2 Control Registers

(1) Interrupt Stack Pointer (ISP) / User Stack Pointer (USP)

The stack pointer (SP) can be either of two types, the interrupt stack pointer (ISP) or the user stack pointer (USP).

Whether the stack pointer operates as the ISP or USP depends on the value of the stack pointer select bit (U) in the processor status word (PSW).

(2) Exception Table Register (EXTB)

The exception table register (EXTB) specifies the address where the exception vector table starts.

(3) Interrupt Table Register (INTB)

The interrupt table register (INTB) specifies the address where the interrupt vector table starts.

(4) Program Counter (PC)

The program counter (PC) indicates the address of the instruction being executed.

(5) Processor Status Word (PSW)

The processor status word (PSW) indicates the results of instruction execution or the state of the CPU.

(6) Backup PC (BPC)

The backup PC (BPC) is provided to speed up response to interrupts.

After a fast interrupt has been generated, the contents of the program counter (PC) are saved in the BPC register.

(7) Backup PSW (BPSW)

The backup PSW (BPSW) is provided to speed up response to interrupts.

After a fast interrupt has been generated, the contents of the processor status word (PSW) are saved in the BPSW. The allocation of bits in the BPSW corresponds to that in the PSW.

(8) Fast Interrupt Vector Register (FINTV)

The fast interrupt vector register (FINTV) is provided to speed up response to interrupts.

The FINTV register specifies a branch destination address when a fast interrupt has been generated.

(9) Floating-Point Status Word (FPSW)

The floating-point status word (FPSW) indicates the results of floating-point operations.

When an exception handling enable bit (Ej) enables the exception handling (Ej = 1), the exception cause can be identified by checking the corresponding Cj flag in the exception handling routine. If the exception handling is masked (Ej = 0), the occurrence of exception can be checked by reading the Fj flag at the end of a series of processing. Once the Fj flag has been set to 1, this value is retained until it is cleared to 0 by software (j = X, U, Z, O, or V).

2.3 Accumulator

The accumulator (ACC0 or ACC1) is a 72-bit register used for DSP instructions. The accumulator is handled as a 96-bit register for reading and writing. At this time, when bits 95 to 72 of the accumulator are read, the value where the value of bit 71 is sign extended is read. Writing to bits 95 to 72 of the accumulator is ignored. ACC0 is also used for the multiply and multiply-and-accumulate instructions; EMUL, EMULU, FMUL, MUL, and RMPA, in which case the prior value in ACC0 is modified by execution of the instruction.

Use the MVTACGU, MVTACHI, and MVTACLO instructions for writing to the accumulator. The MVTACGU, MVTACHI, and MVTACLO instructions write data to bits 95 to 64, the higher-order 32 bits (bits 63 to 32), and the lower-order 32 bits (bits 31 to 0), respectively.

Use the MVFACGU, MVFACHI, MVFACMI, and MVFACLO instructions for reading data from the accumulator. The MVFACGU, MVFACHI, MVFACMI, and MVFACLO instructions read data from the guard bits (bits 95 to 64), higher-order 32 bits (bits 63 to 32), the middle 32 bits (bits 47 to 16), and the lower-order 32 bits (bits 31 to 0), respectively.

3. Address Space

3.1 Address Space

This MCU has a 4-Gbyte address space, consisting of the range of addresses from 0000 0000h to FFFF FFFFh. That is, linear access to an address space of up to 4 Gbytes is possible, and this contains both program and data areas.

Figure 3.1 shows the memory maps in the respective operating modes. Accessible areas will differ according to the operating mode and states of control bits.

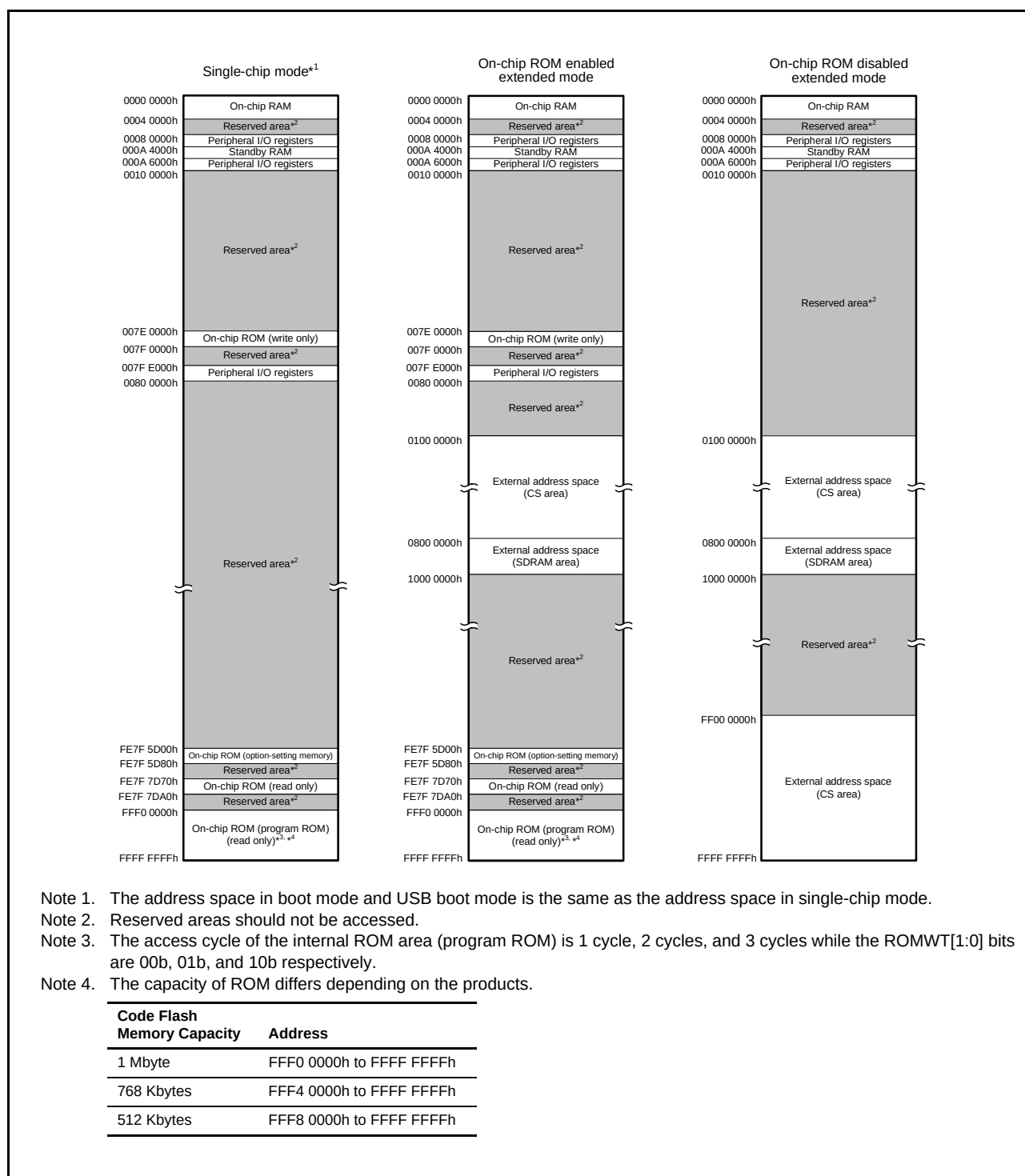
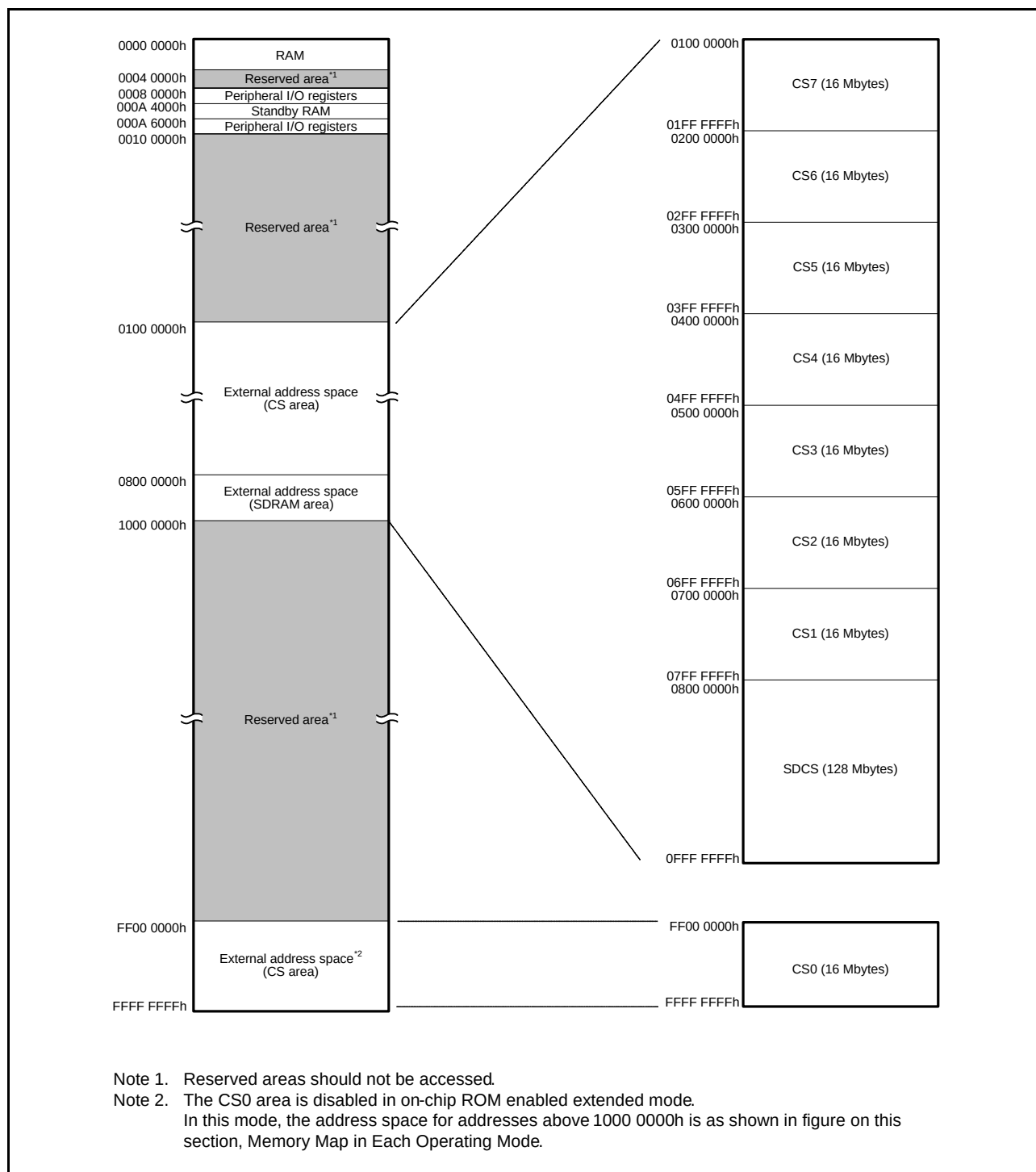


Figure 3.1 Memory Map in Each Operating Mode

3.2 External Address Space

The external address space is divided into CS areas (CS0 to CS7) and SDRAM area (SDCS). The CS areas are divided into up to eight areas (CS0 to CS7), each corresponding to the CSn# signal output from a CSn# (n = 0 to 7) pin.

Figure 3.2 shows the address ranges corresponding to the individual CS areas (CS0 to CS7) and SDRAM areas (SDCS) in on-chip ROM disabled extended mode.



**Figure 3.2 Correspondence between External Address Spaces and CS Areas
(In On-Chip ROM Disabled Extended Mode)**

4. I/O Registers

This section gives information on the on-chip I/O register addresses. The information is given as shown below. Notes on writing to registers are also given at the end.

(1) I/O register addresses (address order)

- Registers are listed from the lower allocation addresses.
- Registers are classified according to module symbols.
- The number of access cycles indicates the number of cycles based on the specified reference clock.
- Among the internal I/O register area, addresses not listed in the list of registers are reserved. Reserved addresses must not be accessed. Do not access these addresses; otherwise, the operation when accessing these bits and subsequent operations cannot be guaranteed.

(2) Notes on writing to I/O registers

When writing to an I/O register, the CPU starts executing the subsequent instruction before completing I/O register write. This may cause the subsequent instruction to be executed before the post-update I/O register value is reflected on the operation.

As described in the following examples, special care is required for the cases in which the subsequent instruction must be executed after the post-update I/O register value is actually reflected.

[Examples of cases requiring special care]

- The subsequent instruction must be executed while an interrupt request is disabled with the IENj bit in IERN of the ICU (interrupt request enable bit) set to 0.
- A WAIT instruction is executed immediately after the preprocessing for causing a transition to the low power consumption state.

In the above cases, after writing to an I/O register, wait until the write operation is completed using the following procedure and then execute the subsequent instruction.

- Write to an I/O register.
- Read the value from the I/O register to a general register.
- Execute the operation using the value read.
- Execute the subsequent instruction.

[Instruction examples]

- Byte-size I/O registers

```
MOV.L #SFR_ADDR, R1
MOV.B #SFR_DATA, [R1]
CMP [R1].UB, R1
;; Next process
```

- Word-size I/O registers

```
MOV.L #SFR_ADDR, R1
MOV.W #SFR_DATA, [R1]
CMP [R1].W, R1
;; Next process
```

- Longword-size I/O registers

```
MOV.L #SFR_ADDR, R1
MOV.L #SFR_DATA, [R1]
CMP [R1].L, R1
;; Next process
```

If multiple registers are written to and a subsequent instruction should be executed after the write operations are entirely completed, only read the I/O register that was last written to and execute the operation using the value; it is not necessary to read or execute operation for all the registers that were written to.

(3) Number of Access Cycles to I/O Registers

For the number of I/O register access cycles, refer to Table 4.1, List of I/O Registers (Address Order).

The number of access cycles to I/O registers is obtained by following equation.*1

Number of access cycles to I/O registers = Number of bus cycles for internal main bus 1 +
 Number of divided clock synchronization cycles +
 Number of bus cycles for internal peripheral busses 1 to 6

The number of bus cycles of internal peripheral bus 1 to 6 differs according to the register to be accessed.

When peripheral functions connected to internal peripheral bus 2 to 6 or registers for the external bus control unit (except for bus error related registers) are accessed, the number of divided clock synchronization cycles is added.

The number of divided clock synchronization cycles differs depending on the frequency ratio between ICLK and PCLK (or FCLK, BCLK) or bus access timing.

In the peripheral function unit, when the frequency ratio of ICLK is equal to or greater than that of PCLK (or FCLK), the sum of the number of bus cycles for internal main bus 1 and the number of the divided clock synchronization cycles will be one cycle of PCLK (or FCLK) at a maximum. Therefore, one PCLK (or FCLK) has been added to the number of access states shown in Table 4.1.

When the frequency ratio of ICLK is lower than that of PCLK (or FCLK), the subsequent bus access is started from the ICLK cycle following the completion of the access to the peripheral functions. Therefore, the access cycles are described on an ICLK basis.

In the external bus control unit, the sum of the number of bus cycles for internal main bus 1 and the number of divided clock synchronization cycles will be one cycle of BCLK at a maximum. Therefore, one BCLK is added to the number of access cycles shown in Table 4.1.

Note 1. This applies to the number of cycles when the access from the CPU does not conflict with the instruction fetching to the external memory or bus access from the different bus master (DMAC or DTC).

(4) Notes on Sleep Mode and Mode Transitions

During sleep mode or mode transitions, do not write to the registers related to system control (indicated by 'SYSTEM' in the Module Symbol column in Table 4.1, List of I/O Registers (Address Order)).

(5) Restrictions in Relation to RMPA and String-Manipulation Instructions

The allocation of data to be handled by RMPA or string-manipulation instructions to I/O registers is prohibited, and operation is not guaranteed if this restriction is not observed.

4.1 I/O Register Addresses (Address Order)

Table 4.1 List of I/O Registers (Address Order) (1 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 0000h	SYSTM	Mode Monitor Register	MDMONR	16	16	3 ICLK		Operating Modes
0008 0006h	SYSTM	System Control Register 0	SYSCR0	16	16	3 ICLK		Operating Modes
0008 0008h	SYSTM	System Control Register 1	SYSCR1	16	16	3 ICLK		Operating Modes
0008 000Ch	SYSTM	Standby Control Register	SBYCR	16	16	3 ICLK		Low Power Consumption
0008 0010h	SYSTM	Module Stop Control Register A	MSTPCRA	32	32	3 ICLK		Low Power Consumption
0008 0014h	SYSTM	Module Stop Control Register B	MSTPCRB	32	32	3 ICLK		Low Power Consumption
0008 0018h	SYSTM	Module Stop Control Register C	MSTPCRC	32	32	3 ICLK		Low Power Consumption
0008 001Ch	SYSTM	Module Stop Control Register D	MSTPCRD	32	32	3 ICLK		Low Power Consumption
0008 0020h	SYSTM	System Clock Control Register	SCKCR	32	32	3 ICLK		Clock Generation Circuit
0008 0024h	SYSTM	System Clock Control Register 2	SCKCR2	16	16	3 ICLK		Clock Generation Circuit
0008 0026h	SYSTM	System Clock Control Register 3	SCKCR3	16	16	3 ICLK		Clock Generation Circuit
0008 0028h	SYSTM	PLL Control Register	PLLCR	16	16	3 ICLK		Clock Generation Circuit
0008 002Ah	SYSTM	PLL Control Register 2	PLLCR2	8	8	3 ICLK		Clock Generation Circuit
0008 0030h	SYSTM	External Bus Clock Control Register	BCKCR	8	8	3 ICLK		Clock Generation Circuit
0008 0032h	SYSTM	Main Clock Oscillator Control Register	MOSCCR	8	8	3 ICLK		Clock Generation Circuit
0008 0033h	SYSTM	Sub-Clock Oscillator Control Register	SOSCCR	8	8	3 ICLK		Clock Generation Circuit
0008 0034h	SYSTM	Low-Speed On-Chip Oscillator Control Register	LOCOCR	8	8	3 ICLK		Clock Generation Circuit
0008 0035h	SYSTM	IWDT-Dedicated On-Chip Oscillator Control Register	ILOCOCR	8	8	3 ICLK		Clock Generation Circuit

Table 4.1 List of I/O Registers (Address Order) (2 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 0036h	SYSTEM	High-Speed On-Chip Oscillator Control Register	HOCOCR	8	8	3 ICLK		Clock Generation Circuit
0008 0037h	SYSTEM	High-Speed On-Chip Oscillator Control Register 2	HOCOCR2	8	8	3 ICLK		Clock Generation Circuit
0008 003Ch	SYSTEM	Oscillation Stabilization Flag Register	OSCOVFSR	8	8	3 ICLK		Clock Generation Circuit
0008 0040h	SYSTEM	Oscillation Stop Detection Control Register	OSTDCR	8	8	3 ICLK		Clock Generation Circuit
0008 0041h	SYSTEM	Oscillation Stop Detection Status Register	OSTDSR	8	8	3 ICLK		Clock Generation Circuit
0008 00A0h	SYSTEM	Operating Power Control Register	OPCCR	8	8	3 ICLK		Low Power Consumption
0008 00A1h	SYSTEM	Sleep Mode Return Clock Source Switching Register	RSTCKCR	8	8	3 ICLK		Low Power Consumption
0008 00A2h	SYSTEM	Main Clock Oscillator Wait Control Register	MOSCWTCR	8	8	3 ICLK		Clock Generation Circuit
0008 00A3h	SYSTEM	Sub-Clock Oscillator Wait Control Register	SOSCWTCR	8	8	3 ICLK		Clock Generation Circuit
0008 00C0h	SYSTEM	Reset Status Register 2	RSTSR2	8	8	3 ICLK		Resets
0008 00C2h	SYSTEM	Software Reset Register	SWRR	16	16	3 ICLK		Resets
0008 00E0h	SYSTEM	Voltage Monitoring 1 Circuit Control Register 1	LVD1CR1	8	8	3 ICLK		LVDA
0008 00E1h	SYSTEM	Voltage Monitoring 1 Circuit Status Register	LVD1SR	8	8	3 ICLK		LVDA
0008 00E2h	SYSTEM	Voltage Monitoring 2 Circuit Control Register 1	LVD2CR1	8	8	3 ICLK		LVDA
0008 00E3h	SYSTEM	Voltage Monitoring 2 Circuit Status Register	LVD2SR	8	8	3 ICLK		LVDA
0008 03FEh	SYSTEM	Protect Register	PRCR	16	16	3 ICLK		Register Write Protection Function
0008 1000h	FLASH	ROM Cache Enable Register	ROMCE	16	16	2 ICLK		Flash
0008 1004h	FLASH	ROM Cache Invalidate Register	ROMCIV	16	16	2 ICLK		Flash
0008 101Ch	SYSTEM	ROM Wait Cycle Setting Register	ROMWT	8	8	2 ICLK		Clock Generation Circuit
0008 1200h	RAM	RAM Operating Mode Control Register	RAMMODE	8	8	2 ICLK		RAM
0008 1201h	RAM	RAM Error Status Register	RAMSTS	8	8	2 ICLK		RAM
0008 1204h	RAM	RAM Protection Register	RAMPRCR	8	8	2 ICLK		RAM
0008 1208h	RAM	RAM Error Address Capture Register	RAMECAD	32	32	2 ICLK		RAM
0008 1300h	BSC	Bus Error Status Clear Register	BERCLR	8	8	2 ICLK		Buses
0008 1304h	BSC	Bus Error Monitoring Enable Register	BEREN	8	8	2 ICLK		Buses
0008 1308h	BSC	Bus Error Status Register 1	BERSR1	8	8	2 ICLK		Buses

Table 4.1 List of I/O Registers (Address Order) (3 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 130Ah	BSC	Bus Error Status Register 2	BERSR2	16	16	2 ICLK		Buses
0008 1310h	BSC	Bus Priority Control Register	BUSPRI	16	16	2 ICLK		Buses
0008 2000h	DMAC0	DMA Source Address Register	DMSAR	32	32	2 ICLK		DMACaA
0008 2004h	DMAC0	DMA Destination Address Register	DMDAR	32	32	2 ICLK		DMACaA
0008 2008h	DMAC0	DMA Transfer Count Register	DMCRA	32	32	2 ICLK		DMACaA
0008 200Ch	DMAC0	DMA Block Transfer Count Register	DMCRB	16	16	2 ICLK		DMACaA
0008 2010h	DMAC0	DMA Transfer Mode Register	DMTMD	16	16	2 ICLK		DMACaA
0008 2013h	DMAC0	DMA Interrupt Setting Register	DMINT	8	8	2 ICLK		DMACaA
0008 2014h	DMAC0	DMA Address Mode Register	DMAMD	16	16	2 ICLK		DMACaA
0008 2018h	DMAC0	DMA Offset Register	DMOFR	32	32	2 ICLK		DMACaA
0008 201Ch	DMAC0	DMA Transfer Enable Register	DMCNT	8	8	2 ICLK		DMACaA
0008 201Dh	DMAC0	DMA Software Start Register	DMREQ	8	8	2 ICLK		DMACaA
0008 201Eh	DMAC0	DMA Status Register	DMSTS	8	8	2 ICLK		DMACaA
0008 201Fh	DMAC0	DMA Request Source Flag Control Register	DMCSL	8	8	2 ICLK		DMACaA
0008 2040h	DMAC1	DMA Source Address Register	DMSAR	32	32	2 ICLK		DMACaA
0008 2044h	DMAC1	DMA Destination Address Register	DMDAR	32	32	2 ICLK		DMACaA
0008 2048h	DMAC1	DMA Transfer Count Register	DMCRA	32	32	2 ICLK		DMACaA
0008 204Ch	DMAC1	DMA Block Transfer Count Register	DMCRB	16	16	2 ICLK		DMACaA
0008 2050h	DMAC1	DMA Transfer Mode Register	DMTMD	16	16	2 ICLK		DMACaA
0008 2053h	DMAC1	DMA Interrupt Setting Register	DMINT	8	8	2 ICLK		DMACaA
0008 2054h	DMAC1	DMA Address Mode Register	DMAMD	16	16	2 ICLK		DMACaA
0008 205Ch	DMAC1	DMA Transfer Enable Register	DMCNT	8	8	2 ICLK		DMACaA
0008 205Dh	DMAC1	DMA Software Start Register	DMREQ	8	8	2 ICLK		DMACaA
0008 205Eh	DMAC1	DMA Status Register	DMSTS	8	8	2 ICLK		DMACaA
0008 205Fh	DMAC1	DMA Request Source Flag Control Register	DMCSL	8	8	2 ICLK		DMACaA
0008 2080h	DMAC2	DMA Source Address Register	DMSAR	32	32	2 ICLK		DMACaA
0008 2084h	DMAC2	DMA Destination Address Register	DMDAR	32	32	2 ICLK		DMACaA
0008 2088h	DMAC2	DMA Transfer Count Register	DMCRA	32	32	2 ICLK		DMACaA
0008 208Ch	DMAC2	DMA Block Transfer Count Register	DMCRB	16	16	2 ICLK		DMACaA
0008 2090h	DMAC2	DMA Transfer Mode Register	DMTMD	16	16	2 ICLK		DMACaA
0008 2093h	DMAC2	DMA Interrupt Setting Register	DMINT	8	8	2 ICLK		DMACaA
0008 2094h	DMAC2	DMA Address Mode Register	DMAMD	16	16	2 ICLK		DMACaA
0008 209Ch	DMAC2	DMA Transfer Enable Register	DMCNT	8	8	2 ICLK		DMACaA
0008 209Dh	DMAC2	DMA Software Start Register	DMREQ	8	8	2 ICLK		DMACaA
0008 209Eh	DMAC2	DMA Status Register	DMSTS	8	8	2 ICLK		DMACaA
0008 209Fh	DMAC2	DMA Request Source Flag Control Register	DMCSL	8	8	2 ICLK		DMACaA
0008 20C0h	DMAC3	DMA Source Address Register	DMSAR	32	32	2 ICLK		DMACaA
0008 20C4h	DMAC3	DMA Destination Address Register	DMDAR	32	32	2 ICLK		DMACaA
0008 20C8h	DMAC3	DMA Transfer Count Register	DMCRA	32	32	2 ICLK		DMACaA
0008 20CCh	DMAC3	DMA Block Transfer Count Register	DMCRB	16	16	2 ICLK		DMACaA
0008 20D0h	DMAC3	DMA Transfer Mode Register	DMTMD	16	16	2 ICLK		DMACaA
0008 20D3h	DMAC3	DMA Interrupt Setting Register	DMINT	8	8	2 ICLK		DMACaA
0008 20D4h	DMAC3	DMA Address Mode Register	DMAMD	16	16	2 ICLK		DMACaA
0008 20DCh	DMAC3	DMA Transfer Enable Register	DMCNT	8	8	2 ICLK		DMACaA
0008 20DDh	DMAC3	DMA Software Start Register	DMREQ	8	8	2 ICLK		DMACaA
0008 20DEh	DMAC3	DMA Status Register	DMSTS	8	8	2 ICLK		DMACaA
0008 20DFh	DMAC3	DMA Request Source Flag Control Register	DMCSL	8	8	2 ICLK		DMACaA
0008 2100h	DMAC4	DMA Source Address Register	DMSAR	32	32	2 ICLK		DMACaA
0008 2104h	DMAC4	DMA Destination Address Register	DMDAR	32	32	2 ICLK		DMACaA
0008 2108h	DMAC4	DMA Transfer Count Register	DMCRA	32	32	2 ICLK		DMACaA

Table 4.1 List of I/O Registers (Address Order) (4 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 210Ch	DMAC4	DMA Block Transfer Count Register	DMCRB	16	16	2	ICLK	DMACaA
0008 2110h	DMAC4	DMA Transfer Mode Register	DMTMD	16	16	2	ICLK	DMACaA
0008 2113h	DMAC4	DMA Interrupt Setting Register	DMINT	8	8	2	ICLK	DMACaA
0008 2114h	DMAC4	DMA Address Mode Register	DMAMD	16	16	2	ICLK	DMACaA
0008 211Ch	DMAC4	DMA Transfer Enable Register	DMCNT	8	8	2	ICLK	DMACaA
0008 211Dh	DMAC4	DMA Software Start Register	DMREQ	8	8	2	ICLK	DMACaA
0008 211Eh	DMAC4	DMA Status Register	DMSTS	8	8	2	ICLK	DMACaA
0008 211Fh	DMAC4	DMA Request Source Flag Control Register	DMCSL	8	8	2	ICLK	DMACaA
0008 2140h	DMAC5	DMA Source Address Register	DMSAR	32	32	2	ICLK	DMACaA
0008 2144h	DMAC5	DMA Destination Address Register	DMDAR	32	32	2	ICLK	DMACaA
0008 2148h	DMAC5	DMA Transfer Count Register	DMCRA	32	32	2	ICLK	DMACaA
0008 214Ch	DMAC5	DMA Block Transfer Count Register	DMCRB	16	16	2	ICLK	DMACaA
0008 2150h	DMAC5	DMA Transfer Mode Register	DMTMD	16	16	2	ICLK	DMACaA
0008 2153h	DMAC5	DMA Interrupt Setting Register	DMINT	8	8	2	ICLK	DMACaA
0008 2154h	DMAC5	DMA Address Mode Register	DMAMD	16	16	2	ICLK	DMACaA
0008 215Ch	DMAC5	DMA Transfer Enable Register	DMCNT	8	8	2	ICLK	DMACaA
0008 215Dh	DMAC5	DMA Software Start Register	DMREQ	8	8	2	ICLK	DMACaA
0008 215Eh	DMAC5	DMA Status Register	DMSTS	8	8	2	ICLK	DMACaA
0008 215Fh	DMAC5	DMA Request Source Flag Control Register	DMCSL	8	8	2	ICLK	DMACaA
0008 2180h	DMAC6	DMA Source Address Register	DMSAR	32	32	2	ICLK	DMACaA
0008 2184h	DMAC6	DMA Destination Address Register	DMDAR	32	32	2	ICLK	DMACaA
0008 2188h	DMAC6	DMA Transfer Count Register	DMCRA	32	32	2	ICLK	DMACaA
0008 218Ch	DMAC6	DMA Block Transfer Count Register	DMCRB	16	16	2	ICLK	DMACaA
0008 2190h	DMAC6	DMA Transfer Mode Register	DMTMD	16	16	2	ICLK	DMACaA
0008 2193h	DMAC6	DMA Interrupt Setting Register	DMINT	8	8	2	ICLK	DMACaA
0008 2194h	DMAC6	DMA Address Mode Register	DMAMD	16	16	2	ICLK	DMACaA
0008 219Ch	DMAC6	DMA Transfer Enable Register	DMCNT	8	8	2	ICLK	DMACaA
0008 219Dh	DMAC6	DMA Software Start Register	DMREQ	8	8	2	ICLK	DMACaA
0008 219Eh	DMAC6	DMA Status Register	DMSTS	8	8	2	ICLK	DMACaA
0008 219Fh	DMAC6	DMA Request Source Flag Control Register	DMCSL	8	8	2	ICLK	DMACaA
0008 21C0h	DMAC7	DMA Source Address Register	DMSAR	32	32	2	ICLK	DMACaA
0008 21C4h	DMAC7	DMA Destination Address Register	DMDAR	32	32	2	ICLK	DMACaA
0008 21C8h	DMAC7	DMA Transfer Count Register	DMCRA	32	32	2	ICLK	DMACaA
0008 21CCh	DMAC7	DMA Block Transfer Count Register	DMCRB	16	16	2	ICLK	DMACaA
0008 21D0h	DMAC7	DMA Transfer Mode Register	DMTMD	16	16	2	ICLK	DMACaA
0008 21D3h	DMAC7	DMA Interrupt Setting Register	DMINT	8	8	2	ICLK	DMACaA
0008 21D4h	DMAC7	DMA Address Mode Register	DMAMD	16	16	2	ICLK	DMACaA
0008 21DCh	DMAC7	DMA Transfer Enable Register	DMCNT	8	8	2	ICLK	DMACaA
0008 21DDh	DMAC7	DMA Software Start Register	DMREQ	8	8	2	ICLK	DMACaA
0008 21DEh	DMAC7	DMA Status Register	DMSTS	8	8	2	ICLK	DMACaA
0008 21DFh	DMAC7	DMA Request Source Flag Control Register	DMCSL	8	8	2	ICLK	DMACaA
0008 2200h	DMAC	DMAC Module Start Register	DMAST	8	8	2	ICLK	DMACaA
0008 2204h	DMAC	DMAC74 Interrupt Status Monitor Register	DMIST	8	8	2	ICLK	DMACaA
0008 2400h	DTC	DTC Control Register	DTCCR	8	8	2	ICLK	DTCb
0008 2404h	DTC	DTC Vector Base Register	DTCVBR	32	32	2	ICLK	DTCb
0008 2408h	DTC	DTC Address Mode Register	DTCADMOD	8	8	2	ICLK	DTCb
0008 240Ch	DTC	DTC Module Start Register	DTCST	8	8	2	ICLK	DTCb
0008 240Eh	DTC	DTC Status Register	DTCSTS	16	16	2	ICLK	DTCb
0008 2410h	DTC	DTC Index Table Base Register	DTCIBR	32	32	2	ICLK	DTCb
0008 2414h	DTC	DTC Operation Register	DTCOR	8	8	2	ICLK	DTCb

Table 4.1 List of I/O Registers (Address Order) (5 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 2416h	DTC	DTC Sequence Transfer Enable Register	DTCSQE	16	16	2 ICLK		DTCb
0008 2418h	DTC	DTC Address Displacement Register	DTCDISP	32	32	2 ICLK		DTCb
0008 2800h	EXDMA C0	EXDMA Source Address Register	EDMSAR	32	32	1, 2 BCLK		EXDMA Ca
0008 2804h	EXDMA C0	EXDMA Destination Address Register	EDMDAR	32	32	1, 2 BCLK		EXDMA Ca
0008 2808h	EXDMA C0	EXDMA Transfer Count Register	EDMCRA	32	32	1, 2 BCLK		EXDMA Ca
0008 280Ch	EXDMA C0	EXDMA Block Transfer Count Register	EDMCRB	16	16	1, 2 BCLK		EXDMA Ca
0008 2810h	EXDMA C0	EXDMA Transfer Mode Register	EDMTMD	16	16	1, 2 BCLK		EXDMA Ca
0008 2812h	EXDMA C0	EXDMA Output Setting Register	EDMOMD	8	8	1, 2 BCLK		EXDMA Ca
0008 2813h	EXDMA C0	EXDMA Interrupt Setting Register	EDMINT	8	8	1, 2 BCLK		EXDMA Ca
0008 2814h	EXDMA C0	EXDMA Address Mode Register	EDMAMD	32	32	1, 2 BCLK		EXDMA Ca
0008 2818h	EXDMA C0	EXDMA Offset Register	EDMOFR	32	32	1, 2 BCLK		EXDMA Ca
0008 281Ch	EXDMA C0	EXDMA Transfer Enable Register	EDMCNT	8	8	1, 2 BCLK		EXDMA Ca
0008 281Dh	EXDMA C0	EXDMA Software Start Register	EDMREQ	8	8	1, 2 BCLK		EXDMA Ca
0008 281Eh	EXDMA C0	EXDMA Status Register	EDMSTS	8	8	1, 2 BCLK		EXDMA Ca
0008 2820h	EXDMA C0	EXDMA External Request Sense Mode Register	EDMRMD	8	8	1, 2 BCLK		EXDMA Ca
0008 2821h	EXDMA C0	EXDMA External Request Flag Register	EDMERF	8	8	1, 2 BCLK		EXDMA Ca
0008 2822h	EXDMA C0	EXDMA Peripheral Request Flag Register	EDMPRF	8	8	1, 2 BCLK		EXDMA Ca
0008 2840h	EXDMA C1	EXDMA Source Address Register	EDMSAR	32	32	1, 2 BCLK		EXDMA Ca
0008 2844h	EXDMA C1	EXDMA Destination Address Register	EDMDAR	32	32	1, 2 BCLK		EXDMA Ca
0008 2848h	EXDMA C1	EXDMA Transfer Count Register	EDMCRA	32	32	1, 2 BCLK		EXDMA Ca
0008 284Ch	EXDMA C1	EXDMA Block Transfer Count Register	EDMCRB	16	16	1, 2 BCLK		EXDMA Ca
0008 2850h	EXDMA C1	EXDMA Transfer Mode Register	EDMTMD	16	16	1, 2 BCLK		EXDMA Ca
0008 2852h	EXDMA C1	EXDMA Output Setting Register	EDMOMD	8	8	1, 2 BCLK		EXDMA Ca
0008 2853h	EXDMA C1	EXDMA Interrupt Setting Register	EDMINT	8	8	1, 2 BCLK		EXDMA Ca
0008 2854h	EXDMA C1	EXDMA Address Mode Register	EDMAMD	32	32	1, 2 BCLK		EXDMA Ca
0008 285Ch	EXDMA C1	EXDMA Transfer Enable Register	EDMCNT	8	8	1, 2 BCLK		EXDMA Ca
0008 285Dh	EXDMA C1	EXDMA Software Start Register	EDMREQ	8	8	1, 2 BCLK		EXDMA Ca
0008 285Eh	EXDMA C1	EXDMA Status Register	EDMSTS	8	8	1, 2 BCLK		EXDMA Ca
0008 2860h	EXDMA C1	EXDMA External Request Sense Mode Register	EDMRMD	8	8	1, 2 BCLK		EXDMA Ca
0008 2861h	EXDMA C1	EXDMA External Request Flag Register	EDMERF	8	8	1, 2 BCLK		EXDMA Ca
0008 2862h	EXDMA C1	EXDMA Peripheral Request Flag Register	EDMPRF	8	8	1, 2 BCLK		EXDMA Ca
0008 2A00h	EXDMA C	EXDMAC Module Start Register	EDMAST	8	8	1, 2 BCLK		EXDMA Ca

Table 4.1 List of I/O Registers (Address Order) (6 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 2BE0h	EXDMA C	Cluster Buffer Register 0	CLSB0R	32	32	1, 2 BCLK		EXDMA Ca
0008 2BE4h	EXDMA C	Cluster Buffer Register 1	CLSB1R	32	32	1, 2 BCLK		EXDMA Ca
0008 2BE8h	EXDMA C	Cluster Buffer Register 2	CLSB2R	32	32	1, 2 BCLK		EXDMA Ca
0008 2BECh	EXDMA C	Cluster Buffer Register 3	CLSB3R	32	32	1, 2 BCLK		EXDMA Ca
0008 2BF0h	EXDMA C	Cluster Buffer Register 4	CLSB4R	32	32	1, 2 BCLK		EXDMA Ca
0008 2BF4h	EXDMA C	Cluster Buffer Register 5	CLSB5R	32	32	1, 2 BCLK		EXDMA Ca
0008 2BF8h	EXDMA C	Cluster Buffer Register 6	CLSB6R	32	32	1, 2 BCLK		EXDMA Ca
0008 2BFCh	EXDMA C	Cluster Buffer Register 7	CLSB7R	32	32	1, 2 BCLK		EXDMA Ca
0008 3002h	BSC	CS0 Mode Register	CS0MOD	16	16	1, 2 BCLK		Buses
0008 3004h	BSC	CS0 Wait Control Register 1	CS0WCR1	32	32	1, 2 BCLK		Buses
0008 3008h	BSC	CS0 Wait Control Register 2	CS0WCR2	32	32	1, 2 BCLK		Buses
0008 3012h	BSC	CS1 Mode Register	CS1MOD	16	16	1, 2 BCLK		Buses
0008 3014h	BSC	CS1 Wait Control Register 1	CS1WCR1	32	32	1, 2 BCLK		Buses
0008 3018h	BSC	CS1 Wait Control Register 2	CS1WCR2	32	32	1, 2 BCLK		Buses
0008 3022h	BSC	CS2 Mode Register	CS2MOD	16	16	1, 2 BCLK		Buses
0008 3024h	BSC	CS2 Wait Control Register 1	CS2WCR1	32	32	1, 2 BCLK		Buses
0008 3028h	BSC	CS2 Wait Control Register 2	CS2WCR2	32	32	1, 2 BCLK		Buses
0008 3032h	BSC	CS3 Mode Register	CS3MOD	16	16	1, 2 BCLK		Buses
0008 3034h	BSC	CS3 Wait Control Register 1	CS3WCR1	32	32	1, 2 BCLK		Buses
0008 3038h	BSC	CS3 Wait Control Register 2	CS3WCR2	32	32	1, 2 BCLK		Buses
0008 3042h	BSC	CS4 Mode Register	CS4MOD	16	16	1, 2 BCLK		Buses
0008 3044h	BSC	CS4 Wait Control Register 1	CS4WCR1	32	32	1, 2 BCLK		Buses
0008 3048h	BSC	CS4 Wait Control Register 2	CS4WCR2	32	32	1, 2 BCLK		Buses
0008 3052h	BSC	CS5 Mode Register	CS5MOD	16	16	1, 2 BCLK		Buses
0008 3054h	BSC	CS5 Wait Control Register 1	CS5WCR1	32	32	1, 2 BCLK		Buses
0008 3058h	BSC	CS5 Wait Control Register 2	CS5WCR2	32	32	1, 2 BCLK		Buses
0008 3062h	BSC	CS6 Mode Register	CS6MOD	16	16	1, 2 BCLK		Buses
0008 3064h	BSC	CS6 Wait Control Register 1	CS6WCR1	32	32	1, 2 BCLK		Buses
0008 3068h	BSC	CS6 Wait Control Register 2	CS6WCR2	32	32	1, 2 BCLK		Buses
0008 3072h	BSC	CS7 Mode Register	CS7MOD	16	16	1, 2 BCLK		Buses
0008 3074h	BSC	CS7 Wait Control Register 1	CS7WCR1	32	32	1, 2 BCLK		Buses
0008 3078h	BSC	CS7 Wait Control Register 2	CS7WCR2	32	32	1, 2 BCLK		Buses
0008 3802h	BSC	CS0 Control Register	CS0CR	16	16	1, 2 BCLK		Buses
0008 380Ah	BSC	CS0 Recovery Cycle Register	CS0REC	16	16	1, 2 BCLK		Buses
0008 3812h	BSC	CS1 Control Register	CS1CR	16	16	1, 2 BCLK		Buses
0008 381Ah	BSC	CS1 Recovery Cycle Register	CS1REC	16	16	1, 2 BCLK		Buses
0008 3822h	BSC	CS2 Control Register	CS2CR	16	16	1, 2 BCLK		Buses
0008 382Ah	BSC	CS2 Recovery Cycle Register	CS2REC	16	16	1, 2 BCLK		Buses
0008 3832h	BSC	CS3 Control Register	CS3CR	16	16	1, 2 BCLK		Buses
0008 383Ah	BSC	CS3 Recovery Cycle Register	CS3REC	16	16	1, 2 BCLK		Buses
0008 3842h	BSC	CS4 Control Register	CS4CR	16	16	1, 2 BCLK		Buses
0008 384Ah	BSC	CS4 Recovery Cycle Register	CS4REC	16	16	1, 2 BCLK		Buses
0008 3852h	BSC	CS5 Control Register	CS5CR	16	16	1, 2 BCLK		Buses
0008 385Ah	BSC	CS5 Recovery Cycle Register	CS5REC	16	16	1, 2 BCLK		Buses
0008 3862h	BSC	CS6 Control Register	CS6CR	16	16	1, 2 BCLK		Buses

Table 4.1 List of I/O Registers (Address Order) (7 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 386Ah	BSC	CS6 Recovery Cycle Register	CS6REC	16	16	1, 2 BCLK		Buses
0008 3872h	BSC	CS7 Control Register	CS7CR	16	16	1, 2 BCLK		Buses
0008 387Ah	BSC	CS7 Recovery Cycle Register	CS7REC	16	16	1, 2 BCLK		Buses
0008 3880h	BSC	CS Recovery Cycle Insertion Enable Register	CSRECEN	16	16	1, 2 BCLK		Buses
0008 3C00h	BSC	SDC Control Register	SDCCR	8	8	1, 2 BCLK		Buses
0008 3C01h	BSC	SDC Mode Register	SDCMOD	8	8	1, 2 BCLK		Buses
0008 3C02h	BSC	SDRAM Access Mode Register	SDAMOD	8	8	1, 2 BCLK		Buses
0008 3C10h	BSC	SDRAM Self-Refresh Control Register	SDSELF	8	8	1, 2 BCLK		Buses
0008 3C14h	BSC	SDRAM Refresh Control Register	SDRFCR	16	16	1, 2 BCLK		Buses
0008 3C16h	BSC	SDRAM Auto-Refresh Control Register	SDRFEN	8	8	1, 2 BCLK		Buses
0008 3C20h	BSC	SDRAM Initialization Sequence Control Register	SDICR	8	8	1, 2 BCLK		Buses
0008 3C24h	BSC	SDRAM Initialization Register	SDIR	16	16	1, 2 BCLK		Buses
0008 3C40h	BSC	SDRAM Address Register	SDADR	8	8	1, 2 BCLK		Buses
0008 3C44h	BSC	SDRAM Timing Register	SDTR	32	32	1, 2 BCLK		Buses
0008 3C48h	BSC	SDRAM Mode Register	SDMOD	16	16	1, 2 BCLK		Buses
0008 3C50h	BSC	SDRAM Status Register	SDSR	8	8	1, 2 BCLK		Buses
0008 6400h	MPU	Region-0 Start Page Number Register	RSPAGE0	32	32	1 ICLK		MPU
0008 6404h	MPU	Region-0 End Page Number Register	REPAGE0	32	32	1 ICLK		MPU
0008 6408h	MPU	Region-1 Start Page Number Register	RSPAGE1	32	32	1 ICLK		MPU
0008 640Ch	MPU	Region-1 End Page Number Register	REPAGE1	32	32	1 ICLK		MPU
0008 6410h	MPU	Region-2 Start Page Number Register	RSPAGE2	32	32	1 ICLK		MPU
0008 6414h	MPU	Region-2 End Page Number Register	REPAGE2	32	32	1 ICLK		MPU
0008 6418h	MPU	Region-3 Start Page Number Register	RSPAGE3	32	32	1 ICLK		MPU
0008 641Ch	MPU	Region-3 End Page Number Register	REPAGE3	32	32	1 ICLK		MPU
0008 6420h	MPU	Region-4 Start Page Number Register	RSPAGE4	32	32	1 ICLK		MPU
0008 6424h	MPU	Region-4 End Page Number Register	REPAGE4	32	32	1 ICLK		MPU
0008 6428h	MPU	Region-5 Start Page Number Register	RSPAGE5	32	32	1 ICLK		MPU
0008 642Ch	MPU	Region-5 End Page Number Register	REPAGE5	32	32	1 ICLK		MPU
0008 6430h	MPU	Region-6 Start Page Number Register	RSPAGE6	32	32	1 ICLK		MPU
0008 6434h	MPU	Region-6 End Page Number Register	REPAGE6	32	32	1 ICLK		MPU
0008 6438h	MPU	Region-7 Start Page Number Register	RSPAGE7	32	32	1 ICLK		MPU
0008 643Ch	MPU	Region-7 End Page Number Register	REPAGE7	32	32	1 ICLK		MPU
0008 6500h	MPU	Memory-Protection Enable Register	MPEN	32	32	1 ICLK		MPU
0008 6504h	MPU	Background Access Control Register	MPBAC	32	32	1 ICLK		MPU
0008 6508h	MPU	Memory-Protection Error Status-Clearing Register	MPECLR	32	32	1 ICLK		MPU
0008 650Ch	MPU	Memory-Protection Error Status Register	MPESTS	32	32	1 ICLK		MPU
0008 6514h	MPU	Data Memory-Protection Error Address Register	MPDEA	32	32	1 ICLK		MPU
0008 6520h	MPU	Region Search Address Register	MPSA	32	32	1 ICLK		MPU
0008 6524h	MPU	Region Search Operation Register	MPOPS	16	16	1 ICLK		MPU
0008 6526h	MPU	Region Invalidation Operation Register	MPOPI	16	16	1 ICLK		MPU
0008 6528h	MPU	Instruction-Hit Region Register	MHITI	32	32	1 ICLK		MPU
0008 652Ch	MPU	Data-Hit Region Register	MHITD	32	32	1 ICLK		MPU
0008 7010h to 0008 70FFh	ICU	Interrupt Request Registers 016 to 255	IR016 to 255	8	8	2 ICLK		ICUB
0008 711Ah to 0008 71FFh	ICU	DTC Transfer Request Enable Registers 026 to 255	DTCE026 to DTCE255	8	8	2 ICLK		ICUB
0008 7202h to 0008 721Fh	ICU	Interrupt Request Enable Registers 02 to 1F	IER02 to IER1F	8	8	2 ICLK		ICUB
0008 72E0h	ICU	Software Interrupt Generation Register	SWINTR	8	8	2 ICLK		ICUB
0008 72E1h	ICU	Software Interrupt 2 Generation Register	SWINT2R	8	8	2 ICLK		ICUB
0008 72F0h	ICU	Fast Interrupt Set Register	FIR	16	16	2 ICLK		ICUB

Table 4.1 List of I/O Registers (Address Order) (8 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 7300h to 0008 73FFh	ICU	Interrupt Source Priority Registers 000 to 255	IPR000 to IPR255	8	8	2 ICLK		ICUB
0008 7400h	ICU	DMAC Trigger Select Register 0	DMRSR0	8	8	2 ICLK		ICUB
0008 7404h	ICU	DMAC Trigger Select Register 1	DMRSR1	8	8	2 ICLK		ICUB
0008 7408h	ICU	DMAC Trigger Select Register 2	DMRSR2	8	8	2 ICLK		ICUB
0008 740Ch	ICU	DMAC Trigger Select Register 3	DMRSR3	8	8	2 ICLK		ICUB
0008 7410h	ICU	DMAC Trigger Select Register 4	DMRSR4	8	8	2 ICLK		ICUB
0008 7414h	ICU	DMAC Trigger Select Register 5	DMRSR5	8	8	2 ICLK		ICUB
0008 7418h	ICU	DMAC Trigger Select Register 6	DMRSR6	8	8	2 ICLK		ICUB
0008 741Ch	ICU	DMAC Trigger Select Register 7	DMRSR7	8	8	2 ICLK		ICUB
0008 7500h to 0008 750Fh	ICU	IRQ Control Registers 0 to 15	IRQCR0 to 15	8	8	2 ICLK		ICUB
0008 7520h	ICU	IRQ Pin Digital Filter Enable Register 0	IRQFLTE0	8	8	2 ICLK		ICUB
0008 7521h	ICU	IRQ Pin Digital Filter Enable Register 1	IRQFLTE1	8	8	2 ICLK		ICUB
0008 7528h	ICU	IRQ Pin Digital Filter Setting Register 0	IRQFLTC0	16	16	2 ICLK		ICUB
0008 752Ah	ICU	IRQ Pin Digital Filter Setting Register 1	IRQFLTC1	16	16	2 ICLK		ICUB
0008 7580h	ICU	Non-Maskable Interrupt Status Register	NMISR	8	8	2 ICLK		ICUB
0008 7581h	ICU	Non-Maskable Interrupt Enable Register	NMIER	8	8	2 ICLK		ICUB
0008 7582h	ICU	Non-Maskable Interrupt Status Clear Register	NMICLR	8	8	2 ICLK		ICUB
0008 7583h	ICU	NMI Pin Interrupt Control Register	NMICR	8	8	2 ICLK		ICUB
0008 7590h	ICU	NMI Pin Digital Filter Enable Register	NMIFLTE	8	8	2 ICLK		ICUB
0008 7594h	ICU	NMI Pin Digital Filter Setting Register	NMIFLTC	8	8	2 ICLK		ICUB
0008 7600h	ICU	Group BE0 Interrupt Request Register	GRPBE0	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7630h	ICU	Group BL0 Interrupt Request Register	GRPBL0	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7634h	ICU	Group BL1 Interrupt Request Register	GRPBL1	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7638h	ICU	Group BL2 Interrupt Request Register	GRPBL2	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7640h	ICU	Group BE0 Interrupt Request Enable Register	GENBE0	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7670h	ICU	Group BL0 Interrupt Request Enable Register	GENBL0	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7674h	ICU	Group BL1 Interrupt Request Enable Register	GENBL1	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7678h	ICU	Group BL2 Interrupt Request Enable Register	GENBL2	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7680h	ICU	Group BE0 Interrupt Clear Register	GCRBE0	32	32	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7700h	ICU	Software Configurable Interrupt B Request Register 0	PIBR0	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7701h	ICU	Software Configurable Interrupt B Request Register 1	PIBR1	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7702h	ICU	Software Configurable Interrupt B Request Register 2	PIBR2	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7703h	ICU	Software Configurable Interrupt B Request Register 3	PIBR3	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7704h	ICU	Software Configurable Interrupt B Request Register 4	PIBR4	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7705h	ICU	Software Configurable Interrupt B Request Register 5	PIBR5	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7706h	ICU	Software Configurable Interrupt B Request Register 6	PIBR6	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7707h	ICU	Software Configurable Interrupt B Request Register 7	PIBR7	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7708h	ICU	Software Configurable Interrupt B Request Register 8	PIBR8	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7709h	ICU	Software Configurable Interrupt B Request Register 9	PIBR9	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 770Ah	ICU	Software Configurable Interrupt B Request Register A	PIBRA	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB

Table 4.1 List of I/O Registers (Address Order) (9 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 7780h	ICU	Software Configurable Interrupt B Source Select Register X128	SLIBXR128	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7781h	ICU	Software Configurable Interrupt B Source Select Register X129	SLIBXR129	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7782h	ICU	Software Configurable Interrupt B Source Select Register X130	SLIBXR130	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7783h	ICU	Software Configurable Interrupt B Source Select Register X131	SLIBXR131	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7784h	ICU	Software Configurable Interrupt B Source Select Register X132	SLIBXR132	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7785h	ICU	Software Configurable Interrupt B Source Select Register X133	SLIBXR133	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7786h	ICU	Software Configurable Interrupt B Source Select Register X134	SLIBXR134	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7787h	ICU	Software Configurable Interrupt B Source Select Register X135	SLIBXR135	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7788h	ICU	Software Configurable Interrupt B Source Select Register X136	SLIBXR136	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7789h	ICU	Software Configurable Interrupt B Source Select Register X137	SLIBXR137	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 778Ah	ICU	Software Configurable Interrupt B Source Select Register X138	SLIBXR138	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 778Bh	ICU	Software Configurable Interrupt B Source Select Register X139	SLIBXR139	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 778Ch	ICU	Software Configurable Interrupt B Source Select Register X140	SLIBXR140	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 778Dh	ICU	Software Configurable Interrupt B Source Select Register X141	SLIBXR141	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 778Eh	ICU	Software Configurable Interrupt B Source Select Register X142	SLIBXR142	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 778Fh	ICU	Software Configurable Interrupt B Source Select Register X143	SLIBXR143	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7790h	ICU	Software Configurable Interrupt B Source Select Register 144	SLIBR144	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7791h	ICU	Software Configurable Interrupt B Source Select Register 145	SLIBR145	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7792h	ICU	Software Configurable Interrupt B Source Select Register 146	SLIBR146	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7793h	ICU	Software Configurable Interrupt B Source Select Register 147	SLIBR147	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7794h	ICU	Software Configurable Interrupt B Source Select Register 148	SLIBR148	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7795h	ICU	Software Configurable Interrupt B Source Select Register 149	SLIBR149	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7796h	ICU	Software Configurable Interrupt B Source Select Register 150	SLIBR150	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7797h	ICU	Software Configurable Interrupt B Source Select Register 151	SLIBR151	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7798h	ICU	Software Configurable Interrupt B Source Select Register 152	SLIBR152	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7799h	ICU	Software Configurable Interrupt B Source Select Register 153	SLIBR153	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 779Ah	ICU	Software Configurable Interrupt B Source Select Register 154	SLIBR154	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 779Bh	ICU	Software Configurable Interrupt B Source Select Register 155	SLIBR155	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 779Ch	ICU	Software Configurable Interrupt B Source Select Register 156	SLIBR156	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 779Dh	ICU	Software Configurable Interrupt B Source Select Register 157	SLIBR157	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 779Eh	ICU	Software Configurable Interrupt B Source Select Register 158	SLIBR158	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB

Table 4.1 List of I/O Registers (Address Order) (10 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 779Fh	ICU	Software Configurable Interrupt B Source Select Register 159	SLIBR159	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A0h	ICU	Software Configurable Interrupt B Source Select Register 160	SLIBR160	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A1h	ICU	Software Configurable Interrupt B Source Select Register 161	SLIBR161	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A2h	ICU	Software Configurable Interrupt B Source Select Register 162	SLIBR162	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A3h	ICU	Software Configurable Interrupt B Source Select Register 163	SLIBR163	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A4h	ICU	Software Configurable Interrupt B Source Select Register 164	SLIBR164	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A5h	ICU	Software Configurable Interrupt B Source Select Register 165	SLIBR165	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A6h	ICU	Software Configurable Interrupt B Source Select Register 166	SLIBR166	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A7h	ICU	Software Configurable Interrupt B Source Select Register 167	SLIBR167	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A8h	ICU	Software Configurable Interrupt B Source Select Register 168	SLIBR168	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77A9h	ICU	Software Configurable Interrupt B Source Select Register 169	SLIBR169	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77AAh	ICU	Software Configurable Interrupt B Source Select Register 170	SLIBR170	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77ABh	ICU	Software Configurable Interrupt B Source Select Register 171	SLIBR171	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77ACh	ICU	Software Configurable Interrupt B Source Select Register 172	SLIBR172	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77ADh	ICU	Software Configurable Interrupt B Source Select Register 173	SLIBR173	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77AEh	ICU	Software Configurable Interrupt B Source Select Register 174	SLIBR174	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77AFh	ICU	Software Configurable Interrupt B Source Select Register 175	SLIBR175	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B0h	ICU	Software Configurable Interrupt B Source Select Register 176	SLIBR176	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B1h	ICU	Software Configurable Interrupt B Source Select Register 177	SLIBR177	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B2h	ICU	Software Configurable Interrupt B Source Select Register 178	SLIBR178	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B3h	ICU	Software Configurable Interrupt B Source Select Register 179	SLIBR179	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B4h	ICU	Software Configurable Interrupt B Source Select Register 180	SLIBR180	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B5h	ICU	Software Configurable Interrupt B Source Select Register 181	SLIBR181	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B6h	ICU	Software Configurable Interrupt B Source Select Register 182	SLIBR182	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B7h	ICU	Software Configurable Interrupt B Source Select Register 183	SLIBR183	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B8h	ICU	Software Configurable Interrupt B Source Select Register 184	SLIBR184	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77B9h	ICU	Software Configurable Interrupt B Source Select Register 185	SLIBR185	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77BAh	ICU	Software Configurable Interrupt B Source Select Register 186	SLIBR186	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77BBh	ICU	Software Configurable Interrupt B Source Select Register 187	SLIBR187	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77BCh	ICU	Software Configurable Interrupt B Source Select Register 188	SLIBR188	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77BDh	ICU	Software Configurable Interrupt B Source Select Register 189	SLIBR189	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB

Table 4.1 List of I/O Registers (Address Order) (11 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 77BEh	ICU	Software Configurable Interrupt B Source Select Register 190	SLIBR190	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77BFh	ICU	Software Configurable Interrupt B Source Select Register 191	SLIBR191	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C0h	ICU	Software Configurable Interrupt B Source Select Register 192	SLIBR192	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C1h	ICU	Software Configurable Interrupt B Source Select Register 193	SLIBR193	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C2h	ICU	Software Configurable Interrupt B Source Select Register 194	SLIBR194	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C3h	ICU	Software Configurable Interrupt B Source Select Register 195	SLIBR195	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C4h	ICU	Software Configurable Interrupt B Source Select Register 196	SLIBR196	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C5h	ICU	Software Configurable Interrupt B Source Select Register 197	SLIBR197	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C6h	ICU	Software Configurable Interrupt B Source Select Register 198	SLIBR198	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C7h	ICU	Software Configurable Interrupt B Source Select Register 199	SLIBR199	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C8h	ICU	Software Configurable Interrupt B Source Select Register 200	SLIBR200	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77C9h	ICU	Software Configurable Interrupt B Source Select Register 201	SLIBR201	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77CAh	ICU	Software Configurable Interrupt B Source Select Register 202	SLIBR202	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77CBh	ICU	Software Configurable Interrupt B Source Select Register 203	SLIBR203	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77CCh	ICU	Software Configurable Interrupt B Source Select Register 204	SLIBR204	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77CDh	ICU	Software Configurable Interrupt B Source Select Register 205	SLIBR205	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77CEh	ICU	Software Configurable Interrupt B Source Select Register 206	SLIBR206	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 77CFh	ICU	Software Configurable Interrupt B Source Select Register 207	SLIBR207	8	8	2 ICLK to 1 PCLKB	2 ICLK	ICUB
0008 7830h	ICU	Group AL0 Interrupt Request Register	GRPAL0	32	32	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7834h	ICU	Group AL1 Interrupt Request Register	GRPAL1	32	32	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7870h	ICU	Group AL0 Interrupt Request Enable Register	GENAL0	32	32	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7874h	ICU	Group AL1 Interrupt Request Enable Register	GENAL1	32	32	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7900h	ICU	Software Configurable Interrupt A Request Register 0	PIAR0	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7901h	ICU	Software Configurable Interrupt A Request Register 1	PIAR1	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7902h	ICU	Software Configurable Interrupt A Request Register 2	PIAR2	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7903h	ICU	Software Configurable Interrupt A Request Register 3	PIAR3	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7904h	ICU	Software Configurable Interrupt A Request Register 4	PIAR4	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7905h	ICU	Software Configurable Interrupt A Request Register 5	PIAR5	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 790Bh	ICU	Software Configurable Interrupt A Request Register B	PIARB	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D0h	ICU	Software Configurable Interrupt A Source Select Register 208	SLIAR208	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D1h	ICU	Software Configurable Interrupt A Source Select Register 209	SLIAR209	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D2h	ICU	Software Configurable Interrupt A Source Select Register 210	SLIAR210	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D3h	ICU	Software Configurable Interrupt A Source Select Register 211	SLIAR211	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB

Table 4.1 List of I/O Registers (Address Order) (12 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLKA	ICLK $<$ PCLKA	
0008 79D4h	ICU	Software Configurable Interrupt A Source Select Register 212	SLIAR212	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D5h	ICU	Software Configurable Interrupt A Source Select Register 213	SLIAR213	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D6h	ICU	Software Configurable Interrupt A Source Select Register 214	SLIAR214	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D7h	ICU	Software Configurable Interrupt A Source Select Register 215	SLIAR215	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D8h	ICU	Software Configurable Interrupt A Source Select Register 216	SLIAR216	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79D9h	ICU	Software Configurable Interrupt A Source Select Register 217	SLIAR217	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79DAh	ICU	Software Configurable Interrupt A Source Select Register 218	SLIAR218	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79DBh	ICU	Software Configurable Interrupt A Source Select Register 219	SLIAR219	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79DCh	ICU	Software Configurable Interrupt A Source Select Register 220	SLIAR220	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79DDh	ICU	Software Configurable Interrupt A Source Select Register 221	SLIAR221	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79DEh	ICU	Software Configurable Interrupt A Source Select Register 222	SLIAR222	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79DFh	ICU	Software Configurable Interrupt A Source Select Register 223	SLIAR223	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E0h	ICU	Software Configurable Interrupt A Source Select Register 224	SLIAR224	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E1h	ICU	Software Configurable Interrupt A Source Select Register 225	SLIAR225	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E2h	ICU	Software Configurable Interrupt A Source Select Register 226	SLIAR226	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E3h	ICU	Software Configurable Interrupt A Source Select Register 227	SLIAR227	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E4h	ICU	Software Configurable Interrupt A Source Select Register 228	SLIAR228	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E5h	ICU	Software Configurable Interrupt A Source Select Register 229	SLIAR229	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E6h	ICU	Software Configurable Interrupt A Source Select Register 230	SLIAR230	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E7h	ICU	Software Configurable Interrupt A Source Select Register 231	SLIAR231	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E8h	ICU	Software Configurable Interrupt A Source Select Register 232	SLIAR232	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79E9h	ICU	Software Configurable Interrupt A Source Select Register 233	SLIAR233	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79EAh	ICU	Software Configurable Interrupt A Source Select Register 234	SLIAR234	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79EBh	ICU	Software Configurable Interrupt A Source Select Register 235	SLIAR235	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79ECh	ICU	Software Configurable Interrupt A Source Select Register 236	SLIAR236	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79EDh	ICU	Software Configurable Interrupt A Source Select Register 237	SLIAR237	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79EEh	ICU	Software Configurable Interrupt A Source Select Register 238	SLIAR238	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79EFh	ICU	Software Configurable Interrupt A Source Select Register 239	SLIAR239	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F0h	ICU	Software Configurable Interrupt A Source Select Register 240	SLIAR240	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F1h	ICU	Software Configurable Interrupt A Source Select Register 241	SLIAR241	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F2h	ICU	Software Configurable Interrupt A Source Select Register 242	SLIAR242	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB

Table 4.1 List of I/O Registers (Address Order) (13 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 79F3h	ICU	Software Configurable Interrupt A Source Select Register 243	SLIAR243	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F4h	ICU	Software Configurable Interrupt A Source Select Register 244	SLIAR244	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F5h	ICU	Software Configurable Interrupt A Source Select Register 245	SLIAR245	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F6h	ICU	Software Configurable Interrupt A Source Select Register 246	SLIAR246	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F7h	ICU	Software Configurable Interrupt A Source Select Register 247	SLIAR247	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F8h	ICU	Software Configurable Interrupt A Source Select Register 248	SLIAR248	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79F9h	ICU	Software Configurable Interrupt A Source Select Register 249	SLIAR249	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79FAh	ICU	Software Configurable Interrupt A Source Select Register 250	SLIAR250	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79FBh	ICU	Software Configurable Interrupt A Source Select Register 251	SLIAR251	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79FCh	ICU	Software Configurable Interrupt A Source Select Register 252	SLIAR252	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79FDh	ICU	Software Configurable Interrupt A Source Select Register 253	SLIAR253	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79FEh	ICU	Software Configurable Interrupt A Source Select Register 254	SLIAR254	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 79FFh	ICU	Software Configurable Interrupt A Source Select Register 255	SLIAR255	8	8	2 ICLK to 1 PCLKA	2 ICLK	ICUB
0008 7A00h	ICU	Software Configurable Interrupt Source Select Register Write Protect Register	SLIPRCR	8	8	2 ICLK to 1 PCLKA/B	2 ICLK	ICUB
0008 7A01h	ICU	EXDMAC Trigger Select Register	SELEXDR	8	8	2 ICLK to 1 PCLKA/B	2 ICLK	ICUB
0008 8000h	CMT	Compare Match Timer Start Register 0	CMSTR0	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8002h	CMT0	Compare Match Timer Control Register	CMCR	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8004h	CMT0	Compare Match Counter	CMCNT	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8006h	CMT0	Compare Match Constant Register	CMCOR	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8008h	CMT1	Compare Match Timer Control Register	CMCR	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 800Ah	CMT1	Compare Match Counter	CMCNT	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 800Ch	CMT1	Compare Match Constant Register	CMCOR	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8010h	CMT	Compare Match Timer Start Register 1	CMSTR1	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8012h	CMT2	Compare Match Timer Control Register	CMCR	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8014h	CMT2	Compare Match Counter	CMCNT	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8016h	CMT2	Compare Match Constant Register	CMCOR	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8018h	CMT3	Compare Match Timer Control Register	CMCR	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 801Ah	CMT3	Compare Match Counter	CMCNT	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 801Ch	CMT3	Compare Match Constant Register	CMCOR	16	16	2, 3 PCLKB	2 ICLK	CMT
0008 8020h	WDT	WDT Refresh Register	WDTRR	8	8	2, 3 PCLKB	2 ICLK	WDTA
0008 8022h	WDT	WDT Control Register	WDTCR	16	16	2, 3 PCLKB	2 ICLK	WDTA
0008 8024h	WDT	WDT Status Register	WDTSR	16	16	2, 3 PCLKB	2 ICLK	WDTA
0008 8026h	WDT	WDT Reset Control Register	WDTRCR	8	8	2, 3 PCLKB	2 ICLK	WDTA
0008 8030h	IWDT	IWDT Refresh Register	IWDTRR	8	8	2, 3 PCLKB	2 ICLK	IWDTa
0008 8032h	IWDT	IWDT Control Register	IWDTCR	16	16	2, 3 PCLKB	2 ICLK	IWDTa
0008 8034h	IWDT	IWDT Status Register	IWDTSR	16	16	2, 3 PCLKB	2 ICLK	IWDTa
0008 8036h	IWDT	IWDT Reset Control Register	IWDTRCR	8	8	2, 3 PCLKB	2 ICLK	IWDTa
0008 8038h	IWDT	IWDT Count Stop Control Register	IWDCSTPR	8	8	2, 3 PCLKB	2 ICLK	IWDTa
0008 8040h	DA	D/A Data Register 0	DADR0	16	16	2, 3 PCLKB	2 ICLK	R12DA
0008 8042h	DA	D/A Data Register 1	DADR1	16	16	2, 3 PCLKB	2 ICLK	R12DA
0008 8044h	DA	D/A Control Register	DACR	8	8	2, 3 PCLKB	2 ICLK	R12DA

Table 4.1 List of I/O Registers (Address Order) (14 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 8045h	DA	DADRM Format Select Register	DADPR	8	8	2, 3 PCLKB	2 ICLK	R12DA
0008 8046h	DA	D/A A/D Synchronous Start Control Register	DAADSCR	8	8	2, 3 PCLKB	2 ICLK	R12DA
0008 8100h	TPUA	Timer Start Register	TSTR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8101h	TPUA	Timer Synchronous Register	TSYR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8108h	TPU0	Noise Filter Control Register	NFCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8109h	TPU1	Noise Filter Control Register	NFCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 810Ah	TPU2	Noise Filter Control Register	NFCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 810Bh	TPU3	Noise Filter Control Register	NFCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 810Ch	TPU4	Noise Filter Control Register	NFCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 810Dh	TPU5	Noise Filter Control Register	NFCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8110h	TPU0	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8111h	TPU0	Timer Mode Register	TMDR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8112h	TPU0	Timer I/O Control Register H	TIORH	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8113h	TPU0	Timer I/O Control Register L	TIORL	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8114h	TPU0	Timer Interrupt Enable Register	TIER	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8115h	TPU0	Timer Status Register	TSR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8116h	TPU0	Timer Counter	TCNT	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8118h	TPU0	Timer General Register A	TGRA	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 811Ah	TPU0	Timer General Register B	TGRB	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 811Ch	TPU0	Timer General Register C	TGRC	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 811Eh	TPU0	Timer General Register D	TGRD	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8120h	TPU1	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8121h	TPU1	Timer Mode Register	TMDR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8122h	TPU1	Timer I/O Control Register	TIOR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8124h	TPU1	Timer Interrupt Enable Register	TIER	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8125h	TPU1	Timer Status Register	TSR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8126h	TPU1	Timer Counter	TCNT	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8128h	TPU1	Timer General Register A	TGRA	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 812Ah	TPU1	Timer General Register B	TGRB	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8130h	TPU2	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8131h	TPU2	Timer Mode Register	TMDR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8132h	TPU2	Timer I/O Control Register	TIOR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8134h	TPU2	Timer Interrupt Enable Register	TIER	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8135h	TPU2	Timer Status Register	TSR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8136h	TPU2	Timer Counter	TCNT	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8138h	TPU2	Timer General Register A	TGRA	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 813Ah	TPU2	Timer General Register B	TGRB	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8140h	TPU3	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8141h	TPU3	Timer Mode Register	TMDR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8142h	TPU3	Timer I/O Control Register H	TIORH	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8143h	TPU3	Timer I/O Control Register L	TIORL	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8144h	TPU3	Timer Interrupt Enable Register	TIER	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8145h	TPU3	Timer Status Register	TSR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8146h	TPU3	Timer Counter	TCNT	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8148h	TPU3	Timer General Register A	TGRA	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 814Ah	TPU3	Timer General Register B	TGRB	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 814Ch	TPU3	Timer General Register C	TGRC	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 814Eh	TPU3	Timer General Register D	TGRD	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8150h	TPU4	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8151h	TPU4	Timer Mode Register	TMDR	8	8	2, 3 PCLKB	2 ICLK	TPUa

Table 4.1 List of I/O Registers (Address Order) (15 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 8152h	TPU4	Timer I/O Control Register	TIOR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8154h	TPU4	Timer Interrupt Enable Register	TIER	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8155h	TPU4	Timer Status Register	TSR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8156h	TPU4	Timer Counter	TCNT	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8158h	TPU4	Timer General Register A	TGRA	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 815Ah	TPU4	Timer General Register B	TGRB	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8160h	TPU5	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8161h	TPU5	Timer Mode Register	TMDR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8162h	TPU5	Timer I/O Control Register	TIOR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8164h	TPU5	Timer Interrupt Enable Register	TIER	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8165h	TPU5	Timer Status Register	TSR	8	8	2, 3 PCLKB	2 ICLK	TPUa
0008 8166h	TPU5	Timer Counter	TCNT	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 8168h	TPU5	Timer General Register A	TGRA	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 816Ah	TPU5	Timer General Register B	TGRB	16	16	2, 3 PCLKB	2 ICLK	TPUa
0008 81E6h	PPG0	PPG Output Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81E7h	PPG0	PPG Output Mode Register	PMR	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81E8h	PPG0	Next Data Enable Registers H	NDERH	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81E9h	PPG0	Next Data Enable Registers L	NDERL	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81EAh	PPG0	Output Data Registers H	PODRH	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81EBh	PPG0	Output Data Registers L	PODRL	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81ECh	PPG0	Next Data Registers H*1	NDRH	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81EDh	PPG0	Next Data Registers L*2	NDRL	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81EEh	PPG0	Next Data Registers H*1	NDRH2	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81EFh	PPG0	Next Data Registers L*2	NDRL2	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81F0h	PPG1	PPG Trigger Select Register	PTRSLR	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81F6h	PPG1	PPG Output Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81F7h	PPG1	PPG Output Mode Register	PMR	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81F8h	PPG1	Next Data Enable Registers H	NDERH	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81F9h	PPG1	Next Data Enable Registers L	NDERL	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81FAh	PPG1	Output Data Registers H	PODRH	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81FBh	PPG1	Output Data Registers L	PODRL	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81FCh	PPG1	Next Data Registers H*3	NDRH	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81FDh	PPG1	Next Data Registers L*4	NDRL	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81FEh	PPG1	Next Data Registers H*3	NDRH2	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 81FFh	PPG1	Next Data Registers L*4	NDRL2	8	8	2, 3 PCLKB	2 ICLK	PPG
0008 8200h	TMR0	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8201h	TMR1	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8202h	TMR0	Timer Control/Status Register	TCSR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8203h	TMR1	Timer Control/Status Register	TCSR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8204h	TMR0	Time Constant Register A	TCORA	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8204h	TMR01	Time Constant Register A	TCORA	16	16	2, 3 PCLKB	2 ICLK	TMR
0008 8205h	TMR1	Time Constant Register A	TCORA	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8206h	TMR0	Time Constant Register B	TCORB	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8206h	TMR01	Time Constant Register B	TCORB	16	16	2, 3 PCLKB	2 ICLK	TMR
0008 8207h	TMR1	Time Constant Register B	TCORB	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8208h	TMR0	Timer Counter	TCNT	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8208h	TMR01	Timer Counter	TCNT	16	16	2, 3 PCLKB	2 ICLK	TMR
0008 8209h	TMR1	Timer Counter	TCNT	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 820Ah	TMR0	Timer Counter Control Register	TCCR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 820Ah	TMR01	Timer Counter Control Register	TCCR	16	16	2, 3 PCLKB	2 ICLK	TMR

Table 4.1 List of I/O Registers (Address Order) (16 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 820Bh	TMR1	Timer Counter Control Register	TCCR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 820Ch	TMR0	Timer Counter Start Register	TCSTR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 820Dh	TMR1	Timer Counter Start Register	TCSTR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8210h	TMR2	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8211h	TMR3	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8212h	TMR2	Timer Control/Status Register	TCSR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8213h	TMR3	Timer Control/Status Register	TCSR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8214h	TMR2	Time Constant Register A	TCORA	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8214h	TMR23	Time Constant Register A	TCORA	16	16	2, 3 PCLKB	2 ICLK	TMR
0008 8215h	TMR3	Time Constant Register A	TCORA	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8216h	TMR2	Time Constant Register B	TCORB	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8216h	TMR23	Time Constant Register B	TCORB	16	16	2, 3 PCLKB	2 ICLK	TMR
0008 8217h	TMR3	Time Constant Register B	TCORB	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8218h	TMR2	Timer Counter	TCNT	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8218h	TMR23	Timer Counter	TCNT	16	16	2, 3 PCLKB	2 ICLK	TMR
0008 8219h	TMR3	Timer Counter	TCNT	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 821Ah	TMR2	Timer Counter Control Register	TCCR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 821Ah	TMR23	Timer Counter Control Register	TCCR	16	16	2, 3 PCLKB	2 ICLK	TMR
0008 821Bh	TMR3	Timer Counter Control Register	TCCR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 821Ch	TMR2	Timer Counter Start Register	TCSTR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 821Dh	TMR3	Timer Counter Start Register	TCSTR	8	8	2, 3 PCLKB	2 ICLK	TMR
0008 8280h	CRC	CRC Control Register	CRCCR	8	8	2, 3 PCLKB	2 ICLK	CRCA
0008 8284h	CRC	CRC Data Input Register	CRCDIR	32	8, 32	2, 3 PCLKB	2 ICLK	CRCA
0008 8288h	CRC	CRC Data Output Register	CRCDOR	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CRCA
0008 8300h	RIIC0	I ² C-Bus Control Register 1	ICCR1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8301h	RIIC0	I ² C-Bus Control Register 2	ICCR2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8302h	RIIC0	I ² C-Bus Mode Register 1	ICMR1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8303h	RIIC0	I ² C-Bus Mode Register 2	ICMR2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8304h	RIIC0	I ² C-Bus Mode Register 3	ICMR3	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8305h	RIIC0	I ² C-Bus Function Enable Register	ICFER	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8306h	RIIC0	I ² C-Bus Status Enable Register	ICSER	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8307h	RIIC0	I ² C-Bus Interrupt Enable Register	ICIER	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8308h	RIIC0	I ² C-Bus Status Register 1	ICSR1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8309h	RIIC0	I ² C-Bus Status Register 2	ICSR2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 830Ah	RIIC0	Slave Address Register L0	SARL0	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 830Bh	RIIC0	Slave Address Register U0	SARU0	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 830Ch	RIIC0	Slave Address Register L1	SARL1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 830Dh	RIIC0	Slave Address Register U1	SARU1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 830Eh	RIIC0	Slave Address Register L2	SARL2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 830Fh	RIIC0	Slave Address Register U2	SARU2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8310h	RIIC0	I ² C-Bus Bit Rate Low-Level Register	ICBRL	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8311h	RIIC0	I ² C-Bus Bit Rate High-Level Register	ICBRH	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8312h	RIIC0	I ² C-Bus Transmit Data Register	ICDRT	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8313h	RIIC0	I ² C-Bus Receive Data Register	ICDRR	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8340h	RIIC2	I ² C-Bus Control Register 1	ICCR1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8341h	RIIC2	I ² C-Bus Control Register 2	ICCR2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8342h	RIIC2	I ² C-Bus Mode Register 1	ICMR1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8343h	RIIC2	I ² C-Bus Mode Register 2	ICMR2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8344h	RIIC2	I ² C-Bus Mode Register 3	ICMR3	8	8	2, 3 PCLKB	2 ICLK	RIICa

Table 4.1 List of I/O Registers (Address Order) (17 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 8345h	RIIC2	I ² C-Bus Function Enable Register	ICFER	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8346h	RIIC2	I ² C-Bus Status Enable Register	ICSER	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8347h	RIIC2	I ² C-Bus Interrupt Enable Register	ICIER	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8348h	RIIC2	I ² C-Bus Status Register 1	ICSR1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8349h	RIIC2	I ² C-Bus Status Register 2	ICSR2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 834Ah	RIIC2	Slave Address Register L0	SARL0	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 834Bh	RIIC2	Slave Address Register U0	SARU0	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 834Ch	RIIC2	Slave Address Register L1	SARL1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 834Dh	RIIC2	Slave Address Register U1	SARU1	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 834Eh	RIIC2	Slave Address Register L2	SARL2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 834Fh	RIIC2	Slave Address Register U2	SARU2	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8350h	RIIC2	I ² C-Bus Bit Rate Low-Level Register	ICBRL	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8351h	RIIC2	I ² C-Bus Bit Rate High-Level Register	ICBRH	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8352h	RIIC2	I ² C-Bus Transmit Data Register	ICDRT	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8353h	RIIC2	I ² C-Bus Receive Data Register	ICDRR	8	8	2, 3 PCLKB	2 ICLK	RIICa
0008 8500h	MMCIF	Command Setting Register	CECMDSET	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8508h	MMCIF	Argument Register	CEARG	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 850Ch	MMCIF	Automatically Issued CMD12 Argument Register	CEARGCMD12	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8510h	MMCIF	Command Control Register	CECMDCTRL	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8514h	MMCIF	Transfer Block Setting Register	CEBLOCKSET	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8518h	MMCIF	Clock Control Register	CECLKCTRL	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 851Ch	MMCIF	Buffer Access Setting Register	CEBUFACC	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8520h	MMCIF	Response Register 3	CERESP3	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8524h	MMCIF	Response Register 2	CERESP2	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8528h	MMCIF	Response Register 1	CERESP1	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 852Ch	MMCIF	Response Register 0	CERESP0	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8530h	MMCIF	Automatically Issued CMD12 Response Register	CERESPCMD12	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8534h	MMCIF	Data Register	CEDATA	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 853Ch	MMCIF	Boot Operation Setting Register	CEBOOT	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8540h	MMCIF	Interrupt status Flag Register	CEINT	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8544h	MMCIF	Interrupt request Enable Register	CEINTEN	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8548h	MMCIF	Status Register 1	CEHOSTSTS1	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 854Ch	MMCIF	Status Register 2	CEHOSTSTS2	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8570h	MMCIF	MMC Detection and Port Control Register	CEDETECT	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 8574h	MMCIF	Special Mode Setting Register	CEADDMODE	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 857Ch	MMCIF	Version Register	CEVERSION	32	32	2, 3 PCLKB	2 ICLK	MMCIF
0008 9000h	S12AD	A/D Control Register	ADCSR	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9004h	S12AD	A/D Channel Select Register A0	ADANSA0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9008h	S12AD	A/D-Converted Value Addition/Average Function Select Register 0	ADADS0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 900Ch	S12AD	A/D-Converted Value Addition/Average Count Select Register	ADADC	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 900Eh	S12AD	A/D Control Extended Register	ADCER	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9010h	S12AD	A/D Conversion Start Trigger Select Register	ADSTRGR	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa

Table 4.1 List of I/O Registers (Address Order) (18 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 9014h	S12AD	A/D Channel Select Register B0	ADANSB0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9018h	S12AD	A/D Data Duplication Register	ADDBLDR	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 901Eh	S12AD	A/D Self-Diagnosis Data Register	ADRD	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9020h	S12AD	A/D Data Register 0	ADDR0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9022h	S12AD	A/D Data Register 1	ADDR1	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9024h	S12AD	A/D Data Register 2	ADDR2	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9026h	S12AD	A/D Data Register 3	ADDR3	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9028h	S12AD	A/D Data Register 4	ADDR4	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 902Ah	S12AD	A/D Data Register 5	ADDR5	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 902Ch	S12AD	A/D Data Register 6	ADDR6	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 902Eh	S12AD	A/D Data Register 7	ADDR7	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9063h	S12AD	A/D Conversion Time Setting Protection Release Register	ADSAMPR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9066h	S12AD	A/D Sample-and-Hold Circuit Control Register	ADSHCR	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 906Eh	S12AD	A/D Conversion Time Setting Register	ADSAM	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 907Ah	S12AD	A/D Disconnection Detection Control Register	ADDISCR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 907Ch	S12AD	A/D Sample-and-Hold Operating Mode Select Register	ADSHMSR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9080h	S12AD	A/D Group Scan Priority Control Register	ADGSPCR	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9084h	S12AD	A/D Data Duplication Register A	ADDBLDRA	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9086h	S12AD	A/D Data Duplication Register B	ADDBLDRB	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 908Ch	S12AD	A/D Comparison Function Window A/B Status Monitoring Register	ADWINMON	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9090h	S12AD	A/D Comparison Function Control Register	ADCMPCR	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9094h	S12AD	A/D Comparison Function Window A Channel Select Register 0	ADCMPSR0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9098h	S12AD	A/D Comparison Function Window A Comparison Condition Setting Register 0	ADCMPLR0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 909Ch	S12AD	A/D Comparison Function Window A Lower Level Setting Register	ADCMPDR0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 909Eh	S12AD	A/D Comparison Function Window A Upper Level Setting Register	ADCMPDR1	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 90A0h	S12AD	A/D Comparison Function Window A Channel Status Register 0	ADCMPSR0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 90A6h	S12AD	A/D Comparison Function Window B Channel Select Register	ADCMPBNSR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 90A8h	S12AD	A/D Comparison Function Window B Lower Level Setting Register	ADWINLLB	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 90AAh	S12AD	A/D Comparison Function Window B Upper Level Setting Register	ADWINULB	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 90ACh	S12AD	A/D Comparison Function Window B Channel Status Register	ADCMPBSR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 90D4h	S12AD	A/D Channel Select Register C0	ADANSC0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa

Table 4.1 List of I/O Registers (Address Order) (19 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 90D9h	S12AD	A/D Group C Trigger Select Register	ADGCTRGR	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 90E0h	S12AD	A/D Sampling State Register 0	ADSSTR0	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 90E1h	S12AD	A/D Sampling State Register 1	ADSSTR1	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 90E2h	S12AD	A/D Sampling State Register 2	ADSSTR2	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 90E3h	S12AD	A/D Sampling State Register 3	ADSSTR3	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 90E4h	S12AD	A/D Sampling State Register 4	ADSSTR4	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 90E5h	S12AD	A/D Sampling State Register 5	ADSSTR5	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 90E6h	S12AD	A/D Sampling State Register 6	ADSSTR6	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 90E7h	S12AD	A/D Sampling State Register 7	ADSSTR7	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9100h	S12AD1	A/D Control Register	ADCSR	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9104h	S12AD1	A/D Channel Select Register A0	ADANSA0	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9106h	S12AD1	A/D Channel Select Register A1	ADANSA1	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9108h	S12AD1	A/D-Converted Value Addition/Average Function Select Register 0	ADADS0	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 910Ah	S12AD1	A/D-Converted Value Addition/Average Function Select Register 1	ADADS1	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 910Ch	S12AD1	A/D-Converted Value Addition/Average Count Select Register	ADADC	8	8	2, 3 PCLK	2 ICLK	S12AD Fa
0008 910Eh	S12AD1	A/D Control Extended Register	ADCER	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9110h	S12AD1	A/D Conversion Start Trigger Select Register	ADSTRGR	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9112h	S12AD1	A/D Conversion Extended Input Control Register	ADEXICR	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9114h	S12AD1	A/D Channel Select Register B0	ADANSB0	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9116h	S12AD1	A/D Channel Select Register B1	ADANSB1	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9118h	S12AD1	A/D Data Duplication Register	ADDBLDR	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 911Ah	S12AD1	A/D Temperature Sensor Data Register	ADTSDR	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 911Ch	S12AD1	A/D Internal Reference Voltage Data Register	ADOCDR	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 911Eh	S12AD1	A/D Self-Diagnosis Data Register	ADRD	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9120h	S12AD1	A/D Data Register 0	ADDR0	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9122h	S12AD1	A/D Data Register 1	ADDR1	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9124h	S12AD1	A/D Data Register 2	ADDR2	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9126h	S12AD1	A/D Data Register 3	ADDR3	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 9128h	S12AD1	A/D Data Register 4	ADDR4	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 912Ah	S12AD1	A/D Data Register 5	ADDR5	16	16	2, 3 PCLK	2 ICLK	S12AD Fa
0008 912Ch	S12AD1	A/D Data Register 6	ADDR6	16	16	2, 3 PCLK	2 ICLK	S12AD Fa

Table 4.1 List of I/O Registers (Address Order) (20 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 912Eh	S12AD1	A/D Data Register 7	ADDR7	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9130h	S12AD1	A/D Data Register 8	ADDR8	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9132h	S12AD1	A/D Data Register 9	ADDR9	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9134h	S12AD1	A/D Data Register 10	ADDR10	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9136h	S12AD1	A/D Data Register 11	ADDR11	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9138h	S12AD1	A/D Data Register 12	ADDR12	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 913Ah	S12AD1	A/D Data Register 13	ADDR13	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 913Ch	S12AD1	A/D Data Register 14	ADDR14	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 913Eh	S12AD1	A/D Data Register 15	ADDR15	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9140h	S12AD1	A/D Data Register 16	ADDR16	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9142h	S12AD1	A/D Data Register 17	ADDR17	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9144h	S12AD1	A/D Data Register 18	ADDR18	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9146h	S12AD1	A/D Data Register 19	ADDR19	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9148h	S12AD1	A/D Data Register 20	ADDR20	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9163h	S12AD1	A/D Conversion Time Setting Protection Release Register	ADSAMPR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 916Eh	S12AD1	A/D Conversion Time Setting Register	ADSAM	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 917Ah	S12AD1	A/D Disconnection Detection Control Register	ADDISCR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9180h	S12AD1	A/D Group Scan Priority Control Register	ADGSPCR	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9184h	S12AD1	A/D Data Duplication Register A	ADDBLDRA	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9186h	S12AD1	A/D Data Duplication Register B	ADDBLDRB	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 918Ch	S12AD1	A/D Comparison Function Window A/B Status Monitoring Register	ADWINMON	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9190h	S12AD1	A/D Comparison Function Control Register	ADCMPCR	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9192h	S12AD1	A/D Comparison Function Window A Extended Input Select Register	ADCMPANSE R	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9193h	S12AD1	A/D Comparison Function Window A Extended Input Comparison Condition Setting Register	ADCMPLER	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9194h	S12AD1	A/D Comparison Function Window A Channel Select Register 0	ADCMPANSR 0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9196h	S12AD1	A/D Comparison Function Window A Channel Select Register 1	ADCMPANSR 1	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9198h	S12AD1	A/D Comparison Function Window A Comparison Condition Setting Register 0	ADCMPLR0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 919Ah	S12AD1	A/D Comparison Function Window A Comparison Condition Setting Register 1	ADCMPLR1	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 919Ch	S12AD1	A/D Comparison Function Window A Lower Level Setting Register	ADCMPLR0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 919Eh	S12AD1	A/D Comparison Function Window A Upper Level Setting Register	ADCMPLR1	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91A0h	S12AD1	A/D Comparison Function Window A Channel Status Register 0	ADCMPSR0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa

Table 4.1 List of I/O Registers (Address Order) (21 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 91A2h	S12AD1	A/D Comparison Function Window A Channel Status Register 1	ADCMPSR1	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91A4h	S12AD1	A/D Comparison Function Window A Extended Input Channel Status Register	ADCMPSER	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91A6h	S12AD1	A/D Comparison Function Window B Channel Select Register	ADCMPSNR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91A8h	S12AD1	A/D Comparison Function Window B Lower Level Setting Register	ADWINLLB	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91AAh	S12AD1	A/D Comparison Function Window B Upper Level Setting Register	ADWINULB	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91ACh	S12AD1	A/D Comparison Function Window B Channel Status Register	ADCMPSR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91D4h	S12AD1	A/D Channel Select Register C0	ADANSC0	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91D6h	S12AD1	A/D Channel Select Register C1	ADANSC1	16	16	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91D8h	S12AD1	A/D Group C Extended Input Control Register	ADGCXCR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91D9h	S12AD1	A/D Group C Trigger Select Register	ADGCTRGR	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91DDh	S12AD1	A/D Sampling State Register L	ADSSTRL	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91DEh	S12AD1	A/D Sampling State Register T	ADSSTRT	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91DFh	S12AD1	A/D Sampling State Register O	ADSSTRO	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E0h	S12AD1	A/D Sampling State Register 0	ADSSTR0	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E1h	S12AD1	A/D Sampling State Register 1	ADSSTR1	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E2h	S12AD1	A/D Sampling State Register 2	ADSSTR2	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E3h	S12AD1	A/D Sampling State Register 3	ADSSTR3	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E4h	S12AD1	A/D Sampling State Register 4	ADSSTR4	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E5h	S12AD1	A/D Sampling State Register 5	ADSSTR5	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E6h	S12AD1	A/D Sampling State Register 6	ADSSTR6	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E7h	S12AD1	A/D Sampling State Register 7	ADSSTR7	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E8h	S12AD1	A/D Sampling State Register 8	ADSSTR8	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91E9h	S12AD1	A/D Sampling State Register 9	ADSSTR9	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91EAh	S12AD1	A/D Sampling State Register 10	ADSSTR10	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91EBh	S12AD1	A/D Sampling State Register 11	ADSSTR11	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91ECh	S12AD1	A/D Sampling State Register 12	ADSSTR12	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91EDh	S12AD1	A/D Sampling State Register 13	ADSSTR13	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91EEh	S12AD1	A/D Sampling State Register 14	ADSSTR14	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 91EFh	S12AD1	A/D Sampling State Register 15	ADSSTR15	8	8	2, 3 PCLKB	2 ICLK	S12AD Fa
0008 9E00h	QSPI	QSPI Control Register	SPCR	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E01h	QSPI	QSPI Slave Select Polarity Register	SSLP	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E02h	QSPI	QSPI Pin Control Register	SPPCR	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E03h	QSPI	QSPI Status Register	SPSR	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI

Table 4.1 List of I/O Registers (Address Order) (22 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 9E04h	QSPI	QSPI Data Register	SPDR	32	8, 16, 32	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E08h	QSPI	QSPI Sequence Control Register	SPSCR	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E09h	QSPI	QSPI Sequence Status Register	SPSSR	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E0Ah	QSPI	QSPI Bit Rate Register	SPBR	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E0Bh	QSPI	QSPI Data Control Register	SPDCR	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E0Ch	QSPI	QSPI Clock Delay Register	SPCKD	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E0Dh	QSPI	QSPI Slave Select Negation Delay Register	SSLND	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E0Eh	QSPI	QSPI Next-Access Delay Register	SPND	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E10h	QSPI	QSPI Command Register 0	SPCMD0	16	16	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E12h	QSPI	QSPI Command Register 1	SPCMD1	16	16	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E14h	QSPI	QSPI Command Register 2	SPCMD2	16	16	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E16h	QSPI	QSPI Command Register 3	SPCMD3	16	16	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E18h	QSPI	QSPI Buffer Control Register	SPBFCR	8	8	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E1Ah	QSPI	QSPI Buffer Data Count Set Register	SPBDCR	16	16	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E1Ch	QSPI	QSPI Transfer Data Length Multiplier Setting Register 0	SPBMUL0	32	32	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E20h	QSPI	QSPI Transfer Data Length Multiplier Setting Register 1	SPBMUL1	32	32	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E24h	QSPI	QSPI Transfer Data Length Multiplier Setting Register 2	SPBMUL2	32	32	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 9E28h	QSPI	QSPI Transfer Data Length Multiplier Setting Register 3	SPBMUL3	32	32	4, 5 PCLKB	2, 3 ICLK	QSPI
0008 A000h	SCIO	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A001h	SCIO	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A002h	SCIO	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A003h	SCIO	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A004h	SCIO	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A005h	SCIO	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A006h	SMCIO	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A007h	SCIO	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A008h	SCIO	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A009h	SCIO	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A00Ah	SCIO	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A00Bh	SCIO	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli
0008 A00Ch	SCIO	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, SCli

Table 4.1 List of I/O Registers (Address Order) (23 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 A00Dh	SCI0	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A00Eh	SCI0	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A00Fh	SCI0	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A00Eh	SCI0	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A010h	SCI0	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A011h	SCI0	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A010h	SCI0	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A012h	SCI0	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A020h	SCI1	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A021h	SCI1	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A022h	SCI1	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A023h	SCI1	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A024h	SCI1	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A025h	SCI1	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A026h	SMC11	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A027h	SCI1	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A028h	SCI1	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A029h	SCI1	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A02Ah	SCI1	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A02Bh	SCI1	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A02Ch	SCI1	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A02Dh	SCI1	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A02Eh	SCI1	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i

Table 4.1 List of I/O Registers (Address Order) (24 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 A02Fh	SCI1	Transmit Data Register L	TDLR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A02Eh	SCI1	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A030h	SCI1	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A031h	SCI1	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A030h	SCI1	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A032h	SCI1	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A040h	SCI2	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A041h	SCI2	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A042h	SCI2	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A043h	SCI2	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A044h	SCI2	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A045h	SCI2	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A046h	SMCI2	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A047h	SCI2	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A048h	SCI2	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A049h	SCI2	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A04Ah	SCI2	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A04Bh	SCI2	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A04Ch	SCI2	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A04Dh	SCI2	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A04Eh	SCI2	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A04Fh	SCI2	Transmit Data Register L	TDLR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A04Eh	SCI2	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SC1h, SC1i

Table 4.1 List of I/O Registers (Address Order) (25 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 A050h	SCI2	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A051h	SCI2	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A050h	SCI2	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A052h	SCI2	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A060h	SCI3	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A061h	SCI3	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A062h	SCI3	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A063h	SCI3	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A064h	SCI3	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A065h	SCI3	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A066h	SMCI3	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A067h	SCI3	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A068h	SCI3	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A069h	SCI3	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A06Ah	SCI3	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A06Bh	SCI3	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A06Ch	SCI3	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A06Dh	SCI3	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A06Eh	SCI3	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A06Fh	SCI3	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A06Eh	SCI3	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A070h	SCI3	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i
0008 A071h	SCI3	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SC1h, SC1i

Table 4.1 List of I/O Registers (Address Order) (26 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 A070h	SCI3	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A072h	SCI3	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A080h	SCI4	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A081h	SCI4	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A082h	SCI4	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A083h	SCI4	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A084h	SCI4	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A085h	SCI4	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A086h	SMCI4	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A087h	SCI4	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A088h	SCI4	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A089h	SCI4	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A08Ah	SCI4	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A08Bh	SCI4	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A08Ch	SCI4	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A08Dh	SCI4	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A08Eh	SCI4	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A08Fh	SCI4	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A08Eh	SCI4	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A090h	SCI4	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A091h	SCI4	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A090h	SCI4	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIH, SCIL
0008 A092h	SCI4	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIH, SCIL

Table 4.1 List of I/O Registers (Address Order) (27 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 A0A0h	SCI5	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A1h	SCI5	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A2h	SCI5	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A3h	SCI5	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A4h	SCI5	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A5h	SCI5	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A6h	SMCI5	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A7h	SCI5	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A8h	SCI5	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0A9h	SCI5	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0AAh	SCI5	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0ABh	SCI5	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0ACh	SCI5	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0ADh	SCI5	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0AEh	SCI5	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0AFh	SCI5	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0AEh	SCI5	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0B0h	SCI5	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0B1h	SCI5	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0B0h	SCI5	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0B2h	SCI5	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C0h	SCI6	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C1h	SCI6	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi

Table 4.1 List of I/O Registers (Address Order) (28 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK $<$ PCLK	
0008 A0C2h	SCI6	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C3h	SCI6	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C4h	SCI6	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C5h	SCI6	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C6h	SMCI6	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C7h	SCI6	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C8h	SCI6	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0C9h	SCI6	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0CAh	SCI6	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0CBh	SCI6	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0CCh	SCI6	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0CDh	SCI6	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0CEh	SCI6	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0CFh	SCI6	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0CEh	SCI6	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0D0h	SCI6	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0D1h	SCI6	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0D0h	SCI6	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0D2h	SCI6	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0E0h	SCI7	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0E1h	SCI7	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0E2h	SCI7	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi
0008 A0E3h	SCI7	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCLi

Table 4.1 List of I/O Registers (Address Order) (29 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 A0E4h	SCI7	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0E5h	SCI7	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0E6h	SMCI7	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0E7h	SCI7	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0E8h	SCI7	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0E9h	SCI7	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0EAh	SCI7	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0EBh	SCI7	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0ECh	SCI7	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0EDh	SCI7	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0EEh	SCI7	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0EFh	SCI7	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0EEh	SCI7	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0F0h	SCI7	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0F1h	SCI7	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0F0h	SCI7	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A0F2h	SCI7	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A100h	SCI8	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A101h	SCI8	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A102h	SCI8	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A103h	SCI8	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A104h	SCI8	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A105h	SCI8	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli

Table 4.1 List of I/O Registers (Address Order) (30 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 A106h	SMCI8	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A107h	SCI8	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A108h	SCI8	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A109h	SCI8	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A10Ah	SCI8	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A10Bh	SCI8	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A10Ch	SCI8	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A10Dh	SCI8	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A10Eh	SCI8	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A10Fh	SCI8	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A10Eh	SCI8	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A110h	SCI8	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A111h	SCI8	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A110h	SCI8	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A112h	SCI8	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A120h	SCI9	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A121h	SCI9	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A122h	SCI9	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A123h	SCI9	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A124h	SCI9	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A125h	SCI9	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A126h	SMCI9	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli
0008 A127h	SCI9	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SClg, SClh, Scli

Table 4.1 List of I/O Registers (Address Order) (31 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 A128h	SCI9	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A129h	SCI9	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A12Ah	SCI9	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A12Bh	SCI9	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A12Ch	SCI9	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A12Dh	SCI9	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A12Eh	SCI9	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A12Fh	SCI9	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A12Eh	SCI9	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A130h	SCI9	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A131h	SCI9	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A130h	SCI9	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 A132h	SCI9	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIg, SCIh, SCli
0008 AC00h	SDHI	Command Register	SDCMD	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC08h	SDHI	Argument Register	SDARG	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC10h	SDHI	Data Stop Register	SDSTOP	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC14h	SDHI	Block Count Register	SDBLKCNT	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC18h	SDHI	Response Register 10	SDRSP10	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC20h	SDHI	Response Register 32	SDRSP32	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC28h	SDHI	Response Register 54	SDRSP54	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC30h	SDHI	Response Register 76	SDRSP76	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC38h	SDHI	SD Status Register 1	SDSTS1	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC3Ch	SDHI	SD Status Register 2	SDSTS2	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC40h	SDHI	SD Interrupt Mask Register 1	SDIMSK1	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC44h	SDHI	SD Interrupt Mask Register 2	SDIMSK2	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC48h	SDHI	SDHI Clock Control Register	SDCLKCR	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC4Ch	SDHI	Transfer Data Size Register	SDSIZE	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC50h	SDHI	Card Access Option Register	SDOPT	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC58h	SDHI	SD Error Status Register 1	SDERSTS1	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC5Ch	SDHI	SD Error Status Register 2	SDERSTS2	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC60h	SDHI	SD Buffer Register	SDBUFR	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC68h	SDHI	SDIO Mode Control Register	SDIOMD	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC6Ch	SDHI	SDIO Status Register	SDIOSTS	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 AC70h	SDHI	SDIO Interrupt Mask Register	SDIOIMSK	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 ADB0h	SDHI	DMA Transfer Enable Register	SDDMAEN	32	32	2, 3 PCLKB	2 ICLK	SDHI

Table 4.1 List of I/O Registers (Address Order) (32 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 ADC0h	SDHI	SDHI Software Reset Register	SDRST	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 ADC4h	SDHI	Version Register	SDVER	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 ADE0h	SDHI	Swap Control Register	SDSWAP	32	32	2, 3 PCLKB	2 ICLK	SDHI
0008 B000h	CAC	CAC Control Register 0	CACR0	8	8	2, 3 PCLKB	2 ICLK	CAC
0008 B001h	CAC	CAC Control Register 1	CACR1	8	8	2, 3 PCLKB	2 ICLK	CAC
0008 B002h	CAC	CAC Control Register 2	CACR2	8	8	2, 3 PCLKB	2 ICLK	CAC
0008 B003h	CAC	CAC Interrupt Request Enable Register	CAICR	8	8	2, 3 PCLKB	2 ICLK	CAC
0008 B004h	CAC	CAC Status Register	CASTR	8	8	2, 3 PCLKB	2 ICLK	CAC
0008 B006h	CAC	CAC Upper-Limit Value Setting Register	CAULVR	16	16	2, 3 PCLKB	2 ICLK	CAC
0008 B008h	CAC	CAC Lower-Limit Value Setting Register	CALLVR	16	16	2, 3 PCLKB	2 ICLK	CAC
0008 B00Ah	CAC	CAC Counter Buffer Register	CACNTBR	16	16	2, 3 PCLKB	2 ICLK	CAC
0008 B080h	DOC	DOC Control Register	DOCR	8	8	2, 3 PCLKB	2 ICLK	DOC
0008 B082h	DOC	DOC Data Input Register	DODIR	16	16	2, 3 PCLKB	2 ICLK	DOC
0008 B084h	DOC	DOC Data Setting Register	DODSR	16	16	2, 3 PCLKB	2 ICLK	DOC
0008 B100h	ELC	Event Link Control Register	ELCR	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B101h	ELC	Event Link Setting Register 0	ELSR0	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B104h	ELC	Event Link Setting Register 3	ELSR3	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B105h	ELC	Event Link Setting Register 4	ELSR4	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B108h	ELC	Event Link Setting Register 7	ELSR7	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B10Bh	ELC	Event Link Setting Register 10	ELSR10	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B10Ch	ELC	Event Link Setting Register 11	ELSR11	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B10Dh	ELC	Event Link Setting Register 12	ELSR12	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B10Eh	ELC	Event Link Setting Register 13	ELSR13	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B110h	ELC	Event Link Setting Register 15	ELSR15	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B111h	ELC	Event Link Setting Register 16	ELSR16	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B113h	ELC	Event Link Setting Register 18	ELSR18	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B114h	ELC	Event Link Setting Register 19	ELSR19	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B115h	ELC	Event Link Setting Register 20	ELSR20	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B116h	ELC	Event Link Setting Register 21	ELSR21	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B117h	ELC	Event Link Setting Register 22	ELSR22	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B118h	ELC	Event Link Setting Register 23	ELSR23	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B119h	ELC	Event Link Setting Register 24	ELSR24	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B11Ah	ELC	Event Link Setting Register 25	ELSR25	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B11Bh	ELC	Event Link Setting Register 26	ELSR26	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B11Ch	ELC	Event Link Setting Register 27	ELSR27	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B11Dh	ELC	Event Link Setting Register 28	ELSR28	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B11Fh	ELC	Event Link Option Setting Register A	ELOPA	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B120h	ELC	Event Link Option Setting Register B	ELOPB	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B121h	ELC	Event Link Option Setting Register C	ELOPC	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B122h	ELC	Event Link Option Setting Register D	ELOPD	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B123h	ELC	Port Group Setting Register 1	PGR1	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B124h	ELC	Port Group Setting Register 2	PGR2	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B125h	ELC	Port Group Control Register 1	PGC1	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B126h	ELC	Port Group Control Register 2	PGC2	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B127h	ELC	Port Buffer Register 1	PDBF1	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B128h	ELC	Port Buffer Register 2	PDBF2	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B129h	ELC	Event Link Port Setting Register 0	PEL0	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B12Ah	ELC	Event Link Port Setting Register 1	PEL1	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B12Bh	ELC	Event Link Port Setting Register 2	PEL2	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B12Ch	ELC	Event Link Port Setting Register 3	PEL3	8	8	2, 3 PCLKB	2 ICLK	ELC

Table 4.1 List of I/O Registers (Address Order) (33 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 B12Dh	ELC	Event Link Software Event Generation Register	ELSEGR	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B131h	ELC	Event Link Setting Register 33	ELSR33	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B133h	ELC	Event Link Setting Register 35	ELSR35	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B134h	ELC	Event Link Setting Register 36	ELSR36	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B135h	ELC	Event Link Setting Register 37	ELSR37	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B136h	ELC	Event Link Setting Register 38	ELSR38	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B13Dh	ELC	Event Link Setting Register 45	ELSR45	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B13Fh	ELC	Event Link Option Setting Register F	ELOPF	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B141h	ELC	Event Link Option Setting Register H	ELOPH	8	8	2, 3 PCLKB	2 ICLK	ELC
0008 B300h	SCI12	Serial Mode Register	SMR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B301h	SCI12	Bit Rate Register	BRR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B302h	SCI12	Serial Control Register	SCR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B303h	SCI12	Transmit Data Register	TDR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B304h	SCI12	Serial Status Register	SSR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B305h	SCI12	Receive Data Register	RDR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B306h	SMCI12	Smart Card Mode Register	SCMR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B307h	SCI12	Serial Extended Mode Register	SEMR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B308h	SCI12	Noise Filter Setting Register	SNFR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B309h	SCI12	I ² C Mode Register 1	SIMR1	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B30Ah	SCI12	I ² C Mode Register 2	SIMR2	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B30Bh	SCI12	I ² C Mode Register 3	SIMR3	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B30Ch	SCI12	I ² C Status Register	SISR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B30Dh	SCI12	SPI Mode Register	SPMR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B30Eh	SCI12	Transmit Data Register H	TDRH	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B30Fh	SCI12	Transmit Data Register L	TDRL	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B30Eh	SCI12	Transmit Data Register HL	TDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIh
0008 B310h	SCI12	Receive Data Register H	RDRH	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B311h	SCI12	Receive Data Register L	RDRL	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B310h	SCI12	Receive Data Register HL	RDRHL	16	16	4, 5 PCLKB	2 ICLK	SCIh
0008 B312h	SCI12	Modulation Duty Register	MDDR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B320h	SCI12	Extended Serial Module Enable Register	ESMER	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B321h	SCI12	Control Register 0	CR0	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B322h	SCI12	Control Register 1	CR1	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B323h	SCI12	Control Register 2	CR2	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B324h	SCI12	Control Register 3	CR3	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B325h	SCI12	Port Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B326h	SCI12	Interrupt Control Register	ICR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B327h	SCI12	Status Register	STR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B328h	SCI12	Status Clear Register	STCR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B329h	SCI12	Control Field 0 Data Register	CF0DR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B32Ah	SCI12	Control Field 0 Compare Enable Register	CF0CR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B32Bh	SCI12	Control Field 0 Receive Data Register	CF0RR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B32Ch	SCI12	Primary Control Field 1 Data Register	PCF1DR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B32Dh	SCI12	Secondary Control Field 1 Data Register	SCF1DR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B32Eh	SCI12	Control Field 1 Compare Enable Register	CF1CR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B32Fh	SCI12	Control Field 1 Receive Data Register	CF1RR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B330h	SCI12	Timer Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B331h	SCI12	Timer Mode Register	TMR	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B332h	SCI12	Timer Prescaler Register	TPRE	8	8	2, 3 PCLKB	2 ICLK	SCIh
0008 B333h	SCI12	Timer Count Register	TCNT	8	8	2, 3 PCLKB	2 ICLK	SCIh

Table 4.1 List of I/O Registers (Address Order) (34 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLKB	ICLK < PCLKB	
0008 C000h	PORT0	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C001h	PORT1	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C002h	PORT2	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C003h	PORT3	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C004h	PORT4	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C005h	PORT5	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C006h	PORT6	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C007h	PORT7	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C008h	PORT8	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C009h	PORT9	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C00Ah	PORTA	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C00Bh	PORTB	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C00Ch	PORTC	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C00Dh	PORTD	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C00Eh	PORTE	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C00Fh	PORTF	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C012h	PORTJ	Port Direction Register	PDR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C020h	PORT0	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C021h	PORT1	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C022h	PORT2	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C023h	PORT3	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C024h	PORT4	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C025h	PORT5	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C026h	PORT6	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C027h	PORT7	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C028h	PORT8	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C029h	PORT9	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C02Ah	PORTA	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C02Bh	PORTB	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C02Ch	PORTC	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C02Dh	PORTD	Port Output Data Register	PODR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports

Table 4.1 List of I/O Registers (Address Order) (35 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 C02Eh	PORTE	Port Output Data Register	PODR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C02Fh	PORTF	Port Output Data Register	PODR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C032h	PORTJ	Port Output Data Register	PODR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C040h	PORT0	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C041h	PORT1	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C042h	PORT2	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C043h	PORT3	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C044h	PORT4	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C045h	PORT5	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C046h	PORT6	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C047h	PORT7	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C048h	PORT8	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C049h	PORT9	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C04Ah	PORTA	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C04Bh	PORTB	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C04Ch	PORTC	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C04Dh	PORTD	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C04Eh	PORTE	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C04Fh	PORTF	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C052h	PORTJ	Port Input Register	PIDR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C060h	PORT0	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C061h	PORT1	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C062h	PORT2	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C063h	PORT3	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C064h	PORT4	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C065h	PORT5	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C066h	PORT6	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C067h	PORT7	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C068h	PORT8	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C069h	PORT9	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports
0008 C06Ah	PORTA	Port Mode Register	PMR	8	8	2, 3 PCLK	2 ICLK	I/O Ports

Table 4.1 List of I/O Registers (Address Order) (36 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 C06Bh	PORTB	Port Mode Register	PMR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C06Ch	PORTC	Port Mode Register	PMR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C06Dh	PORTD	Port Mode Register	PMR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C06Eh	PORTE	Port Mode Register	PMR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C06Fh	PORTF	Port Mode Register	PMR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C072h	PORTJ	Port Mode Register	PMR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C080h	PORT0	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C081h	PORT0	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C082h	PORT1	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C083h	PORT1	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C084h	PORT2	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C085h	PORT2	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C086h	PORT3	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C087h	PORT3	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C088h	PORT4	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C089h	PORT4	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C08Ah	PORT5	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C08Bh	PORT5	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C08Ch	PORT6	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C08Dh	PORT6	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C08Eh	PORT7	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C08Fh	PORT7	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C090h	PORT8	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C091h	PORT8	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C092h	PORT9	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C094h	PORTA	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C095h	PORTA	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C096h	PORTB	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C097h	PORTB	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C098h	PORTC	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C099h	PORTC	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports

Table 4.1 List of I/O Registers (Address Order) (37 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLK	ICLK < PCLK	
0008 C09Ah	PORTD	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C09Bh	PORTD	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C09Ch	PORTE	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C09Dh	PORTE	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C09Fh	PORTF	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0A4h	PORTJ	Open-Drain Control Register 0	ODR0	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0A5h	PORTJ	Open-Drain Control Register 1	ODR1	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C0h	PORT0	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C1h	PORT1	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C2h	PORT2	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C3h	PORT3	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C4h	PORT4	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C5h	PORT5	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C6h	PORT6	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C7h	PORT7	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C8h	PORT8	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0C9h	PORT9	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0CAh	PORTA	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0CBh	PORTB	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0CCh	PORTC	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0CDh	PORTD	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0CEh	PORTE	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0CFh	PORTF	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0D2h	PORTJ	Pull-Up Resistor Control Register	PCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0E0h	PORT0	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0E2h	PORT2	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0E5h	PORT5	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0E9h	PORT9	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0EAh	PORTA	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0EBh	PORTB	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0ECh	PORTC	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports

Table 4.1 List of I/O Registers (Address Order) (38 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 C0EDh	PORTD	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C0EEh	PORTE	Drive Capacity Control Register	DSCR	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C100h	MPC	CS Output Enable Register	PFCSE	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C102h	MPC	CS Output Pin Select Register 0	PFCSS0	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C103h	MPC	CS Output Pin Select Register 1	PFCSS1	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C104h	MPC	Address Output Enable Register 0	PFAOE0	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C105h	MPC	Address Output Enable Register 1	PFAOE1	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C106h	MPC	External Bus Control Register 0	PFBCR0	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C107h	MPC	External Bus Control Register 1	PFBCR1	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C10Eh	MPC	Ethernet Control Register	PFENET	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C11Fh	MPC	Write-Protect Register	PWPR	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C128h	PORT0	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C12Ah	PORT2	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C12Bh	PORT3	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C12Dh	PORT5	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C12Fh	PORT7	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C130h	PORT8	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C131h	PORT9	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C132h	PORTA	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C133h	PORTB	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C134h	PORTC	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C135h	PORTD	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C136h	PORTE	Drive Capacity Control Register 2	DSCR2	8	8	2, 3 PCLKB	2 ICLK	I/O Ports
0008 C140h	MPC	P00 Pin Function Control Register	P00PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C141h	MPC	P01 Pin Function Control Register	P01PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C142h	MPC	P02 Pin Function Control Register	P02PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C143h	MPC	P03 Pin Function Control Register	P03PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C145h	MPC	P05 Pin Function Control Register	P05PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C147h	MPC	P07 Pin Function Control Register	P07PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C14Ah	MPC	P12 Pin Function Control Register	P12PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C14Bh	MPC	P13 Pin Function Control Register	P13PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C14Ch	MPC	P14 Pin Function Control Register	P14PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C14Dh	MPC	P15 Pin Function Control Register	P15PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C14Eh	MPC	P16 Pin Function Control Register	P16PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C14Fh	MPC	P17 Pin Function Control Register	P17PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C150h	MPC	P20 Pin Function Control Register	P20PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C151h	MPC	P21 Pin Function Control Register	P21PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C152h	MPC	P22 Pin Function Control Register	P22PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C153h	MPC	P23 Pin Function Control Register	P23PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C154h	MPC	P24 Pin Function Control Register	P24PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C155h	MPC	P25 Pin Function Control Register	P25PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C156h	MPC	P26 Pin Function Control Register	P26PFS	8	8	2, 3 PCLKB	2 ICLK	MPC

Table 4.1 List of I/O Registers (Address Order) (39 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 C157h	MPC	P27 Pin Function Control Register	P27PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C158h	MPC	P30 Pin Function Control Register	P30PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C159h	MPC	P31 Pin Function Control Register	P31PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C15Ah	MPC	P32 Pin Function Control Register	P32PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C15Bh	MPC	P33 Pin Function Control Register	P33PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C15Ch	MPC	P34 Pin Function Control Register	P34PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C160h	MPC	P40 Pin Function Control Register	P40PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C161h	MPC	P41 Pin Function Control Register	P41PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C162h	MPC	P42 Pin Function Control Register	P42PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C163h	MPC	P43 Pin Function Control Register	P43PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C164h	MPC	P44 Pin Function Control Register	P44PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C165h	MPC	P45 Pin Function Control Register	P45PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C166h	MPC	P46 Pin Function Control Register	P46PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C167h	MPC	P47 Pin Function Control Register	P47PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C168h	MPC	P50 Pin Function Control Register	P50PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C169h	MPC	P51 Pin Function Control Register	P51PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C16Ah	MPC	P52 Pin Function Control Register	P52PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C16Ch	MPC	P54 Pin Function Control Register	P54PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C16Dh	MPC	P55 Pin Function Control Register	P55PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C16Eh	MPC	P56 Pin Function Control Register	P56PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C176h	MPC	P66 Pin Function Control Register	P66PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C177h	MPC	P67 Pin Function Control Register	P67PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C179h	MPC	P71 Pin Function Control Register	P71PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C17Ah	MPC	P72 Pin Function Control Register	P72PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C17Bh	MPC	P73 Pin Function Control Register	P73PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C17Ch	MPC	P74 Pin Function Control Register	P74PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C17Dh	MPC	P75 Pin Function Control Register	P75PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C17Eh	MPC	P76 Pin Function Control Register	P76PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C17Fh	MPC	P77 Pin Function Control Register	P77PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C180h	MPC	P80 Pin Function Control Register	P80PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C181h	MPC	P81 Pin Function Control Register	P81PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C182h	MPC	P82 Pin Function Control Register	P82PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C183h	MPC	P83 Pin Function Control Register	P83PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C186h	MPC	P86 Pin Function Control Register	P86PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C187h	MPC	P87 Pin Function Control Register	P87PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C188h	MPC	P90 Pin Function Control Register	P90PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C189h	MPC	P91 Pin Function Control Register	P91PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C18Ah	MPC	P92 Pin Function Control Register	P92PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C18Bh	MPC	P93 Pin Function Control Register	P93PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C190h	MPC	PA0 Pin Function Control Register	PA0PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C191h	MPC	PA1 Pin Function Control Register	PA1PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C192h	MPC	PA2 Pin Function Control Register	PA2PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C193h	MPC	PA3 Pin Function Control Register	PA3PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C194h	MPC	PA4 Pin Function Control Register	PA4PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C195h	MPC	PA5 Pin Function Control Register	PA5PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C196h	MPC	PA6 Pin Function Control Register	PA6PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C197h	MPC	PA7 Pin Function Control Register	PA7PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C198h	MPC	PB0 Pin Function Control Register	PB0PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C199h	MPC	PB1 Pin Function Control Register	PB1PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C19Ah	MPC	PB2 Pin Function Control Register	PB2PFS	8	8	2, 3 PCLKB	2 ICLK	MPC

Table 4.1 List of I/O Registers (Address Order) (40 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 C19Bh	MPC	PB3 Pin Function Control Register	PB3PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C19Ch	MPC	PB4 Pin Function Control Register	PB4PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C19Dh	MPC	PB5 Pin Function Control Register	PB5PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C19Eh	MPC	PB6 Pin Function Control Register	PB6PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C19Fh	MPC	PB7 Pin Function Control Register	PB7PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A0h	MPC	PC0 Pin Function Control Register	PC0PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A1h	MPC	PC1 Pin Function Control Register	PC1PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A2h	MPC	PC2 Pin Function Control Register	PC2PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A3h	MPC	PC3 Pin Function Control Register	PC3PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A4h	MPC	PC4 Pin Function Control Register	PC4PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A5h	MPC	PC5 Pin Function Control Register	PC5PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A6h	MPC	PC6 Pin Function Control Register	PC6PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A7h	MPC	PC7 Pin Function Control Register	PC7PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A8h	MPC	PD0 Pin Function Control Register	PD0PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1A9h	MPC	PD1 Pin Function Control Register	PD1PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1AAh	MPC	PD2 Pin Function Control Register	PD2PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1ABh	MPC	PD3 Pin Function Control Register	PD3PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1Ach	MPC	PD4 Pin Function Control Register	PD4PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1Adh	MPC	PD5 Pin Function Control Register	PD5PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1AEh	MPC	PD6 Pin Function Control Register	PD6PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1AFh	MPC	PD7 Pin Function Control Register	PD7PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1B0h	MPC	PE0 Pin Function Control Register	PE0PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1B1h	MPC	PE1 Pin Function Control Register	PE1PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1B2h	MPC	PE2 Pin Function Control Register	PE2PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1B3h	MPC	PE3 Pin Function Control Register	PE3PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1B4h	MPC	PE4 Pin Function Control Register	PE4PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1B5h	MPC	PE5 Pin Function Control Register	PE5PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1B6h	MPC	PE6 Pin Function Control Register	PE6PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1B7h	MPC	PE7 Pin Function Control Register	PE7PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1BDh	MPC	PF5 Pin Function Control Register	PF5PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1D3h	MPC	PJ3 Pin Function Control Register	PJ3PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C1D5h	MPC	PJ5 Pin Function Control Register	PJ5PFS	8	8	2, 3 PCLKB	2 ICLK	MPC
0008 C280h	SYSTEM	Deep Standby Control Register	DPSBYCR	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C282h	SYSTEM	Deep Standby Interrupt Enable Register 0	DPSIER0	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C283h	SYSTEM	Deep Standby Interrupt Enable Register 1	DPSIER1	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C284h	SYSTEM	Deep Standby Interrupt Enable Register 2	DPSIER2	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C285h	SYSTEM	Deep Standby Interrupt Enable Register 3	DPSIER3	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C286h	SYSTEM	Deep Standby Interrupt Flag Register 0	DPSIFR0	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption

Table 4.1 List of I/O Registers (Address Order) (41 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK $\geq$ PCLKB	ICLK < PCLKB	
0008 C287h	SYSTEM	Deep Standby Interrupt Flag Register 1	DPSIFR1	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C288h	SYSTEM	Deep Standby Interrupt Flag Register 2	DPSIFR2	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C289h	SYSTEM	Deep Standby Interrupt Flag Register 3	DPSIFR3	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C28Ah	SYSTEM	Deep Standby Interrupt Edge Register 0	DPSIEGR0	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C28Bh	SYSTEM	Deep Standby Interrupt Edge Register 1	DPSIEGR1	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C28Ch	SYSTEM	Deep Standby Interrupt Edge Register 2	DPSIEGR2	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C28Dh	SYSTEM	Deep Standby Interrupt Edge Register 3	DPSIEGR3	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C290h	SYSTEM	Reset Status Register 0	RSTSR0	8	8	4, 5 PCLKB	2, 3 ICLK	Resets
0008 C291h	SYSTEM	Reset Status Register 1	RSTSR1	8	8	4, 5 PCLKB	2, 3 ICLK	Resets
0008 C293h	SYSTEM	Main Clock Oscillator Forced Oscillation Control Register	MOFCR	8	8	4, 5 PCLKB	2, 3 ICLK	Clock Generation Circuit
0008 C294h	SYSTEM	High-Speed On-Chip Oscillator Power Supply Control Register	HOCOPCR	8	8	4, 5 PCLKB	2, 3 ICLK	Clock Generation Circuit
0008 C296h	FLASH	Flash P/E Protect Register	FWEPROR	8	8	2 ICLK		Flash
0008 C297h	SYSTEM	Voltage Monitoring Circuit Control Register	LVCMPCR	8	8	4, 5 PCLKB	2, 3 ICLK	LVDA
0008 C298h	SYSTEM	Voltage Detection Level Select Register	LVDLVL	8	8	4, 5 PCLKB	2, 3 ICLK	LVDA
0008 C29Ah	SYSTEM	Voltage Monitoring 1 Circuit Control Register 0	LVD1CR0	8	8	4, 5 PCLKB	2, 3 ICLK	LVDA
0008 C29Bh	SYSTEM	Voltage Monitoring 2 Circuit Control Register 0	LVD2CR0	8	8	4, 5 PCLKB	2, 3 ICLK	LVDA
0008 C2A0h to 0008 C2BFh	SYSTEM	Deep Standby Backup Registers 0 to 31	DPSBKR0 to 31	8	8	4, 5 PCLKB	2, 3 ICLK	Low Power Consumption
0008 C400h	RTC	64-Hz Counter	R64CNT	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C402h	RTC	Second Counter	RSECCNT	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C402h	RTC	Binary Counter 0	BCNT0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C404h	RTC	Minute Counter	RMINCNT	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C404h	RTC	Binary Counter 1	BCNT1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C406h	RTC	Hour Counter	RHRCNT	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C406h	RTC	Binary Counter 2	BCNT2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C408h	RTC	Day-of-Week Counter	RWKCNT	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C408h	RTC	Binary Counter 3	BCNT3	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C40Ah	RTC	Date Counter	RDAYCNT	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C40Ch	RTC	Month Counter	RMONCNT	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C40Eh	RTC	Year Counter	RYRCNT	16	16	2, 3 PCLKB	2 ICLK	RTCd

Table 4.1 List of I/O Registers (Address Order) (42 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 C410h	RTC	Second Alarm Register	RSECAR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C410h	RTC	Binary Counter 0 Alarm Register	BCNT0AR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C412h	RTC	Minute Alarm Register	RMINAR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C412h	RTC	Binary Counter 1 Alarm Register	BCNT1AR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C414h	RTC	Hour Alarm Register	RHRAR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C414h	RTC	Binary Counter 2 Alarm Register	BCNT2AR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C416h	RTC	Day-of-Week Alarm Register	RWKAR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C416h	RTC	Binary Counter 3 Alarm Register	BCNT3AR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C418h	RTC	Date Alarm Register	RDAYAR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C418h	RTC	Binary Counter 0 Alarm Enable Register	BCNT0AER	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C41Ah	RTC	Month Alarm Register	RMONAR	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C41Ah	RTC	Binary Counter 1 Alarm Enable Register	BCNT1AER	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C41Ch	RTC	Year Alarm Register	RYRAR	16	16	2, 3 PCLKB	2 ICLK	RTCd
0008 C41Ch	RTC	Binary Counter 2 Alarm Enable Register	BCNT2AER	16	16	2, 3 PCLKB	2 ICLK	RTCd
0008 C41Eh	RTC	Year Alarm Enable Register	RYRAREN	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C41Eh	RTC	Binary Counter 3 Alarm Enable Register	BCNT3AER	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C422h	RTC	RTC Control Register 1	RCR1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C424h	RTC	RTC Control Register 2	RCR2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C426h	RTC	RTC Control Register 3	RCR3	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C428h	RTC	RTC Control Register 4	RCR4	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C42Ah	RTC	Frequency Register H	RFRH	16	16	2, 3 PCLKB	2 ICLK	RTCd
0008 C42Ch	RTC	Frequency Register L	RFRL	16	16	2, 3 PCLKB	2 ICLK	RTCd
0008 C42Eh	RTC	Time Error Adjustment Register	RADJ	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C440h	RTC	Time Capture Control Register 0	RTCCR0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C442h	RTC	Time Capture Control Register 1	RTCCR1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C444h	RTC	Time Capture Control Register 2	RTCCR2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C452h	RTC	Second Capture Register 0	RSECCP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C452h	RTC	BCNT0 Capture Register 0	BCNT0CP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C454h	RTC	Minute Capture Register 0	RMINCP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C454h	RTC	BCNT1 Capture Register 0	BCNT1CP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C456h	RTC	Hour Capture Register 0	RHRCP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C456h	RTC	BCNT2 Capture Register 0	BCNT2CP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C45Ah	RTC	Date Capture Register 0	RDAYCP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C45Ah	RTC	BCNT3 Capture Register 0	BCNT3CP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C45Ch	RTC	Month Capture Register 0	RMONCP0	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C462h	RTC	Second Capture Register 1	RSECCP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C462h	RTC	BCNT0 Capture Register 1	BCNT0CP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C464h	RTC	Minute Capture Register 1	RMINCP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C464h	RTC	BCNT1 Capture Register 1	BCNT1CP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C466h	RTC	Hour Capture Register 1	RHRCP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C466h	RTC	BCNT2 Capture Register 1	BCNT2CP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C46Ah	RTC	Date Capture Register 1	RDAYCP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C46Ah	RTC	BCNT3 Capture Register 1	BCNT3CP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C46Ch	RTC	Month Capture Register 1	RMONCP1	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C472h	RTC	Second Capture Register 2	RSECCP2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C472h	RTC	BCNT0 Capture Register 2	BCNT0CP2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C474h	RTC	Minute Capture Register 2	RMINCP2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C474h	RTC	BCNT1 Capture Register 2	BCNT1CP2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C476h	RTC	Hour Capture Register 2	RHRCP2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C476h	RTC	BCNT2 Capture Register 2	BCNT2CP2	8	8	2, 3 PCLKB	2 ICLK	RTCd

Table 4.1 List of I/O Registers (Address Order) (43 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0008 C47Ah	RTC	Date Capture Register 2	RDAYCP2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C47Ah	RTC	BCNT3 Capture Register 2	BCNT3CP2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C47Ch	RTC	Month Capture Register 2	RMONCP2	8	8	2, 3 PCLKB	2 ICLK	RTCd
0008 C4C0h	POE3	Input Level Control/Status Register 1	ICSR1	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4C2h	POE3	Output Level Control/Status Register 1	OCSR1	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4C4h	POE3	Input Level Control/Status Register 2	ICSR2	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4C6h	POE3	Output Level Control/Status Register 2	OCSR2	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4C8h	POE3	Input Level Control/Status Register 3	ICSR3	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4CAh	POE3	Software Port Output Enable Register	SPOER	8	8	2, 3 PCLKB	2 ICLK	POE3a
0008 C4CBh	POE3	Port Output Enable Control Register 1	POECR1	8	8	2, 3 PCLKB	2 ICLK	POE3a
0008 C4CCh	POE3	Port Output Enable Control Register 2	POECR2	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4D0h	POE3	Port Output Enable Control Register 4	POECR4	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4D2h	POE3	Port Output Enable Control Register 5	POECR5	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4D6h	POE3	Input Level Control/Status Register 4	ICSR4	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4D8h	POE3	Input Level Control/Status Register 5	ICSR5	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4DAh	POE3	Active Level Setting Register 1	ALR1	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4DCh	POE3	Input Level Control/Status Register 6	ICSR6	16	16	2, 3 PCLKB	2 ICLK	POE3a
0008 C4E4h	POE3	MTU0 Pin Select Register 1	M0SELR1	8	8	2, 3 PCLKB	2 ICLK	POE3a
0008 C4E5h	POE3	MTU0 Pin Select Register 2	M0SELR2	8	8	2, 3 PCLKB	2 ICLK	POE3a
0008 C4E6h	POE3	MTU3 Pin Select Register	M3SELR	8	8	2, 3 PCLKB	2 ICLK	POE3a
0008 C4E7h	POE3	MTU4 Pin Select Register 1	M4SELR1	8	8	2, 3 PCLKB	2 ICLK	POE3a
0008 C4E8h	POE3	MTU4 Pin Select Register 2	M4SELR2	8	8	2, 3 PCLKB	2 ICLK	POE3a
0008 C500h	TEMPS	Temperature Sensor Control Register	TSCR	8	8	2, 3 PCLKB	2 ICLK	TEMPS
0008 C5C0h	DA	D/A A/D Synchronous Unit Select Register	DAADUSR	8	8	2, 3 PCLKB	2 ICLK	R12DA
0009 0200h to 0009 03FFh	CAN0	Mailbox Registers 0 to 31	MB0 to 31	128	8, 16, 32*6	2, 3 PCLKB	2 ICLK	CAN
0009 0400h to 0009 041Fh	CAN0	Mask Registers 0 to 7	MKR0 to 7	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 0420h	CAN0	FIFO Received ID Compare Register 0	FIDCR0	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 0424h	CAN0	FIFO Received ID Compare Register 1	FIDCR1	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 0428h	CAN0	Mask Invalid Register	MKIVLR	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 042Ch	CAN0	Mailbox Interrupt Enable Register	MIER	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 0820h to 0009 083Fh	CAN0	Message Control Registers 0 to 31	MCTL0 to 31	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 0840h	CAN0	Control Register	CTLR	16	8, 16	2, 3 PCLKB	2 ICLK	CAN
0009 0842h	CAN0	Status Register	STR	16	8, 16	2, 3 PCLKB	2 ICLK	CAN
0009 0844h	CAN0	Bit Configuration Register	BCR	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 0848h	CAN0	Receive FIFO Control Register	RFCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 0849h	CAN0	Receive FIFO Pointer Control Register	RFPCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 084Ah	CAN0	Transmit FIFO Control Register	TFCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 084Bh	CAN0	Transmit FIFO Pointer Control Register	TFPCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 084Ch	CAN0	Error Interrupt Enable Register	EIER	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 084Dh	CAN0	Error Interrupt Factor Judge Register	EIFR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 084Eh	CAN0	Receive Error Count Register	RECR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 084Fh	CAN0	Transmit Error Count Register	TECR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 0850h	CAN0	Error Code Store Register	ECSR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 0851h	CAN0	Channel Search Support Register	CSSR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 0852h	CAN0	Mailbox Search Status Register	MSSR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 0853h	CAN0	Mailbox Search Mode Register	MSMR	8	8	2, 3 PCLKB	2 ICLK	CAN

Table 4.1 List of I/O Registers (Address Order) (44 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0009 0854h	CAN0	Time Stamp Register	TSR	16	16	2, 3 PCLKB	2 ICLK	CAN
0009 0856h	CAN0	Acceptance Filter Support Register	AFSR	16	16	2, 3 PCLKB	2 ICLK	CAN
0009 0858h	CAN0	Test Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 1200h to 0009 13FFh	CAN1	Mailbox Registers 0 to 31	MB0 to 31	128	8, 16, 32*6	2, 3 PCLKB	2 ICLK	CAN
0009 1400h to 0009 141Fh	CAN1	Mask Registers 0 to 7	MKR0 to 7	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 1420h	CAN1	FIFO Received ID Compare Register 0	FIDCR0	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 1424h	CAN1	FIFO Received ID Compare Register 1	FIDCR1	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 1428h	CAN1	Mask Invalid Register	MKIVLR	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 142Ch	CAN1	Mailbox Interrupt Enable Register	MIER	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 1820h to 0009 183Fh	CAN1	Message Control Registers 0 to 31	MCTL0 to 31	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 1840h	CAN1	Control Register	CTLR	16	8, 16	2, 3 PCLKB	2 ICLK	CAN
0009 1842h	CAN1	Status Register	STR	16	8, 16	2, 3 PCLKB	2 ICLK	CAN
0009 1844h	CAN1	Bit Configuration Register	BCR	32	8, 16, 32	2, 3 PCLKB	2 ICLK	CAN
0009 1848h	CAN1	Receive FIFO Control Register	RFCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 1849h	CAN1	Receive FIFO Pointer Control Register	RFPCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 184Ah	CAN1	Transmit FIFO Control Register	TFCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 184Bh	CAN1	Transmit FIFO Pointer Control Register	TFPCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 184Ch	CAN1	Error Interrupt Enable Register	EIER	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 184Dh	CAN1	Error Interrupt Factor Judge Register	EIFR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 184Eh	CAN1	Receive Error Count Register	RECR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 184Fh	CAN1	Transmit Error Count Register	TECR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 1850h	CAN1	Error Code Store Register	ECSR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 1851h	CAN1	Channel Search Support Register	CSSR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 1852h	CAN1	Mailbox Search Status Register	MSSR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 1853h	CAN1	Mailbox Search Mode Register	MSMR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 1854h	CAN1	Time Stamp Register	TSR	16	16	2, 3 PCLKB	2 ICLK	CAN
0009 1856h	CAN1	Acceptance Filter Support Register	AFSR	16	16	2, 3 PCLKB	2 ICLK	CAN
0009 1858h	CAN1	Test Control Register	TCR	8	8	2, 3 PCLKB	2 ICLK	CAN
0009 4200h	CMTW0	Timer Start Register	CMWSTR	16	16	2, 3 PCLKB	2 ICLK	CMTW
0009 4204h	CMTW0	Timer Control Register	CMWCR	16	16	2, 3 PCLKB	2 ICLK	CMTW
0009 4208h	CMTW0	Timer I/O Control Register	CMWIOR	16	16	2, 3 PCLKB	2 ICLK	CMTW
0009 4210h	CMTW0	Timer Counter	CMWCNT	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 4214h	CMTW0	Compare Match Constant Register	CMWCOR	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 4218h	CMTW0	Input Capture Register 0	CMWICR0	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 421Ch	CMTW0	Input Capture Register 1	CMWICR1	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 4220h	CMTW0	Output Compare Register 0	CMWOCR0	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 4224h	CMTW0	Output Compare Register 1	CMWOCR1	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 4280h	CMTW1	Timer Start Register	CMWSTR	16	16	2, 3 PCLKB	2 ICLK	CMTW
0009 4284h	CMTW1	Timer Control Register	CMWCR	16	16	2, 3 PCLKB	2 ICLK	CMTW
0009 4288h	CMTW1	Timer I/O Control Register	CMWIOR	16	16	2, 3 PCLKB	2 ICLK	CMTW
0009 4290h	CMTW1	Timer Counter	CMWCNT	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 4294h	CMTW1	Compare Match Constant Register	CMWCOR	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 4298h	CMTW1	Input Capture Register 0	CMWICR0	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 429Ch	CMTW1	Input Capture Register 1	CMWICR1	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 42A0h	CMTW1	Output Compare Register 0	CMWOCR0	32	32	2, 3 PCLKB	2 ICLK	CMTW
0009 42A4h	CMTW1	Output Compare Register 1	CMWOCR1	32	32	2, 3 PCLKB	2 ICLK	CMTW

Table 4.1 List of I/O Registers (Address Order) (45 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
0009 5000h	SDSI	FN1 Access Control Register	FN1ACCR	32	32	10, 11 PCLKB	2 to 6 ICLK	SDSI
0009 5004h	SDSI	Interrupt Enable Control Register 1	INTENCR1	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 5005h	SDSI	Interrupt Status Register 1	INTSR1	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 5006h	SDSI	SD Command Control Register	SDCMDCR	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 5007h	SDSI	SD Command Access Address 0 Register	SDCADD0R	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 5008h	SDSI	SD Command Access Address 1 Register	SDCADD1R	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 5009h	SDSI	SD Command Access Address 2 Register	SDCADD2R	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 500Ah	SDSI	SDSI Control Register 1	SDSICR1	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 500Bh	SDSI	DMA Control Register 1	DMACR1	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 500Ch	SDSI	Block Counter	BLKCNT	16	16	8, 9 PCLKB	2 to 5 ICLK	SDSI
0009 500Eh	SDSI	Byte Counter	BYTCNT	16	16	8, 9 PCLKB	2 to 5 ICLK	SDSI
0009 5010h	SDSI	DMA Transfer Address Register	DMATRADDR	32	32	10, 11 PCLKB	2 to 6 ICLK	SDSI
0009 5100h	SDSI	SDSI Control Register 2	SDSICR2	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5104h	SDSI	SDSI Control Register 3	SDSICR3	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5108h	SDSI	Interrupt Enable Control Register 2	INTENCR2	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 510Ch	SDSI	Interrupt Status Register 2	INTSR2	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5110h	SDSI	DMA Control Register 2	DMACR2	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5200h to 0009 526Bh	SDSI	CIS Data Register 0 to 26	CISDAR0 to 26	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5270h	SDSI	FBR Setting Register 1	FBR1	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5274h	SDSI	FBR Setting Register 2	FBR2	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5278h	SDSI	FBR Setting Register 3	FBR3	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 527Ch	SDSI	FBR Setting Register 4	FBR4	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5280h	SDSI	FBR Setting Register 5	FBR5	32	32	2, 3 PCLKB	2 ICLK	SDSI
0009 5800h to 0009 58FFh	SDSI	FN1 Data Register 10 to 163	FN1DAR10 to 163	32	32	10, 11 PCLKB	2 to 6 ICLK	SDSI
0009 5900h to 0009 59FFh	SDSI	FN1 Data Register 20 to 263	FN1DAR20 to 263	32	32	10, 11 PCLKB	2 to 6 ICLK	SDSI
0009 5A00h to 0009 5AFFh	SDSI	FN1 Data Register 30 to 363	FN1DAR30 to 363	32	32	10, 11 PCLKB	2 to 6 ICLK	SDSI
0009 5B00h	SDSI	FN1 Interrupt Vector Register	FN1INTVECR	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 5B01h	SDSI	FN1 Interrupt Clear Register	FN1INTCLR	8	8	7, 8 PCLKB	2 to 5 ICLK	SDSI
0009 5C00h to 0009 5FFFh	SDSI	FN1 Data Register 50 to 5255	FN1DAR50 to 5255	32	32	7, 8 PCLKB	2 to 5 ICLK	SDSI
000A 0000h	USB0	System Configuration Control Register	SYSCFG	16	16	3, 4 PCLKB	2 ICLK	USBb
000A 0004h	USB0	System Configuration Status Register 0	SYSSTS0	16	16	9 PCLKB or more	Rounded up to the nearest integer greater than $1 + 9 \times (\text{frequency ratio of ICLK/PCLKB})^5$	USBb
000A 0008h	USB0	Device State Control Register 0	DVSTCTR0	16	16	9 PCLKB or more	Rounded up to the nearest integer greater than $1 + 9 \times (\text{frequency ratio of ICLK/PCLKB})^5$	USBb
000A 0014h	USB0	CFIFO Port Register	CFIFO	16	8, 16	3, 4 PCLKB	2 ICLK	USBb
000A 0018h	USB0	D0FIFO Port Register	D0FIFO	16	8, 16	3, 4 PCLKB	2 ICLK	USBb
000A 001Ch	USB0	D1FIFO Port Register	D1FIFO	16	8, 16	3, 4 PCLKB	2 ICLK	USBb
000A 0020h	USB0	CFIFO Port Select Register	CFIFOSEL	16	16	3, 4 PCLKB	2 ICLK	USBb
000A 0022h	USB0	CFIFO Port Control Register	CFIFOCTR	16	16	3, 4 PCLKB	2 ICLK	USBb
000A 0028h	USB0	D0FIFO Port Select Register	D0FIFOSEL	16	16	3, 4 PCLKB	2 ICLK	USBb
000A 002Ah	USB0	D0FIFO Port Control Register	D0FIFOCTR	16	16	3, 4 PCLKB	2 ICLK	USBb
000A 002Ch	USB0	D1FIFO Port Select Register	D1FIFOSEL	16	16	3, 4 PCLKB	2 ICLK	USBb
000A 002Eh	USB0	D1FIFO Port Control Register	D1FIFOCTR	16	16	3, 4 PCLKB	2 ICLK	USBb

Table 4.1 List of I/O Registers (Address Order) (46 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000A 0030h	USB0	Interrupt Enable Register 0	INTENB0	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0032h	USB0	Interrupt Enable Register 1	INTENB1	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0036h	USB0	BRDY Interrupt Enable Register	BRDYENB	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0038h	USB0	NRDY Interrupt Enable Register	NRDYENB	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 003Ah	USB0	BEMP Interrupt Enable Register	BEMPENB	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 003Ch	USB0	SOF Output Configuration Register	SOFCFG	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0040h	USB0	Interrupt Status Register 0	INTSTS0	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0042h	USB0	Interrupt Status Register 1	INTSTS1	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0046h	USB0	BRDY Interrupt Status Register	BRDYSTS	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0048h	USB0	NRDY Interrupt Status Register	NRDYSTS	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 004Ah	USB0	BEMP Interrupt Status Register	BEMPSTS	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 004Ch	USB0	Frame Number Register	FRMNUM	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 004Eh	USB0	Device State Change Register	DVCHGR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0050h	USB0	USB Address Register	USBADDR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0054h	USB0	USB Request Type Register	USBREQ	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0056h	USB0	USB Request Value Register	USBVAL	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0058h	USB0	USB Request Index Register	USBINDX	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 005Ah	USB0	USB Request Length Register	USBLENG	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 005Ch	USB0	DCP Configuration Register	DCPCFG	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 005Eh	USB0	DCP Maximum Packet Size Register	DCPMAXP	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0060h	USB0	DCP Control Register	DCPCTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0064h	USB0	Pipe Window Select Register	PIPESEL	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0068h	USB0	Pipe Configuration Register	PIPECFG	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 006Ch	USB0	Pipe Maximum Packet Size Register	PIPEMAXP	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 006Eh	USB0	Pipe Cycle Control Register	PIPEPERI	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0070h	USB0	PIPE1 Control Register	PIPE1CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb

Table 4.1 List of I/O Registers (Address Order) (47 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000A 0072h	USB0	PIPE2 Control Register	PIPE2CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0074h	USB0	PIPE3 Control Register	PIPE3CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0076h	USB0	PIPE4 Control Register	PIPE4CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0078h	USB0	PIPE5 Control Register	PIPE5CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 007Ah	USB0	PIPE6 Control Register	PIPE6CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 007Ch	USB0	PIPE7 Control Register	PIPE7CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 007Eh	USB0	PIPE8 Control Register	PIPE8CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0080h	USB0	PIPE9 Control Register	PIPE9CTR	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0090h	USB0	Pipe1 Transaction Counter Enable Register	PIPE1TRE	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0092h	USB0	Pipe1 Transaction Counter Register	PIPE1TRN	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0094h	USB0	Pipe2 Transaction Counter Enable Register	PIPE2TRE	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0096h	USB0	Pipe2 Transaction Counter Register	PIPE2TRN	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0098h	USB0	Pipe3 Transaction Counter Enable Register	PIPE3TRE	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 009Ah	USB0	Pipe3 Transaction Counter Register	PIPE3TRN	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 009Ch	USB0	Pipe4 Transaction Counter Enable Register	PIPE4TRE	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 009Eh	USB0	Pipe4 Transaction Counter Register	PIPE4TRN	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00A0h	USB0	Pipe5 Transaction Counter Enable Register	PIPE5TRE	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00A2h	USB0	Pipe5 Transaction Counter Register	PIPE5TRN	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00D0h	USB0	Device Address 0 Configuration Register	DEVADD0	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00D2h	USB0	Device Address 1 Configuration Register	DEVADD1	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00D4h	USB0	Device Address 2 Configuration Register	DEVADD2	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00D6h	USB0	Device Address 3 Configuration Register	DEVADD3	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00D8h	USB0	Device Address 4 Configuration Register	DEVADD4	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00DAh	USB0	Device Address 5 Configuration Register	DEVADD5	16	16	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 00F0h	USB0	PHY Cross Point Adjustment Register	PHYSLEW	32	32	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb
000A 0400h	USB	Deep Standby USB Transceiver Control/Pin Monitoring Register	DPUSR0R	32	32	9 PCLKB or more	Frequency with 1 + 9 × (frequency ratio of ICLK/PCLKB) ^{*5}	USBb

Table 4.1 List of I/O Registers (Address Order) (48 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000A 0404h	USB	Deep Standby USB Suspend/Resume Interrupt Register	DPUSR1R	32	32	9 PCLKB or more	Frequency with $1 + 9 \times (\text{frequency ratio of ICLK/PCLKB})^{+5}$	USBb
000A 0500h	PDC	PDC Control Register 0	PCCR0	32	32	2, 3 PCLKB	2 ICLK	PDC
000A 0504h	PDC	PDC Control Register 1	PCCR1	32	32	2, 3 PCLKB	2 ICLK	PDC
000A 0508h	PDC	PDC Status Register	PCSR	32	32	2, 3 PCLKB	2 ICLK	PDC
000A 050Ch	PDC	PDC Pin Monitor Register	PCMONR	32	32	2, 3 PCLKB	2 ICLK	PDC
000A 0510h	PDC	PDC Receive Data Register	PCDR	32	32	2, 3 PCLKB	2 ICLK	PDC
000A 0514h	PDC	Vertical Capture Register	VCR	32	32	2, 3 PCLKB	2 ICLK	PDC
000A 0518h	PDC	Horizontal Capture Register	HCR	32	32	2, 3 PCLKB	2 ICLK	PDC
000C 0000h	EDMAC 0	EDMAC Mode Register	EDMR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0008h	EDMAC 0	EDMAC Transmit Request Register	EDTRR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0010h	EDMAC 0	EDMAC Receive Request Register	EDRRR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0018h	EDMAC 0	Transmit Descriptor List Start Address Register	TDLAR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0020h	EDMAC 0	Receive Descriptor List Start Address Register	RDLAR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0028h	EDMAC 0	ETHERC/EDMAC Status Register	EESR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0030h	EDMAC 0	ETHERC/EDMAC Status Interrupt Enable Register	EESIPR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0038h	EDMAC 0	ETHERC/EDMAC Transmit/Receive Status Copy Enable Register	TRSCER	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0040h	EDMAC 0	Missed-Frame Counter Register	RMFCR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0048h	EDMAC 0	Transmit FIFO Threshold Register	TFTR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0050h	EDMAC 0	FIFO Depth Register	FDR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0058h	EDMAC 0	Receive Method Control Register	RMCR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0064h	EDMAC 0	Transmit FIFO Underflow Counter	TFUCR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0068h	EDMAC 0	Receive FIFO Overflow Counter	RFOCR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 006Ch	EDMAC 0	Independent Output Signal Setting Register	IOSR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0070h	EDMAC 0	Flow Control Start FIFO Threshold Setting Register	FCFTR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0078h	EDMAC 0	Receive Data Padding Insert Register	RPADIR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 007Ch	EDMAC 0	Transmit Interrupt Setting Register	TRIMD	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 00C8h	EDMAC 0	Receive Buffer Write Address Register	RBWAR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 00CCh	EDMAC 0	Receive Descriptor Fetch Address Register	RDFAR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 00D4h	EDMAC 0	Transmit Buffer Read Address Register	TBRAR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 00D8h	EDMAC 0	Transmit Descriptor Fetch Address Register	TDFAR	32	32	4, 5 PCLKA	1 to 3 ICLK	EDMAC a
000C 0100h	ETHER C0	ETHERC Mode Register	ECMR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0108h	ETHER C0	Receive Frame Maximum Length Register	RFLR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0110h	ETHER C0	ETHERC Status Register	ECSR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0118h	ETHER C0	ETHERC Interrupt Enable Register	ECSIPR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C

Table 4.1 List of I/O Registers (Address Order) (49 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000C 0120h	ETHER C0	PHY Interface Register	PIR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0128h	ETHER C0	PHY Status Register	PSR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0140h	ETHER C0	Random Number Generation Counter Limit Setting Register	RDMLR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0150h	ETHER C0	Interpacket Gap Register	IPGR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0154h	ETHER C0	Automatic PAUSE Frame Register	APR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0158h	ETHER C0	Manual PAUSE Frame Register	MPR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0160h	ETHER C0	Received PAUSE Frame Counter	RFCF	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0164h	ETHER C0	PAUSE Frame Retransmit Count Setting Register	TPAUSER	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 0168h	ETHER C0	PAUSE Frame Retransmit Counter	TPAUSECR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 016Ch	ETHER C0	Broadcast Frame Receive Count Setting Register	BCFRR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01C0h	ETHER C0	MAC Address Upper Bit Register	MAHR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01C8h	ETHER C0	MAC Address Lower Bit Register	MALR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01D0h	ETHER C0	Transmit Retry Over Counter Register	TROCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01D4h	ETHER C0	Late Collision Detect Counter Register	CDCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01D8h	ETHER C0	Lost Carrier Counter Register	LCCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01DCh	ETHER C0	Carrier Not Detect Counter Register	CNDCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01E4h	ETHER C0	CRC Error Frame Receive Counter Register	CEFCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01E8h	ETHER C0	Frame Receive Error Counter Register	FRECR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01ECh	ETHER C0	Too-Short Frame Receive Counter Register	TSFRRCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01F0h	ETHER C0	Too-Long Frame Receive Counter Register	TLFRRCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01F4h	ETHER C0	Received Alignment Error Frame Counter Register	RFCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 01F8h	ETHER C0	Multicast Address Frame Receive Counter Register	MAFCR	32	32	13, 14 PCLKA	1 to 7 ICLK	ETHER C
000C 1200h	MTU3	Timer Control Register	TCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1201h	MTU4	Timer Control Register	TCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1202h	MTU3	Timer Mode Register 1	TMDR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1203h	MTU4	Timer Mode Register 1	TMDR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1204h	MTU3	Timer I/O Control Register H	TIORH	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1205h	MTU3	Timer I/O Control Register L	TIORL	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1206h	MTU4	Timer I/O Control Register H	TIORH	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1207h	MTU4	Timer I/O Control Register L	TIORL	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1208h	MTU3	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1209h	MTU4	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 120Ah	MTU	Timer Output Master Enable Register A	TOERA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 120Dh	MTU	Timer Gate Control Register A	TGCRA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 120Eh	MTU	Timer Output Control Register 1A	TOCR1A	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 120Fh	MTU	Timer Output Control Register 2A	TOCR2A	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1210h	MTU3	Timer Counter	TCNT	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a

Table 4.1 List of I/O Registers (Address Order) (50 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000C 1212h	MTU4	Timer Counter	TCNT	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1214h	MTU	Timer Cycle Data Register A	TCDRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1216h	MTU	Timer Dead Time Data Register A	TDDRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1218h	MTU3	Timer General Register A	TGRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 121Ah	MTU3	Timer General Register B	TGRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 121Ch	MTU4	Timer General Register A	TGRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 121Eh	MTU4	Timer General Register B	TGRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1220h	MTU	Timer Subcounter A	TCNTSA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1222h	MTU	Timer Cycle Buffer Register A	TCBRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1224h	MTU3	Timer General Register C	TGRC	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1226h	MTU3	Timer General Register D	TGRD	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1228h	MTU4	Timer General Register C	TGRC	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 122Ah	MTU4	Timer General Register D	TGRD	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 122Ch	MTU3	Timer Status Register	TSR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 122Dh	MTU4	Timer Status Register	TSR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1230h	MTU	Timer Interrupt Skipping Set Register 1A	TITCR1A	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1231h	MTU	Timer Interrupt Skipping Counter 1A	TITCNT1A	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1232h	MTU	Timer Buffer Transfer Set Register A	TBTERA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1234h	MTU	Timer Dead Time Enable Register A	TDERA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1236h	MTU	Timer Output Level Buffer Register A	TOLBRA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1238h	MTU3	Timer Buffer Operation Transfer Mode Register	TBTM	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1239h	MTU4	Timer Buffer Operation Transfer Mode Register	TBTM	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 123Ah	MTU	Timer Interrupt Skipping Mode Register A	TITMRA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 123Bh	MTU	Timer Interrupt Skipping Set Register 2A	TITCR2A	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 123Ch	MTU	Timer Interrupt Skipping Counter 2A	TITCNT2A	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1240h	MTU4	Timer A/D Converter Start Request Control Register	TADCR	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1244h	MTU4	Timer A/D Converter Start Request Cycle Set Register A	TADCORA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1246h	MTU4	Timer A/D Converter Start Request Cycle Set Register B	TADCORB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1248h	MTU4	Timer A/D Converter Start Request Cycle Set Buffer Register A	TADCOBRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 124Ah	MTU4	Timer A/D Converter Start Request Cycle Set Buffer Register B	TADCOBRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 124Ch	MTU3	Timer Control Register 2	TCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 124Dh	MTU4	Timer Control Register 2	TCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1260h	MTU	Timer Waveform Control Register A	TWCRA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1270h	MTU	Timer Mode Register 2A	TMDR2A	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1272h	MTU3	Timer General Register E	TGRE	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1274h	MTU4	Timer General Register E	TGRE	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1276h	MTU4	Timer General Register F	TGRF	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1280h	MTU	Timer Start Register A	TSTRA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1281h	MTU	Timer Synchronous Register A	TSYRA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1282h	MTU	Timer Counter Synchronous Start Register	TCSYSTR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1284h	MTU	Timer Read/Write Enable Register A	TRWERA	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1290h	MTU0	Noise Filter Control Register 0	NFCR0	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1291h	MTU1	Noise Filter Control Register 1	NFCR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1292h	MTU2	Noise Filter Control Register 2	NFCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1293h	MTU3	Noise Filter Control Register 3	NFCR3	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1294h	MTU4	Noise Filter Control Register 4	NFCR4	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1298h	MTU8	Noise Filter Control Register 8	NFCR8	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a

Table 4.1 List of I/O Registers (Address Order) (51 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000C 1299h	MTU0	Noise Filter Control Register C	NFCRC	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1300h	MTU0	Timer Control Register	TCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1301h	MTU0	Timer Mode Register 1	TMDR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1302h	MTU0	Timer I/O Control Register H	TIORH	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1303h	MTU0	Timer I/O Control Register L	TIORL	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1304h	MTU0	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1306h	MTU0	Timer Counter	TCNT	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1308h	MTU0	Timer General Register A	TGRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 130Ah	MTU0	Timer General Register B	TGRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 130Ch	MTU0	Timer General Register C	TGRC	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 130Eh	MTU0	Timer General Register D	TGRD	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1320h	MTU0	Timer General Register E	TGRE	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1322h	MTU0	Timer General Register F	TGRF	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1324h	MTU0	Timer Interrupt Enable Register 2	TIER2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1326h	MTU0	Timer Buffer Operation Transfer Mode Register	TBTM	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1328h	MTU0	Timer Control Register 2	TCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1380h	MTU1	Timer Control Register	TCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1381h	MTU1	Timer Mode Register 1	TMDR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1382h	MTU1	Timer I/O Control Register	TIOR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1384h	MTU1	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1385h	MTU1	Timer Status Register	TSR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1386h	MTU1	Timer Counter	TCNT	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1388h	MTU1	Timer General Register A	TGRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 138Ah	MTU1	Timer General Register B	TGRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1390h	MTU1	Timer Input Capture Control Register	TICCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1391h	MTU1	Timer Mode Register 3	TMDR3	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1394h	MTU1	Timer Control Register 2	TCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 13A0h	MTU1	Timer Longword Counter	TCNTLW	32	32	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 13A4h	MTU1	Timer Longword General Register	TGRALW	32	32	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 13A8h	MTU1	Timer Longword General Register	TGRBLW	32	32	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1400h	MTU2	Timer Control Register	TCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1401h	MTU2	Timer Mode Register 1	TMDR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1402h	MTU2	Timer I/O Control Register	TIOR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1404h	MTU2	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1405h	MTU2	Timer Status Register	TSR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1406h	MTU2	Timer Counter	TCNT	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1408h	MTU2	Timer General Register A	TGRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 140Ah	MTU2	Timer General Register B	TGRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 140Ch	MTU2	Timer Control Register 2	TCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1600h	MTU8	Timer Control Register	TCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1601h	MTU8	Timer Mode Register 1	TMDR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1602h	MTU8	Timer I/O Control Register H	TIORH	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1603h	MTU8	Timer I/O Control Register L	TIORL	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1604h	MTU8	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1606h	MTU8	Timer Control Register 2	TCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1608h	MTU8	Timer Counter	TCNT	32	32	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 160Ch	MTU8	Timer General Register A	TGRA	32	32	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1610h	MTU8	Timer General Register B	TGRB	32	32	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1614h	MTU8	Timer General Register C	TGRC	32	32	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1618h	MTU8	Timer General Register D	TGRD	32	32	4, 5 PCLKA	1, 2 ICLK	MTU3a

Table 4.1 List of I/O Registers (Address Order) (52 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000C 1A00h	MTU6	Timer Control Register	TCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A01h	MTU7	Timer Control Register	TCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A02h	MTU6	Timer Mode Register 1	TMDR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A03h	MTU7	Timer Mode Register 1	TMDR1	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A04h	MTU6	Timer I/O Control Register H	TIORH	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A05h	MTU6	Timer I/O Control Register L	TIORL	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A06h	MTU7	Timer I/O Control Register H	TIORH	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A07h	MTU7	Timer I/O Control Register L	TIORL	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A08h	MTU6	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A09h	MTU7	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A0Ah	MTU	Timer Output Master Enable Register B	TOERB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A0Eh	MTU	Timer Output Control Register 1B	TOCR1B	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A0Fh	MTU	Timer Output Control Register 2B	TOCR2B	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A10h	MTU6	Timer Counter	TCNT	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A12h	MTU7	Timer Counter	TCNT	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A14h	MTU	Timer Cycle Data Register B	TCDRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A16h	MTU	Timer Dead Time Data Register B	TDDRb	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A18h	MTU6	Timer General Register A	TGRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A1Ah	MTU6	Timer General Register B	TGRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A1Ch	MTU7	Timer General Register A	TGRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A1Eh	MTU7	Timer General Register B	TGRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A20h	MTU	Timer Subcounter B	TCNTSB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A22h	MTU	Timer Cycle Buffer Register B	TCBRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A24h	MTU6	Timer General Register C	TGRC	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A26h	MTU6	Timer General Register D	TGRD	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A28h	MTU7	Timer General Register C	TGRC	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A2Ah	MTU7	Timer General Register D	TGRD	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A2Ch	MTU6	Timer Status Register	TSR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A2Dh	MTU7	Timer Status Register	TSR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A30h	MTU	Timer Interrupt Skipping Set Register 1B	TITCR1B	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A31h	MTU	Timer Interrupt Skipping Counter 1B	TITCNT1B	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A32h	MTU	Timer Buffer Transfer Set Register B	TBTERB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A34h	MTU	Timer Dead Time Enable Register B	TDERB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A36h	MTU	Timer Output Level Buffer Register B	TOLBRB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A38h	MTU6	Timer Buffer Operation Transfer Mode Register	TBTM	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A39h	MTU7	Timer Buffer Operation Transfer Mode Register	TBTM	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A3Ah	MTU	Timer Interrupt Skipping Mode Register B	TITMRB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A3Bh	MTU	Timer Interrupt Skipping Set Register 2B	TITCR2B	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A3Ch	MTU	Timer Interrupt Skipping Counter 2B	TITCNT2B	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A40h	MTU7	Timer A/D Converter Start Request Control Register	TADCR	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A44h	MTU7	Timer A/D Converter Start Request Cycle Set Register A	TADCORA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A46h	MTU7	Timer A/D Converter Start Request Cycle Set Register B	TADCORB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A48h	MTU7	Timer A/D Converter Start Request Cycle Set Buffer Register A	TADCOBRA	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A4Ah	MTU7	Timer A/D Converter Start Request Cycle Set Buffer Register B	TADCOBRB	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A4Ch	MTU6	Timer Control Register 2	TCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A4Dh	MTU7	Timer Control Register 2	TCR2	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A50h	MTU6	Timer Synchronous Clear Register	TSYCR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a

Table 4.1 List of I/O Registers (Address Order) (53 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000C 1A60h	MTU	Timer Waveform Control Register B	TWCRB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A70h	MTU	Timer Mode Register 2B	TMDR2B	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A72h	MTU6	Timer General Register E	TGRE	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A74h	MTU7	Timer General Register E	TGRE	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A76h	MTU7	Timer General Register F	TGRF	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A80h	MTU	Timer Start Register B	TSTRB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A81h	MTU	Timer Synchronous Register B	TSYRB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A84h	MTU	Timer Read/Write Enable Register B	TRWERB	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A93h	MTU6	Noise Filter Control Register 6	NFCR6	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A94h	MTU7	Noise Filter Control Register 7	NFCR7	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1A95h	MTU5	Noise Filter Control Register 5	NFCR5	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C80h	MTU5	Timer Counter U	TCNTU	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C82h	MTU5	Timer General Register U	TGRU	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C84h	MTU5	Timer Control Register U	TCRU	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C85h	MTU5	Timer Control Register 2	TCR2U	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C86h	MTU5	Timer I/O Control Register U	TIORU	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C90h	MTU5	Timer Counter V	TCNTV	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C92h	MTU5	Timer General Register V	TGRV	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C94h	MTU5	Timer Control Register V	TCRV	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C95h	MTU5	Timer Control Register 2	TCR2V	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1C96h	MTU5	Timer I/O Control Register V	TIORV	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1CA0h	MTU5	Timer Counter W	TCNTW	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1CA2h	MTU5	Timer General Register W	TGRW	16	16	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1CA4h	MTU5	Timer Control Register W	TCRW	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1CA5h	MTU5	Timer Control Register 2	TCR2W	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1CA6h	MTU5	Timer I/O Control Register W	TIORW	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1CB2h	MTU5	Timer Interrupt Enable Register	TIER	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1CB4h	MTU5	Timer Start Register	TSTR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000C 1CB6h	MTU5	Timer Compare Match Clear Register	TCNTCMPCLR	8	8	4, 5 PCLKA	1, 2 ICLK	MTU3a
000D 0040h	SCI10	Serial Mode Register	SMR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0041h	SCI10	Bit Rate Register	BRR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0042h	SCI10	Serial Control Register	SCR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0043h	SCI10	Transmit Data Register	TDR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0044h	SCI10	Serial Status Register	SSR/SSRFIFO	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0045h	SCI10	Receive Data Register	RDR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0046h	SMCI10	Smart Card Mode Register	SCMR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0047h	SCI10	Serial Extended Mode Register	SEMR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0048h	SCI10	Noise Filter Setting Register	SNFR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 0049h	SCI10	I ² C Mode Register 1	SIMR1	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Ah	SCI10	I ² C Mode Register 2	SIMR2	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Bh	SCI10	I ² C Mode Register 3	SIMR3	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Ch	SCI10	I ² C Status Register	SISR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Dh	SCI10	SPI Mode Register	SPMR	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Eh	SCI10	Transmit Data Register H	TDRH	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Fh	SCI10	Transmit Data Register L	TDRL	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Eh	SCI10	Transmit Data Register HL	TDRHL	16	16	5, 6 PCLKA	1 to 3 ICLK	SCIi
000D 004Eh	SCI10	Transmit FIFO Data Register	FTDR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Fh	SCI10	Transmit FIFO Data Register	FTDR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCIi
000D 004Eh	SCI10	Transmit FIFO Data Register	FTDR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCIi

Table 4.1 List of I/O Registers (Address Order) (54 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000D 0050h	SCI10	Receive Data Register H	RDRH	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0051h	SCI10	Receive Data Register L	RDRL	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0050h	SCI10	Receive Data Register HL	RDRHL	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0050h	SCI10	Receive FIFO Data Register	FRDR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0051h	SCI10	Receive FIFO Data Register	FRDR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0050h	SCI10	Receive FIFO Data Register	FRDR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0052h	SCI10	Modulation Duty Register	MDDR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0053h	SCI10	Data Comparison Control Register	DCCR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0054h	SCI10	FIFO Control Register	FCR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0055h	SCI10	FIFO Control Register	FCR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0054h	SCI10	FIFO Control Register	FCR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0056h	SCI10	FIFO Data Count Register	FDR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0057h	SCI10	FIFO Data Count Register	FDR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0056h	SCI10	FIFO Data Count Register	FDR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0058h	SCI10	Line Status Register	LSR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0059h	SCI10	Line Status Register	LSR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0058h	SCI10	Line Status Register	LSR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 005Ah	SCI10	Comparison Data Register	CDR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 005Bh	SCI10	Comparison Data Register	CDR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 005Ah	SCI10	Comparison Data Register	CDR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 005Ch	SCI10	Serial Port Register	SPTR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0060h	SCI11	Serial Mode Register	SMR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0061h	SCI11	Bit Rate Register	BRR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0062h	SCI11	Serial Control Register	SCR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0063h	SCI11	Transmit Data Register	TDR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0064h	SCI11	Serial Status Register	SSR/ SSRFIFO	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0065h	SCI11	Receive Data Register	RDR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0066h	SMCI11	Smart Card Mode Register	SCMR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0067h	SCI11	Serial Extended Mode Register	SEMR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0068h	SCI11	Noise Filter Setting Register	SNFR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0069h	SCI11	I ² C Mode Register 1	SIMR1	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Ah	SCI11	I ² C Mode Register 2	SIMR2	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Bh	SCI11	I ² C Mode Register 3	SIMR3	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Ch	SCI11	I ² C Status Register	SISR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Dh	SCI11	SPI Mode Register	SPMR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Eh	SCI11	Transmit Data Register H	TDRH	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Fh	SCI11	Transmit Data Register L	TDRL	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Eh	SCI11	Transmit Data Register HL	TDRHL	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 006Eh	SCI11	Transmit FIFO Data Register	FTDR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Fh	SCI11	Transmit FIFO Data Register	FTDR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 006Eh	SCI11	Transmit FIFO Data Register	FTDR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0070h	SCI11	Receive Data Register H	RDRH	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0071h	SCI11	Receive Data Register L	RDRL	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0070h	SCI11	Receive Data Register HL	RDRHL	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0070h	SCI11	Receive FIFO Data Register	FRDR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0071h	SCI11	Receive FIFO Data Register	FRDR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0070h	SCI11	Receive FIFO Data Register	FRDR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0072h	SCI11	Modulation Duty Register	MDDR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0073h	SCI11	Data Comparison Control Register	DCCR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli

Table 4.1 List of I/O Registers (Address Order) (55 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000D 0074h	SCI11	FIFO Control Register	FCR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0075h	SCI11	FIFO Control Register	FCR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0074h	SCI11	FIFO Control Register	FCR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0076h	SCI11	FIFO Data Count Register	FDR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0077h	SCI11	FIFO Data Count Register	FDR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0076h	SCI11	FIFO Data Count Register	FDR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 0078h	SCI11	Line Status Register	LSR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0079h	SCI11	Line Status Register	LSR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0078h	SCI11	Line Status Register	LSR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 007Ah	SCI11	Comparison Data Register	CDR.H	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 007Bh	SCI11	Comparison Data Register	CDR.L	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 007Ah	SCI11	Comparison Data Register	CDR	16	16	5, 6 PCLKA	1 to 3 ICLK	SCli
000D 007Ch	SCI11	Serial Port Register	SPTR	8	8	3, 4 PCLKA	1, 2 ICLK	SCli
000D 0100h	RSPI0	RSPI Control Register	SPCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0101h	RSPI0	RSPI Slave Select Polarity Register	SSLP	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0102h	RSPI0	RSPI Pin Control Register	SPPCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0103h	RSPI0	RSPI Status Register	SPSR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0104h	RSPI0	RSPI Data Register	SPDR	32	8, 16, 32	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0108h	RSPI0	RSPI Sequence Control Register	SPSCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0109h	RSPI0	RSPI Sequence Status Register	SPSSR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 010Ah	RSPI0	RSPI Bit Rate Register	SPBR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 010Bh	RSPI0	RSPI Data Control Register	SPDCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 010Ch	RSPI0	RSPI Clock Delay Register	SPCKD	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 010Dh	RSPI0	RSPI Slave Select Negation Delay Register	SSLND	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 010Eh	RSPI0	RSPI Next-Access Delay Register	SPND	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 010Fh	RSPI0	RSPI Control Register 2	SPCR2	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0110h	RSPI0	RSPI Command Register 0	SPCMD0	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0112h	RSPI0	RSPI Command Register 1	SPCMD1	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0114h	RSPI0	RSPI Command Register 2	SPCMD2	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0116h	RSPI0	RSPI Command Register 3	SPCMD3	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0118h	RSPI0	RSPI Command Register 4	SPCMD4	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 011Ah	RSPI0	RSPI Command Register 5	SPCMD5	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 011Ch	RSPI0	RSPI Command Register 6	SPCMD6	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 011Eh	RSPI0	RSPI Command Register 7	SPCMD7	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0120h	RSPI0	RSPI Data Control Register 2	SPDCR2	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0140h	RSPI1	RSPI Control Register	SPCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0141h	RSPI1	RSPI Slave Select Polarity Register	SSLP	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0142h	RSPI1	RSPI Pin Control Register	SPPCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0143h	RSPI1	RSPI Status Register	SPSR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0144h	RSPI1	RSPI Data Register	SPDR	32	8, 16, 32	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0148h	RSPI1	RSPI Sequence Control Register	SPSCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0149h	RSPI1	RSPI Sequence Status Register	SPSSR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 014Ah	RSPI1	RSPI Bit Rate Register	SPBR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 014Bh	RSPI1	RSPI Data Control Register	SPDCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 014Ch	RSPI1	RSPI Clock Delay Register	SPCKD	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 014Dh	RSPI1	RSPI Slave Select Negation Delay Register	SSLND	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 014Eh	RSPI1	RSPI Next-Access Delay Register	SPND	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 014Fh	RSPI1	RSPI Control Register 2	SPCR2	8	8	3, 4 PCLKA	1, 2 ICLK	RSPIc
000D 0150h	RSPI1	RSPI Command Register 0	SPCMD0	16	16	3, 4 PCLKA	1, 2 ICLK	RSPIc

Table 4.1 List of I/O Registers (Address Order) (56 / 56)

Address	Module Symbol	Register Name	Register Symbol	Number of Bits	Access Size	Number of Access Cycles		Related Function
						ICLK ≥ PCLK	ICLK < PCLK	
000D 0152h	RSP1	RSP1 Command Register 1	SPCMD1	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0154h	RSP1	RSP1 Command Register 2	SPCMD2	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0156h	RSP1	RSP1 Command Register 3	SPCMD3	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0158h	RSP1	RSP1 Command Register 4	SPCMD4	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 015Ah	RSP1	RSP1 Command Register 5	SPCMD5	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 015Ch	RSP1	RSP1 Command Register 6	SPCMD6	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 015Eh	RSP1	RSP1 Command Register 7	SPCMD7	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0160h	RSP1	RSP1 Data Control Register 2	SPDCR2	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0300h	RSP12	RSP1 Control Register	SPCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0301h	RSP12	RSP1 Slave Select Polarity Register	SSLP	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0302h	RSP12	RSP1 Pin Control Register	SPPCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0303h	RSP12	RSP1 Status Register	SPSR	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0304h	RSP12	RSP1 Data Register	SPDR	32	8, 16, 32	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0308h	RSP12	RSP1 Sequence Control Register	SPSCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0309h	RSP12	RSP1 Sequence Status Register	SPSSR	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 030Ah	RSP12	RSP1 Bit Rate Register	SPBR	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 030Bh	RSP12	RSP1 Data Control Register	SPDCR	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 030Ch	RSP12	RSP1 Clock Delay Register	SPCKD	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 030Dh	RSP12	RSP1 Slave Select Negation Delay Register	SSLND	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 030Eh	RSP12	RSP1 Next-Access Delay Register	SPND	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 030Fh	RSP12	RSP1 Control Register 2	SPCR2	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0310h	RSP12	RSP1 Command Register 0	SPCMD0	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0312h	RSP12	RSP1 Command Register 1	SPCMD1	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0314h	RSP12	RSP1 Command Register 2	SPCMD2	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0316h	RSP12	RSP1 Command Register 3	SPCMD3	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0318h	RSP12	RSP1 Command Register 4	SPCMD4	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 031Ah	RSP12	RSP1 Command Register 5	SPCMD5	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 031Ch	RSP12	RSP1 Command Register 6	SPCMD6	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 031Eh	RSP12	RSP1 Command Register 7	SPCMD7	16	16	3, 4 PCLKA	1, 2 ICLK	RSP1c
000D 0320h	RSP12	RSP1 Data Control Register 2	SPDCR2	8	8	3, 4 PCLKA	1, 2 ICLK	RSP1c
FE7F 7D7Ch	TEMPS	Temperature Sensor Calibration Data Register	TSCDR	32	32	1 to 3 ICLK		TEMPS
FE7F 7D90h	FLASH	Unique ID Register 0	UIDR0	32	32	1 to 3 ICLK		Flash
FE7F 7D94h	FLASH	Unique ID Register 1	UIDR1	32	32	1 to 3 ICLK		Flash
FE7F 7D98h	FLASH	Unique ID Register 2	UIDR2	32	32	1 to 3 ICLK		Flash
FE7F 7D9Ch	FLASH	Unique ID Register 3	UIDR3	32	32	1 to 3 ICLK		Flash

Note 1. When the same output trigger is specified for pulse output groups 2 and 3 by the PPG0.PCR setting, the PPG0.NDRH address is 0008 81ECh. When different output triggers are specified, the PPG0.NDRH addresses for pulse output groups 2 and 3 are 0008 81EEh and 0008 81ECh, respectively.

Note 2. When the same output trigger is specified for pulse output groups 0 and 1 by the PPG0.PCR setting, the PPG0.NDRL address is 0008 81EDh. When different output triggers are specified, the PPG0.NDRL addresses for pulse output groups 0 and 1 are 0008 81EFh and 0008 81EDh, respectively.

Note 3. When the same output trigger is specified for pulse output groups 6 and 7 by the PPG1.PCR setting, the PPG1.NDRH address is 0008 81FCh. When different output triggers are specified, the PPG1.NDRH addresses for pulse output groups 6 and 7 are 0008 81FEh and 0008 81FCh, respectively.

Note 4. When the same output trigger is specified for pulse output groups 4 and 5 by the PPG1.PCR setting, the PPG1.NDRL address is 0008 81FDh. When different output triggers are specified, the PPG1.NDRL addresses for pulse output groups 4 and 5 are 0008 81FFh and 0008 81FDh, respectively.

Note 5. When the register is accessed while the USB is operating, a delay may be generated in accessing.

Note 6. The address must end with 0h, 4h, 8h, or Ch when access is made in 32-bit units. The address must end with 0h, 2h, 4h, 6h, 8h, Ah, Ch, or Eh when access is made in 16-bit units.

5. Electrical Characteristics

5.1 Absolute Maximum Ratings

Table 5.1 Absolute Maximum Rating

Conditions: $V_{SS} = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0\text{ V}$

Item	Symbol	Value	Unit
Power supply voltage	V_{CC}, V_{CC_USB}	-0.3 to +4.0	V
V_{BATT} power supply voltage	V_{BATT}	-0.3 to +4.0	V
Input voltage (except for ports for 5 V tolerant*1)	V_{in}	-0.3 to $V_{CC} + 0.3$ (up to 4.0)	V
Input voltage (ports for 5 V tolerant*1)	V_{in}	-0.3 to $V_{CC} + 4.0$ (up to 5.8)	V
Reference power supply voltage	$VREFH0$	-0.3 to $AVCC0 + 0.3$ (up to 4.0)	V
Analog power supply voltage	$AVCC0, AVCC1^{*2}$	-0.3 to +4.0	V
Analog input voltage	V_{AN}	-0.3 to $AVCC + 0.3$ (up to 4.0)	V
Storage temperature	T_{stg}	-55 to +125	°C

Caution: Permanent damage to the LSI may result if absolute maximum ratings are exceeded.

Note 1. Ports 07, 12 to 17, 20, 21, 30 to 33, 67, and C0 to C3 are 5 V tolerant.

Note 2. Connect the $AVCC0$, $AVCC1$, and V_{CC_USB} pins to V_{CC} , and the $AVSS0$, $AVSS1$, and VSS_USB pins to VSS , respectively. Do not leave these pins open.

When the A/D converter unit 0 is not to be used, connect the $VREFH0$ pin to V_{CC} and the $VREFL0$ pin to VSS , respectively.

Do not leave these pins open. Insert capacitors of high frequency characteristics between the $AVCC0$ and $AVSS0$ pins, or $AVCC1$ and $AVSS1$ pins. Place capacitors of about 0.1 μF as close as possible to every power supply pin and use the shortest and heaviest possible traces.

Table 5.2 Recommended operating conditions

Item	Symbol	Min.	Typ.	Max.	Unit
Power supply voltage*1	V_{CC}	2.7	—	3.6	V
	V_{SS}	—	0	—	V
V_{BATT} power supply voltage	V_{BATT}	2.0	—	3.6	V
USB power supply voltage	V_{CC_USB}	—	V_{CC}	—	V
	V_{SS_USB}	—	0	—	V
Analog power supply voltage*1, *2	$AVCC0$	—	V_{CC}	—	V
	$AVSS0$	—	0	—	V
	$AVCC1$	—	V_{CC}	—	V
	$AVSS1$	—	0	—	V
	$VREFH0$	2.7	—	$AVCC0$	V
	$VREFL0$	—	0	—	V
Input voltage (except for 5 V tolerant ports, except for ports 03 to 05 and 40 to 47)*3	V_{in}	-0.3	—	$V_{CC} + 0.3$	V
Input voltage (ports 03 to 05 and 40 to 47)	V_{in}	-0.3	—	$AVCC + 0.3$	V
Input voltage (5V tolerant ports 12 to 17, ports 20 and 21, ports 30 to 33, port 67, and ports C0 to C3)*4	V_{in}	-0.3	—	$V_{CC} + 3.6$ (up to 5.5)	V
Input voltage (5V tolerant port 7)	V_{in}	-0.3	—	$AVCC + 3.6$ (up to 5.5)	V
Operating temperature	T_{opr}	-40	—	85	°C
Operating temperature (high-temperature products)	T_{opr}	-40	—	105 (Under planning)	°C

Note 1. Comply with the following potential condition:

$V_{CC} = AVCC0 = AVCC1 = V_{CC_USB}$

Note 2. For details, see section 50.6.10, Voltage Range of Analog Power Supply Pins in the User's Manual: Hardware.

Note 3. Ports 07, 12 to 17, 20, 21, 30 to 33, 67, and C0 to C3 are 5 V tolerant.

Note 4. For P32, P31, and P30, input as follows when the V_{BATT} power supply is selected.

V_{in} Min. = -0.3, Max. = $V_{BATT} + 0.3$ ($V_{BATT} = 2.0$ to 3.6 V)

5.2 DC Characteristics

Table 5.3 DC Characteristics (1)

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Schmitt trigger input voltage	IRQ input pin*1 MTU input pin*1 POE3 input pin*1 TPU input pin*1 TMR input pin*1 CMT2 input pin*1 SCI input pin*1 CAN input pin*1 CAC input pin*1 ADTRG# input pin*1 QSPI input pin*1 RES#, NMI, TCK	V _{IH}	VCC × 0.8	—	—	V	
		V _{IL}	—	—	VCC × 0.2		
		ΔV _T	VCC × 0.06	—	—		
	RIIC input pin (except for SMBus)	V _{IH}	VCC × 0.7	—	—		
		V _{IL}	—	—	VCC × 0.3		
		ΔV _T	VCC × 0.05	—	—		
Ports for 5 V tolerant*2	V _{IH}	VCC × 0.8	—	—			
	V _{IL}	—	—	VCC × 0.2			
Other input pins excluding ports for 5 V tolerant*3	V _{IH}	VCC × 0.8	—	—			
	V _{IL}	—	—	VCC × 0.2			
Input high voltage (except for Schmitt trigger input pin)	MD pin, EMLE	V _{IH}	VCC × 0.9	—	—	V	
	EXTAL, RSPI input pin, EXDMAC input pin, WAIT#, SDHI input pin, MMC input pin, PDC input pin, SDSI input pin		VCC × 0.8	—	—		
	ETHERC input pin		2.3	—	—		
	D0 to D15		VCC × 0.7	—	—		
	RIIC (SMBus)		2.1	—	—		
Input low voltage (except for Schmitt trigger input pin)	MD pin, EMLE	V _{IL}	—	—	VCC × 0.1	V	
	EXTAL, RSPI input pin, ETHERC input pin, EXDMAC input pin, WAIT#, SDHI input pin, MMC input pin, PDC input pin, SDSI input pin		—	—	VCC × 0.2		
	D0 to D15		—	—	VCC × 0.3		
	RIIC (SMBus)		—	—	0.8		

Note 1. This does not include the pins, which are multiplexed as ports for 5 V tolerant.

Note 2. Ports 07, 12 to 17, 20, 21, 30 to 33, 67, and C0 to C3 are 5 V tolerant.

Note 3. For P32, P31, and P30, input as follows when the V_{BATT} power supply is selected.

V_{IH} Min. = $V_{BATT} \times 0.8$, V_{IL} Max. = $V_{BATT} \times 0.2$ ($V_{BATT} = 2.0$ to 3.6 V)

Table 5.4 DC Characteristics (2)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Output high voltage	All output pins	V _{OH}	VCC – 0.5	—	—	V	I _{OH} = –1 mA
Output low voltage	All output pins (except for RIIC pins and ETHERC output pin)	V _{OL}	—	—	0.5	V	I _{OL} = 1.0 mA
	RIIC output pin		—	—	0.4		I _{OL} = 3.0 mA
			—	—	0.6		I _{OL} = 6.0 mA
	RIIC output pin (only P12 and P13 in channel 0)	V _{OL}	—	—	0.4	V	I _{OL} = 15.0 mA (ICFER.FMPE = 1)
			—	0.4	—		I _{OL} = 20.0 mA (ICFER.FMPE = 1)
	ETHERC output pin	V _{OL}	—	—	0.4	V	I _{OL} = 1.0 mA
Input leakage current	RES#, MD pin, EMLE*1, BSCANP*1, NMI	I _{in}	—	—	1.0	μA	V _{in} = 0 V V _{in} = VCC
Three-state leakage current (off state)	Other than ports for 5 V tolerant	I _{TSI}	—	—	1.0	μA	V _{in} = 0 V V _{in} = VCC
	Ports for 5 V tolerant		—	—	5.0		V _{in} = 0 V V _{in} = 5.5 V
Input pull-up MOS current	Other than P35	I _p	–300	—	–10	μA	VCC = 2.7 to 3.6 V V _{in} = 0 V
Input pull-down MOS current	EMLE, BSCANP	I _p	10	—	300	μA	V _{in} = VCC
Input capacitance	All input pins (except for ports 03, 05, 12, 13, 16, 17, EMLE, BSCANP, USB0_DP, and USB0_DM)	C _{in}	—	—	8	pF	V _{bias} = 0 V V _{amp} = 20 mV f = 1 MHz T _a = 25°C
	Ports 03, 05, 12, 13, 16, 17, EMLE, BSCANP, USB0_DP, and USB0_DM		—	—	16		

Note 1. The input leakage current value at the EMLE and BSCANP pins are only when $V_{in} = 0$ V.

Table 5.5 DC Characteristics (3)Conditions: $V_{CC} = AVCC0 = AVCC1 = VREFH0 = VCC_USB = 2.7$ to 3.6 V, $2.7 \leq VREFH0 \leq AVCC0$, $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V, $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit	Test Conditions		
Supply current*1	High-speed operating mode	Max.*2		I _{CC} *3	—	—	40	mA	ICLK = 120 MHz PCLKA = 120 MHz PCLKB = 60 MHz PCLKC = 60 MHz PCLKD = 60 MHz FCLK = 60 MHz BCLK = 120 MHz BCLK pin = 60 MHz	
		Normal	Peripheral function clock signal supplied*4		—	22	—			
			Peripheral function clock signal stopped*4		—	12	—			
		Coremark	Peripheral function clock signal stopped*4		—	15	—			
		Sleep mode: Supply of the clock signal to peripheral modules is stopped*4				16	24			
		All-module-clock-stop mode (reference value)			—	8	15			
		Low-speed operating mode 1: Supply of the clock signal to peripheral modules is stopped*4			—	1.1	—			All clocks 1 MHz
		Low-speed operating mode 2: Supply of the clock signal to peripheral modules is stopped*4			—	1.1	—			All clocks 32.768 kHz
		Software standby mode			—	1.6	6.4			
	Deep software standby mode	Power supplied to standby RAM and USB resume detecting unit (USB0 only)		—	15.5	51	μA			
		Power not supplied to standby RAM and USB resume detecting unit (USB0 only)	Power-on reset circuit and low-power consumption function disabled*5	—	11.5	29				
			Power-on reset circuit and low-power consumption function enabled*6	—	4.9	20				
		Increased by RTC operation	When a crystal oscillator for low clock loads is in use	—	1	—				
			When a crystal oscillator for standard clock loads is in use	—	2	—				
		RTC operating while VCC is off (with the battery backup function, only the RTC and sub-clock oscillator operate)	When a crystal oscillator for low clock loads is in use	—	0.9	—		V _{BATT} = 2.0 V, VCC = 0 V		
				—	1.6	—		V _{BATT} = 3.3 V, VCC = 0 V		
			When a crystal oscillator for standard clock loads is in use	—	1.7	—		V _{BATT} = 2.0 V, VCC = 0 V		
				—	3.3	—		V _{BATT} = 3.3 V, VCC = 0 V		
		Inrush current on returning from deep software standby mode	Inrush current*7	I _{RUSH}	—	—		70	mA	
			Energy of inrush current*7	E _{RUSH}	—	—		1.0		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripheral modules is stopped in this state.

Note 3. I_{CC} depends on f (ICLK) as follows. (ICLK/PCLKA:PCLKB/PCLKC/PCLKD:BCLK:BCLK pin = 10:5:10:5 when EXTAL = 12 MHz) I_{CC} Max. = $0.31 \times f + 6.5$ (max. operation in high-speed operating mode) I_{CC} Typ. = $0.16 \times f + 2.8$ (ICLK 1 MHz max) (normal operation in high-speed operating mode) I_{CC} Typ. = $0.4 \times f + 1.1$ (low-speed operating mode 1) I_{CC} Max. = $0.15 \times f + 6.5$ (sleep mode)

Note 4. Whether supply of the clock signal to peripheral modules continues or is stopped only depends on the state determined by the settings of the bits in module stop control registers A to D.

The setting for the peripheral module clock stopped state is FCLK = BCLK = PCLKA = PCLKB = PCLKC = PCLKD = BCLK pin = 3.75 MHz (division by 64).

Note 5. The low power consumption function is disabled and DEEPCUT[1:0] = 01b.

Note 6. The low power consumption function is enabled and DEEPCUT[1:0] = 11b.

Note 7. Reference value

Table 5.6 DC Characteristics (4)

Conditions: $V_{CC} = AVCC0 = AVCC1 = VREFH0 = VCC_USB = 2.7$ to 3.6 V, $2.7 \leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Analog power supply current*1	During 12-bit A/D conversion (unit 0)		I _{AlCC}	—	0.8	1	mA	IAVCC0_AD
	During 12-bit A/D conversion (unit 0) with the channel-dedicated sample-and-hold circuits for 3 channels operating			—	1.7	2.5	mA	IAVCC0_AD+SH
	During 12-bit A/D conversion (unit 1)			—	0.6	1	mA	IAVCC1_AD
	During 12-bit A/D conversion (unit 1) with the temperature sensor operating			—	0.7	1.1	mA	IAVCC1_AD+TEMP
	During D/A conversion (per channel)			—	0.25	0.4	mA	IAVCC1_DA
	Waiting for A/D, D/A, or temperature sensor conversion (all units)			—	0.9	1.4	mA	IAVCC0 + IAVCC1
	A/D, D/A converter, temperature sensor in standby mode (all units)			—	1.4	4.5	μA	IAVCC0 + IAVCC1
Reference power supply current	During 12-bit A/D conversion (unit 0)		I _{AlREFH}	—	25	40	μA	IVREFH0
	Waiting for 12-bit A/D conversion (unit 0)			—	0.07	0.4	μA	IVREFH0
	12-bit A/D converter in standby mode (unit 0)			—	0.07	0.2	μA	IVREFH0
USB operating current	Low speed	USB0	I _{CCUSBLS}	—	3.7	6.5	mA	VCC_USB
	Full speed	USB0	I _{CCUSBFS}	—	4.2	10	mA	VCC_USB
RAM hold voltage			V _{RAM}	2.7	—	—	V	
VCC rising gradient			SrVCC	8.4	—	20000	μs/V	
VCC falling gradient*2			SfVCC	8.4	—	—	μs/V	

Note 1. The reference power supply current is included in the power supply current value for 12-bit A/D conversion (unit 1) and D/A conversion.

Note 2. This applies when V_{BATT} is used.

Table 5.7 Permissible Output Currents

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit
Permissible output low current (average value per pin)	All output pins*1	Normal drive	I_{OL}	—	—	2.0	mA
	All output pins*2	High drive	I_{OL}	—	—	3.8	mA
	All output pins*3	High-speed interface high-drive	I_{OL}	—	—	7.5	mA
Permissible output low current (max. value per pin)	All output pins*1	Normal drive	I_{OL}	—	—	4.0	mA
	All output pins*2	High drive	I_{OL}	—	—	7.6	mA
	All output pins*3	High-speed interface high-drive	I_{OL}	—	—	15	mA
Permissible output low current (total)	Total of all output pins		ΣI_{OL}	—	—	80	mA
Permissible output high current (average value per pin)	All output pins*1	Normal drive	I_{OH}	—	—	-2.0	mA
	All output pins*2	High drive	I_{OH}	—	—	-3.8	mA
	All output pins*3	High-speed interface high-drive	I_{OH}	—	—	-7.5	mA
Permissible output high current (max. value per pin)	All output pins*1	Normal drive	I_{OH}	—	—	-4.0	mA
	All output pins*2	High drive	I_{OH}	—	—	-7.6	mA
	All output pins*3	High-speed interface high-drive	I_{OH}	—	—	-15	mA
Permissible output high current (total)	Total of all output pins		ΣI_{OH}	—	—	-80	mA

Caution: To protect the LSI's reliability, the output current values should not exceed the values in this table.

Note 1. This is the value when normal driving ability is set with a pin for which normal driving ability is selectable.

Note 2. This is the value when high driving ability is set with a pin for which normal driving ability is selectable or the value of the pin to which high driving ability is fixed.

Note 3. This is the value when high-speed interface high-driving ability is set with a pin for which high-speed interface high-driving ability is selectable.

5.3 AC Characteristics

Table 5.8 Operating Frequency (High-Speed Operating Mode)

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit
Operating frequency	System clock (ICLK)		f	—	—	120	MHz
	Peripheral module clock (PCLKA)			—	—	120	
	Peripheral module clock (PCLKB)			—	—	60	
	Peripheral module clock (PCLKC)			—	—	60	
	Peripheral module clock (PCLKD)			—	—	60	
	Flash-IF clock (FCLK)			—*1	—	60	
	External bus clock (BCLK)	Other than 100-pin package		—	—	120	
		100-pin package		—	—	60	
	BCLK pin output	Other than 100-pin package		—	—	60	
		100-pin package		—	—	30	
	SDRAM clock (SDCLK)	Other than 100-pin package		—	—	60	
	SDCLK pin output	Other than 100-pin package		—	—	60	

Note 1. The FCLK must run at a frequency of at least 4 MHz when changing the flash memory contents.

Table 5.9 Operating Frequency (Low-Speed Operating Mode 1)

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit
Operating frequency	System clock (ICLK)		f	—	—	1	MHz
	Peripheral module clock (PCLKA)			—	—	1	
	Peripheral module clock (PCLKB)			—	—	1	
	Peripheral module clock (PCLKC)*1			—	—	1	
	Peripheral module clock (PCLKD)*1			—	—	1	
	Flash-IF clock (FCLK)			—	—	1	
	External bus clock (BCLK)	Other than 100-pin package		—	—	1	
		100-pin package		—	—	1	
	BCLK pin output	Other than 100-pin package		—	—	1	
		100-pin package		—	—	1	
	SDRAM clock (SDCLK)	Other than 100-pin package		—	—	1	
	SDCLK pin output	Other than 100-pin package		—	—	1	

Note 1. When the 12-bit A/D converter is used, the frequency must be set to at least 1 MHz.

Table 5.10 Operating Frequency (Low-Speed Operating Mode 2)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit
Operating frequency	System clock (ICLK)		f	32	—	264	kHz
	Peripheral module clock (PCLKA)			—	—	264	
	Peripheral module clock (PCLKB)			—	—	264	
	Peripheral module clock (PCLKC)*1			—	—	264	
	Peripheral module clock (PCLKD)*1			—	—	264	
	Flash-IF clock (FCLK)			32	—	264	
	External bus clock (BCLK)	Other than 100-pin package		—	—	264	
		100-pin package		—	—	264	
	BCLK pin output	Other than 100-pin package		—	—	264	
		100-pin package		—	—	264	
	SDRAM clock (SDCLK)	Other than 100-pin package		—	—	264	
	SDCLK pin output	Other than 100-pin package		—	—	264	

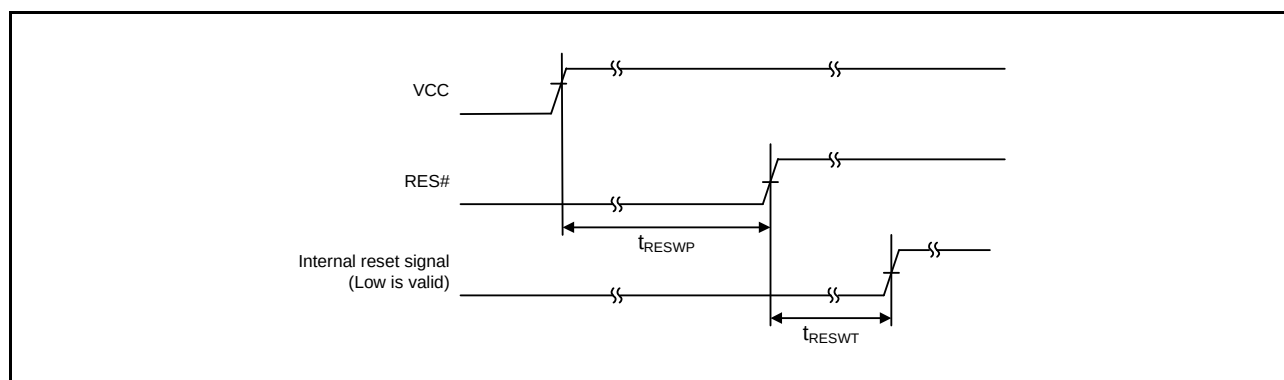
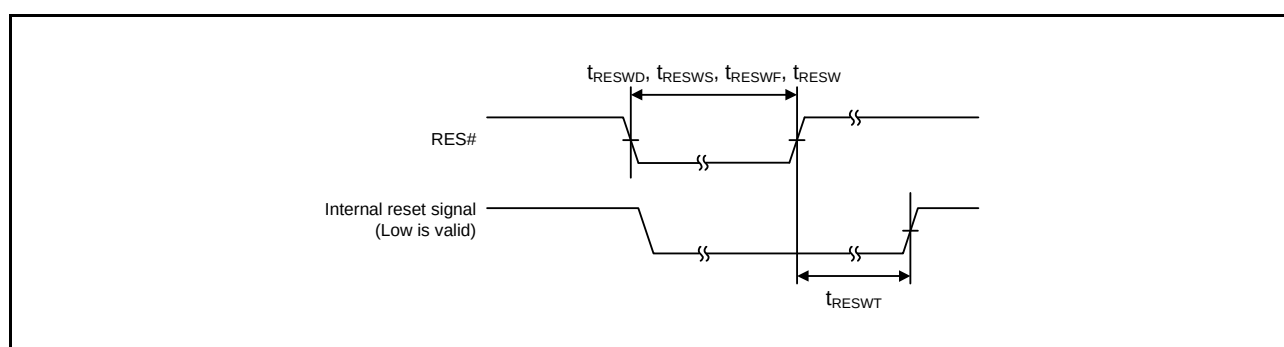
Note 1. The 12-bit A/D converter cannot be used.

5.3.1 Reset Timing

Table 5.11 Reset Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
RES# pulse width	Power-on	t_{RESWP}	1	—	—	ms	Figure 5.1
	Deep software standby mode	t_{RESWD}	0.6	—	—	ms	Figure 5.2
	Software standby mode, low-speed operating mode 2	t_{RESWS}	0.3	—	—	ms	
	Programming or erasure of the code flash memory	t_{RESWF}	200	—	—	μ s	
	Other than above	t_{RESW}	200	—	—	μ s	
Waiting time after release from the RES# pin reset		t_{RESWT}	54	—	55	t_{Lcyc}	Figure 5.1
Internal reset time (independent watchdog timer reset, watchdog timer reset, software reset)		t_{RESW2}	100	—	108	t_{Lcyc}	

**Figure 5.1 Reset Input Timing at Power-On****Figure 5.2 Reset Input Timing**

5.3.2 Clock Timing

Table 5.12 BCLK Pin Output, SDCLK Pin Output Clock Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
BCLK pin output cycle time	Package with 144 pins	t_{Bcyc}	16.6	—	—	ns	Figure 5.3
	Package with 100 pins		33.2	—	—	ns	
BCLK pin output high pulse width		t_{CH}	3.3	—	—	ns	
BCLK pin output low pulse width		t_{CL}	3.3	—	—	ns	
BCLK pin output rising time		t_{Cr}	—	—	5	ns	
BCLK pin output falling time		t_{Cf}	—	—	5	ns	
SDCLK pin output cycle time	Package with 144 pins	t_{Bcyc}	16.6	—	—	ns	
SDCLK pin output high pulse width		t_{CH}	3.3	—	—	ns	
SDCLK pin output low pulse width		t_{CL}	3.3	—	—	ns	
SDCLK pin output rising time		t_{Cr}	—	—	5	ns	
SDCLK pin output falling time		t_{Cf}	—	—	5	ns	

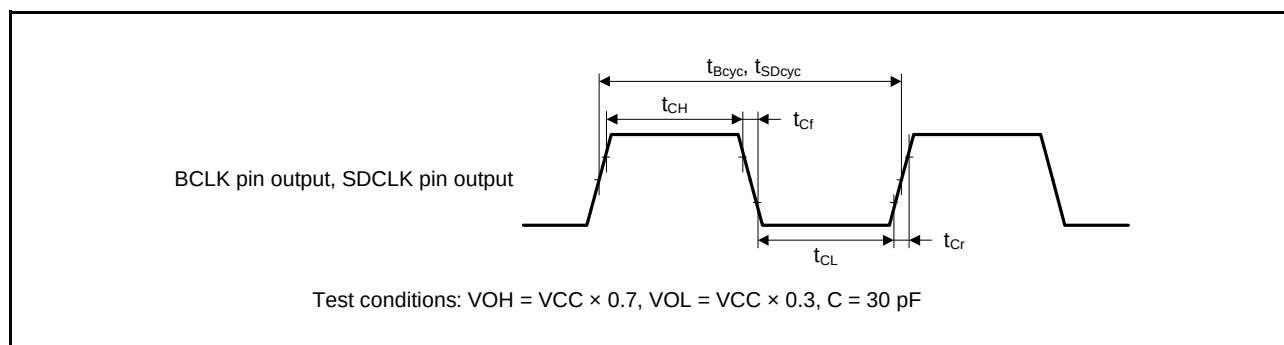
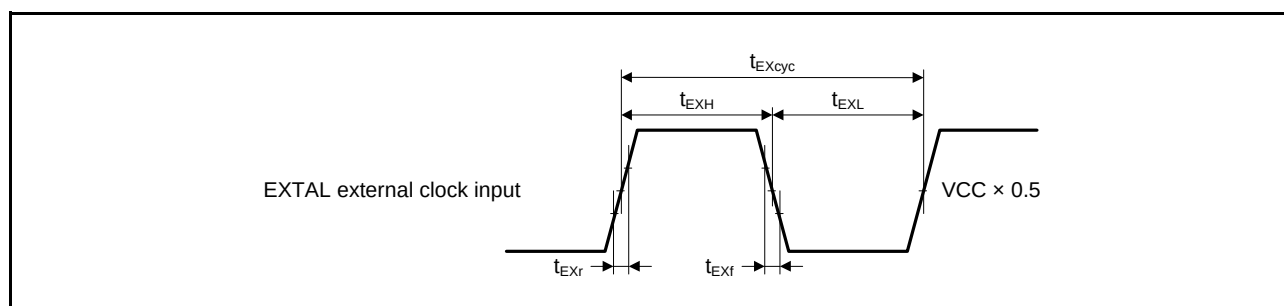
**Figure 5.3 BCLK Pin and SDCLK Pin Output Timing**

Table 5.13 EXTAL Clock Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
EXTAL external clock input cycle time	t_{EXcyc}	41.66	—	—	ns	Figure 5.4
EXTAL external clock input frequency	f_{EXMAIN}	—	—	24	MHz	
EXTAL external clock input high pulse width	t_{EXH}	15.83	—	—	ns	
EXTAL external clock input low pulse width	t_{EXL}	15.83	—	—	ns	
EXTAL external clock rising time	t_{EXr}	—	—	5	ns	
EXTAL external clock falling time	t_{EXf}	—	—	5	ns	

**Figure 5.4 EXTAL External Clock Input Timing****Table 5.14 Main Clock Timing**

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Main clock oscillation frequency	f_{MAIN}	8	—	24	MHz	Figure 5.5
Main clock oscillator stabilization time (crystal)	$t_{MAINOSC}$	—	—	—*1	ms	
Main clock oscillation stabilization wait time (crystal)	$t_{MAINOSCW}$	—	—	—*2	ms	

Note 1. When using a main clock, ask the manufacturer of the oscillator to evaluate its oscillation. Refer to the results of evaluation provided by the manufacturer for the oscillation stabilization time.

Note 2. The number of cycles selected by the value of the MOSCWTCR.MSTS[7:0] bits determines the main clock oscillation stabilization wait time in accord with the formula below.

$$t_{MAINOSCW} = [(MSTS[7:0] \text{ bits} \times 32) + 10] / f_{LOCO}$$

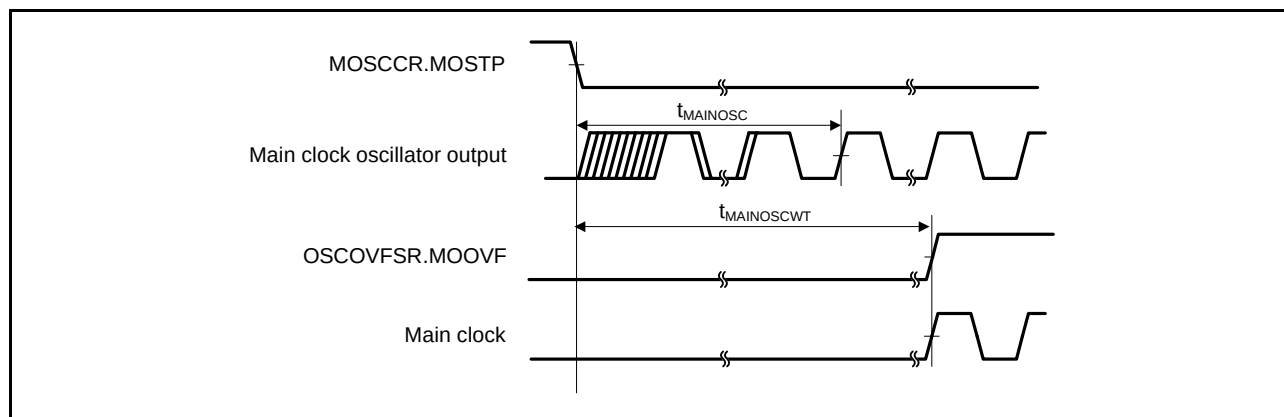
**Figure 5.5 Main Clock Oscillation Start Timing**

Table 5.15 LOCO and IWDT-Dedicated Low-Speed Clock Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
LOCO clock cycle time	t_{Lcyc}	4.63	4.16	3.78	μs	
LOCO clock oscillation frequency	f_{LOCO}	216	240	264	kHz	
LOCO clock oscillation stabilization wait time	t_{LOCOWT}	—	—	44	μs	Figure 5.6
IWDT-dedicated low-speed clock cycle time	t_{iLcyc}	9.26	8.33	7.57	μs	
IWDT-dedicated low-speed clock oscillation frequency	f_{iLOCO}	108	120	132	kHz	
IWDT-dedicated low-speed clock oscillation stabilization wait time	$t_{iLOCOWT}$	—	142	190	μs	Figure 5.7

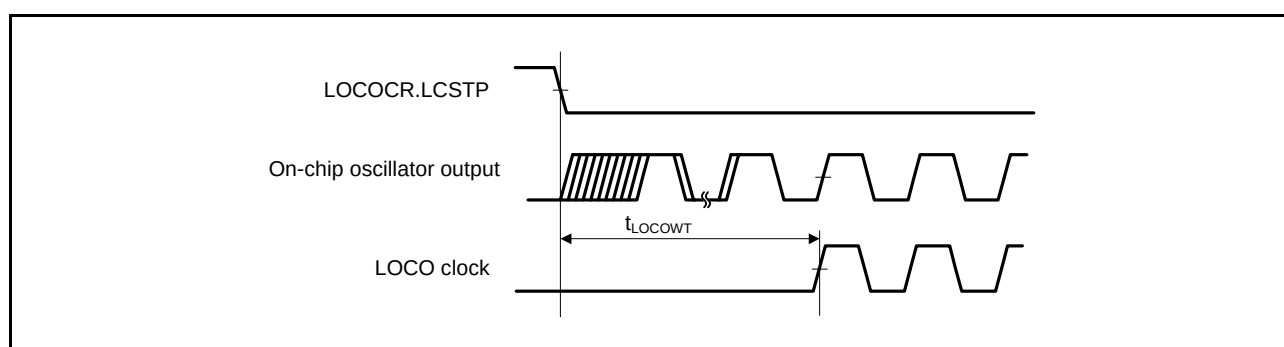
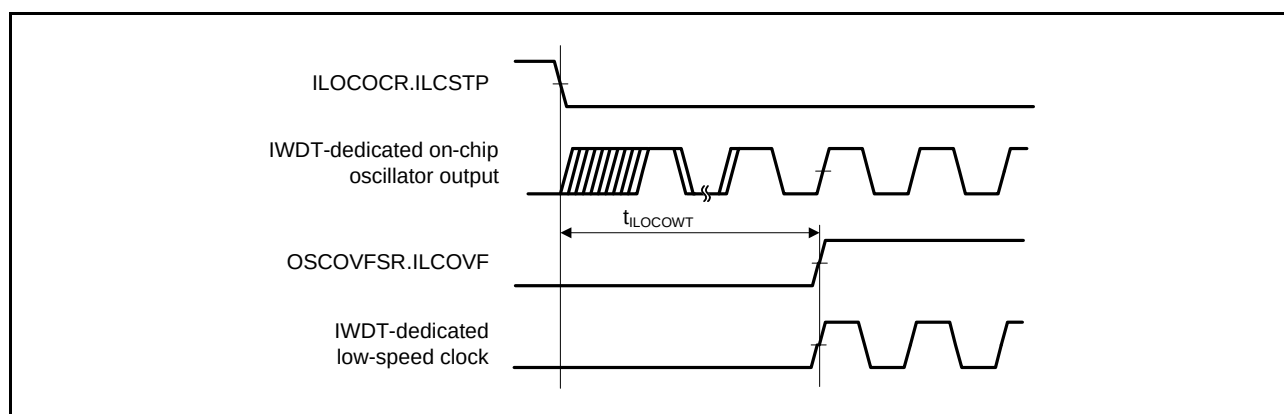
**Figure 5.6 LOCO Clock Oscillation Start Timing****Figure 5.7 IWDT-dedicated Low-Speed Clock Oscillation Start Timing**

Table 5.16 HOCO Clock Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
HOCO clock oscillation frequency	f_{HOCO}	15.61	16	16.39	MHz	$-20^{\circ}\text{C} \leq T_a \leq 85^{\circ}\text{C}$
		17.56	18	18.44	MHz	
		19.52	20	20.48	MHz	
		15.52	16	16.48	MHz	$-40^{\circ}\text{C} \leq T_a < -20^{\circ}\text{C}$
		17.46	18	18.54	MHz	
		19.40	20	20.60	MHz	
HOCO clock oscillation stabilization wait time	t_{HOCOWT}	—	105	149	μs	Figure 5.8
HOCO clock power supply stabilization time	t_{HOCOP}	—	—	150	μs	Figure 5.9

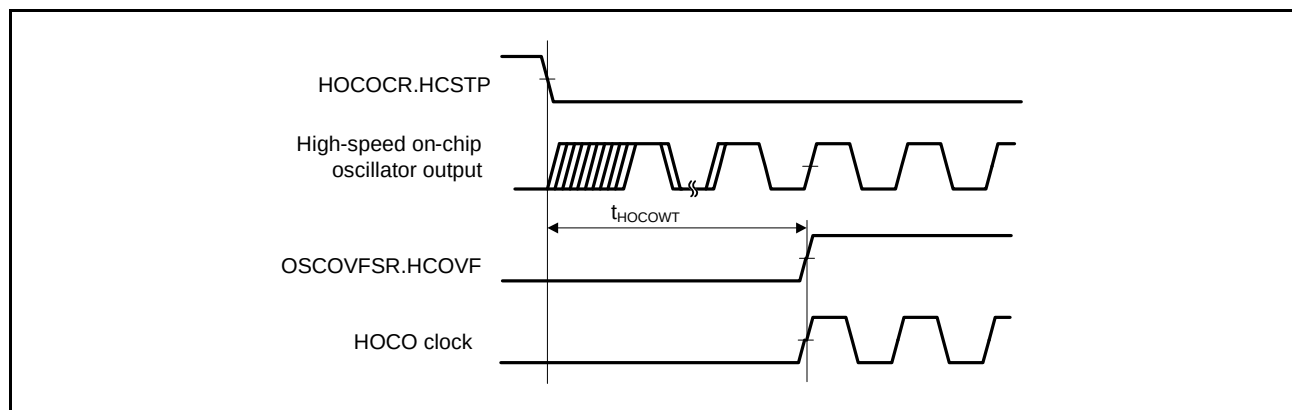
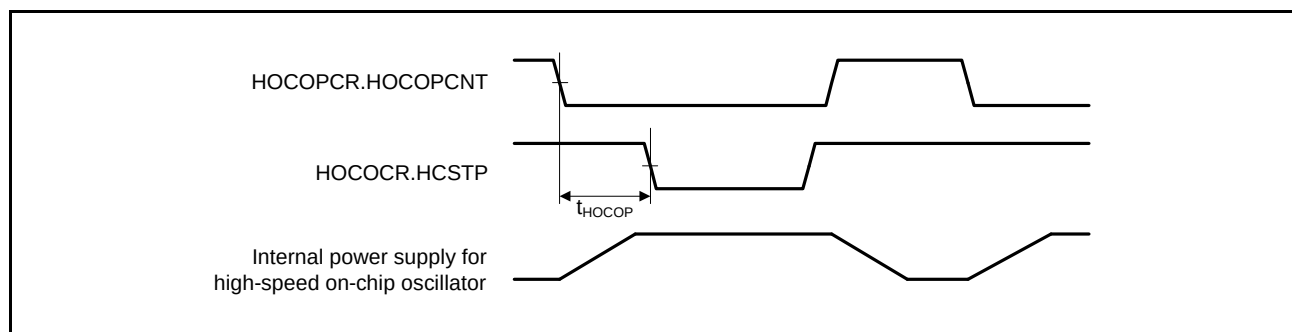
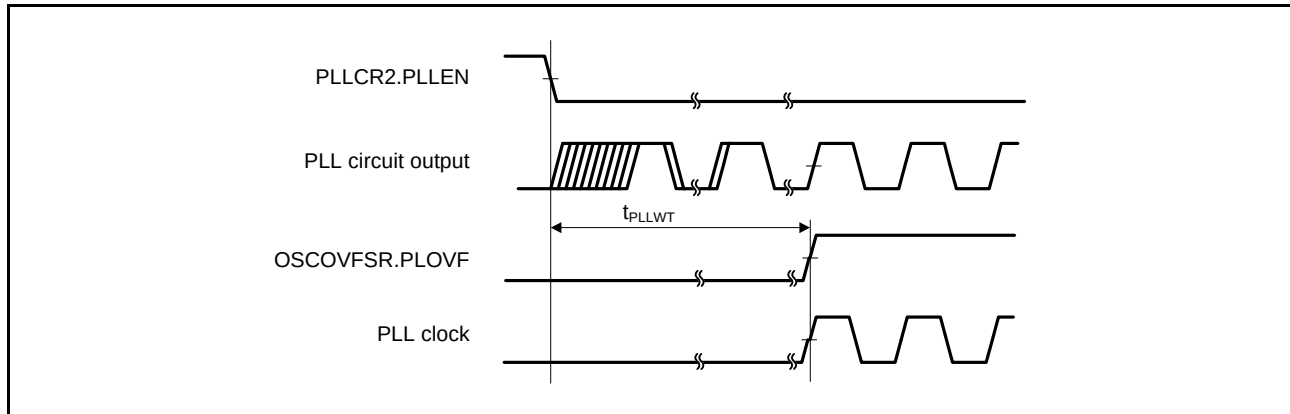
**Figure 5.8 HOCO Clock Oscillation Start Timing (Oscillation is Started by Setting the HOCOCR.HCSTP Bit)****Figure 5.9 High-Speed On-Chip Oscillator Power Supply Control Timing**

Table 5.17 PLL Clock Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
PLL clock oscillation frequency	f_{PLL}	120	—	240	MHz	
PLL clock oscillation stabilization wait time	t_{PLLWT}	—	259	320	μ s	Figure 5.10

**Figure 5.10 PLL Clock Oscillation Start Timing****Table 5.18 Sub-Clock Timing**

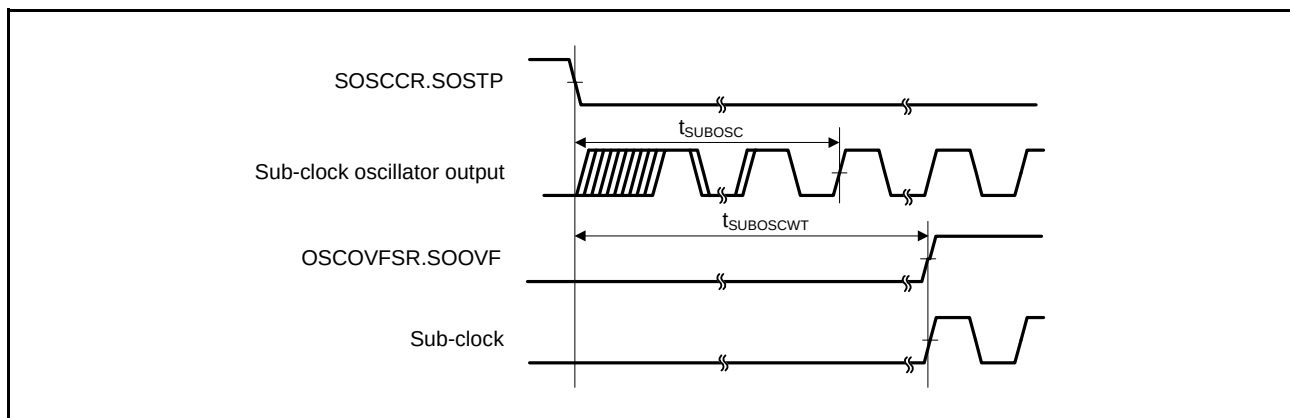
Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $V_{BATT} = 2.0$ to 3.6 V, $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Sub-clock oscillation frequency	f_{SUB}	—	32.768	—	kHz	
Sub-clock oscillation stabilization time	t_{SUBOSC}	—	—	*1	s	Figure 5.11
Sub-clock oscillation stabilization wait time	$t_{SUBOSCWT}$	—	—	*2	s	

Note 1. When using a sub-clock, ask the manufacturer of the oscillator to evaluate its oscillation. Refer to the results of evaluation provided by the manufacturer for the oscillation stabilization time.

Note 2. The number of cycles selected by the value of the $SOSCWTCCR.SSTS[7:0]$ bits determines the sub-clock oscillation stabilization wait time in accord with the formula below.

$$t_{SUBOSCWT} = [(SSTS[7:0] \text{ bits} \times 16384) + 10] / f_{Loco}$$

**Figure 5.11 Sub-Clock Oscillation Start Timing**

5.3.3 Timing of Recovery from Low Power Consumption Modes

Table 5.19 Timing of Recovery from Low Power Consumption Modes (1)

Conditions: $VCC = AVCC0 = AVCC1 = VCC_USB = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.		Unit	Test Conditions
						$t_{SBYOSCWT}^{*2}$	t_{SBYSEQ}^{*3}		
Recovery time after cancellation of software standby mode*1	Crystal resonator connected to main clock oscillator	Main clock oscillator operating	t_{SBYMC}	—	—	$\{(MSTS[7:0] \text{ bits} \times 32) + 76\} / 0.216$	$100 \mu s + 7 / f_{CLK} + 2n / f_{MAIN}$	μs	Figure 5.12
		Main clock oscillator and PLL circuit operating	t_{SBYPC}			$\{(MSTS[7:0] \text{ bits} \times 32) + 138\} / 0.216$	$100 \mu s + 7 / f_{CLK} + 2n / f_{PLL}$		
	External clock input to main clock oscillator	Main clock oscillator operating	t_{SBYEX}			352	$100 \mu s + 7 / f_{CLK} + 2n / f_{EXMAIN}$		
		Main clock oscillator and PLL circuit operating	t_{SBYPE}			639	$100 \mu s + 7 / f_{CLK} + 2n / f_{PLL}$		
	Sub-clock oscillator operating		t_{SBYSC}			$\{(SSTS[7:0] \text{ bits} \times 6384 + 13) / 0.216 + 10 / f_{FCLK}\}$	$100 \mu s + 4 / f_{CLK} + 2n / f_{SUB}$		
	High-speed on-chip oscillator operating	High-speed on-chip oscillator operating	t_{SBYHO}			454	$100 \mu s + 7 / f_{CLK} + 2n / f_{HOCO}$		
		High-speed on-chip oscillator operating and PLL circuit operating	t_{SBYPH}			741	$100 \mu s + 7 / f_{CLK} + 2n / f_{PLL}$		
	Low-speed on-chip oscillator operating*4		t_{SBYLO}			338	$100 \mu s + 7 / f_{CLK} + 2n / f_{LOCO}$		

Note 1. The time for return after release from software standby is determined by the value obtained by adding the oscillation stabilization waiting time ($t_{SBYOSCWT}$) and the time required for operations by the software standby release sequencer (t_{SBYSEQ}).

Note 2. When several oscillators were running before the transition to software standby, the greatest value of the oscillation stabilization waiting time $t_{SBYOSCWT}$ is selected.

Note 3. For n, the greatest value is selected from among the internal clock division settings.

Note 4. This condition applies when $f_{CLK}:f_{FCLK} = 1:1, 2:1, \text{ or } 4:1$.

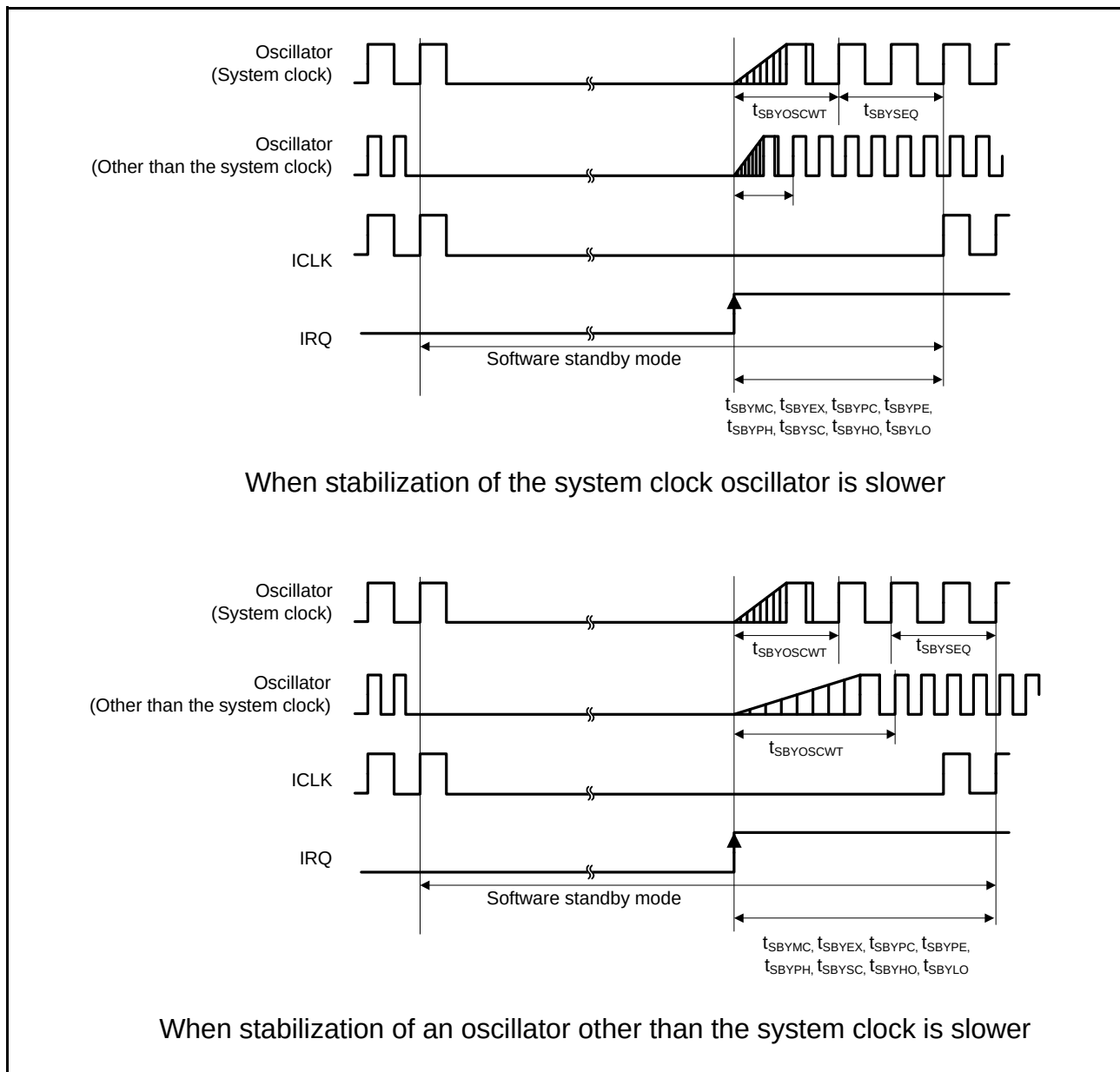
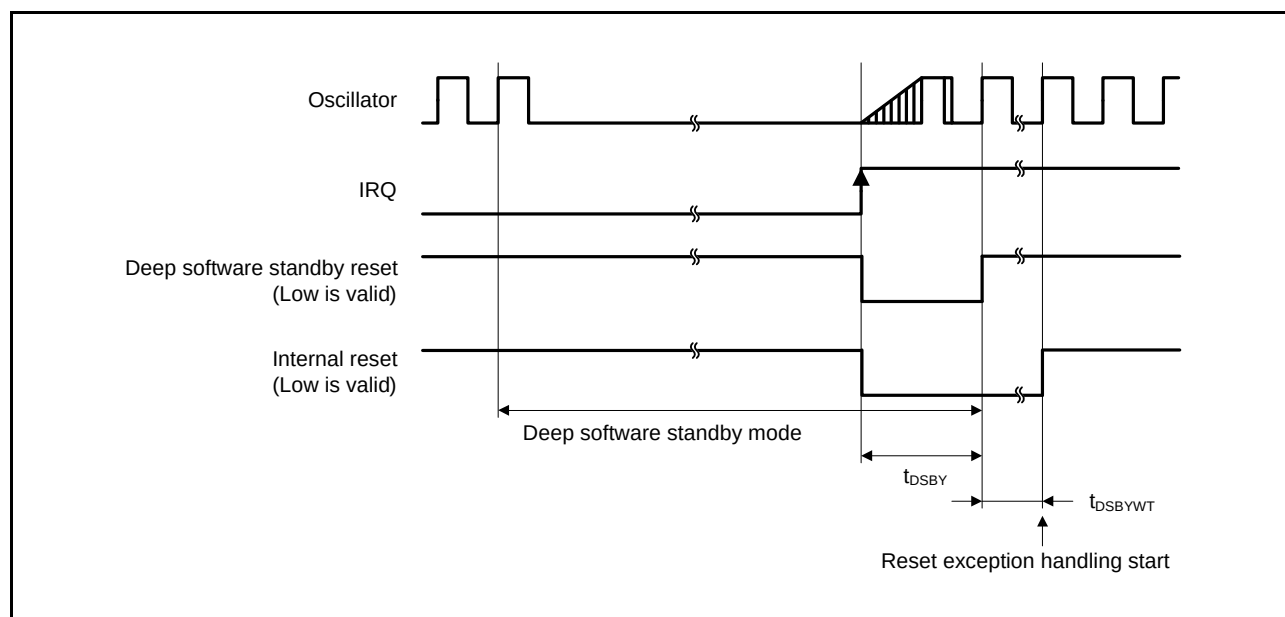


Figure 5.12 Software Standby Mode Cancellation Timing

Table 5.20 Timing of Recovery from Low Power Consumption Modes (2)

Conditions: $V_{CC} = AVCC0 = AVCC1 = VCC_USB = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Recovery time after cancellation of deep software standby mode	t_{DSBY}	—	—	0.9	ms	Figure 5.13
Wait time after cancellation of deep software standby mode	t_{DSBYWT}	23	—	24	t_{Lcyc}	

**Figure 5.13 Deep Software Standby Mode Cancellation Timing**

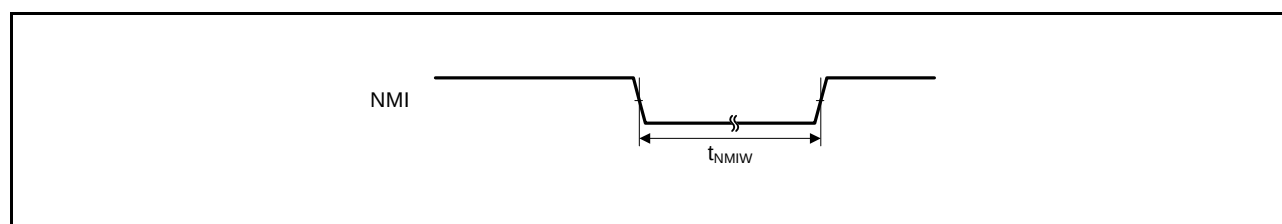
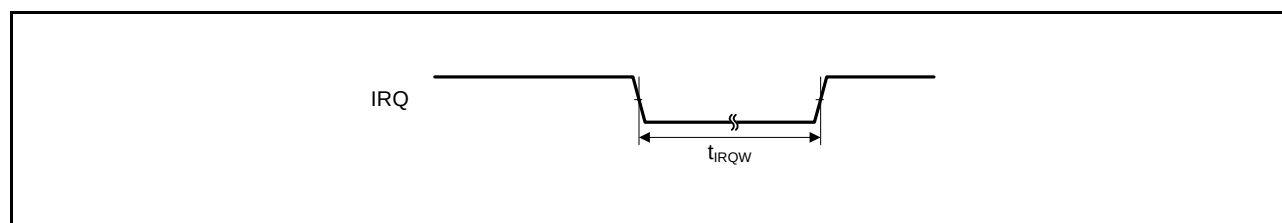
5.3.4 Control Signal Timing

Table 5.21 Control Signal Timing

Conditions: $VCC = AVCC0 = AVCC1 = VCC_USB = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$

Item	Symbol	Min.*1	Typ.	Max.	Unit	Test Conditions*1
NMI pulse width	t_{NMIW}	200	—	—	ns	$t_{PBcyc} \times 2 \leq 200$ ns, Figure 5.14
		$t_{PBcyc} \times 2$	—	—	ns	$t_{PBcyc} \times 2 > 200$ ns, Figure 5.14
IRQ pulse width	t_{IRQW}	200	—	—	ns	$t_{PBcyc} \times 2 \leq 200$ ns, Figure 5.15
		$t_{PBcyc} \times 2$	—	—	ns	$t_{PBcyc} \times 2 > 200$ ns, Figure 5.15

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 5.14 NMI Interrupt Input Timing****Figure 5.15 IRQ Interrupt Input Timing**

5.3.5 Bus Timing

Table 5.22 Bus Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $ICLK = PCLKA = 8$ to 120 MHz, $PCLKB = BCLK = SDCLK = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
 High-drive output is selected by the driving ability control register.

Item	Symbol	Min.	Max.	Unit	Test Conditions
Address delay time	t_{AD}	—	12.5	ns	Figure 5.16 to Figure 5.21
Byte control delay time	t_{BCD}	—	12.5	ns	
CS# delay time	t_{CSD}	—	12.5	ns	
ALE delay time	t_{ALED}	—	12.5	ns	
RD# delay time	t_{RSD}	—	12.5	ns	
Read data setup time	t_{RDS}	12.5	—	ns	
Read data hold time	t_{RDH}	0	—	ns	
WR# delay time	t_{WRD}	—	12.5	ns	
Write data delay time	t_{WDD}	—	12.5	ns	
Write data hold time	t_{WDH}	0	—	ns	
WAIT# setup time	t_{WTS}	12.5	—	ns	Figure 5.22
WAIT# hold time	t_{WTH}	0	—	ns	
Address delay time 2 (SDRAM)	t_{AD2}	1	12.5	ns	Figure 5.23
CS# delay time 2 (SDRAM)	t_{CSD2}	1	12.5	ns	
DQM delay time (SDRAM)	t_{DQMD}	1	12.5	ns	
CKE delay time (SDRAM)	t_{CKED}	1	12.5	ns	
Read data setup time 2 (SDRAM)	t_{RDS2}	10	—	ns	
Read data hold time 2 (SDRAM)	t_{RDH2}	0	—	ns	
Write data delay time 2 (SDRAM)	t_{WDD2}	—	12.5	ns	
Write data hold time 2 (SDRAM)	t_{WDH2}	1	—	ns	
WE# delay time (SDRAM)	t_{WED}	1	12.5	ns	
RAS# delay time (SDRAM)	t_{RASD}	1	12.5	ns	
CAS# delay time (SDRAM)	t_{CASD}	1	12.5	ns	

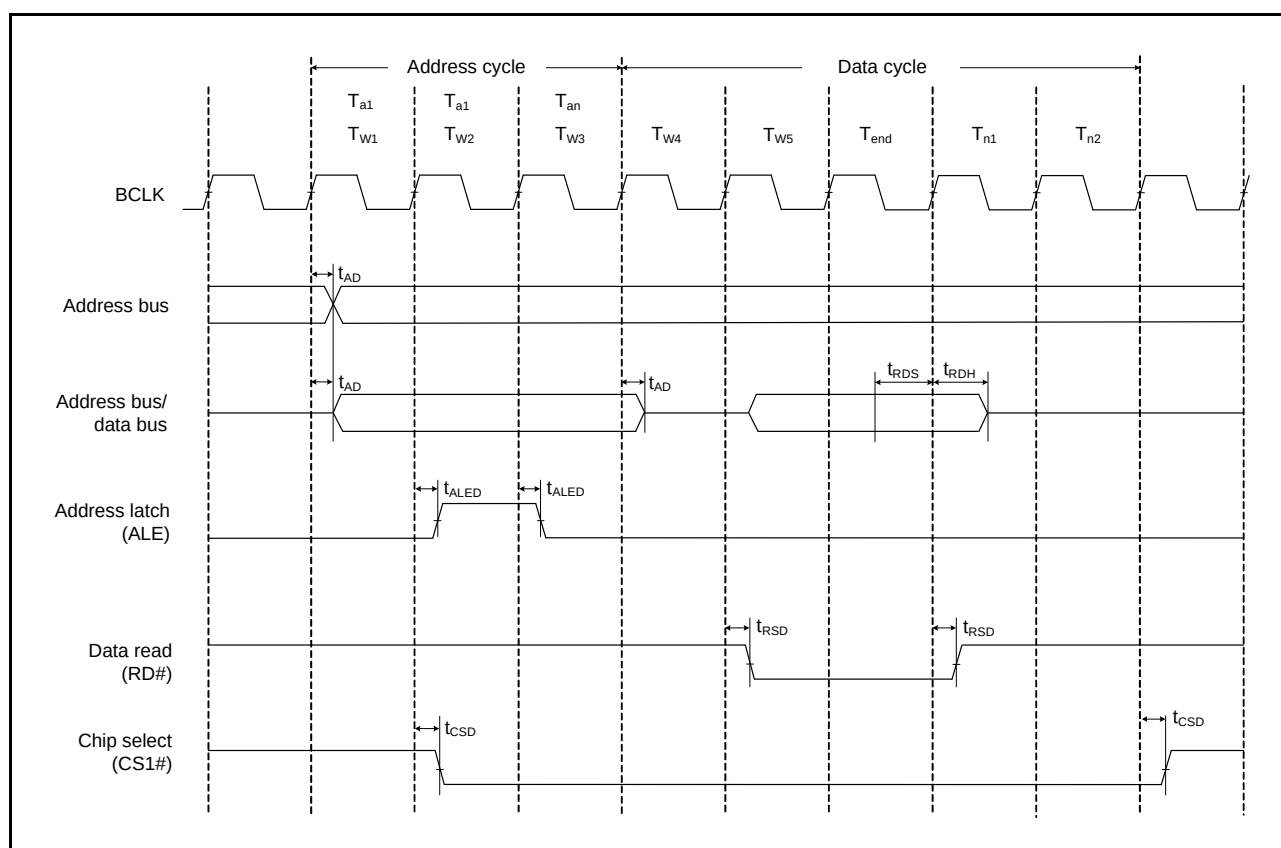


Figure 5.16 Address/Data Multiplexed Bus Read Access Timing

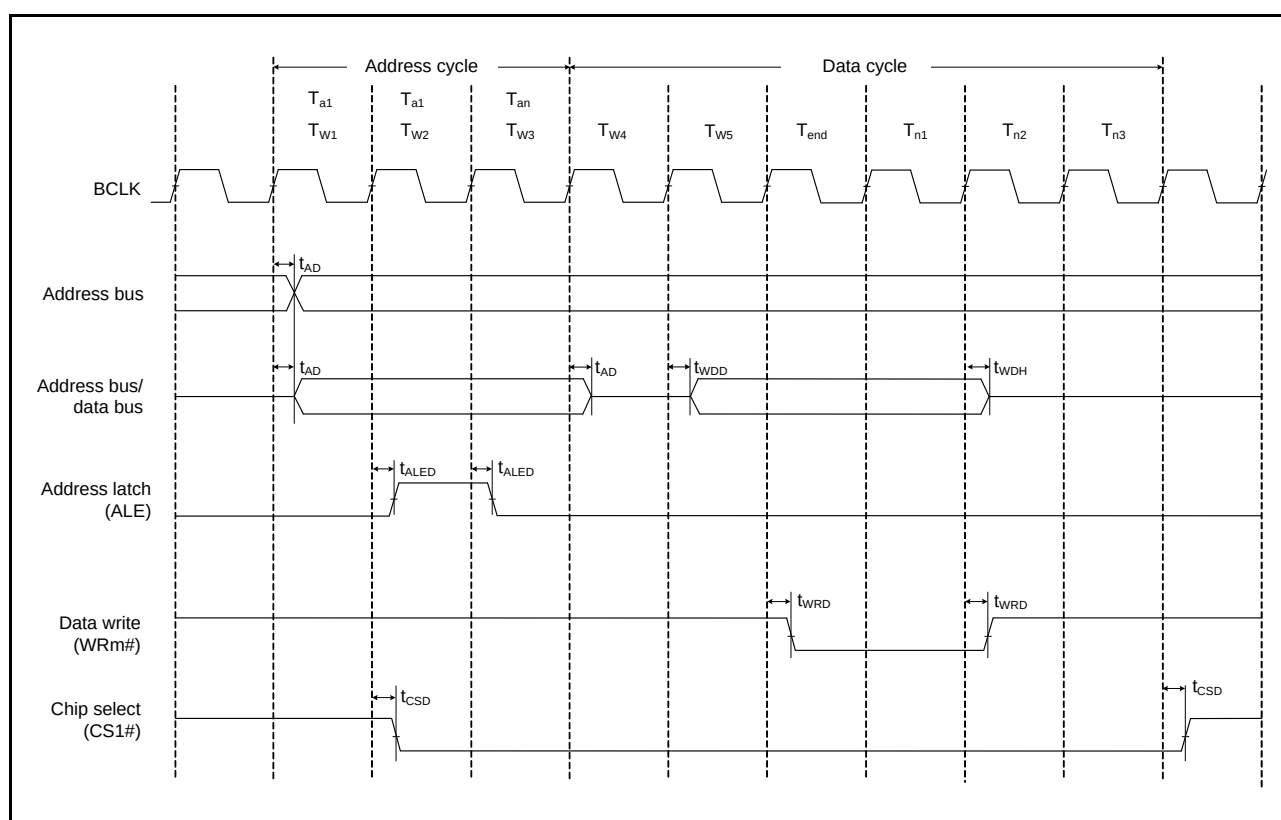


Figure 5.17 Address/Data Multiplexed Bus Write Access Timing

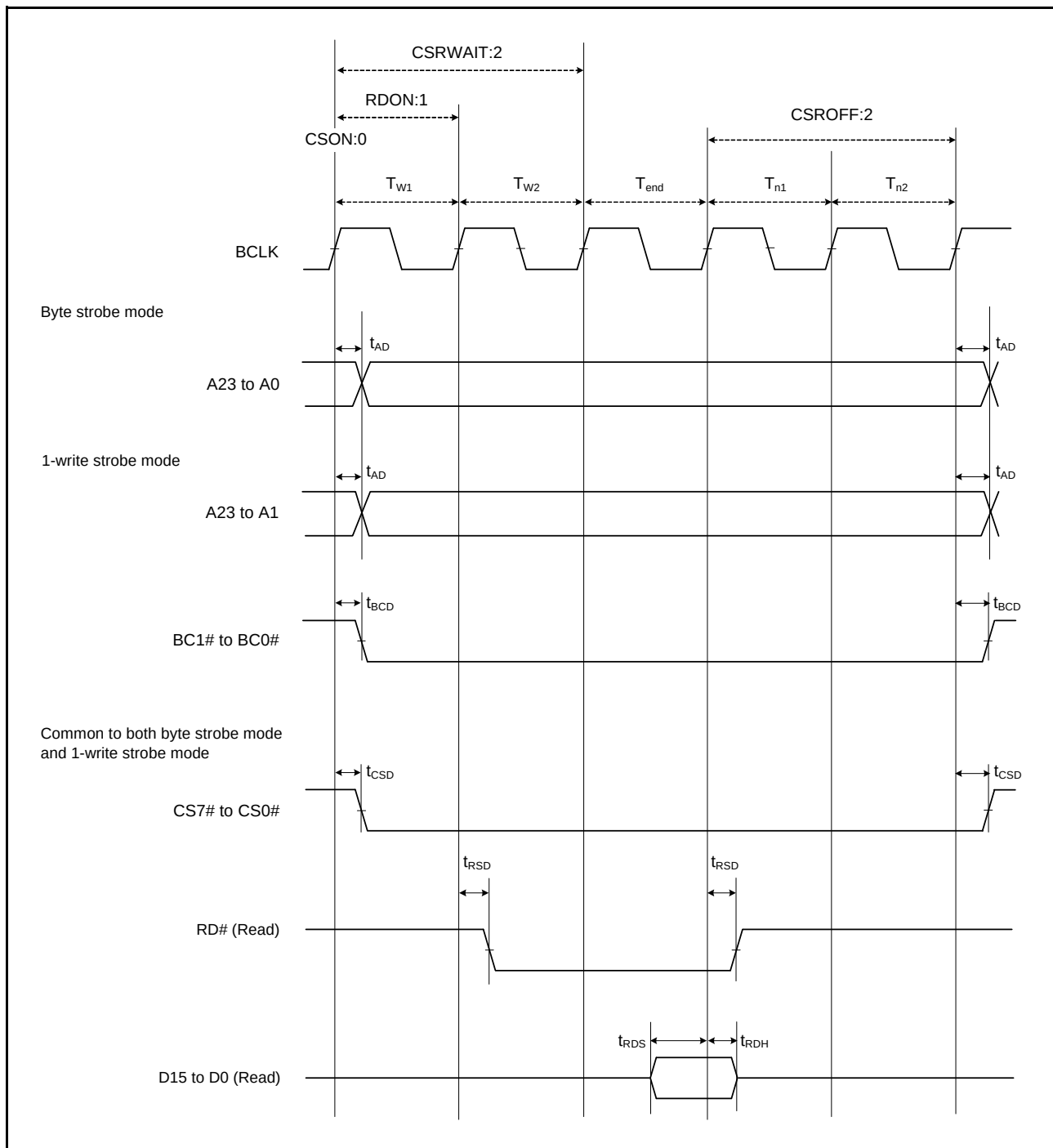


Figure 5.18 External Bus Timing/Normal Read Cycle (Bus Clock Synchronized)

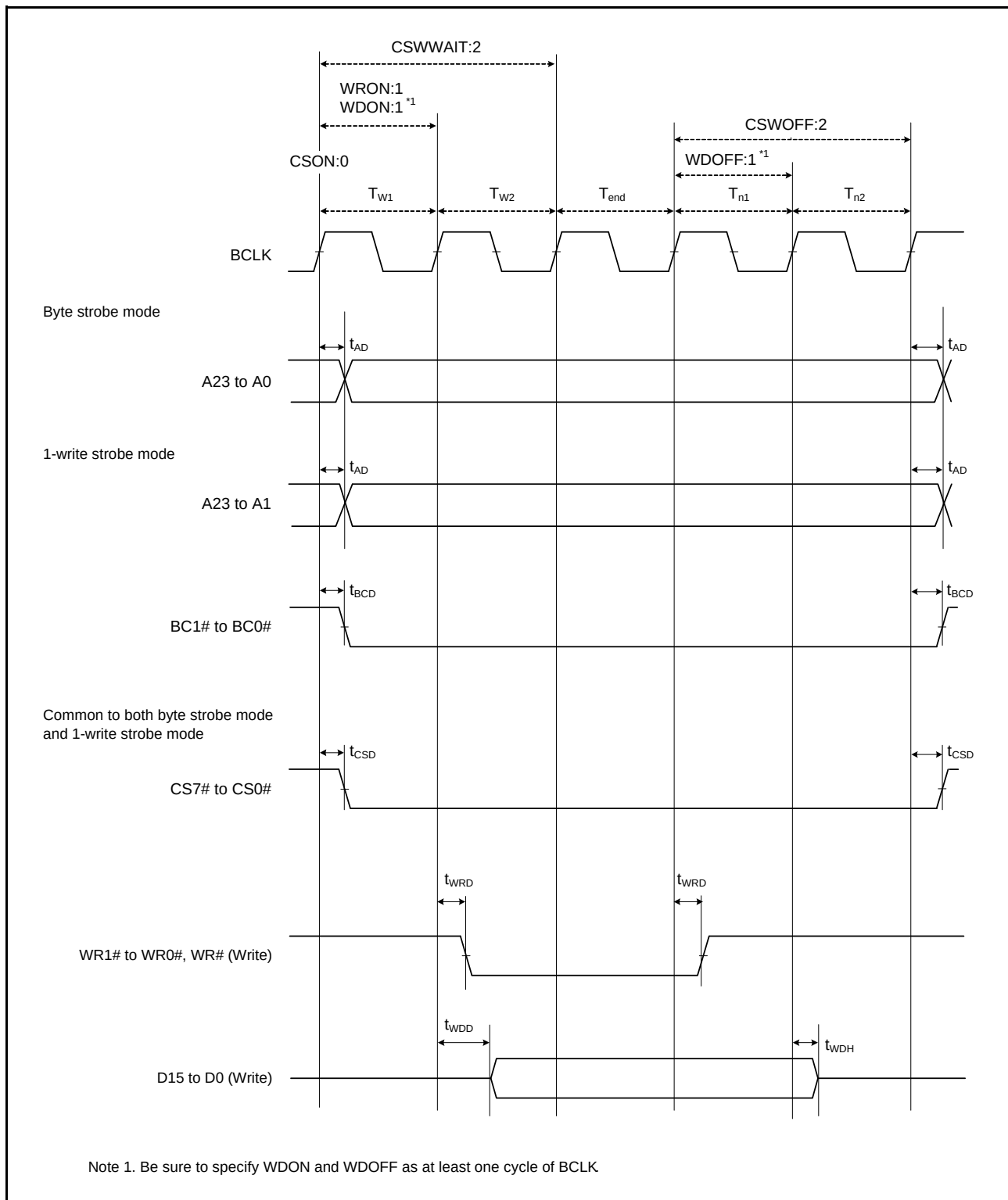


Figure 5.19 External Bus Timing/Normal Write Cycle (Bus Clock Synchronized)

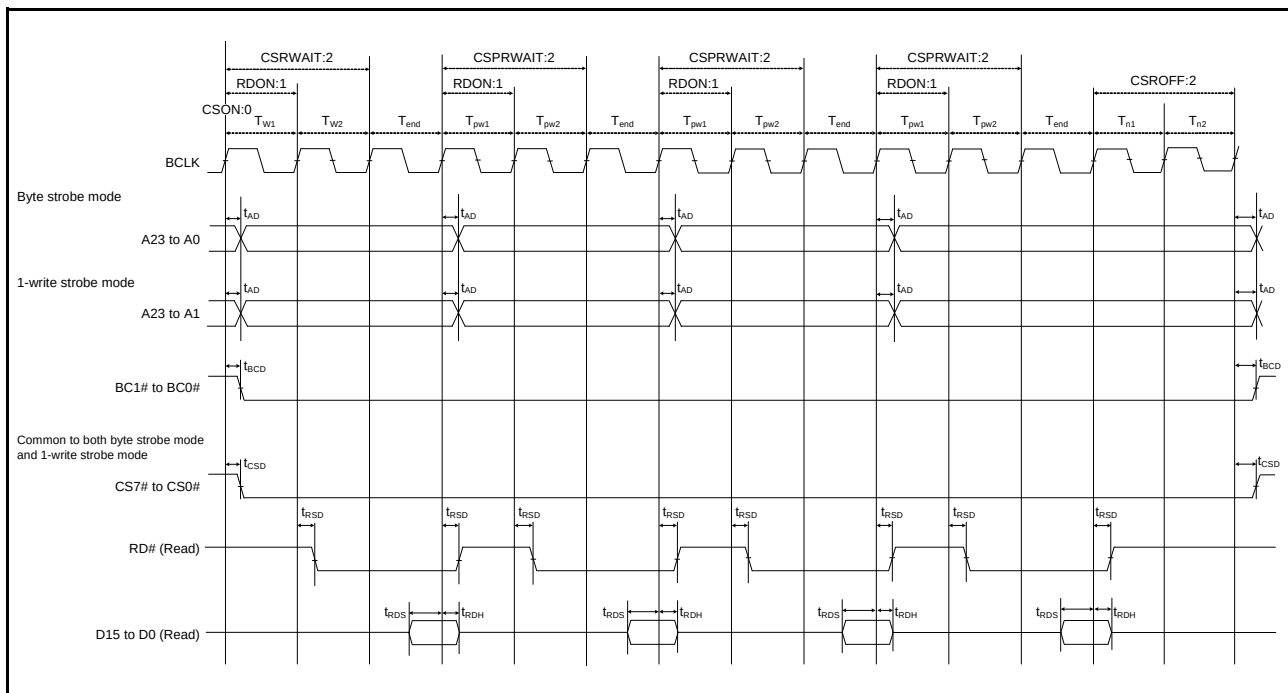


Figure 5.20 External Bus Timing/Page Read Cycle (Bus Clock Synchronized)

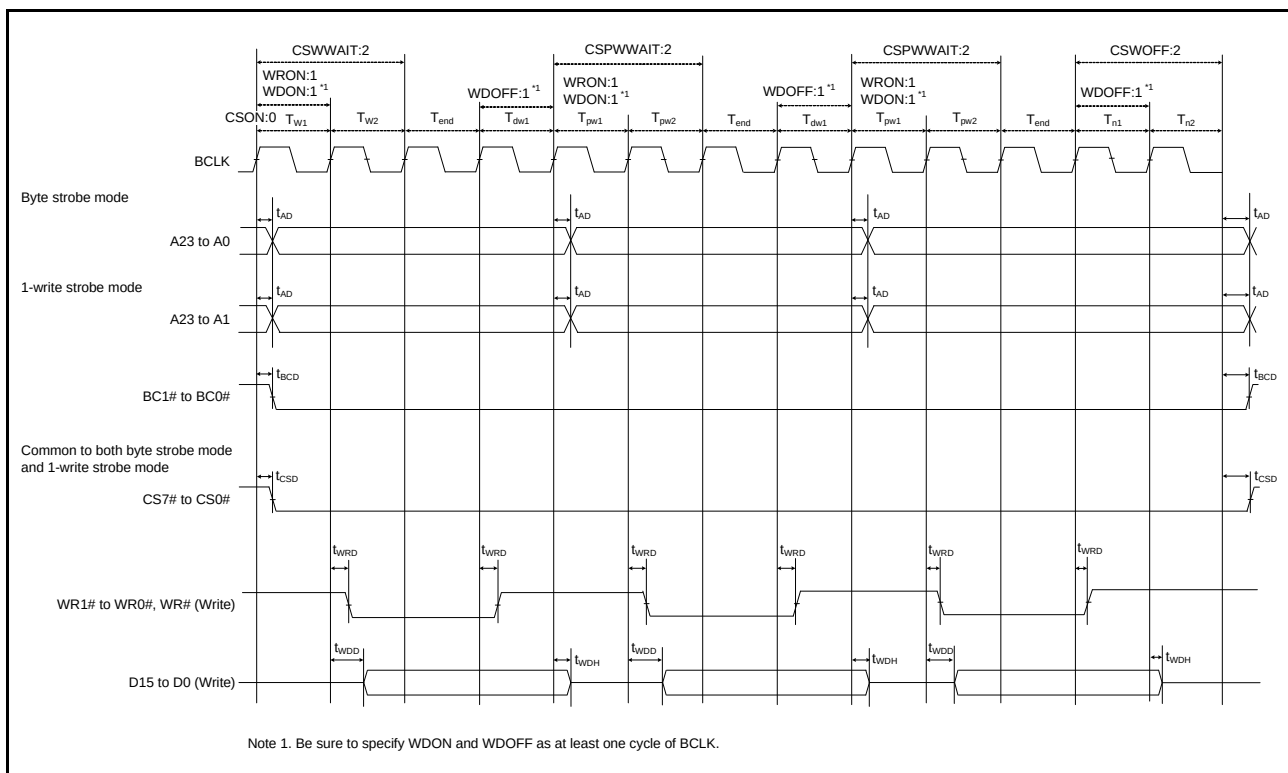


Figure 5.21 External Bus Timing/Page Write Cycle (Bus Clock Synchronized)

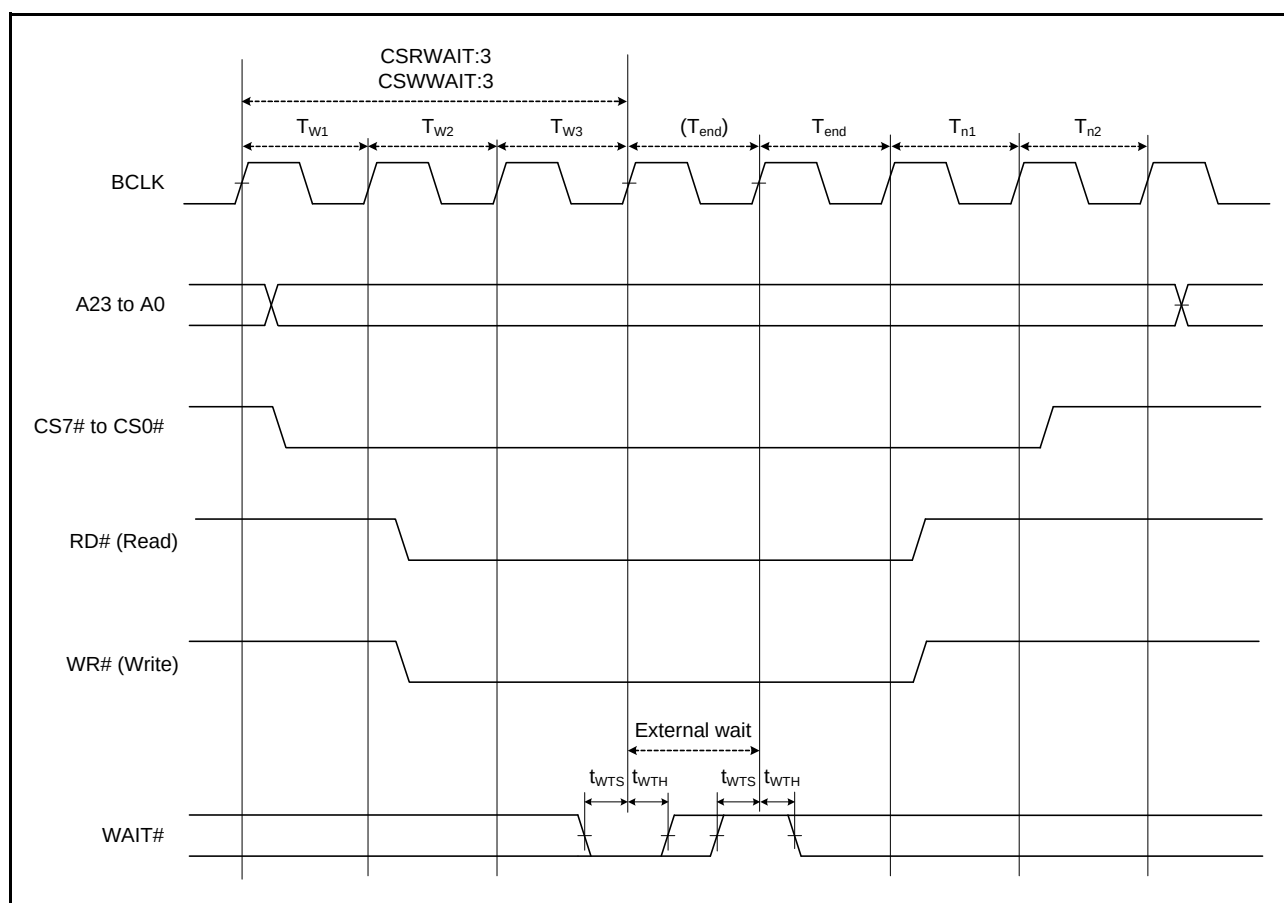


Figure 5.22 External Bus Timing/External Wait Control

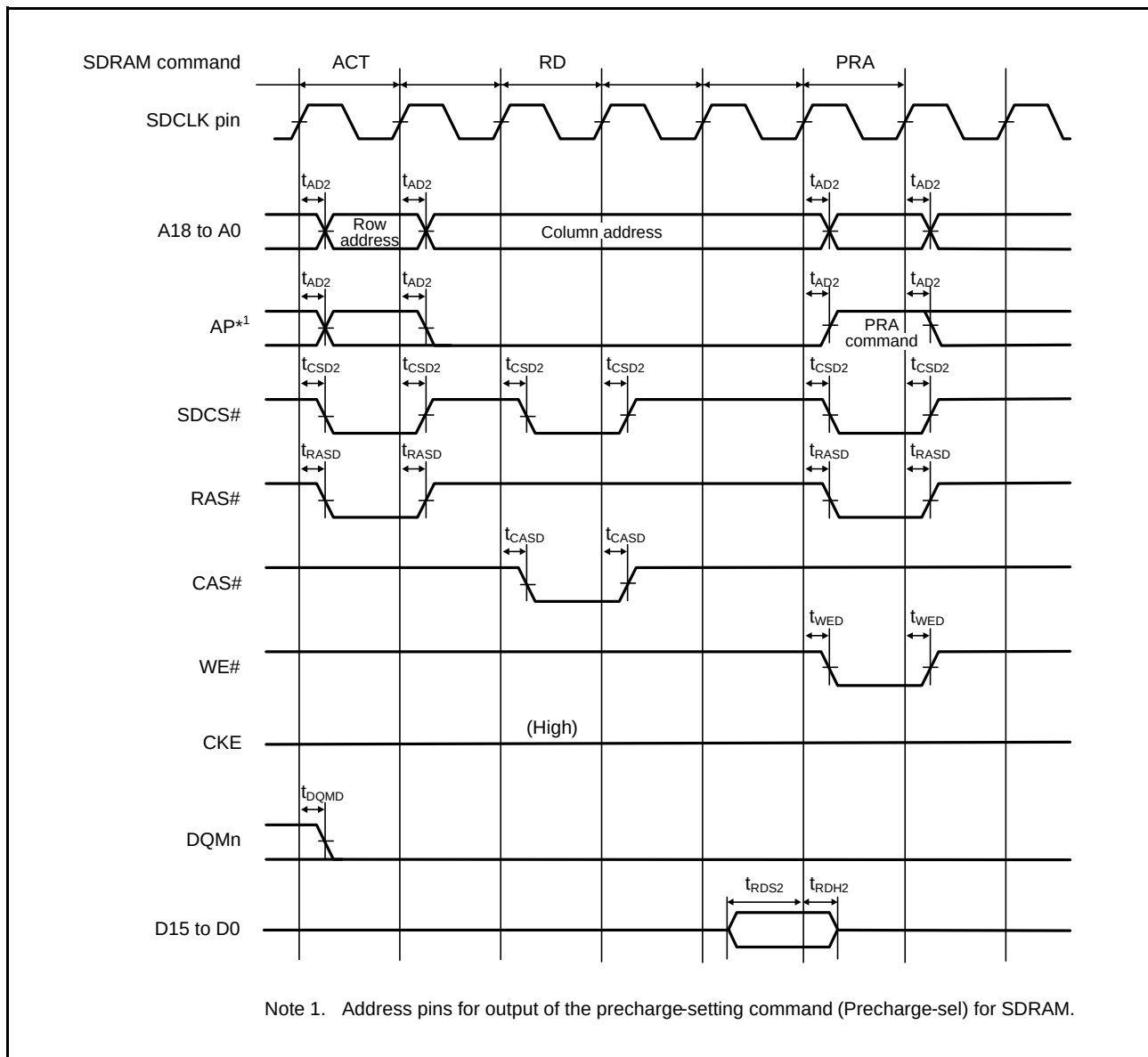


Figure 5.23 SDRAM Space Single Read Bus Timing

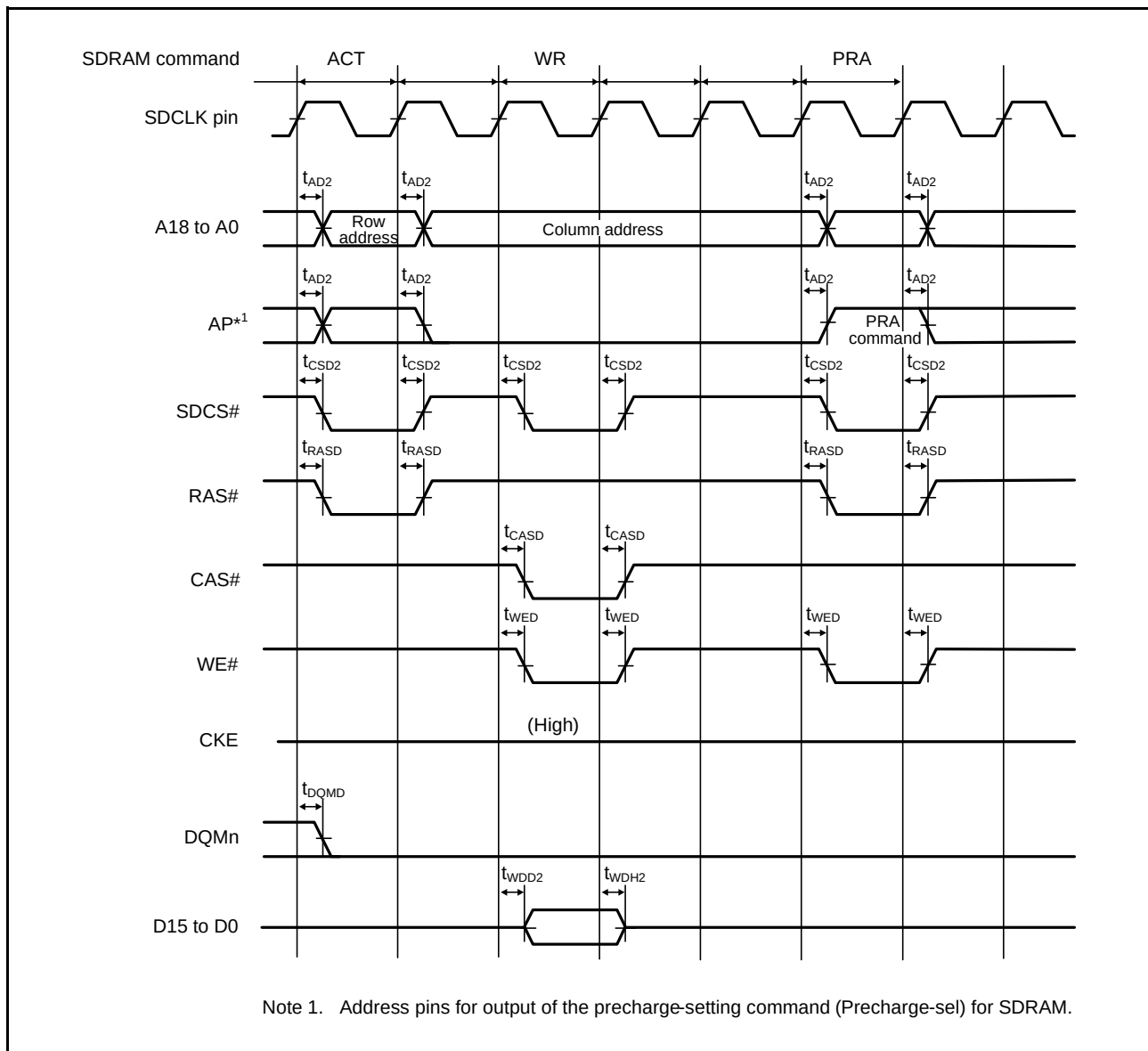
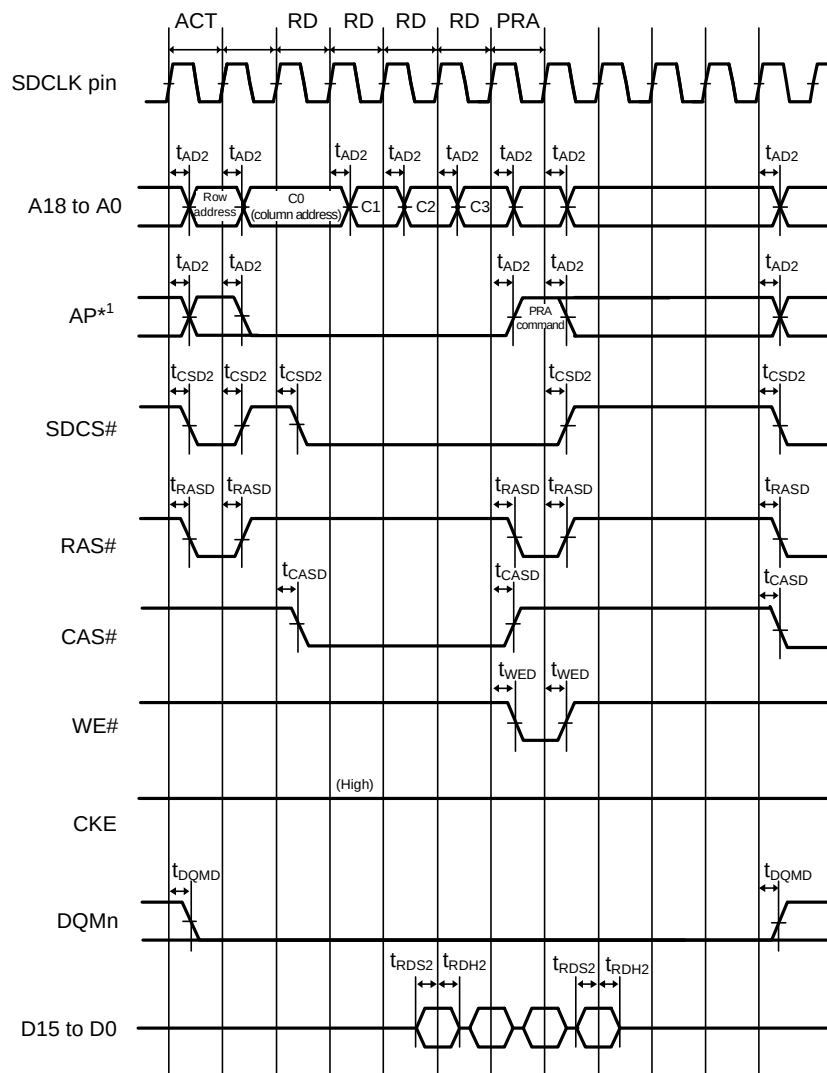
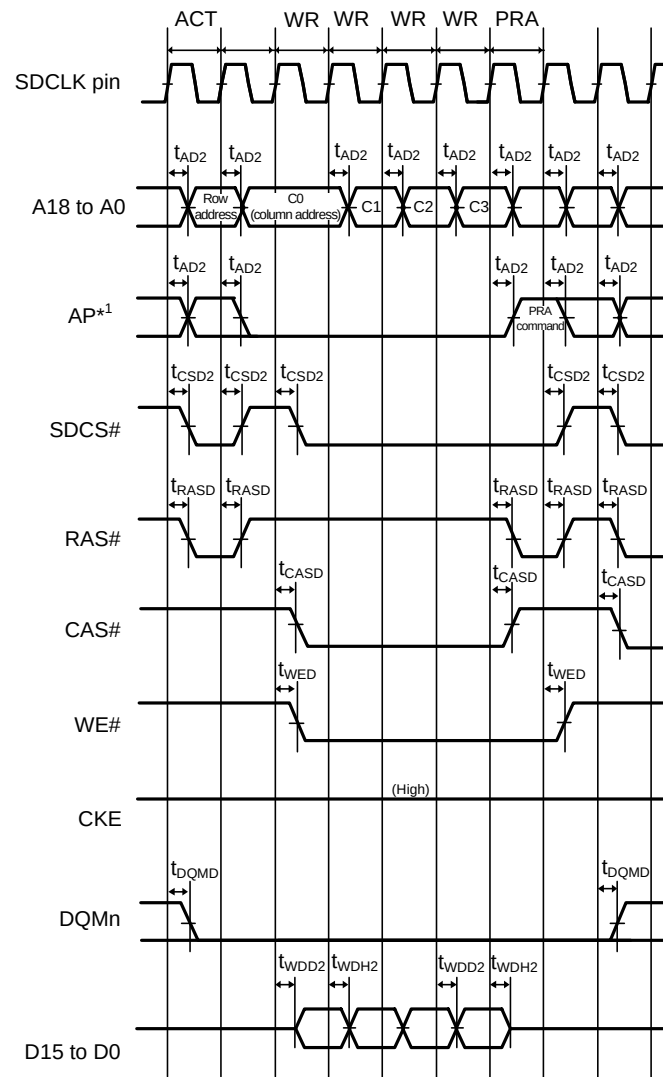


Figure 5.24 SDRAM Space Single Write Bus Timing



Note 1. Address pins for output of the precharge-setting command (Precharge-sel) for SDRAM.

Figure 5.25 SDRAM Space Multiple Read Bus Timing



Note 1. Address pins for output of the precharge-setting command (Precharge-sel) for SDRAM.

Figure 5.26 SDRAM Space Multiple Write Bus Timing

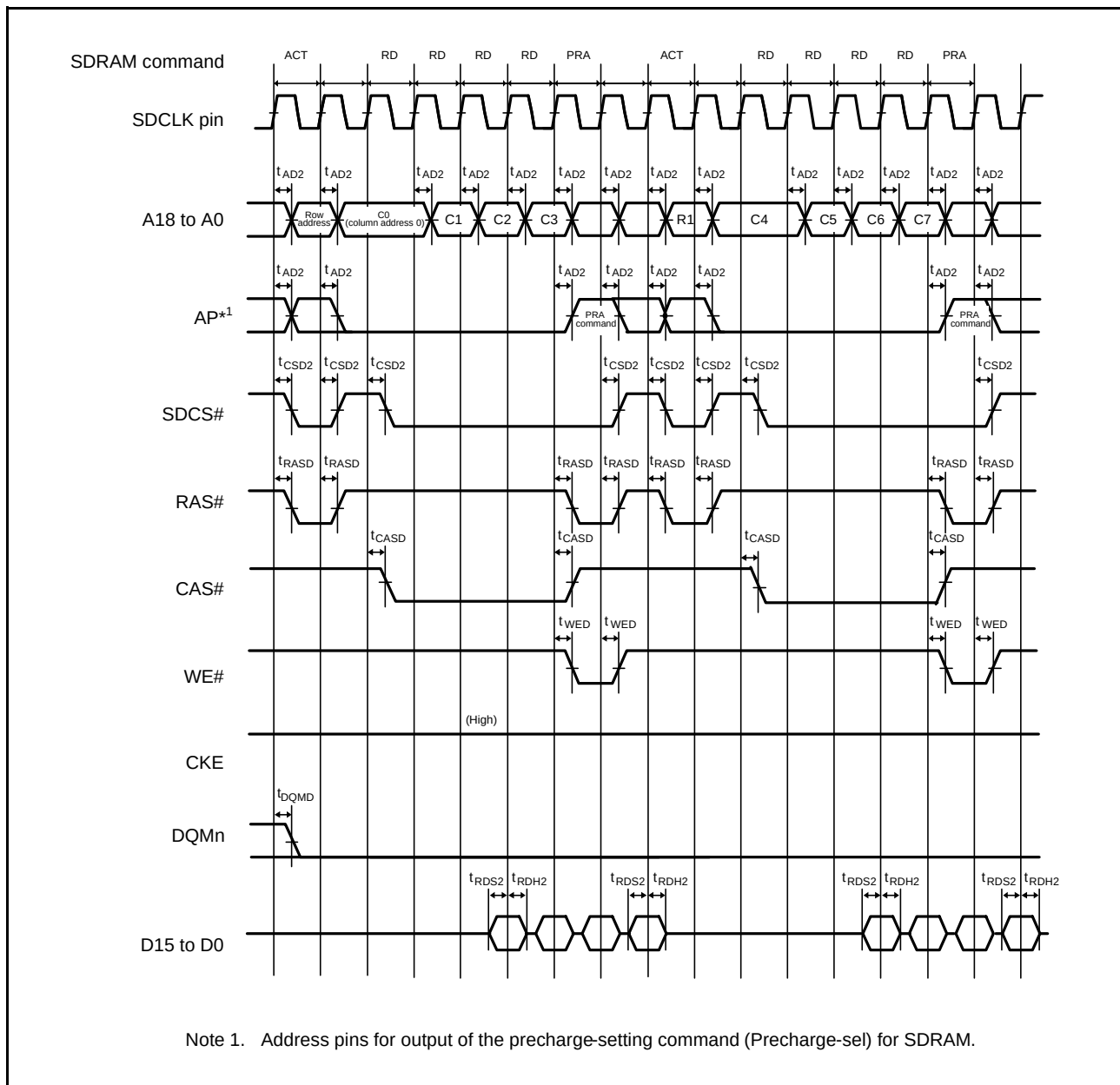
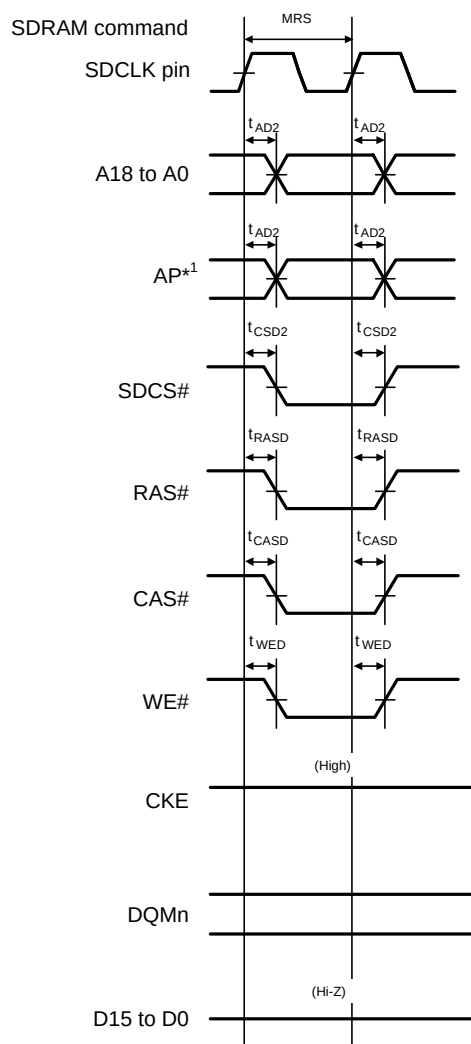
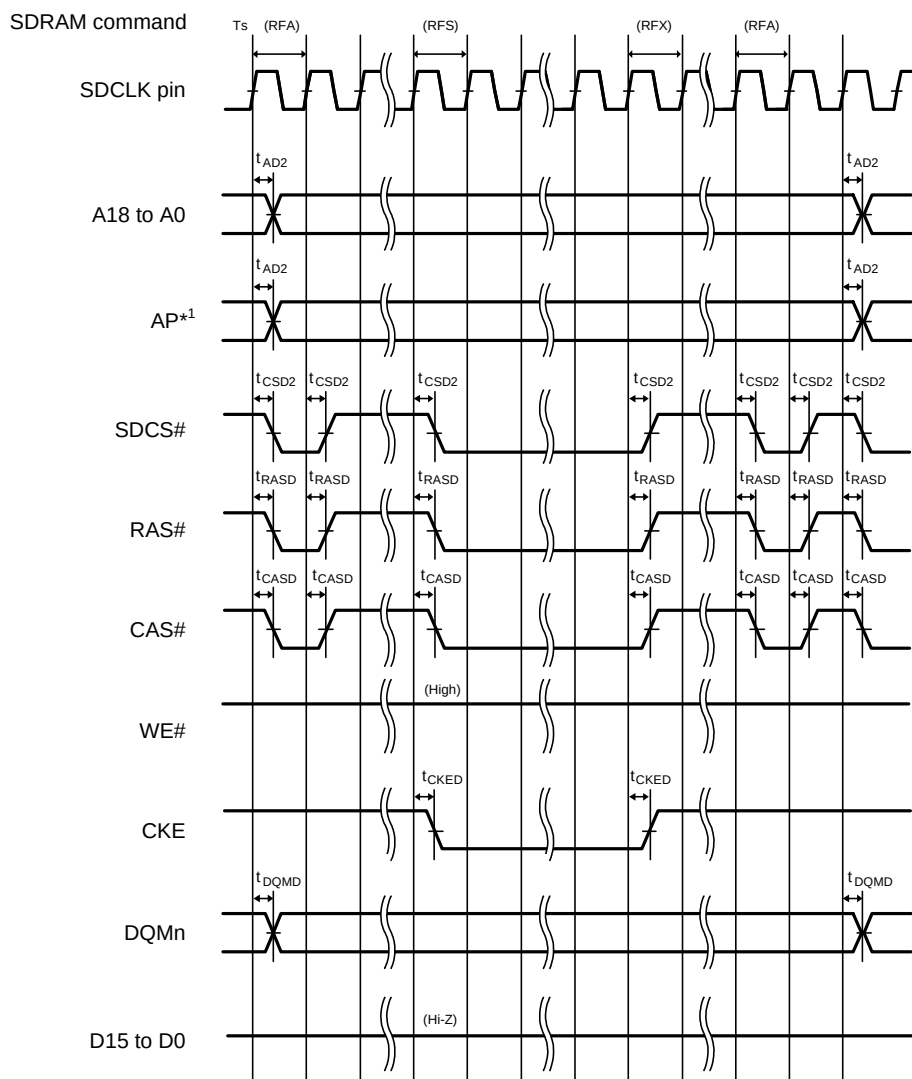


Figure 5.27 SDRAM Space Multiple Read Line Stride Bus Timing



Note 1. Address pins for output of the precharge-setting command (Precharge-sel) for SDRAM.

Figure 5.28 SDRAM Space Mode Register Set Bus Timing



Note 1. Address pins for output of the precharge-setting command (Precharge-sel) for SDRAM.

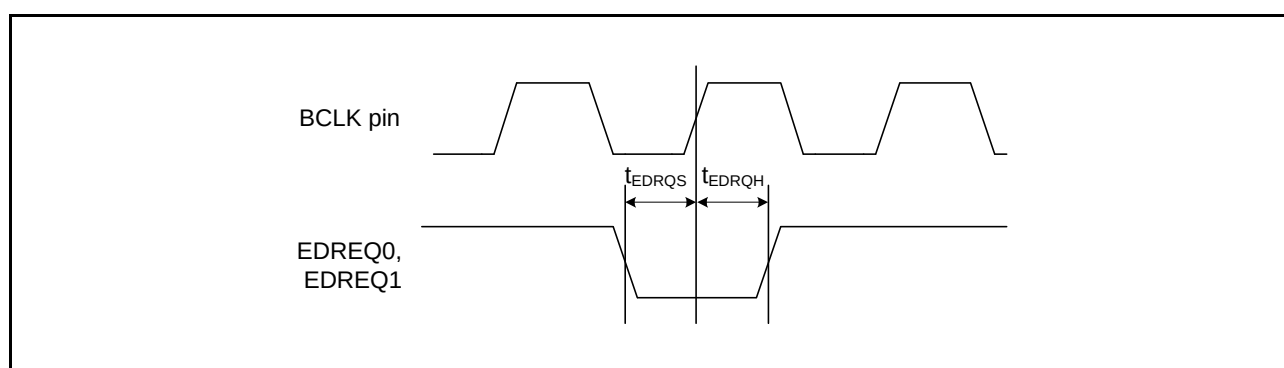
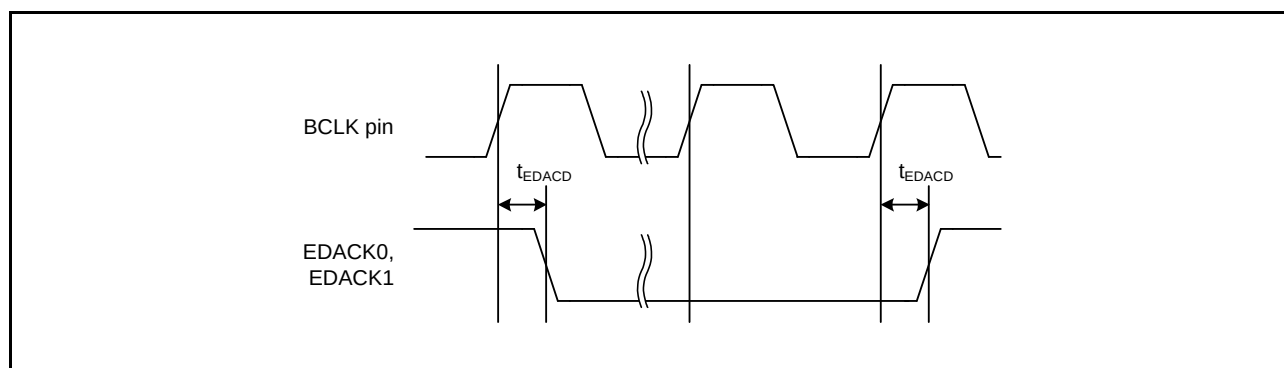
Figure 5.29 SDRAM Space Self-Refresh Bus Timing

5.3.6 EXDMAC Timing

Table 5.23 EXDMAC Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $ICLK = PCLKA = 8$ to 120 MHz, $PCLKB = BCLK = SDCLK = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
 High-drive output is selected by the driving ability control register.

	Item	Symbol	Min.	Max.	Unit	Test Conditions
EXDMAC	EDREQ setup time	t_{EDRQS}	13	—	ns	Figure 5.30
	EDREQ hold time	t_{EDRQH}	2	—	ns	
	EDACK delay time	t_{EDACD}	—	13	ns	Figure 5.31, Figure 5.32

**Figure 5.30 EDREQ0 and EDREQ1 Input Timing****Figure 5.31 EDACK0 and EDACK1 Single-Address Transfer Timing (for a CS Area)**

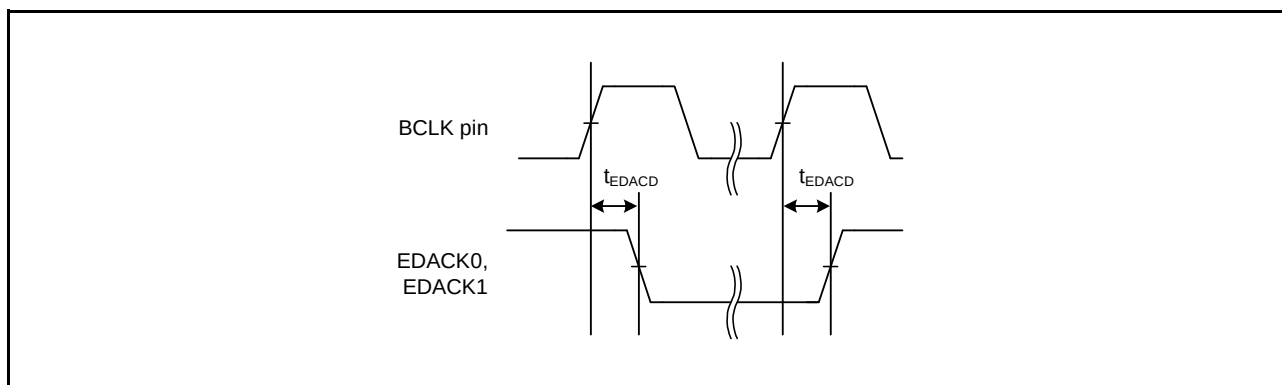


Figure 5.32 EDACK0 and EDACK1 Single-Address Transfer Timing (for SDRAM)

5.3.7 Timing of On-Chip Peripheral Modules

Table 5.24 I/O Port Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
 High-drive output is selected by the driving ability control register.

Item		Symbol	Min.	Max.	Unit*1	Test Conditions
I/O ports	Input data pulse width	t_{PRW}	1.5	—	t_{PBcyc}	Figure 5.33

Note 1. t_{PBcyc} : PCLKB cycle

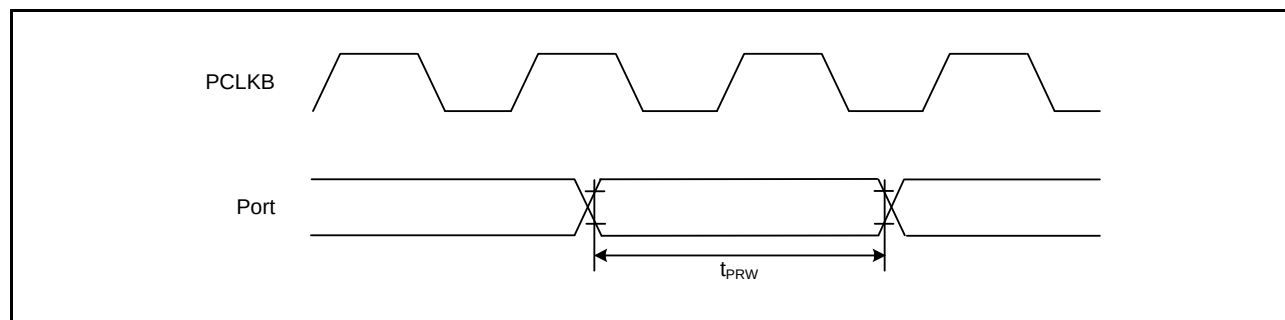
**Figure 5.33 I/O Port Input Timing**

Table 5.25 TPU Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
TPU	Input capture input pulse width	Single-edge setting	t_{TICW}	1.5	—	t_{PBcyc}	Figure 5.34
		Both-edge setting		2.5	—		
	Timer clock pulse width	Single-edge setting	t_{TCKWH} , t_{TCKWL}	1.5	—	t_{PBcyc}	Figure 5.35
		Both-edge setting		2.5	—		
		Phase counting mode		2.5	—		

Note 1. t_{PBcyc} : PCLKB cycle

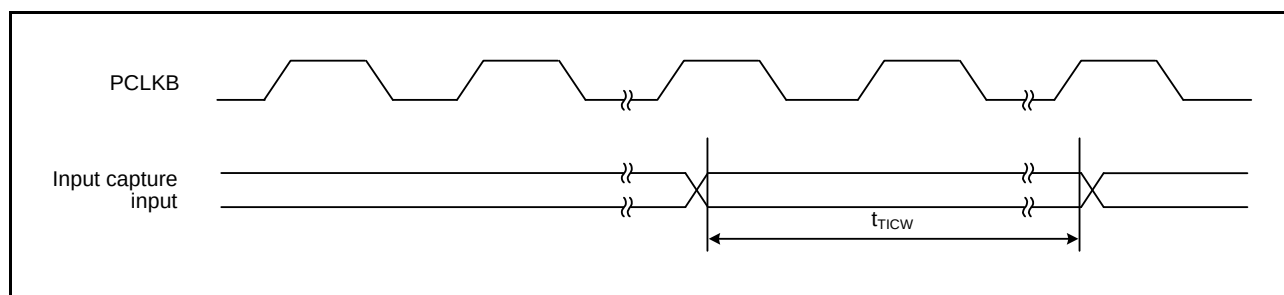
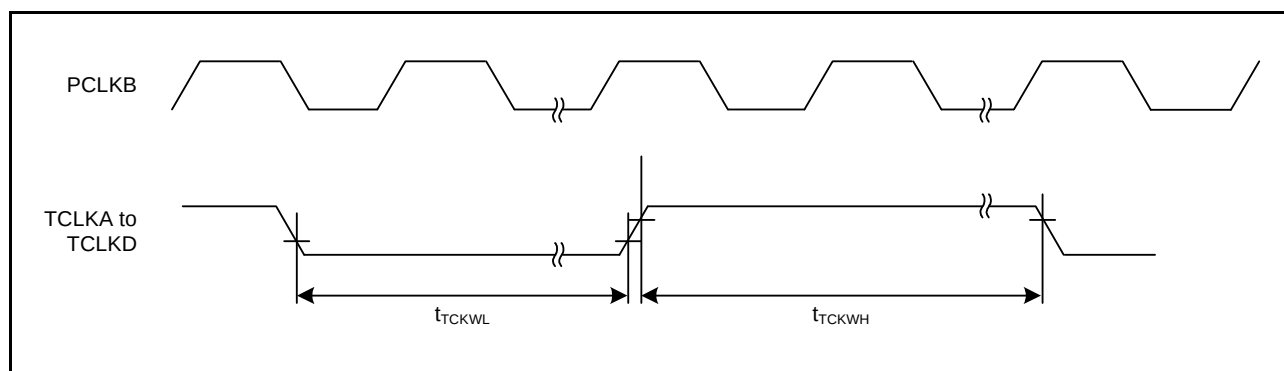
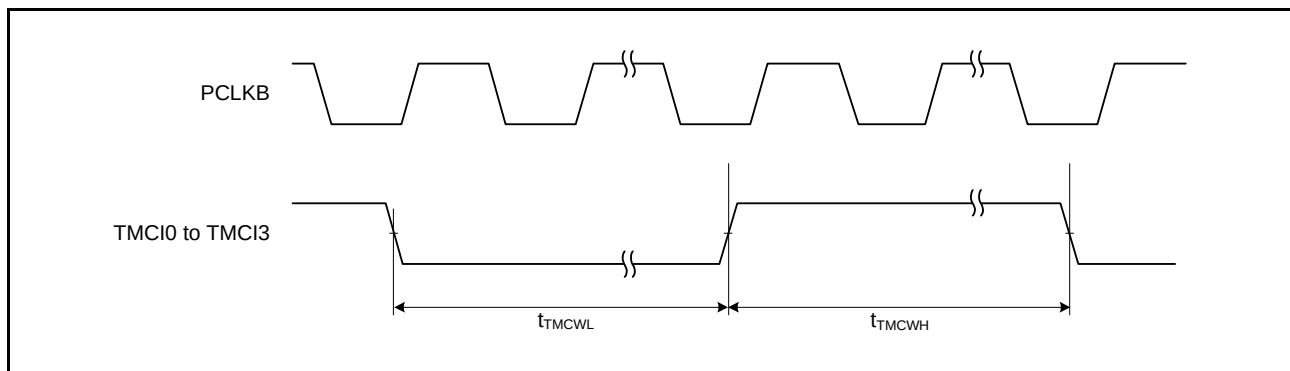
**Figure 5.34 TPU Input Capture Input Timing****Figure 5.35 TPU Clock Input Timing**

Table 5.26 TMR Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
TMR	Timer clock pulse width	Single-edge setting	t_{TMCWH} , t_{TMCWL}	1.5	—	t_{PBcyc}	Figure 5.36
		Both-edge setting		2.5	—		

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 5.36 TMR Clock Input Timing****Table 5.27 CMTW Timing**

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
CMTW	Input capture input pulse width	Single-edge setting	$t_{CMTWTICW}$	1.5	—	t_{PBcyc}	Figure 5.37
		Both-edge setting		2.5	—		

Note 1. t_{PBcyc} : PCLKB cycle

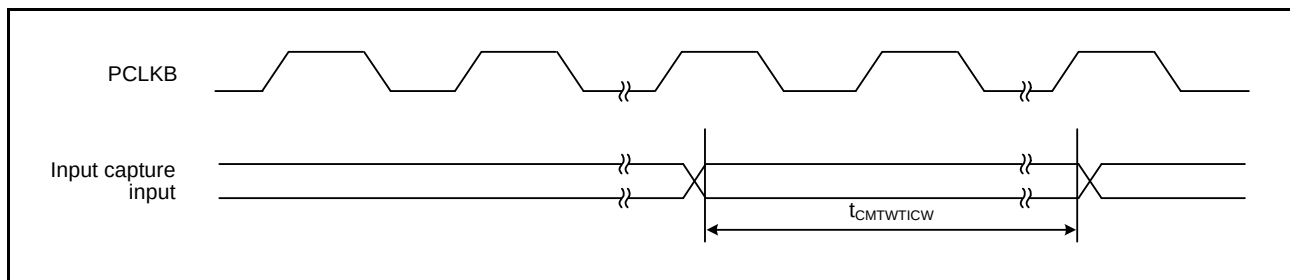
**Figure 5.37 CMTW Input Capture Input Timing**

Table 5.28 MTU3 Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
MTU3	Input capture input pulse width	Single-edge setting	t_{MTICW}	1.5	—	t_{PACyc}	Figure 5.38
		Both-edge setting		2.5	—		
	Timer clock pulse width	Single-edge setting	t_{MTCKWH} , t_{MTCKWL}	1.5	—	t_{PACyc}	Figure 5.39
		Both-edge setting		2.5	—		
		Phase counting mode		2.5	—		

Note 1. t_{PACyc} : PCLKA cycle

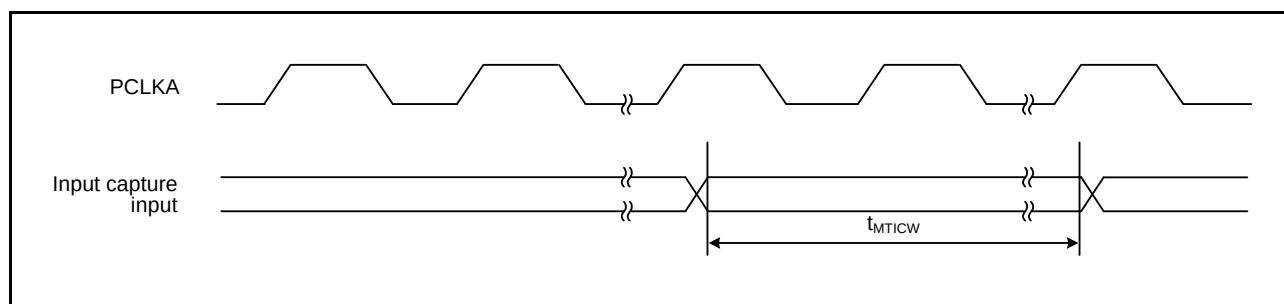
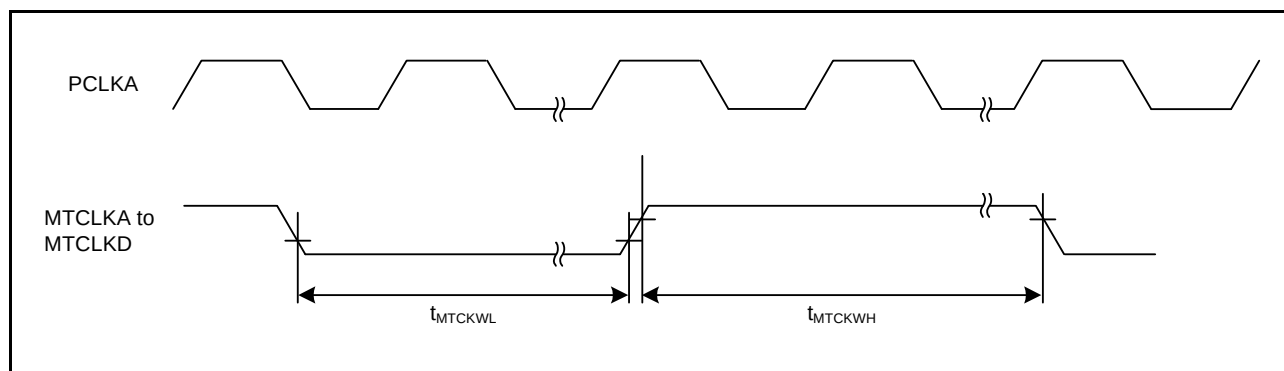
**Figure 5.38 MTU3 Input Capture Input Timing****Figure 5.39 MTU3 Clock Input Timing**

Table 5.29 POE3 Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
 High-drive output is selected by the driving ability control register.

Item		Symbol	Min.	Max.	Unit*1	Test Conditions
POE	POE# input pulse width	t_{POEW}	1.5	—	t_{PBcyc}	Figure 5.40

Note 1. t_{PBcyc} : PCLKB cycle

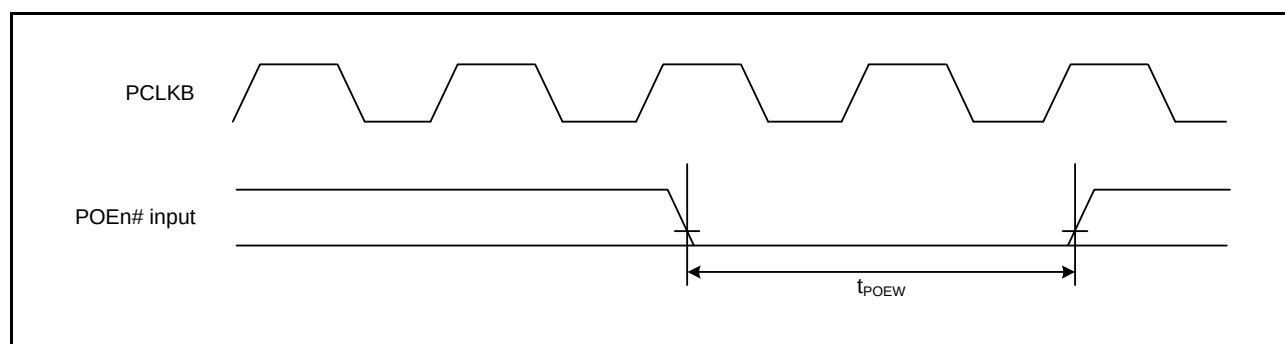
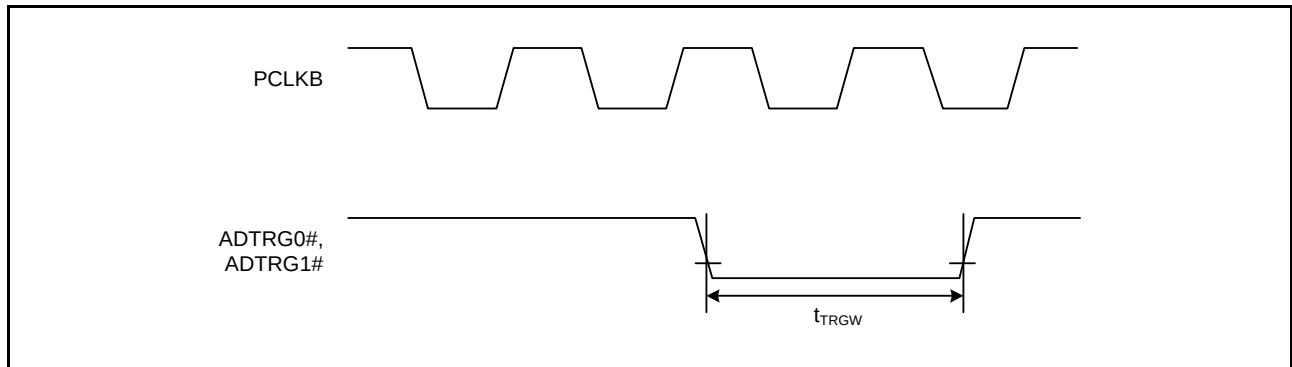
**Figure 5.40 POE# Input Timing**

Table 5.30 A/D Converter Trigger Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
 High-drive output is selected by the driving ability control register.

Item		Symbol	Min.	Max.	Unit*1	Test Conditions
A/D converter	A/D converter trigger input pulse width	t_{TRGW}	1.5	—	t_{PBcyc}	Figure 5.41

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 5.41 A/D Converter Trigger Input Timing****Table 5.31 CAC Timing**

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
 High-drive output is selected by the driving ability control register.

Item*1, *2			Symbol	Min.*1	Max.	Unit*1	Test Conditions
CAC	CACREF input pulse width	$t_{PBcyc} \leq t_{cac}$	t_{CACREF}	$4.5 t_{cac} + 3 t_{PBcyc}$	—	ns	
		$t_{PBcyc} > t_{cac}$		$5 t_{cac} + 6.5 t_{PBcyc}$	—		

Note 1. t_{PBcyc} : PCLKB cycle

Note 2. t_{cac} : CAC count clock source cycle

Table 5.32 SCIg, SCIH, and SCLi Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

Item			Symbol	Min.*1	Max.*1	Unit*1	Test Conditions
SCIg, SCIH	Input clock cycle	Asynchronous	t_{Scyc}	4	—	t_{PBcyc}	Figure 5.42
		Clock synchronous		6	—		
	Input clock pulse width		t_{SCKW}	0.4	0.6	t_{Scyc}	
	Input clock rise time		t_{SCKr}	—	5	ns	
	Input clock fall time		t_{SCKf}	—	5	ns	
	Output clock cycle	Asynchronous*2	t_{Scyc}	8	—	t_{PBcyc}	
		Clock synchronous		4	—		
	Output clock pulse width		t_{SCKW}	0.4	0.6	t_{Scyc}	
	Output clock rise time		t_{SCKr}	—	5	ns	
	Output clock fall time		t_{SCKf}	—	5	ns	
	Transmit data delay time	Clock synchronous	t_{TXD}	—	28	ns	Figure 5.43
	Receive data setup time	Clock synchronous	t_{RXS}	15	—	ns	
	Receive data hold time	Clock synchronous	t_{RXH}	5	—	ns	
SCLi	Input clock cycle	Asynchronous	t_{Scyc}	4	—	t_{PAcyc}	Figure 5.42
		Clock synchronous		12	—		
	Input clock pulse width		t_{SCKW}	0.4	0.6	t_{Scyc}	
	Input clock rise time		t_{SCKr}	—	5	ns	
	Input clock fall time		t_{SCKf}	—	5	ns	
	Output clock cycle	Asynchronous*2	t_{Scyc}	8	—	t_{PAcyc}	
		Clock synchronous		8	—		
	Output clock pulse width		t_{SCKW}	0.4	0.6	t_{Scyc}	
	Output clock rise time		t_{SCKr}	—	5	ns	
	Output clock fall time		t_{SCKf}	—	5	ns	
	Transmit data delay time	Master	t_{TXD}	—	15	ns	Figure 5.43
		Slave		—	28		
	Receive data setup time	Clock synchronous	t_{RXS}	20	—	ns	
Receive data hold time	Clock synchronous	t_{RXH}	5	—			

Note 1. t_{PBcyc} : PCLKB cycle; t_{PAcyc} : PCLKA cycle

Note 2. When the SEMR.ABCS and SEMR.BGDM bits are set to 1

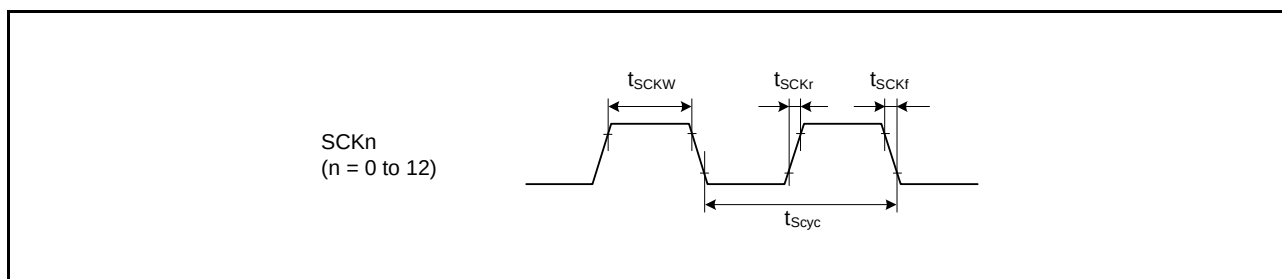


Figure 5.42 SCK Clock Input Timing

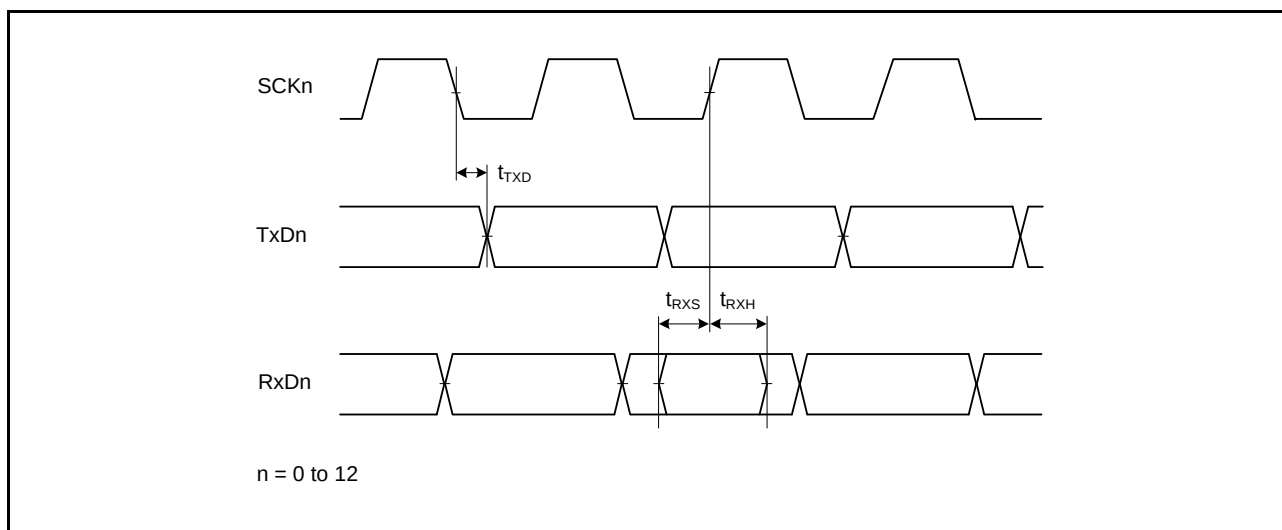


Figure 5.43 SCI Input/Output Timing: Clock Synchronous Mode

Table 5.33 RSPI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

Item				Symbol	Min.*1	Max.*1	Unit*1	Test Conditions*2		
RSPI	RSPCK clock cycle		Master	t_{SPcyc}	2	4096	t_{pAcyc}	Figure 5.44		
			Slave		4	4096				
	RSPCK clock high pulse width		Master	t_{SPCKWH}	$(t_{SPcyc} - t_{SCKr} - t_{SCKf}) / 2 - 3$	—	ns			
			Slave		$(t_{SPcyc} - t_{SCKr} - t_{SCKf}) / 2$	—				
	RSPCK clock low pulse width		Master	t_{SPCKWL}	$(t_{SPcyc} - t_{SCKr} - t_{SCKf}) / 2 - 3$	—	ns			
			Slave		$(t_{SPcyc} - t_{SCKr} - t_{SCKf}) / 2$	—				
	RSPCK clock rise/fall time		Output	t_{SPCKr}, t_{SPCKf}	—	5	ns			
			Input		—	1	μs			
	Data input setup time		Master	t_{SU}	6	—	ns		Figure 5.45 to Figure 5.50	
			Slave		8.3	—				
	Data input hold time		Master	PCLKA division ratio set to 1/2	t_{HF}	0	—			ns
						PCLKA division ratio set to a value other than 1/2	t_H			
			Slave		8.3			—		
	SSL setup time		Master	t_{LEAD}	1	8	t_{SPcyc}			
			Slave		6	—	t_{pAcyc}			
	SSL hold time		Master	t_{LAG}	1	8	t_{SPcyc}			
			Slave		6	—	t_{pAcyc}			
	Data output delay time		Master	t_{OD}	—	6.3	ns			
			Slave		—	28				
	Data output hold time		Master	t_{OH}	0	—	ns			
			Slave		0	—				
	Successive transmission delay time		Master	t_{TD}	$t_{SPcyc} + 2 \times t_{pAcyc}$	$8 \times t_{SPcyc} + 2 \times t_{pAcyc}$	ns			
			Slave		$4 \times t_{pAcyc}$	—				
	MOSI and MISO rise/fall time		Output	t_{Dr}, t_{Df}	—	5	ns			
			Input		—	1	μs			
	SSL rise/fall time		Output	t_{SSLr}, t_{SSLf}	—	5	ns			
			Input		—	1	μs			
	Slave access time				t_{SA}	—	$2 \times t_{pAcyc} + 28$	t_{pAcyc}	Figure 5.49, Figure 5.50	
	Slave output release time				t_{REL}	—	$2 \times t_{pAcyc} + 28$	t_{pAcyc}		

Note 1. t_{pAcyc} : PCLKA cycle

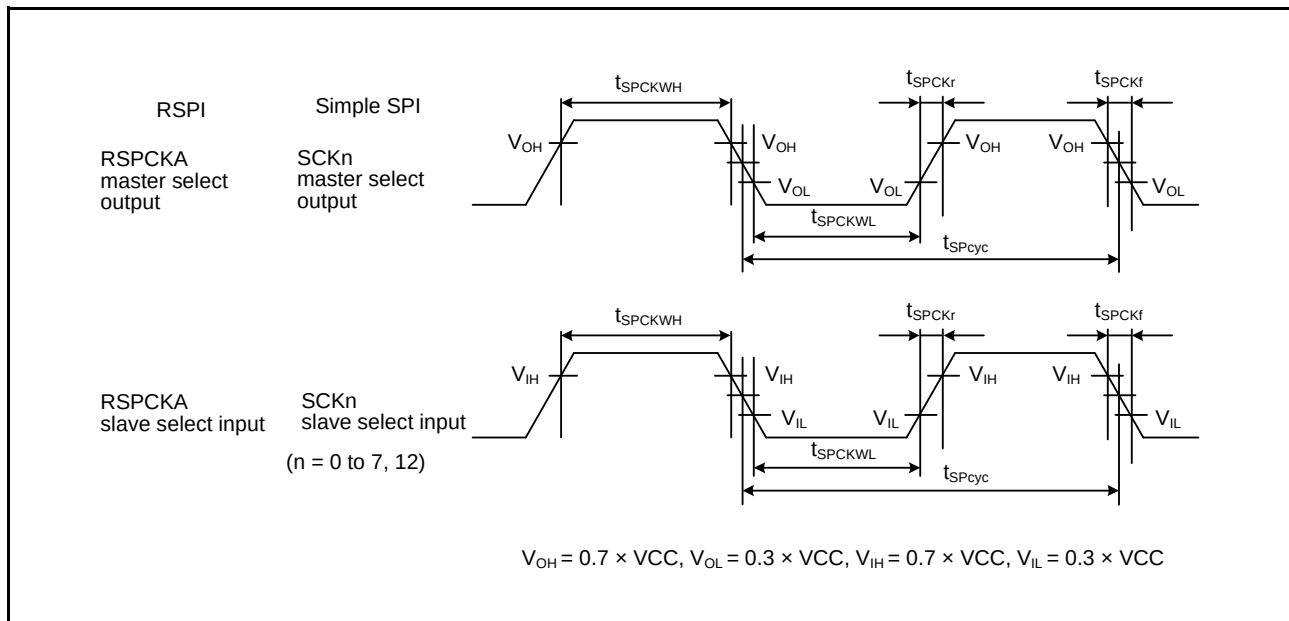
Note 2. We recommend using pins that have a letter ("A", "B", etc.) to indicate group membership appended to their names as groups. For the RSPI interface, the AC portion of the electrical characteristics is measured for each group.

Table 5.34 Simple SPI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

Item		Symbol	Min.	Max.	Unit*1	Test Conditions
Simple SPI	SCK clock cycle output (master)	t_{SPCyc}	4	65536	t_{PACyc}	Figure 5.44
	SCK clock cycle input (slave)		8	65536		
	SCK clock high pulse width	t_{SPCKWH}	0.4	0.6	t_{SPCyc}	
	SCK clock low pulse width	t_{SPCKWL}	0.4	0.6	t_{SPCyc}	
	SCK clock rise/fall time	t_{SPCKr} , t_{SPCKf}	—	20	ns	
	Data input setup time	t_{SU}	33.3	—	ns	Figure 5.45 to Figure 5.50
	Data input hold time	t_H	33.3	—	ns	
	SS input setup time	t_{LEAD}	1	—	t_{SPCyc}	
	SS input hold time	t_{LAG}	1	—	t_{SPCyc}	
	Data output delay time	t_{OD}	—	33.3	ns	
	Data output hold time	t_{OH}	−10	—	ns	
	Data rise/fall time	t_{Dr} , t_{Df}	—	16.6	ns	
	SS input rise/fall time	t_{SSLr} , t_{SSLf}	—	16.6	ns	
	Slave access time	t_{SA}	—	5	t_{PBcyc}	Figure 5.49, Figure 5.50
	Slave output release time	t_{REL}	—	5	t_{PBcyc}	

Note 1. t_{PACyc} : PCLKA cycle, t_{PBcyc} : PCLKB cycle

**Figure 5.44 RSPI Clock Timing and Simple SPI Clock Timing**

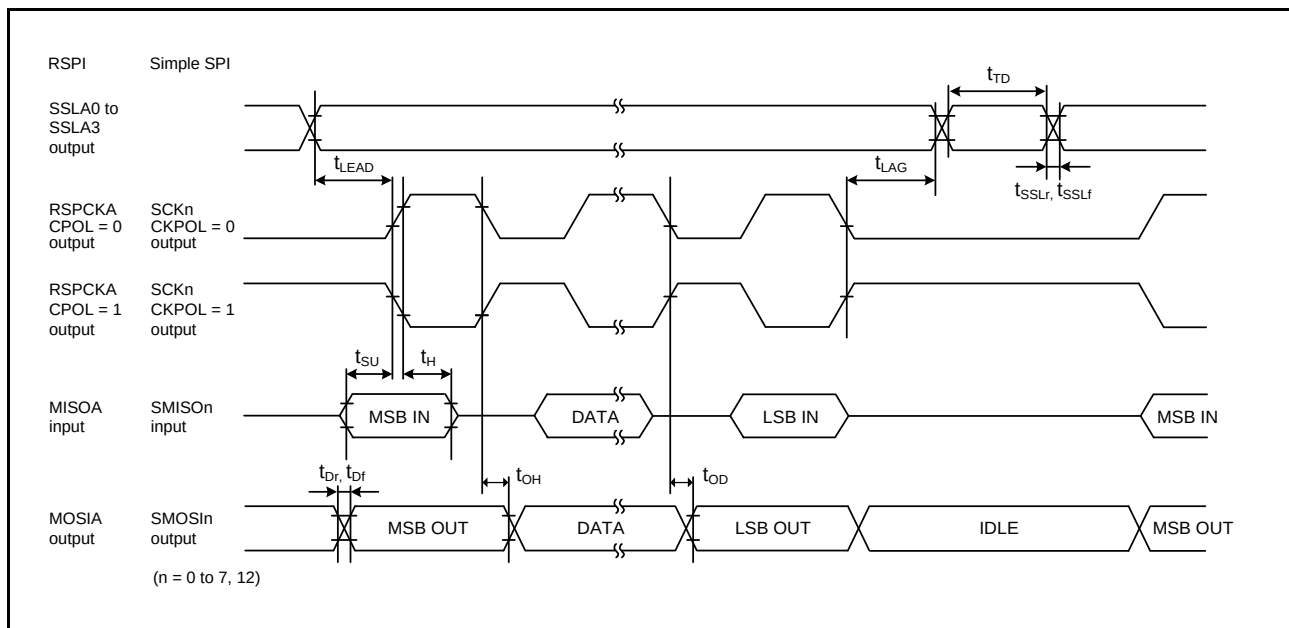


Figure 5.45 RSPI Timing (Master, CPHA = 0) (Bit Rate: PCLKA Division Ratio Set to a Value Other Than 1/2) and Simple SPI Timing (Master, CKPH = 1)

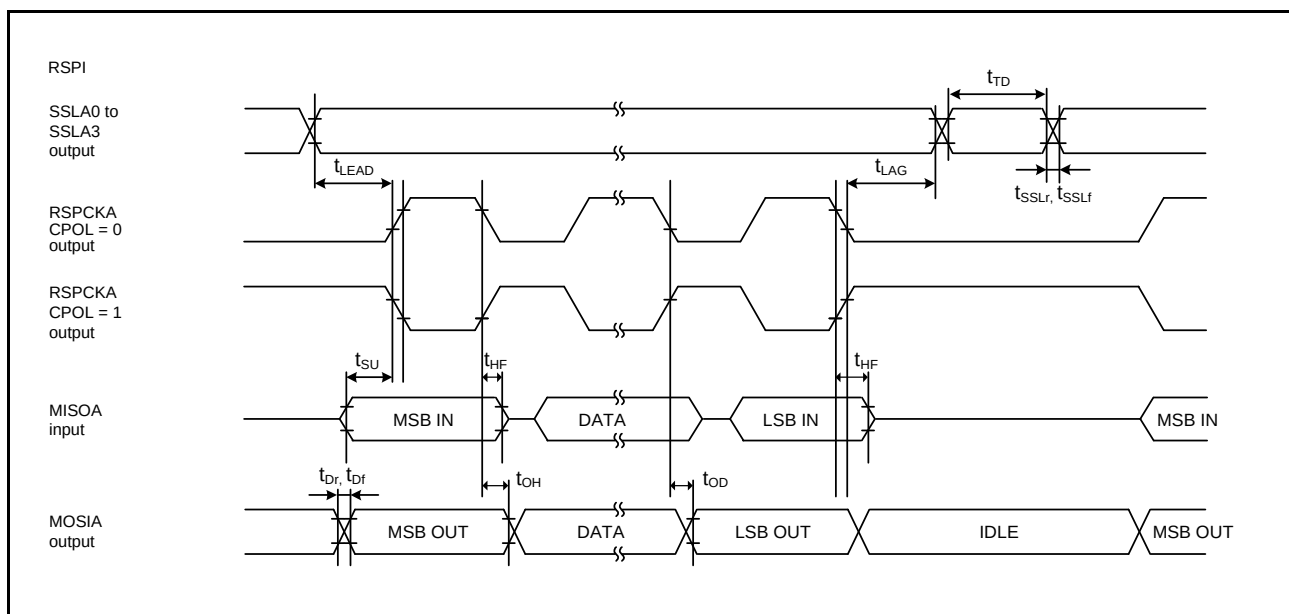


Figure 5.46 RSPI Timing (Master, CPHA = 0) (Bit Rate: PCLKA Division Ratio Set to 1/2)

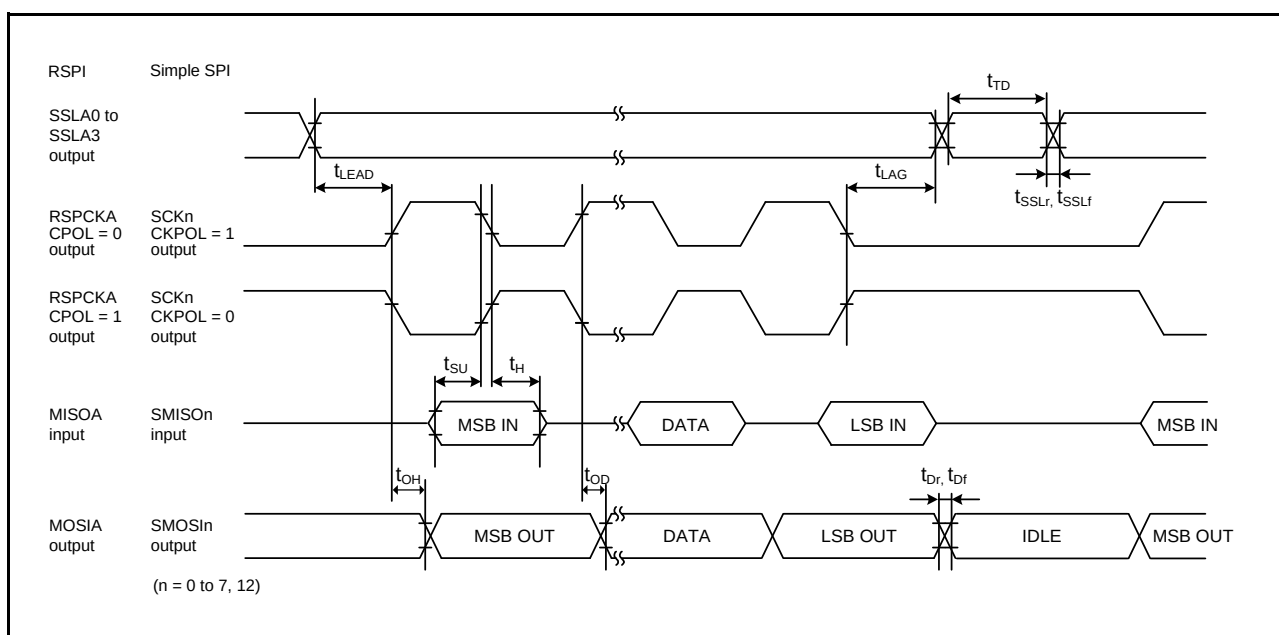


Figure 5.47 RSPI Timing (Master, CPHA = 1) (Bit Rate: PCLKA Division Ratio Set to a Value Other Than 1/2) and Simple SPI Timing (Master, CKPH = 0)

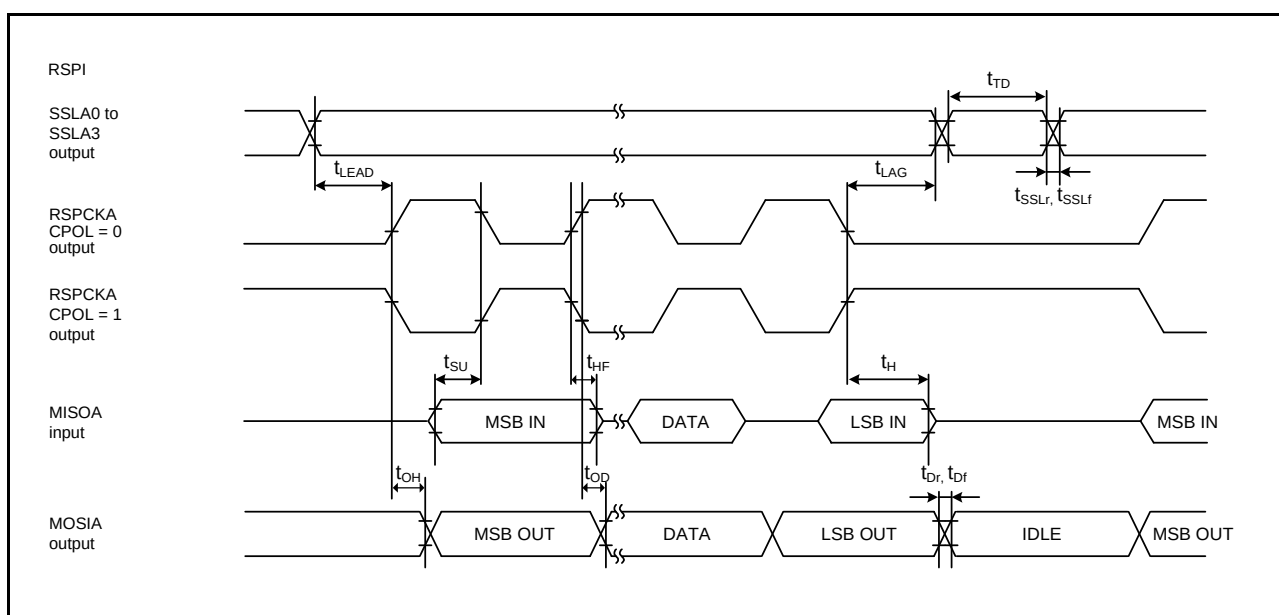


Figure 5.48 RSPI Timing (Master, CPHA = 1) (Bit Rate: PCLKA Division Ratio Set to 1/2)

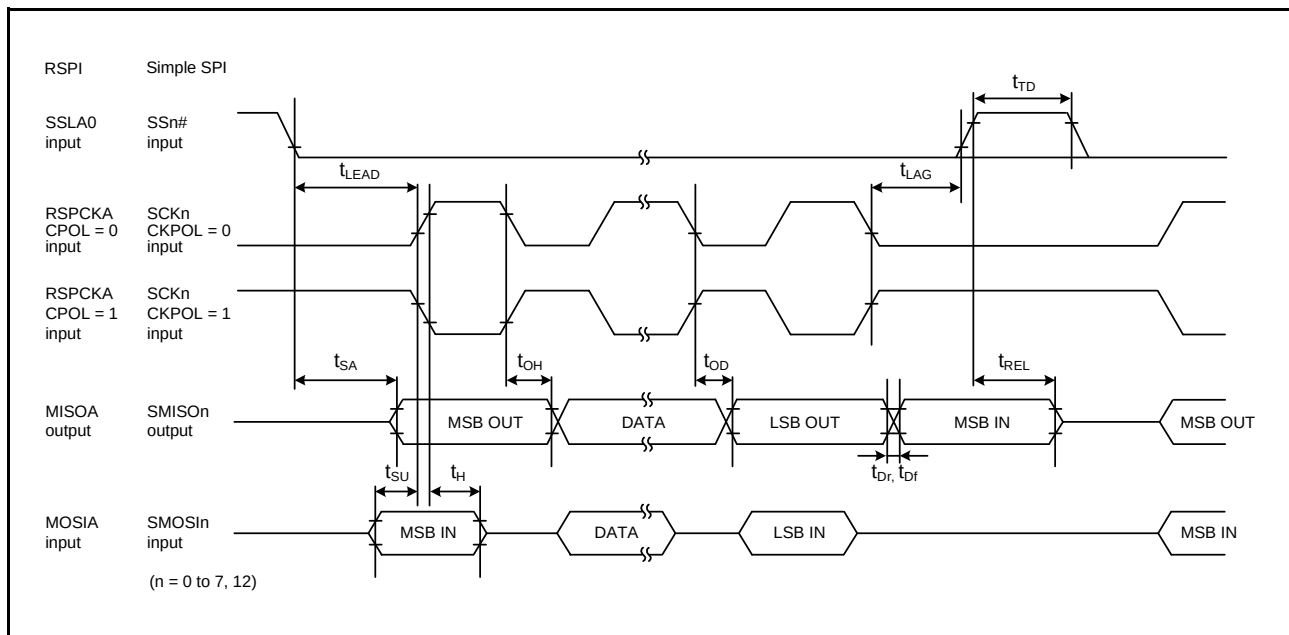


Figure 5.49 RSPI Timing (Slave, CPHA = 0) and Simple SPI Timing (Slave, CKPH = 1)

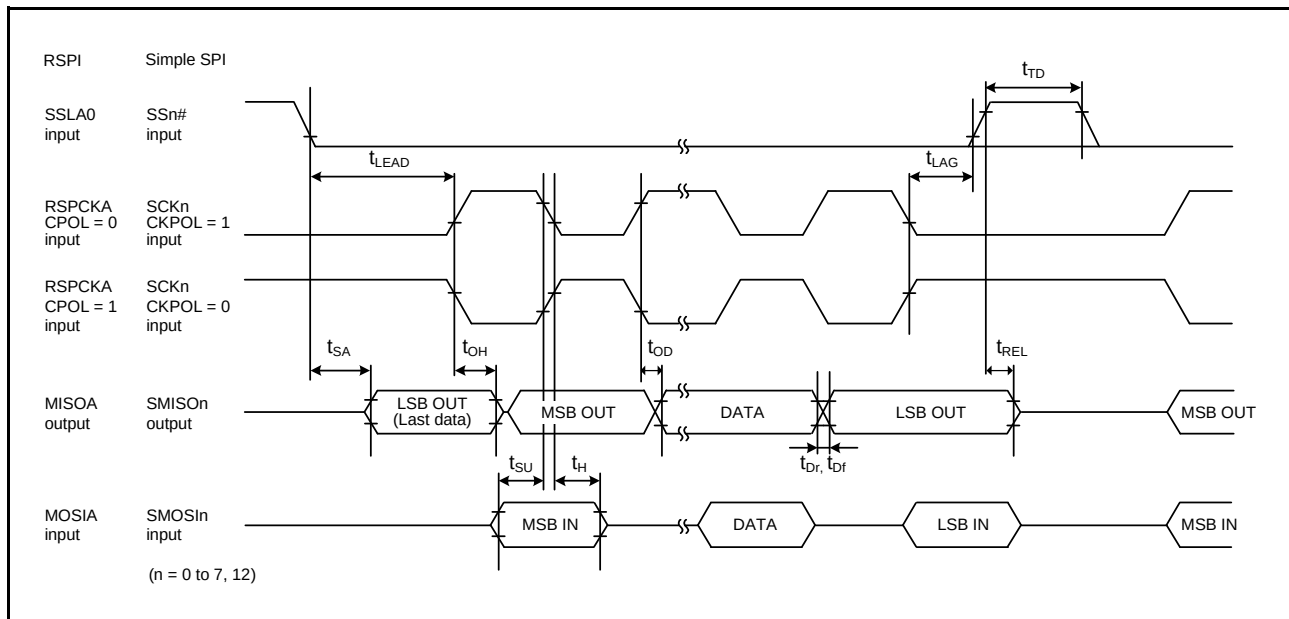


Figure 5.50 RSPI Timing (Slave, CPHA = 1) and Simple SPI Timing (Slave, CKPH = 0)

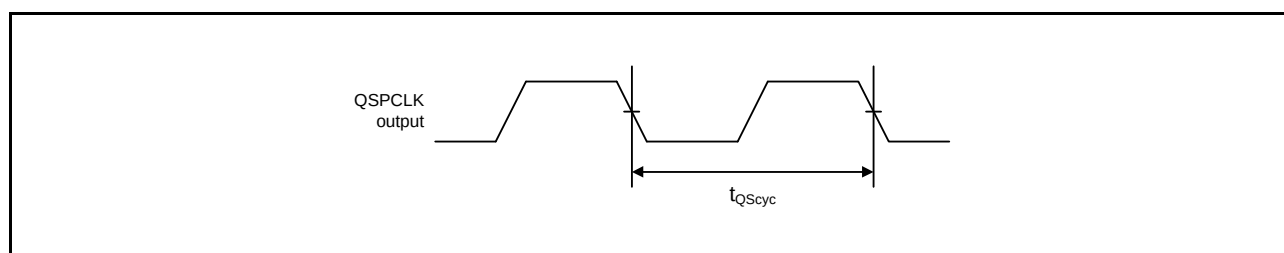
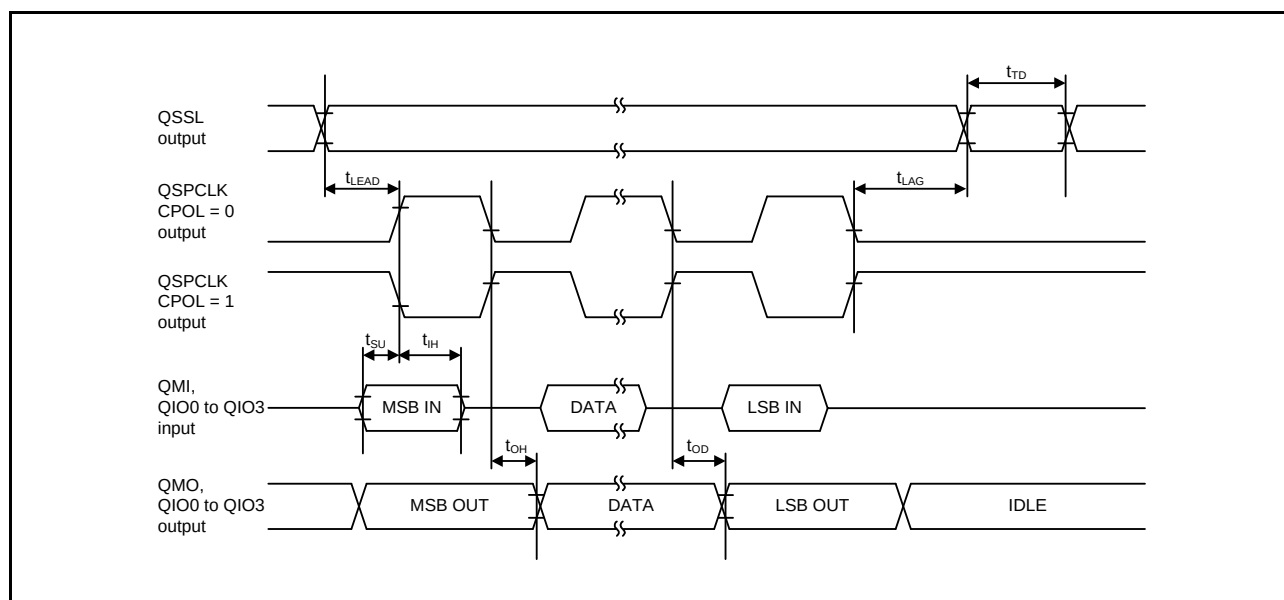
Table 5.35 QSPI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
 High-drive output is selected by the driving ability control register.

	Item	Symbol	Min.	Max.	Unit*1	Test Conditions*2
QSPI	QSPCLK clock cycle	t_{QScyc}	2	4080	t_{PBcyc}	Figure 5.51
	Data input setup time	t_{SU}	6.5	—	ns	Figure 5.52, Figure 5.53
	Data input hold time	t_{IH}	5	—	ns	
	SS setup time	t_{LEAD}	1.5	8.5	t_{QScyc}	
	SS hold time	t_{LAG}	1	8	t_{QScyc}	
	Data output delay time	t_{OD}	—	10.0	ns	
	Data output hold time	t_{OH}	—5	—	ns	
	Successive transmission delay time	t_{TD}	1	8	t_{QScyc}	

Note 1. t_{PBcyc} : PCLKB cycle

Note 2. We recommend using pins that have a letter ("A", "B", etc.) to indicate group membership appended to their names as groups.
 For the QSPI interface, the AC portion of the electrical characteristics is measured for each group.

**Figure 5.51 QSPI Clock Timing****Figure 5.52 Transmit/Receive Timing (CPHA = 0)**

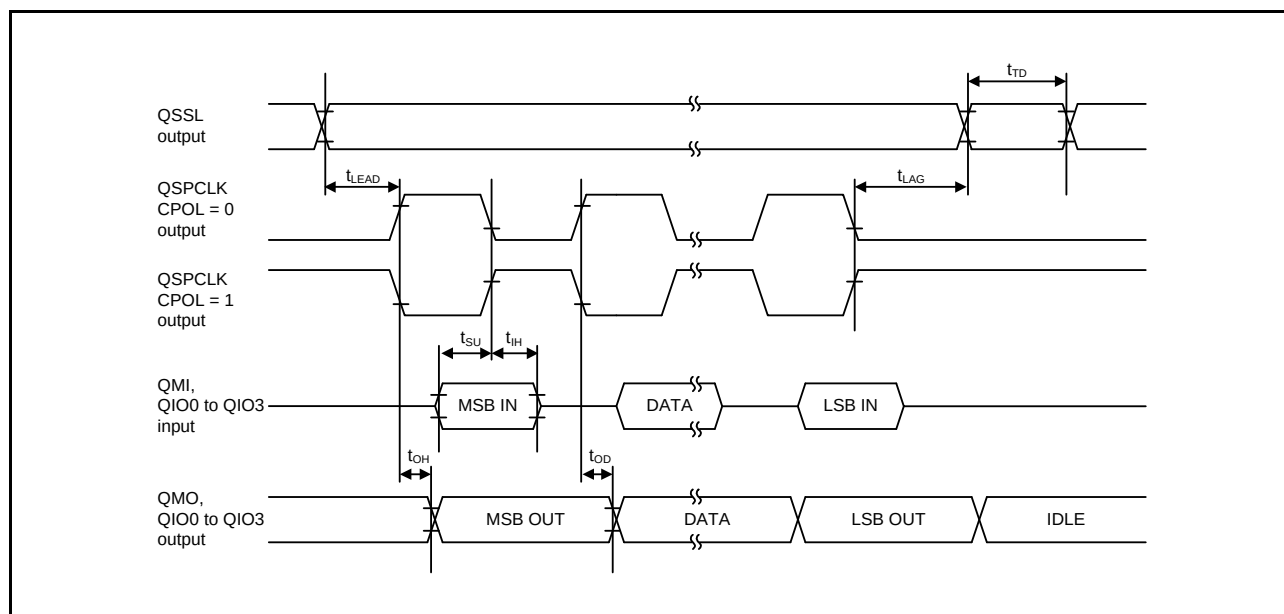


Figure 5.53 Transmit/Receive Timing (CPHA = 1)

Table 5.36 RIIC Timing (1)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 High-drive output is selected by the driving ability control register.

Item		Symbol	Min.*1, *2	Max.	Unit	Test Conditions
RIIC (Standard-mode, SMBus) ICFER.FMPE = 0	SCL input cycle time	t_{SCL}	$6(12) \times t_{IICcyc} + 1300$	—	ns	Figure 5.54
	SCL input high pulse width	t_{SCLH}	$3(6) \times t_{IICcyc} + 300$	—	ns	
	SCL input low pulse width	t_{SCLL}	$3(6) \times t_{IICcyc} + 300$	—	ns	
	SCL, SDA input rise time	t_{Sr}	—	1000	ns	
	SCL, SDA input fall time	t_{Sf}	—	300	ns	
	SCL, SDA input spike pulse removal time	t_{SP}	0	$1(4) \times t_{IICcyc}$	ns	
	SDA input bus free time	t_{BUF}	$3(6) \times t_{IICcyc} + 300$	—	ns	
	Start condition input hold time	t_{STAH}	$t_{IICcyc} + 300$	—	ns	
	Restart condition input setup time	t_{STAS}	1000	—	ns	
	Stop condition input setup time	t_{STOS}	1000	—	ns	
	Data input setup time	t_{SDAS}	$t_{IICcyc} + 50$	—	ns	
	Data input hold time	t_{SDAH}	0	—	ns	
	SCL, SDA capacitive load	C_b	—	400	pF	
RIIC (Fast-mode) ICFER.FMPE = 0	SCL input cycle time	t_{SCL}	$6(12) \times t_{IICcyc} + 600$	—	ns	
	SCL input high pulse width	t_{SCLH}	$3(6) \times t_{IICcyc} + 300$	—	ns	
	SCL input low pulse width	t_{SCLL}	$3(6) \times t_{IICcyc} + 300$	—	ns	
	SCL, SDA input rise time	t_{Sr}	$20 \times (\text{External pull-up voltage}/5.5V)$	300	ns	
	SCL, SDA input fall time	t_{Sf}	$20 \times (\text{External pull-up voltage}/5.5V)$	300	ns	
	SCL, SDA input spike pulse removal time	t_{SP}	0	$1(4) \times t_{IICcyc}$	ns	
	SDA input bus free time	t_{BUF}	$3(6) \times t_{IICcyc} + 300$	—	ns	
	Start condition input hold time	t_{STAH}	$t_{IICcyc} + 300$	—	ns	
	Restart condition input setup time	t_{STAS}	300	—	ns	
	Stop condition input setup time	t_{STOS}	300	—	ns	
	Data input setup time	t_{SDAS}	$t_{IICcyc} + 50$	—	ns	
	Data input hold time	t_{SDAH}	0	—	ns	
	SCL, SDA capacitive load	C_b	—	400	pF	

Note: t_{IICcyc} : RIIC internal reference clock (IIC ϕ) cycle

Note 1. The value within parentheses is applicable when the value of the ICMR3.NF[1:0] bits is 11b while the digital filter is enabled by the setting ICFER.NFE = 1.

Note 2. C_b is the total capacitance of the bus lines.

Table 5.37 RIIC Timing (2)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 High-drive output is selected by the driving ability control register.

	Item	Symbol	Min.*1, *2	Max.	Unit	Test Conditions
RIIC (Fast-mode+) ICFER.FMPE = 1	SCL input cycle time	t_{SCL}	$6(12) \times t_{IICcyc} + 240$	—	ns	Figure 5.54
	SCL input high pulse width	t_{SCLH}	$3(6) \times t_{IICcyc} + 120$	—	ns	
	SCL input low pulse width	t_{SCLL}	$3(6) \times t_{IICcyc} + 120$	—	ns	
	SCL, SDA input rise time	t_{Sr}	—	120	ns	
	SCL, SDA input fall time	t_{Sf}	—	120	ns	
	SCL, SDA input spike pulse removal time	t_{SP}	0	$1(4) \times t_{IICcyc}$	ns	
	SDA input bus free time	t_{BUF}	$3(6) \times t_{IICcyc} + 120$	—	ns	
	Start condition input hold time	t_{STAH}	$t_{IICcyc} + 120$	—	ns	
	Restart condition input setup time	t_{STAS}	120	—	ns	
	Stop condition input setup time	t_{STOS}	120	—	ns	
	Data input setup time	t_{SDAS}	$t_{IICcyc} + 20$	—	ns	
	Data input hold time	t_{SDAH}	0	—	ns	
	SCL, SDA capacitive load	C_b	—	550	pF	
Simple IIC (Standard-mode)	SDA input rise time	t_{Sr}	—	1000	ns	
	SDA input fall time	t_{Sf}	—	300	ns	
	SDA input spike pulse removal time	t_{SP}	0	$4 \times t_{PBcyc}$	ns	
	Data input setup time	t_{SDAS}	250	—	ns	
	Data input hold time	t_{SDAH}	0	—	ns	
	SCL, SDA capacitive load	C_b	—	400	pF	
Simple IIC (Fast-mode)	SCL, SDA input rise time	t_{Sr}	—	300	ns	
	SCL, SDA input fall time	t_{Sf}	—	300	ns	
	SCL, SDA input spike pulse removal time	t_{SP}	0	$4 \times t_{PBcyc}$	ns	
	Data input setup time	t_{SDAS}	100	—	ns	
	Data input hold time	t_{SDAH}	0	—	ns	
	SCL, SDA capacitive load	C_b	—	400	pF	

Note: t_{IICcyc} : RIIC internal reference clock (IIC ϕ) cycle, t_{PBcyc} : PCLKB cycle

Note 1. The value within parentheses is applicable when the value of the ICMR3.NF[1:0] bits is 11b while the digital filter is enabled by the setting ICFER.NFE = 1.

Note 2. C_b is the total capacitance of the bus lines.

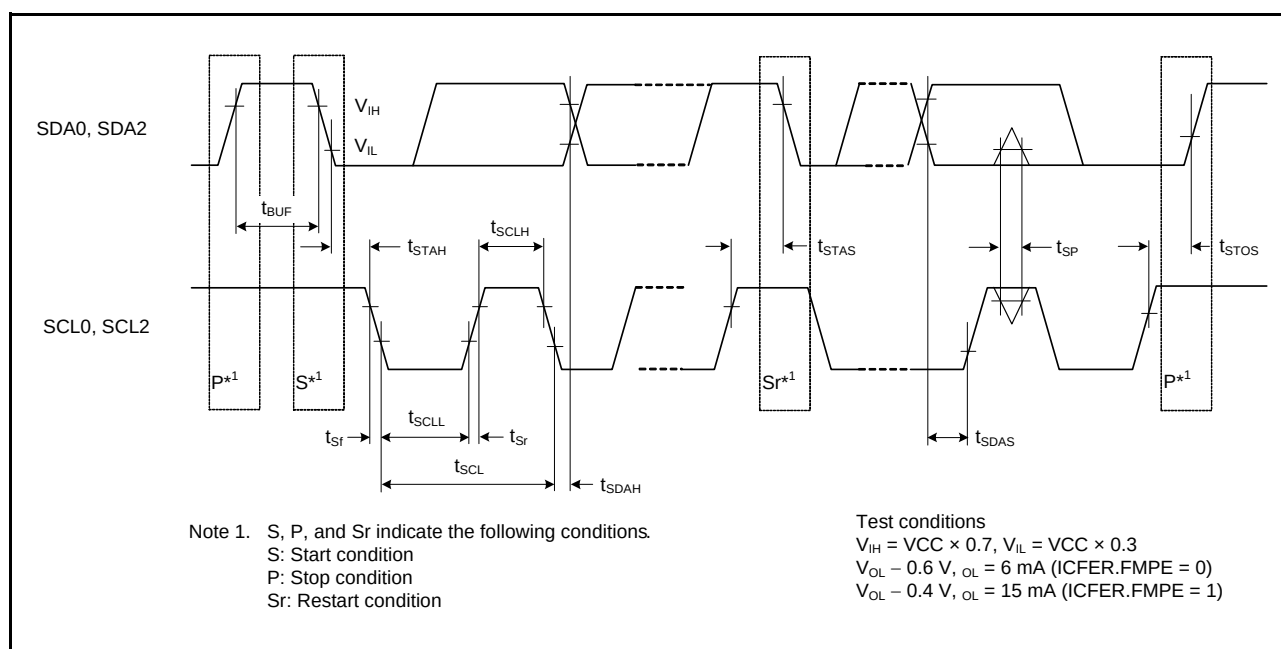


Figure 5.54 RIIC Bus Interface Input/Output Timing and Simple IIC Bus Interface Input/Output Timing

Table 5.38 MMC Host Interface Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

	Item	Symbol	Min.*1	Max.	Unit	Test Conditions*2
MMCIF	MMC_CLK clock cycle	t_{MMCPP}	$2 \times t_{PBcyc}$	—	ns	Figure 5.55
	MMC_CLK clock high level width	t_{MMCWH}	6.5	—	ns	
	MMC_CLK clock low level width	t_{MMCWL}	6.5	—	ns	
	MMC_CLK clock rising time	t_{MMCLH}	—	3	ns	
	MMC_CLK clock falling time	t_{MMCHL}	—	3	ns	
	MMC_CMD, MMC_D7 to MMC_D0 output data delay (data transfer mode)	$t_{MMCODLY}$	-6.6	6.6	ns	
	MMC_CMD, MMC_D7 to MMC_D0 input data setup	t_{MMCISU}	8	—	ns	
	MMC_CMD, MMC_D7 to MMC_D0 input data hold	t_{MMCIH}	2.5	—	ns	

Note 1. t_{PBcyc} : PCLKB cycle

Note 2. We recommend using pins that have a letter ("A", "B", etc.) to indicate group membership appended to their names as groups.
For the MMC interface, the AC portion of the electrical characteristics is measured for each group.

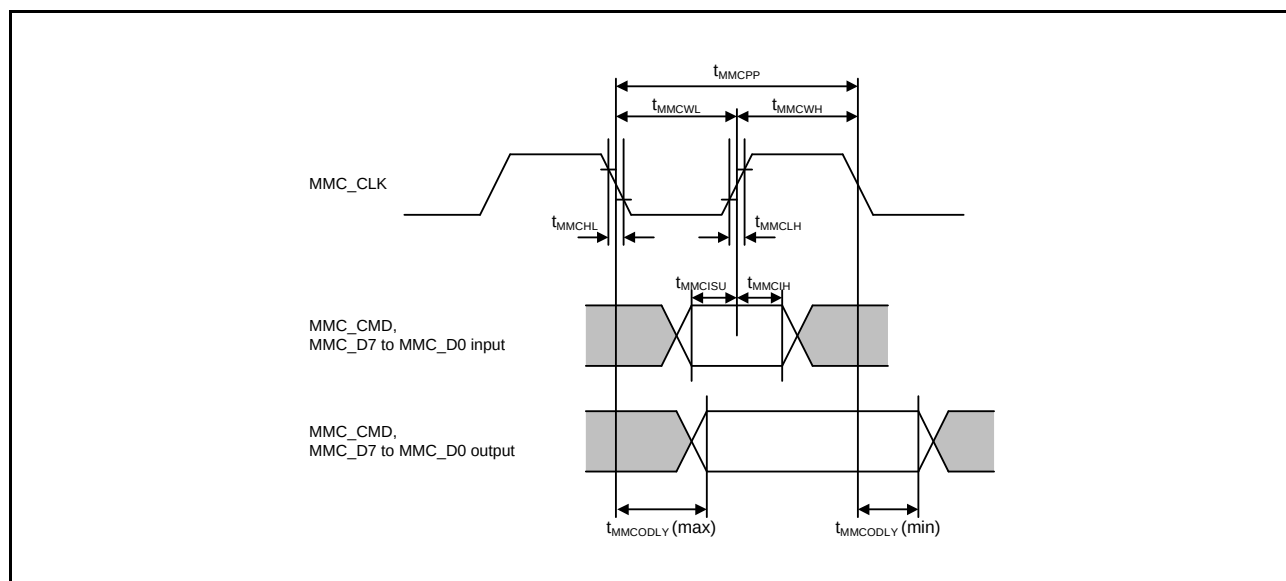
**Figure 5.55 MMC Interface**

Table 5.39 ETHERC Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
High-drive output is selected by the driving ability control register.

	Item	Symbol	Min.	Max.	Unit	Test Conditions
ETHERC (RMII)	REF50CK cycle time	T_{ck}	20	—	ns	Figure 5.56 to Figure 5.58
	REF50CK frequency Typ. 50 MHz	—	—	50 + 100 ppm	MHz	
	REF50CK duty	—	35	65	%	
	REF50CK rise/fall time	$T_{ckr/ckf}$	0.5	3.5	ns	
	RMII0_xxx*1 output delay time	T_{co}	2.5	15.0	ns	
	RMII0_xxx*2 setup time	T_{su}	3	—	ns	
	RMII0_xxx*2 hold time	T_{hd}	1	—	ns	
	RMII0_xxx*1, *2 rise/fall time	T_r/T_f	0.5	5	ns	
	ET0_WOL output delay time	t_{WOLd}	1	23.5	ns	Figure 5.60
ETHERC (MII)	ET0_TX_CLK cycle time	t_{Tcyc}	40	—	ns	—
	ET0_TX_EN output delay time	t_{TEND}	1	20	ns	Figure 5.61
	ET0_ETXD0 to ET0_ETXD3 output delay time	t_{MTDd}	1	20	ns	
	ET0_CRS setup time	t_{CRSs}	10	—	ns	
	ET0_CRS hold time	t_{CRSh}	10	—	ns	
	ET0_COL setup time	t_{COLs}	10	—	ns	Figure 5.62
	ET0_COL hold time	t_{COLh}	10	—	ns	
	ET0_RX_CLK cycle time	t_{TRcyc}	40	—	ns	—
	ET0_RX_DV setup time	t_{RDVs}	10	—	ns	Figure 5.63
	ET0_RX_DV hold time	t_{RDVh}	10	—	ns	
	ET0_ERXD0 to ET0_ERXD3 setup time	t_{MRDs}	10	—	ns	
	ET0_ERXD0 to ET0_ERXD3 hold time	t_{MRDh}	10	—	ns	
	ET0_RX_ER setup time	t_{RERs}	10	—	ns	Figure 5.64
	ET0_RX_ER hold time	t_{RERh}	10	—	ns	
	ET0_WOL output delay time	t_{WOLd}	1	23.5	ns	Figure 5.65

Note 1. RMII0_TXD_EN, RMII0_TXD1, RMII0_TXD0

Note 2. RMII0_CRS_DV, RMII0_RXD1, RMII0_RXD0, RMII0_RX_ER

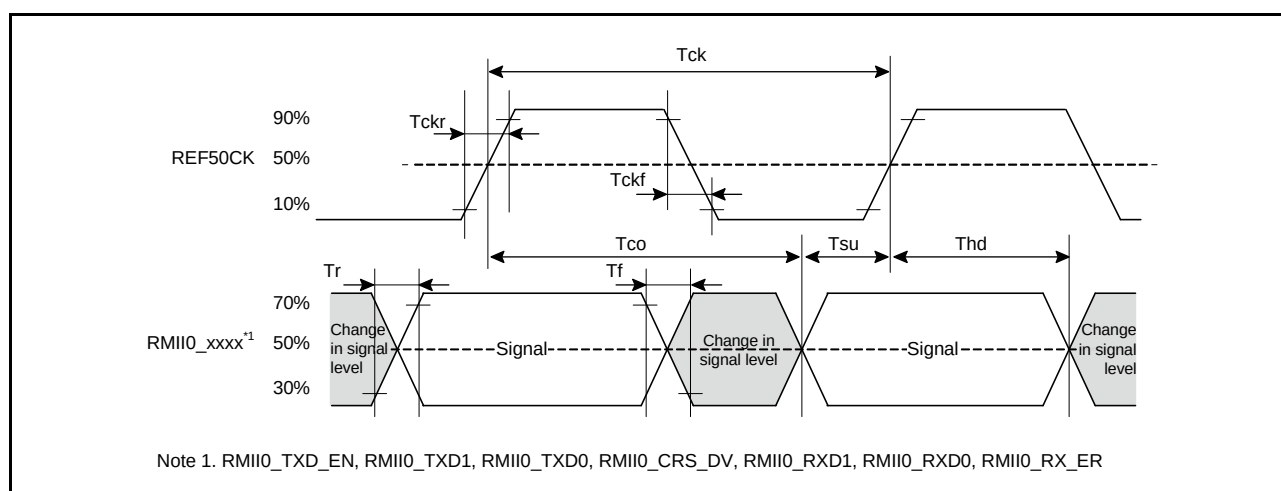


Figure 5.56 Timing with the REF50CK and RMII Signals

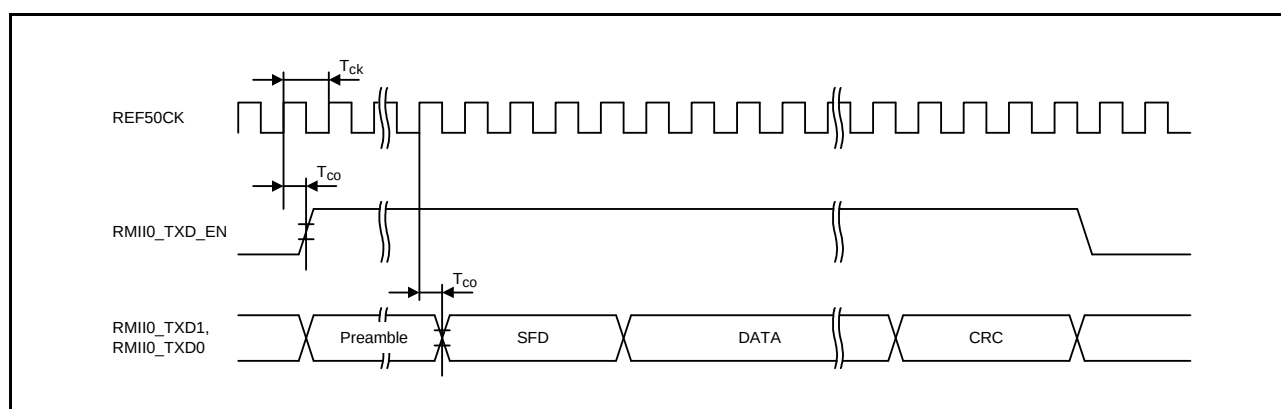


Figure 5.57 RMII Transmission Timing

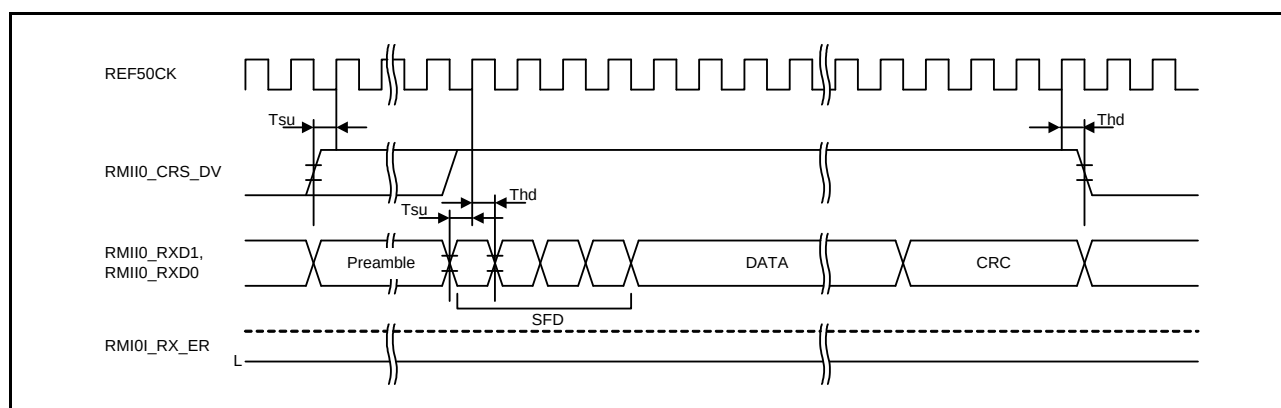


Figure 5.58 RMII Reception Timing (Normal Operation)

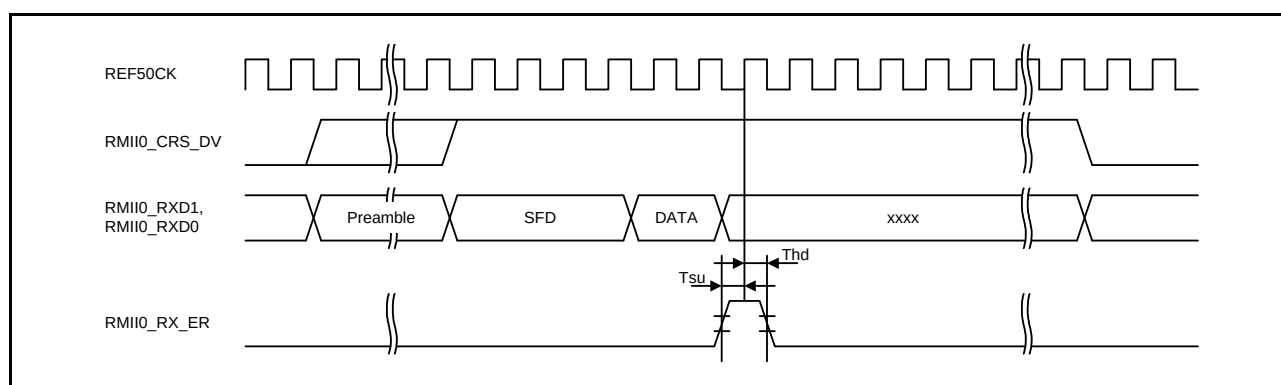


Figure 5.59 RMII Reception Timing (Error Occurrence)

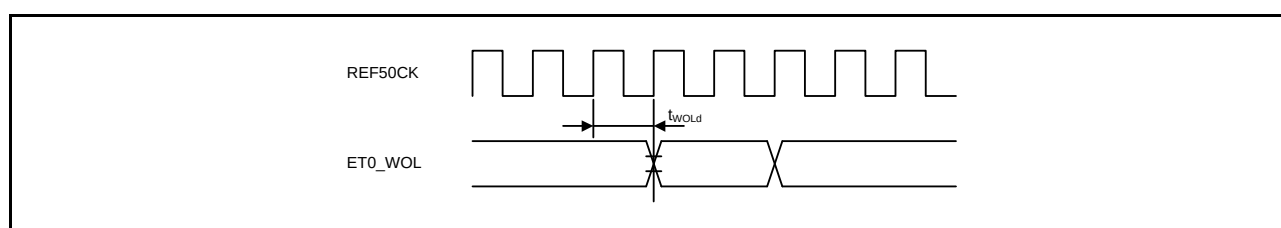


Figure 5.60 WOL Output Timing (RMII)

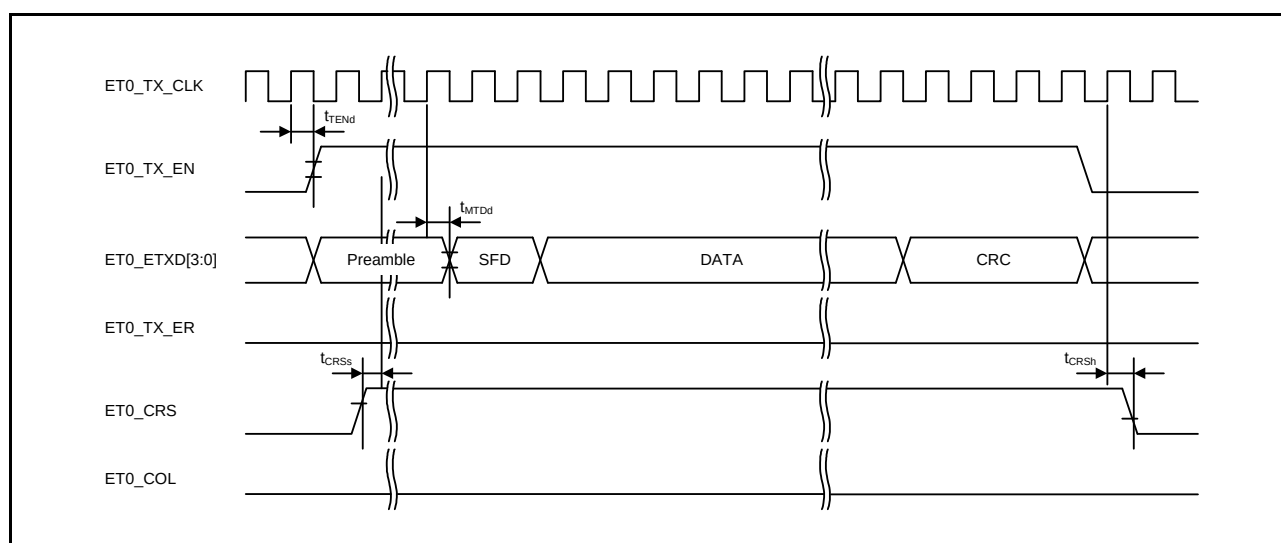


Figure 5.61 MII Transmission Timing (Normal Operation)

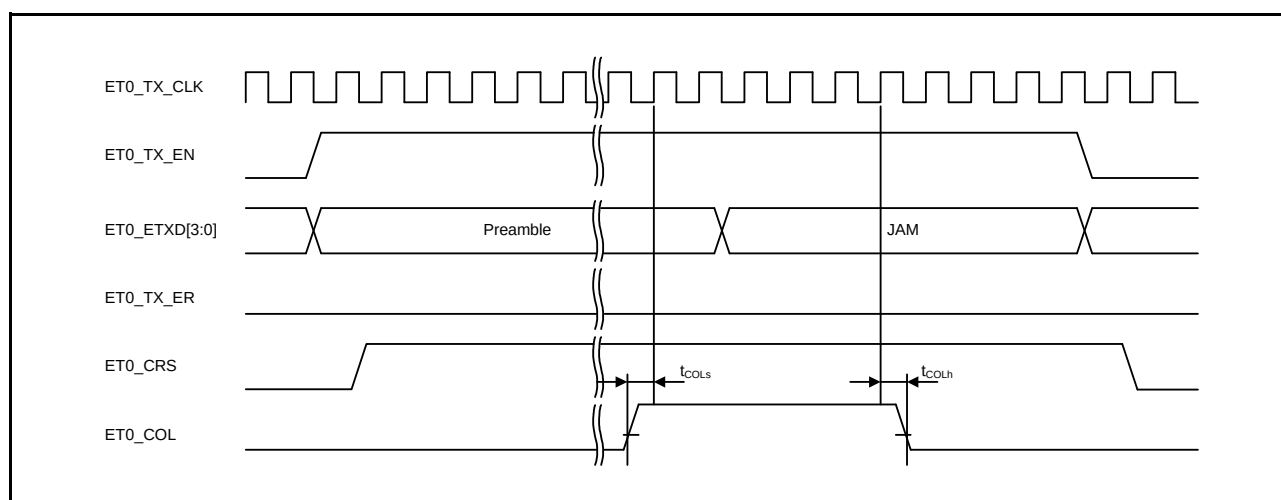


Figure 5.62 MII Transmission Timing (Conflict Occurrence)

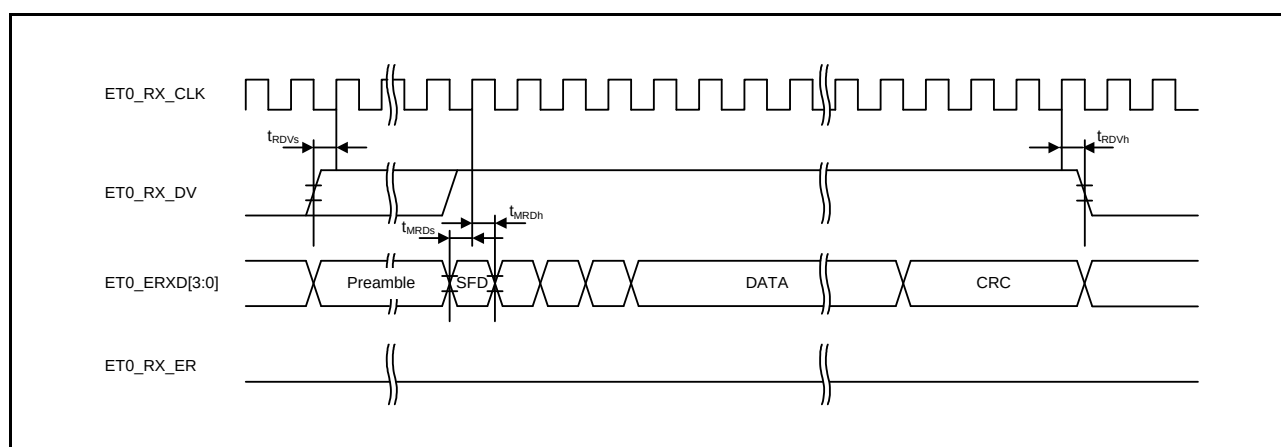


Figure 5.63 MII Reception Timing (Normal Operation)

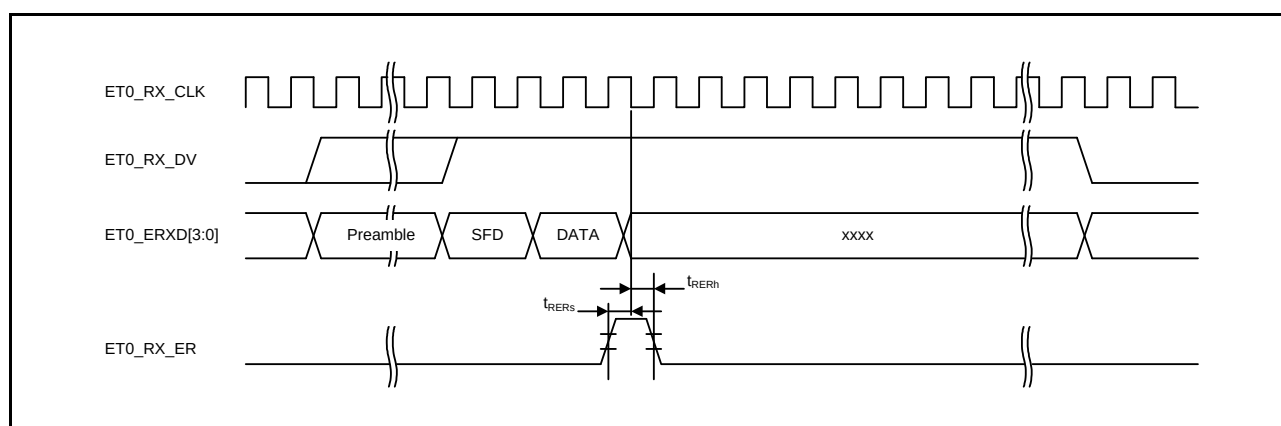


Figure 5.64 MII Reception Timing (Error Occurrence)

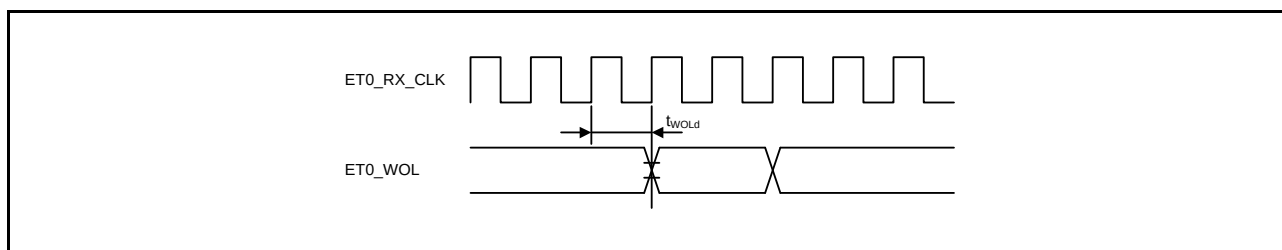
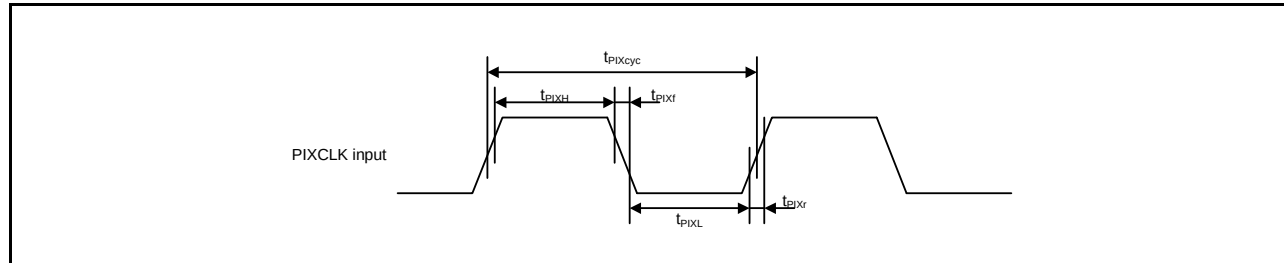
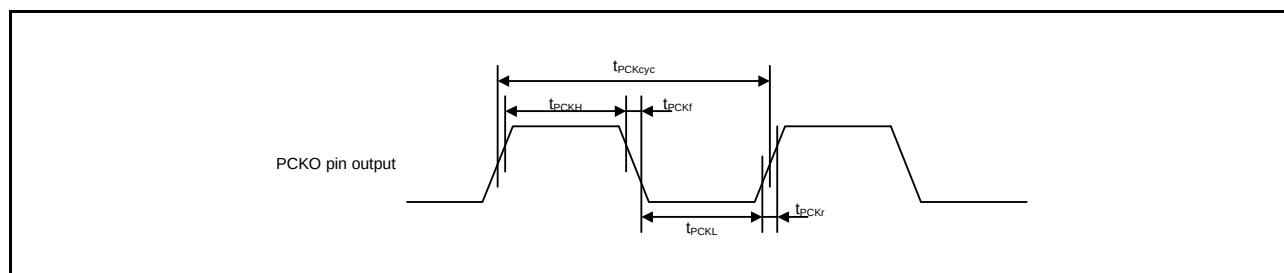
**Figure 5.65 WOL Output Timing (MII)**

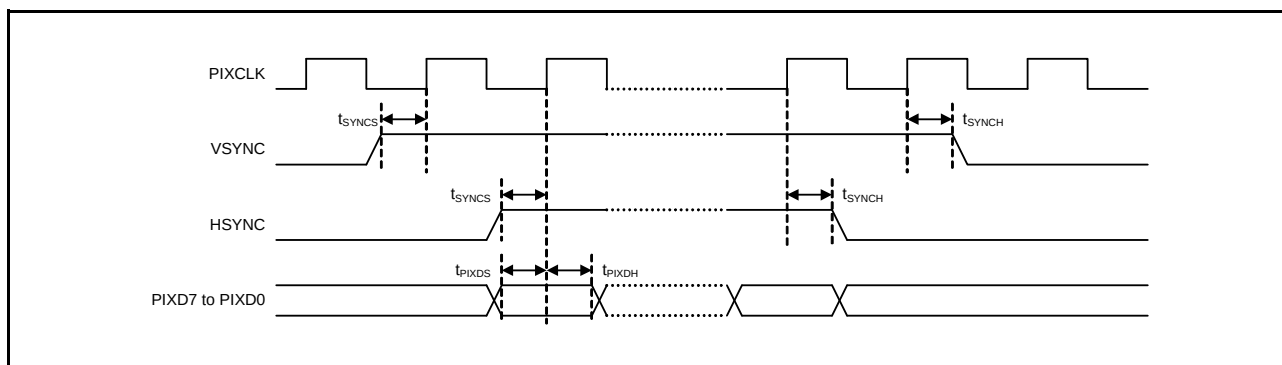
Table 5.40 PDC Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,
 High-drive output is selected by the driving ability control register.

	Item	Symbol	Min.*1	Max.	Unit	Test Conditions
PDC	PIXCLK input cycle time	t_{PIXcyc}	37	—	ns	Figure 5.66
	PIXCLK input high pulse width	t_{PIXH}	10	—	ns	
	PIXCLK input low pulse width	t_{PIXL}	10	—	ns	
	PIXCLK rising time	t_{PIXr}	—	5	ns	
	PIXCLK falling time	t_{PIXf}	—	5	ns	
	PCKO output cycle time	t_{PCKcyc}	$2 \times t_{PBcyc}$	—	ns	Figure 5.67
	PCKO output high pulse width	t_{PCKH}	$(t_{PCKcyc} - t_{PCKr} - t_{PCKf})/2 - 3$	—	ns	
	PCKO output low pulse width	t_{PCKL}	$(t_{PCKcyc} - t_{PCKr} - t_{PCKf})/2 - 3$	—	ns	
	PCKO rising time	t_{PCKr}	—	5	ns	
	PCKO falling time	t_{PCKf}	—	5	ns	
	VSYNC/HSYNC input setup time	t_{SYNCS}	10	—	ns	Figure 5.68
	VSYNC/HSYNC input hold time	t_{SYNCH}	5	—	ns	
	PIXD input setup time	t_{PIXDS}	10	—	ns	
	PIXD input hold time	t_{PIXDH}	5	—	ns	

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 5.66 PDC Input Clock Timing****Figure 5.67 PDC Output Clock Timing**

**Figure 5.68 PDC AC Timing**

5.4 USB Characteristics

Table 5.41 On-Chip USB Low Speed (Host Only) Characteristics (DP and DM Pin Characteristics)

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 3.0$ to 3.6 V, $3.0 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $U_{CLK} = 48$ MHz,
 $P_{CLKA} = 8$ to 120 MHz, $P_{CLKB} = 8$ to 60 MHz, $T_a = T_{opr}$

	Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Input characteristics	Input high level voltage	V_{IH}	2.0	—	—	V	
	Input low level voltage	V_{IL}	—	—	0.8	V	
	Differential input sensitivity	V_{DI}	0.2	—	—	V	DP – DM
	Differential common mode range	V_{CM}	0.8	—	2.5	V	
Output characteristics	Output high level voltage	V_{OH}	2.8	—	3.6	V	$I_{OH} = -200 \mu A$
	Output low level voltage	V_{OL}	0.0	—	0.3	V	$I_{OL} = 2$ mA
	Cross-over voltage	V_{CRS}	1.3	—	2.0	V	Figure 5.69
	Rise time	t_{LR}	75	—	300	ns	
	Fall time	t_{LF}	75	—	300	ns	
	Rise/fall time ratio	t_{LR} / t_{LF}	80	—	125	%	t_{LR} / t_{LF}
Pull-down characteristics	DP/DM pull-down resistance (when the host controller function is selected)	R_{pd}	14.25	—	24.80	k Ω	

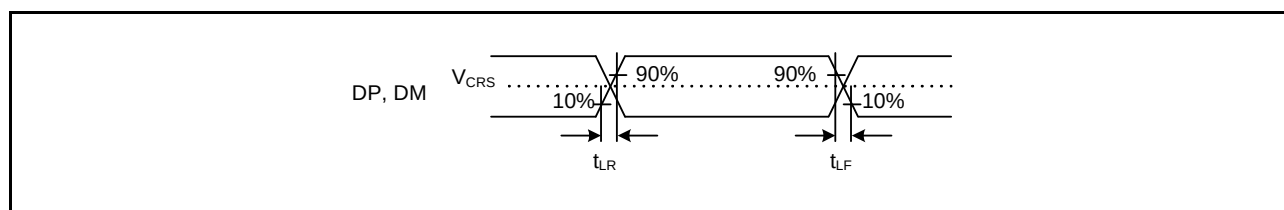
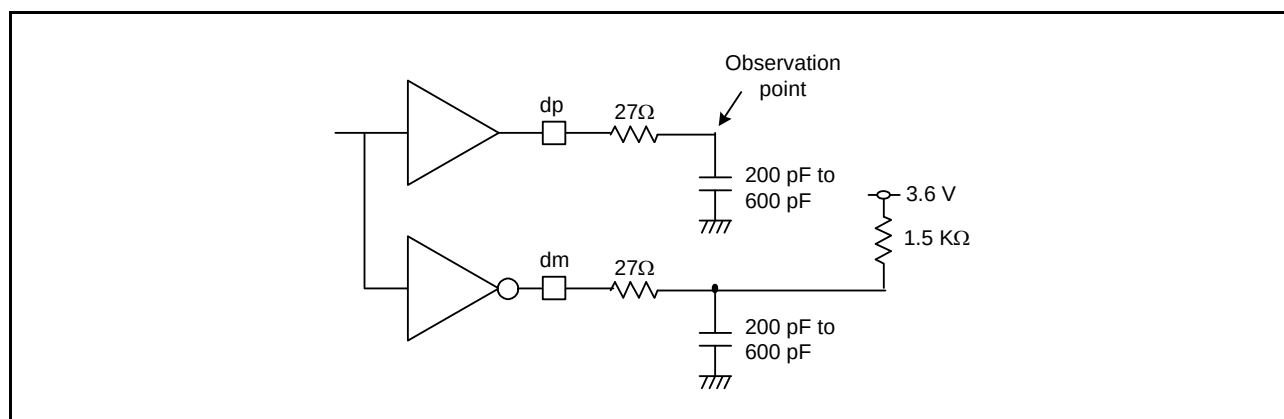
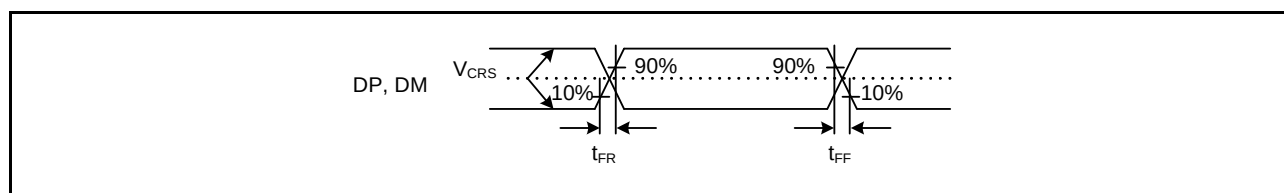
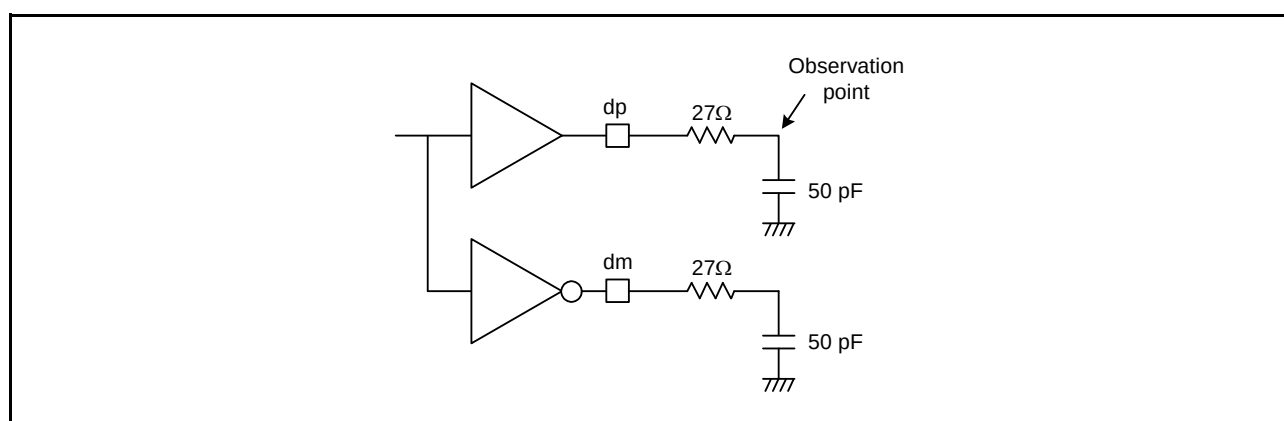
**Figure 5.69 DP and DM Output Timing (Low Speed)****Figure 5.70 Test Circuit (Low Speed)**

Table 5.42 On-Chip USB Full-Speed Characteristics (DP and DM Pin Characteristics)

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 3.0$ to 3.6 V, $3.0 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $U_{CLK} = 48$ MHz,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Input characteristics	Input high level voltage	V_{IH}	2.0	—	—	V
	Input low level voltage	V_{IL}	—	—	0.8	V
	Differential input sensitivity	V_{DI}	0.2	—	—	V DP – DM
	Differential common mode range	V_{CM}	0.8	—	2.5	V
Output characteristics	Output high level voltage	V_{OH}	2.8	—	3.6	V $I_{OH} = -200 \mu A$
	Output low level voltage	V_{OL}	0.0	—	0.3	V $I_{OL} = 2$ mA
	Cross-over voltage	V_{CRS}	1.3	—	2.0	V Figure 5.71
	Rise time	t_{FR}	4	—	20	ns
	Fall time	t_{FF}	4	—	20	ns
	Rise/fall time ratio	t_{FR} / t_{FF}	90	—	111.11	% t_{FR} / t_{FF}
	Output resistance	Z_{DRV}	28	—	44	Ω $R_s = 27 \Omega$ included
Pull-up and pull-down characteristics	DP pull-up resistance (when the function controller function is selected)	R_{pu}	0.900	—	1.575	k Ω Idle state
			1.425	—	3.090	k Ω At transmission and reception
	DP/DM pull-down resistance (when the host controller function is selected)	R_{pd}	14.25	—	24.80	k Ω

**Figure 5.71 DP and DM Output Timing (Full-Speed)****Figure 5.72 Test Circuit (Full-Speed)**

5.5 A/D Conversion Characteristics

Table 5.43 12-Bit A/D (Unit 0) Conversion Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKB = PCLKC = 1$ MHz to 60 MHz, $T_a = T_{opr}$

Item		Min.	Typ.	Max.	Unit	Test Conditions
Resolution		8	—	12	Bit	
Analog input capacitance		—	—	30	pF	
Channel-dedicated sample-and-hold circuits in use (AN000 to AN002)	Conversion time*1 (Operation at PCLK = 60 MHz) Permissible signal source impedance (max.) = 1.0 kΩ	1.06 (0.40 + 0.25) *2	—	—	μs	- Sampling of channel-dedicated sample-and-hold circuits in 24 states - Sampling in 15 states
	Offset error	—	±1.5	±3.5	LSB	AN000 to AN002 = 0.25 V
	Full-scale error	—	±1.5	±3.5	LSB	AN000 to AN002 = VREFH0 - 0.25 V
	Quantization error	—	±0.5	—	LSB	
	Absolute accuracy	—	±3.0	±5.5	LSB	
	DNL differential nonlinearity error	—	±1.0	±2.0	LSB	
	INL integral nonlinearity error	—	±1.5	±3.0	LSB	
	Holding characteristics of sample-and-hold circuits	—	—	20	μs	
	Dynamic range	0.25	—	VREFH0 - 0.25	V	
Channel-dedicated sample-and-hold circuits not in use (AN000 to AN007)	Conversion time*1 (Operation at PCLK = 60 MHz) Permissible signal source impedance (max.) = 1.0 kΩ	0.48 (0.267)*2	—	—	μs	Sampling in 16 states
	Offset error	—	±1.0	±2.5	LSB	
	Full-scale error	—	±1.0	±2.5	LSB	
	Quantization error	—	±0.5	—	LSB	
	Absolute accuracy	—	±2.5	±4.5	LSB	
	DNL differential nonlinearity error	—	±0.5	±1.5	LSB	
	INL integral nonlinearity error	—	±1.0	±2.5	LSB	

Note: The above specification values apply when there is no access to the external bus during A/D conversion. If access proceeds during A/D conversion, values may not fall within the above ranges.

Note 1. The conversion time includes the sampling time and the comparison time. As the test conditions, the number of sampling states is indicated.

Note 2. The value in parentheses indicates the sampling time.

Table 5.44 12-Bit A/D (Unit 1) Conversion Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKB = PCLKD = 1$ MHz to 60 MHz, $T_a = T_{opr}$

Item		Min.	Typ.	Max.	Unit	Test Conditions
Resolution		8	—	12	Bit	
Conversion time*1 (Operation at PCLK = 60 MHz)	Permissible signal source impedance (max.) = 1.0 kΩ	0.88 (0.633)*2	—	—	μs	Sampling in 38 states (ADSAM.SAM = 1)
Conversion time*1 (Operation at PCLK = 30 MHz)		1 (0.500)*2	—	—	μs	Sampling in 15 states (ADSAM.SAM = 1)
Analog input capacitance		—	—	30	pF	
Offset error		—	±2.0	±3.5	LSB	
Full-scale error		—	±2.0	±3.5	LSB	
Quantization error		—	±0.5	—	LSB	
Absolute accuracy		—	±4.0	±6.0	LSB	
DNL differential nonlinearity error (Operation at PCLK = 60 MHz)		—	±1.5	±4.0	LSB	
DNL differential nonlinearity error (Operation at PCLK = 30 MHz)		—	±1.5	±2.5	LSB	
INL integral nonlinearity error (Operation at PCLK = 60 MHz)		—	±2.0	±4.0	LSB	
INL integral nonlinearity error (Operation at PCLK = 30 MHz)		—	±2.0	±3.5	LSB	

Note: The above specification values apply when there is no access to the external bus during A/D conversion. If access proceeds during A/D conversion, values may not fall within the above ranges.

Note 1. The conversion time includes the sampling time and the comparison time. As the test conditions, the number of sampling states is indicated.

Note 2. The value in parentheses indicates the sampling time.

Table 5.45 A/D Internal Reference Voltage Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKB = PCLKD = 60$ MHz, $T_a = T_{opr}$

Item	Min.	Typ.	Max.	Unit	Test Conditions
A/D internal reference voltage	1.13	1.18	1.23	V	

5.6 D/A Conversion Characteristics

Table 5.46 D/A Conversion Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V,
 $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Min.	Typ.	Max.	Unit	Test Conditions
Resolution	12	12	12	Bit	
Absolute accuracy	—	—	± 6.0	LSB	2-M Ω resistive load 10-bit conversion
DNL differential nonlinearity error	—	± 1.0	± 2.0	LSB	2-M Ω resistive load
RO output resistance	—	8.6	—	k Ω	
Conversion time	—	—	3	μ s	20-pF capacitive load

5.7 Temperature Sensor Characteristics

Table 5.47 Temperature Sensor Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Min.	Typ.	Max.	Unit	Test Conditions
Relative accuracy	—	± 1	—	$^{\circ}$ C	
Temperature slope	—	4	—	mV/ $^{\circ}$ C	
Output voltage (at 25 $^{\circ}$ C)	—	1.21	—	V	
Temperature sensor start time	—	—	30	μ s	
Sampling time*1	4.15	—	—	μ s	

Note 1. Set the S12AD1.ADSSTRT register such that the sampling time of the 12-bit A/D converter satisfies this specification.

5.8 Power-on Reset Circuit and Voltage Detection Circuit Characteristics

Table 5.48 Power-on Reset Circuit and Voltage Detection Circuit Characteristics

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Voltage detection level	Power-on reset (POR)	Low power consumption function disabled*1	V _{POR}	2.5	2.6	2.7	V	Figure 5.73
		Low power consumption function enabled*2		1.8	2.25	2.7		
	Voltage detection circuit (LVD0)		V _{det0_1}	2.84	2.94	3.04		Figure 5.74
			V _{det0_2}	2.77	2.87	2.97		
			V _{det0_3}	2.70	2.80	2.90		
	Voltage detection circuit (LVD1)		V _{det1_1}	2.89	2.99	3.09		Figure 5.75
			V _{det1_2}	2.82	2.92	3.02		
			V _{det1_3}	2.75	2.85	2.95		
	Voltage detection circuit (LVD2)		V _{det2_1}	2.89	2.99	3.09		Figure 5.76
			V _{det2_2}	2.82	2.92	3.02		
			V _{det2_3}	2.75	2.85	2.95		
Internal reset time	Power-on reset time		t _{POR}	—	4.6	—	ms	Figure 5.73
	LVD0 reset time		t _{LVD0}	—	0.70	—		Figure 5.74
	LVD1 reset time		t _{LVD1}	—	0.57	—		Figure 5.75
	LVD2 reset time		t _{LVD2}	—	0.57	—		Figure 5.76
Minimum VCC down time			t _{VOFF}	200	—	—	μs	Figure 5.73, Figure 5.74
Response delay time			t _{det}	—	—	200	μs	Figure 5.73 to Figure 5.76
LVD operation stabilization time (after LVD is enabled)			T _{d(E-A)}	—	—	10	μs	Figure 5.75, Figure 5.76
Hysteresis width (LVD1 and LVD2)			V _{LVH}	—	70	—	mV	

Note: The minimum VCC down time indicates the time when VCC is below the minimum value of voltage detection levels V_{POR} , V_{det1} , and V_{det2} for the POR/ LVD.

Note 1. The low power consumption function is disabled and DEEPCUT[1:0] = 00b or 01b.

Note 2. The low power consumption function is enabled and DEEPCUT[1:0] = 11b.

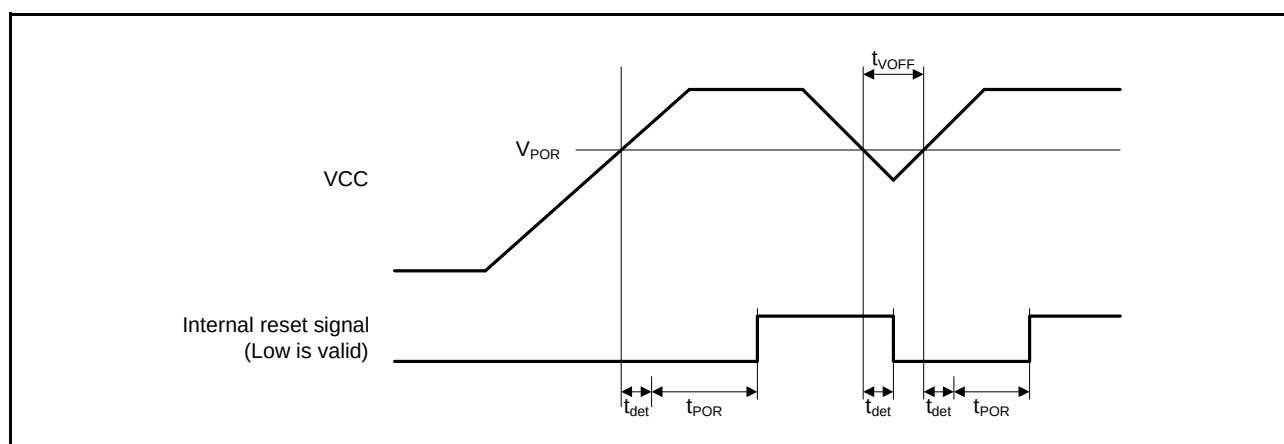
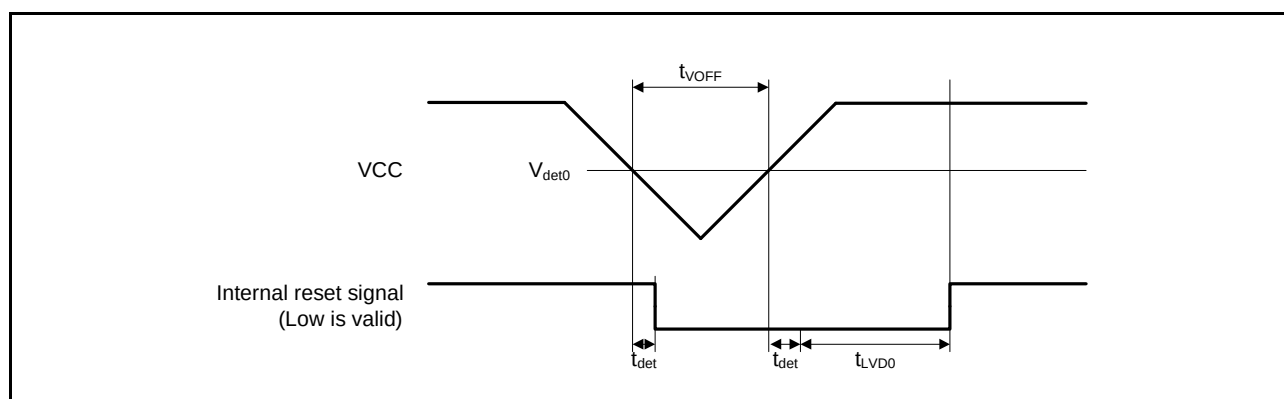
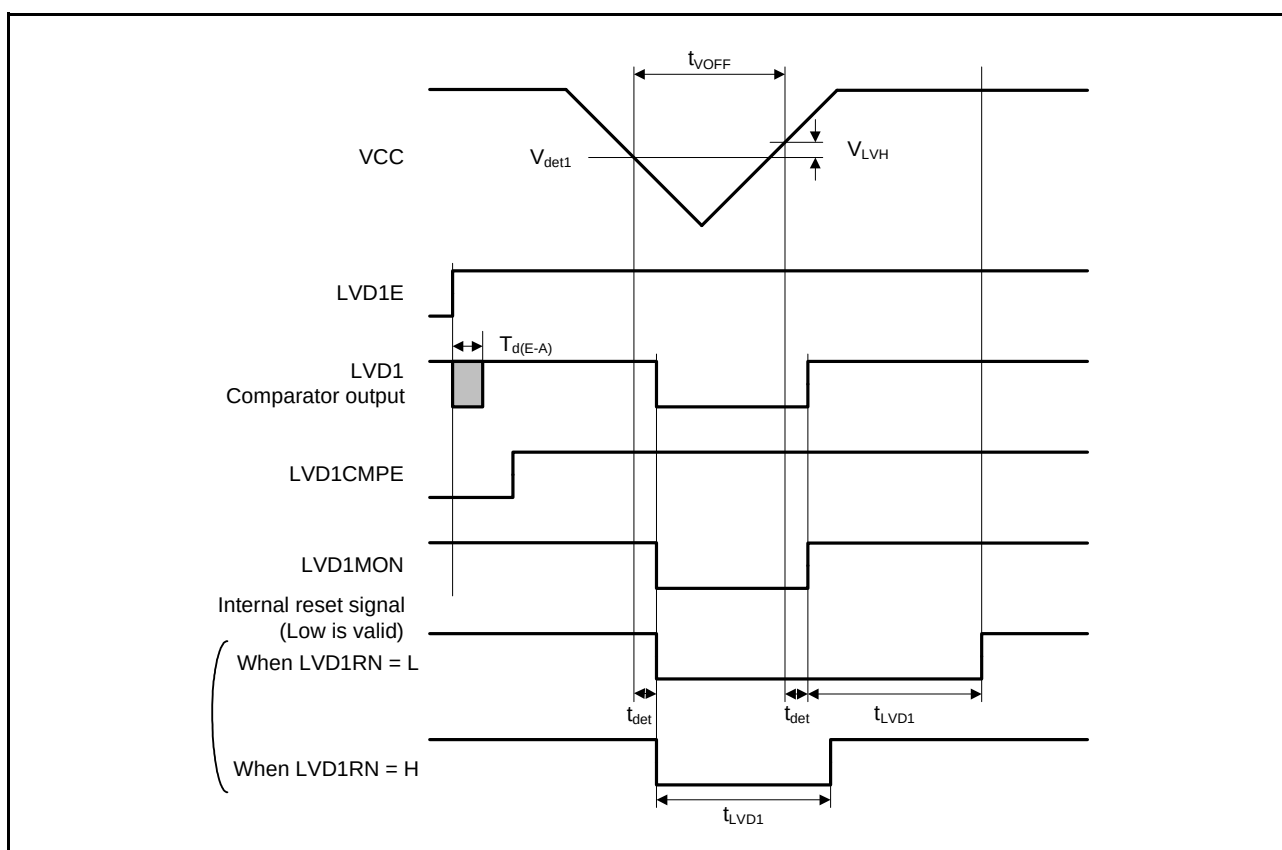
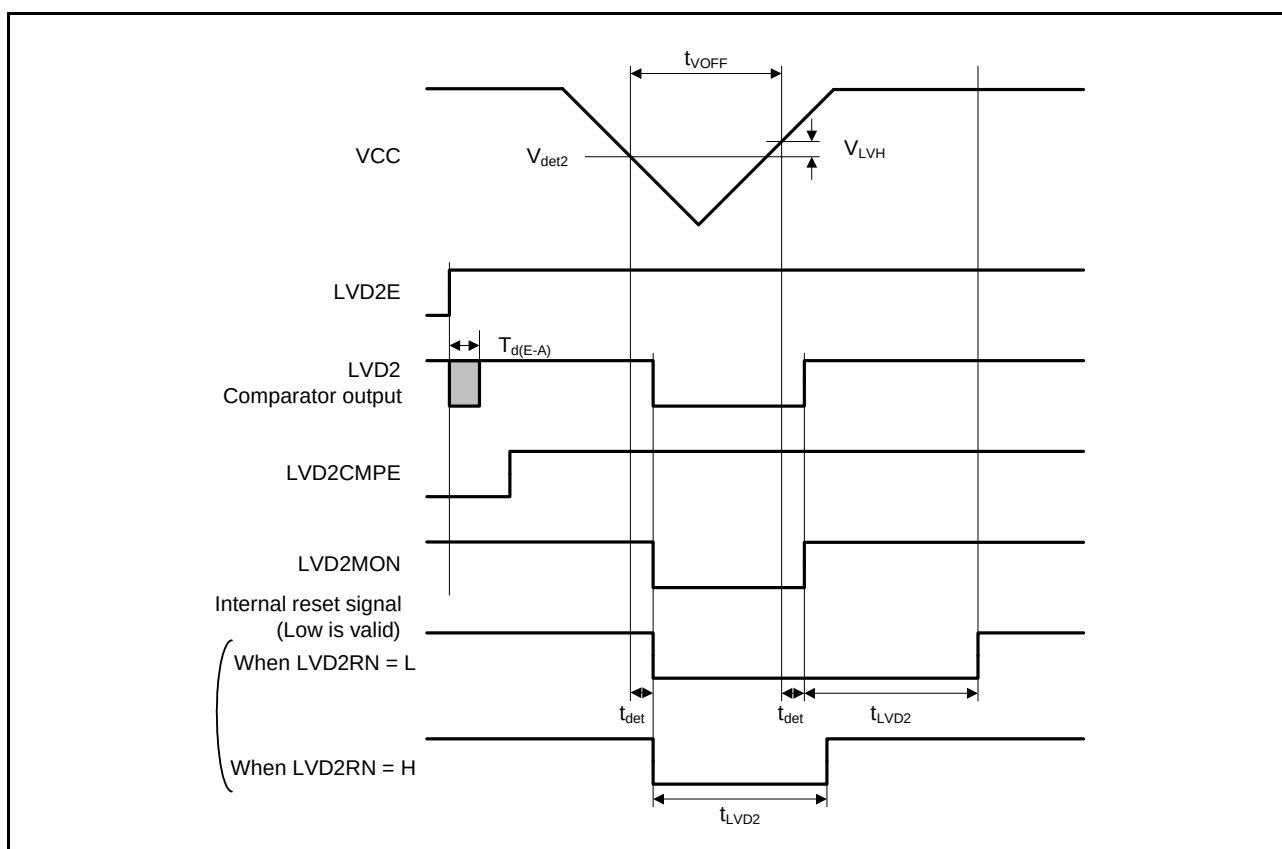


Figure 5.73 Power-on Reset Timing

Figure 5.74 Voltage Detection Circuit Timing (V_{det0})

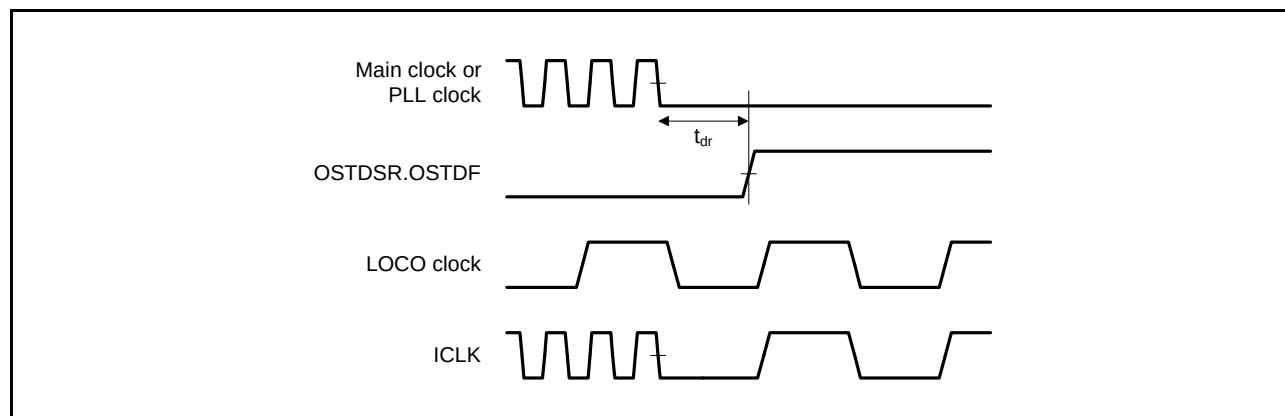
Figure 5.75 Voltage Detection Circuit Timing (V_{det1})Figure 5.76 Voltage Detection Circuit Timing (V_{det2})

5.9 Oscillation Stop Detection Timing

Table 5.49 Oscillation Stop Detection Circuit Characteristics

Conditions: $VCC = AVCC0 = AVCC1 = VCC_USB = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Detection time	t_{dr}	—	—	1	ms	Figure 5.77

**Figure 5.77 Oscillation Stop Detection Timing**

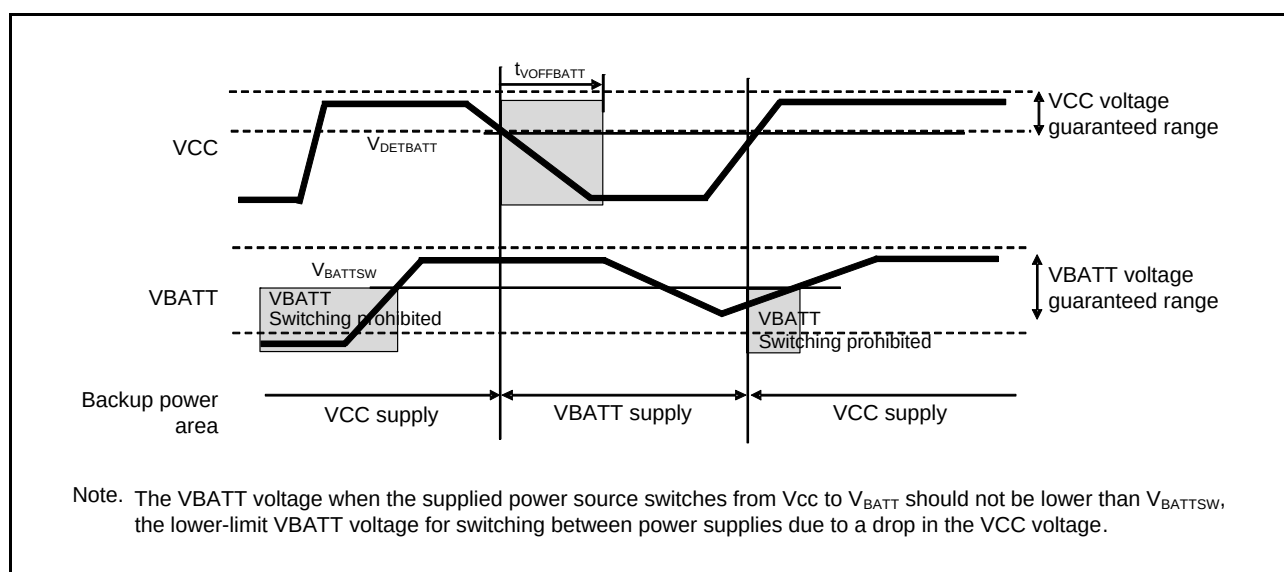
5.10 Battery Backup Function Characteristics

Table 5.50 Battery Backup Function Characteristics

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $V_{BATT} = 2.0$ to 3.6 V, $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Voltage level for switching to battery backup	$V_{DETBATT}$	2.50	2.60	2.70	V	Figure 5.78
Lower-limit V_{BATT} voltage for power supply switching due to VCC voltage drop	V_{BATTSW}	2.70	—	—		
VCC-off period for starting power supply switching	$t_{VOFFBATT}$	200	—	—	μs	

Note: The VCC-off period for starting power supply switching indicates the period in which VCC is below the minimum value of the voltage level for switching to battery backup ($V_{DETBATT}$).

**Figure 5.78 Battery Backup Function Characteristics**

5.11 Flash Memory Characteristics

Table 5.51 Code Flash Memory Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 Temperature range for programming/erasure: $T_a = T_{opr}$

Item		Symbol	FCLK = 4 MHz			20 MHz $\leq$ FCLK $\leq$ 60 MHz			Unit
			Min.	Typ.	Max.	Min.	Typ.	Max.	
Programming time $N_{PEC} \leq 100$ times	128 bytes	t_{P128}	—	1.1	13.2	—	0.52	6	ms
	8 Kbytes	t_{P8K}	—	75	176	—	34	80	ms
	32 Kbytes	t_{P32K}	—	299	704	—	136	320	ms
Programming time $N_{PEC} > 100$ times	128 bytes	t_{P128}	—	1.4	15.8	—	0.62	7.2	ms
	8 Kbytes	t_{P8K}	—	90	212	—	41	96	ms
	32 Kbytes	t_{P32K}	—	359	848	—	163	384	ms
Erasure time $N_{PEC} \leq 100$ times	8 Kbytes	t_{E8K}	—	92	216	—	51	120	ms
	32 Kbytes	t_{E32K}	—	329	864	—	183	480	ms
Erasure time $N_{PEC} > 100$ times	8 Kbytes	t_{E8K}	—	110	260	—	61	144	ms
	32 Kbytes	t_{E32K}	—	396	1040	—	220	576	ms
Reprogramming/erasure cycle*1		N_{PEC}	10000*2	—	—	10000*2	—	—	Times
Suspend delay time during programming		t_{SPD}	—	—	264	—	—	120	μ s
First suspend delay time during erasing (in suspend priority mode)		t_{SESD1}	—	—	216	—	—	120	μ s
Second suspend delay time during erasure (in suspend priority mode)		t_{SESD2}	—	—	1.7	—	—	1.7	ms
Suspend delay time during erasure (in erasure priority mode)		t_{SEED}	—	—	1.7	—	—	1.7	ms
Forced stop command		t_{FD}	—	—	32	—	—	20	μ s
Data hold time*3		t_{DRP}	10	—	—	10	—	—	Year

Note 1. Definition of reprogram/erase cycle:

The reprogram/erase cycle is the number of erasing for each block. When the reprogram/erase cycle is n times ($n = 1000$), erasing can be performed n times for each block. For instance, when 128-byte programming is performed 64 times for different addresses in 8-Kbyte block and then the entire block is erased, the reprogram/erase cycle is counted as one. However, programming the same address for several times as one erasing is not enabled (overwriting is prohibited).

Note 2. This is the minimum number of times to guarantee all the characteristics after reprogramming (guaranteed range is from 1 to the value of the minimum value).

Note 3. This shows the characteristics when reprogramming is performed within the specified range, including the minimum value.

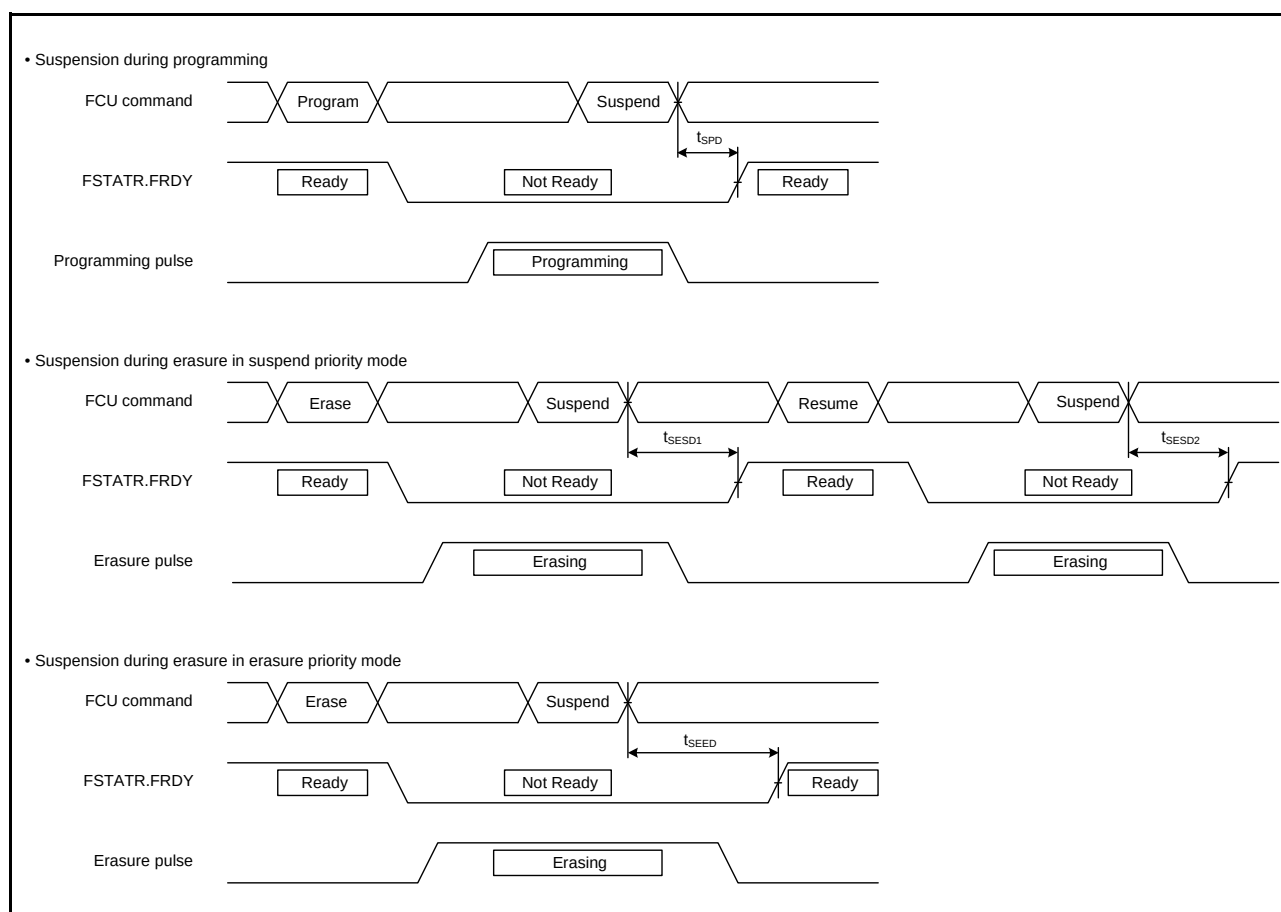


Figure 5.79 Flash Memory Programming/Erasure Suspension Timing

5.12 Boundary Scan

Table 5.52 Boundary Scan Characteristics

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $2.7 \leq V_{REFH0} \leq AV_{CC0}$,

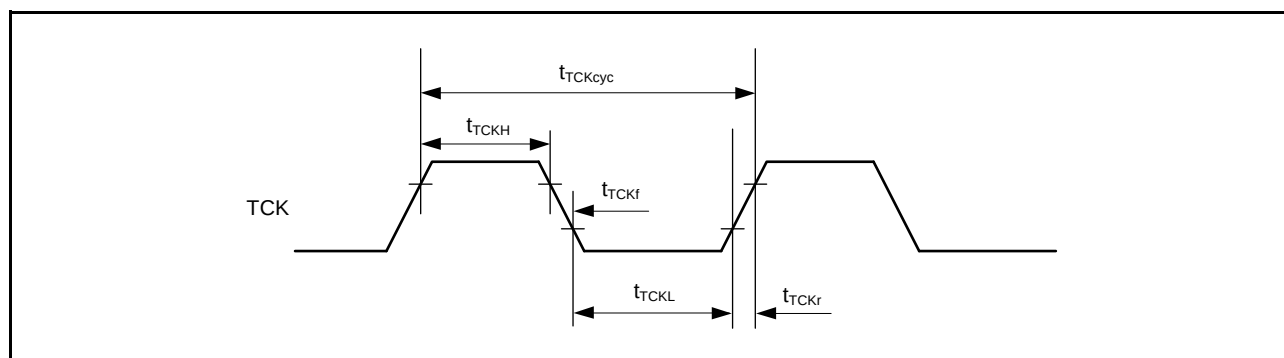
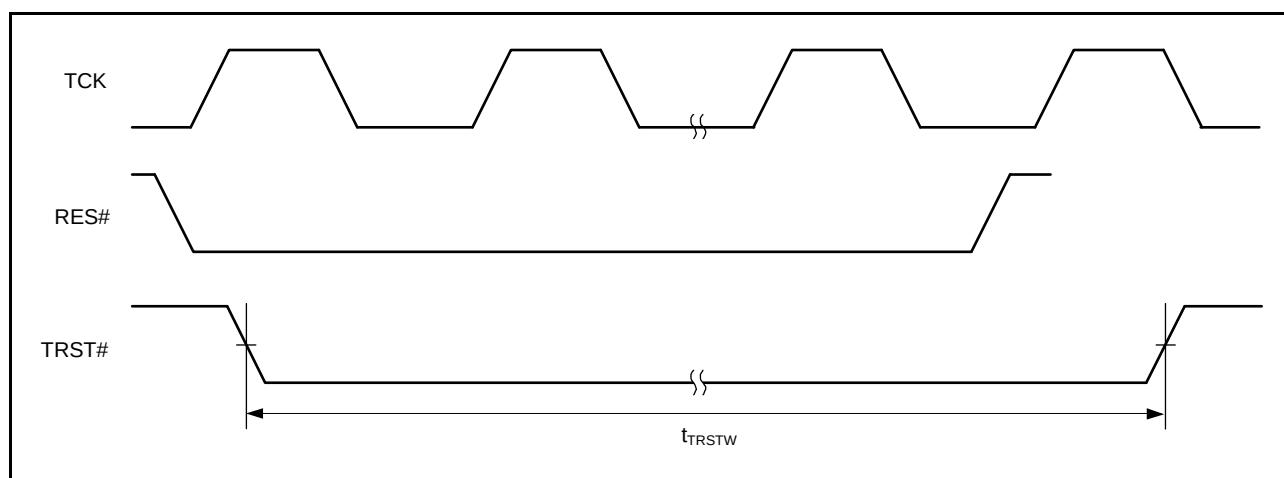
$V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,

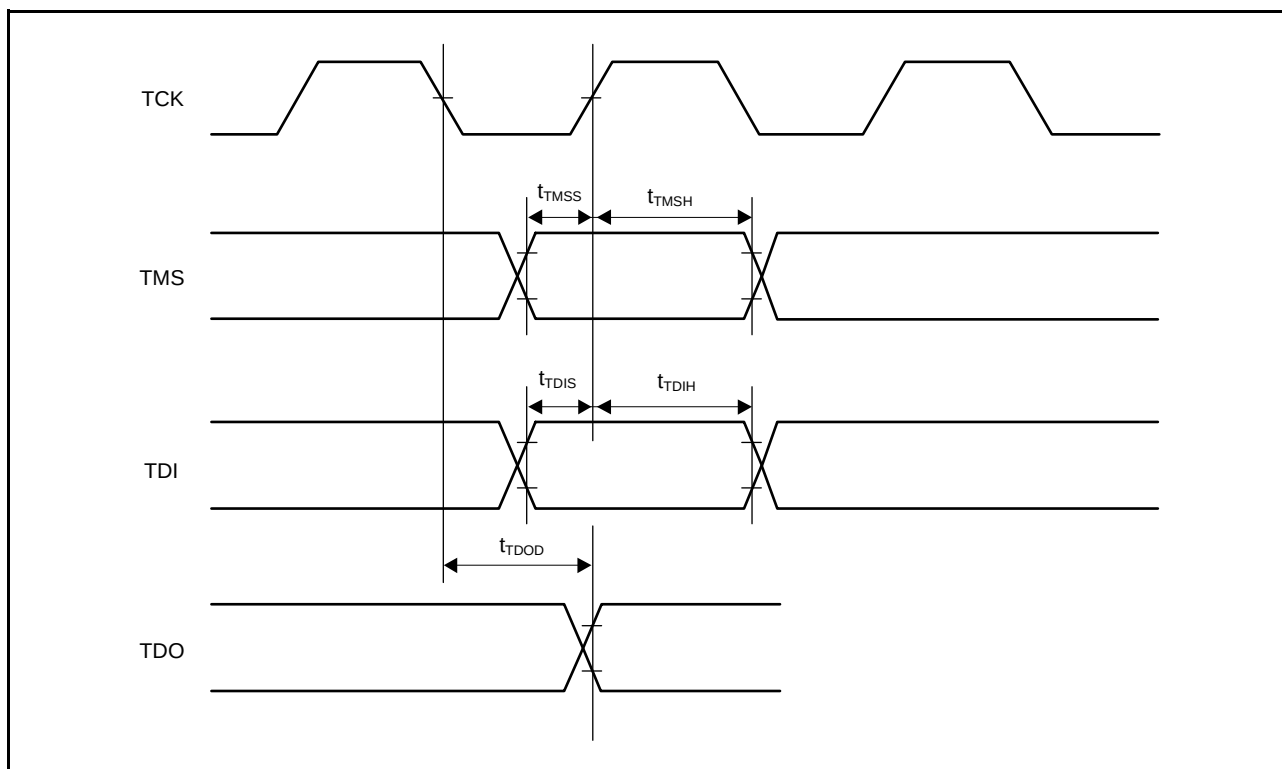
$T_a = T_{opr}$,

Output load conditions: $V_{OH} = V_{CC} \times 0.5$, $V_{OL} = V_{CC} \times 0.5$, $C = 30$ pF,

High-drive output is selected by the driving ability control register.

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
TCK clock cycle time	t_{TCKcyc}	100	—	—	ns	Figure 5.80
TCK clock high pulse width	t_{TCKH}	45	—	—	ns	
TCK clock low pulse width	t_{TCKL}	45	—	—	ns	
TCK clock rise time	t_{TCKr}	—	—	5	ns	
TCK clock fall time	t_{TCKf}	—	—	5	ns	
TRST# pulse width	t_{TRSTW}	20	—	—	t_{TCKcyc}	Figure 5.81
TMS setup time	t_{TMSS}	20	—	—	ns	Figure 5.82
TMS hold time	t_{TMSH}	20	—	—	ns	
TDI setup time	t_{TDIS}	20	—	—	ns	
TDI hold time	t_{TDIH}	20	—	—	ns	
TDO data delay time	t_{TDOD}	—	—	40	ns	

**Figure 5.80 Boundary Scan TCK Timing****Figure 5.81 Boundary Scan TRST# Timing**

**Figure 5.82** Boundary Scan Input/Output Timing

Appendix 1. Package Dimensions

Information on the latest version of the package dimensions or mountings has been displayed in “Packages” on Renesas Electronics Corporation website.

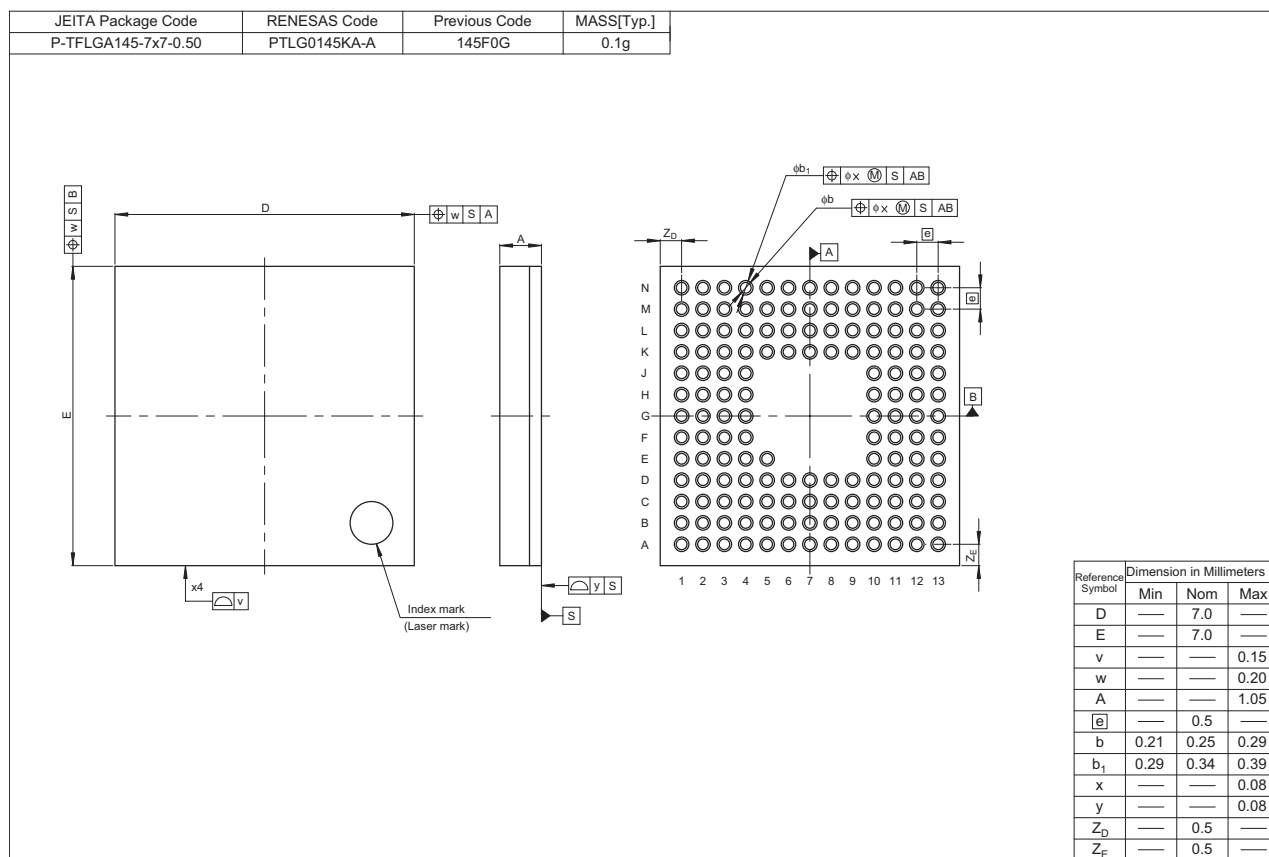
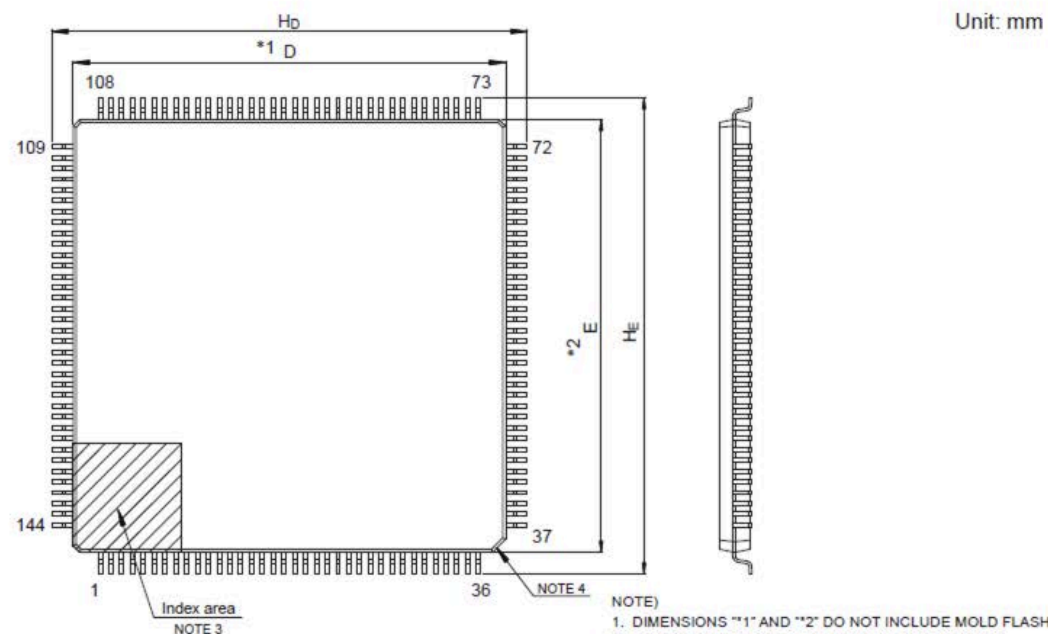
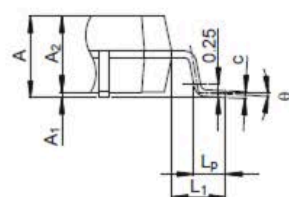
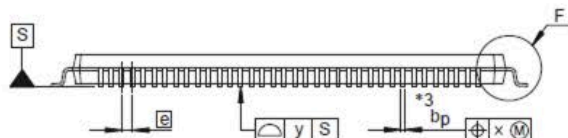


Figure A 145-Pin TFLGA (PTLG0145KA-A)

JEITA Package Code	RENESAS Code	Previous Code	MASS (Typ) [g]
P-LFQFP144-20x20-0.50	PLQP0144KA-B	—	1.2



- NOTE)
1. DIMENSIONS "1" AND "2" DO NOT INCLUDE MOLD FLASH.
 2. DIMENSION "3" DOES NOT INCLUDE TRIM OFFSET.
 3. PIN 1 VISUAL INDEX FEATURE MAY VARY, BUT MUST BE LOCATED WITHIN THE HATCHED AREA.
 4. CHAMFERS AT CORNERS ARE OPTIONAL, SIZE MAY VARY.



Detail F

Reference Symbol	Dimensions in millimeters		
	Min	Nom	Max
D	19.9	20.0	20.1
E	19.9	20.0	20.1
A ₂	—	1.4	—
H _D	21.8	22.0	22.2
H _E	21.8	22.0	22.2
A	—	—	1.7
A ₁	0.05	—	0.15
b _p	0.17	0.20	0.27
c	0.09	—	0.20
θ	0°	3.5°	8°
e	—	0.5	—
x	—	—	0.08
y	—	—	0.08
L _p	0.45	0.6	0.75
L ₁	—	1.0	—

Figure B 144-Pin LFQFP (PLQP0144KA-B)

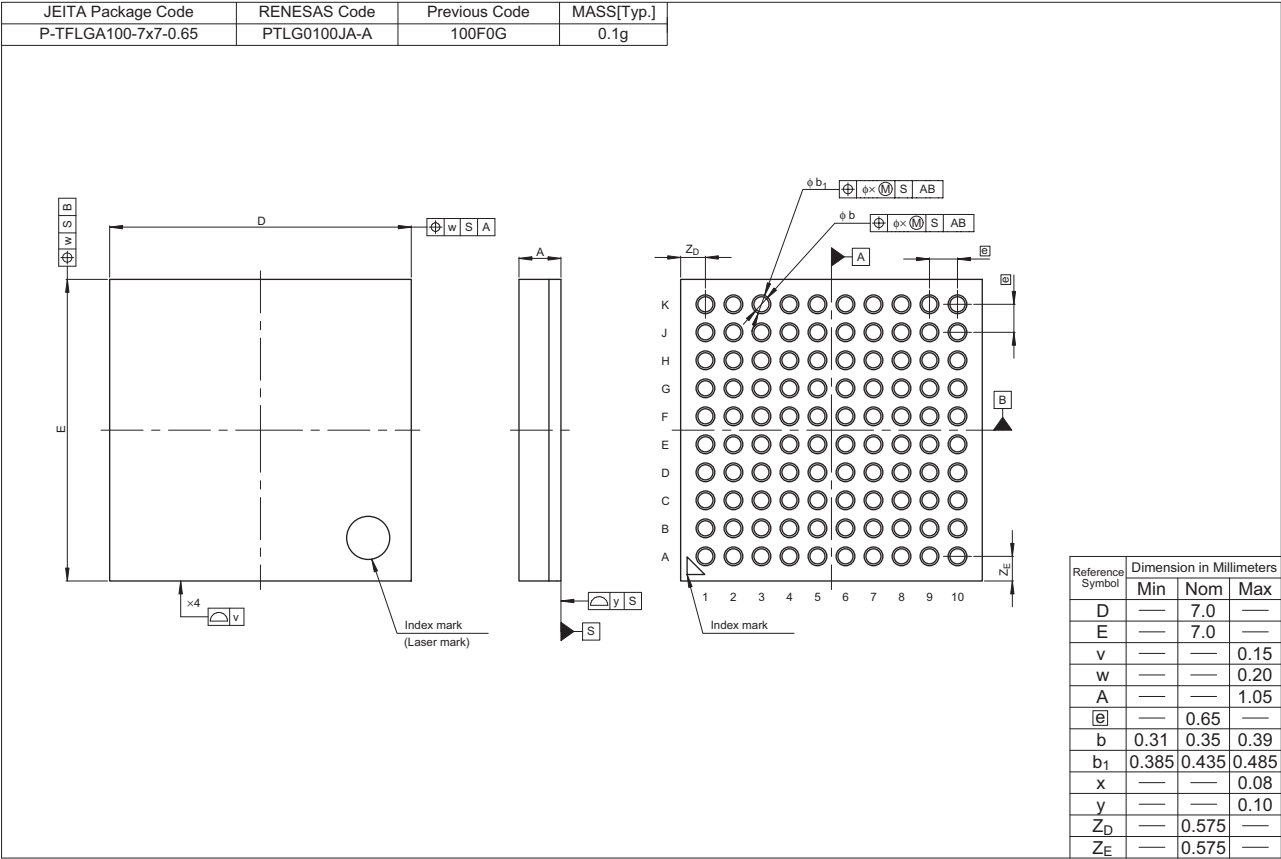


Figure C 100-Pin TFLGA (PTLG0100JA-A)

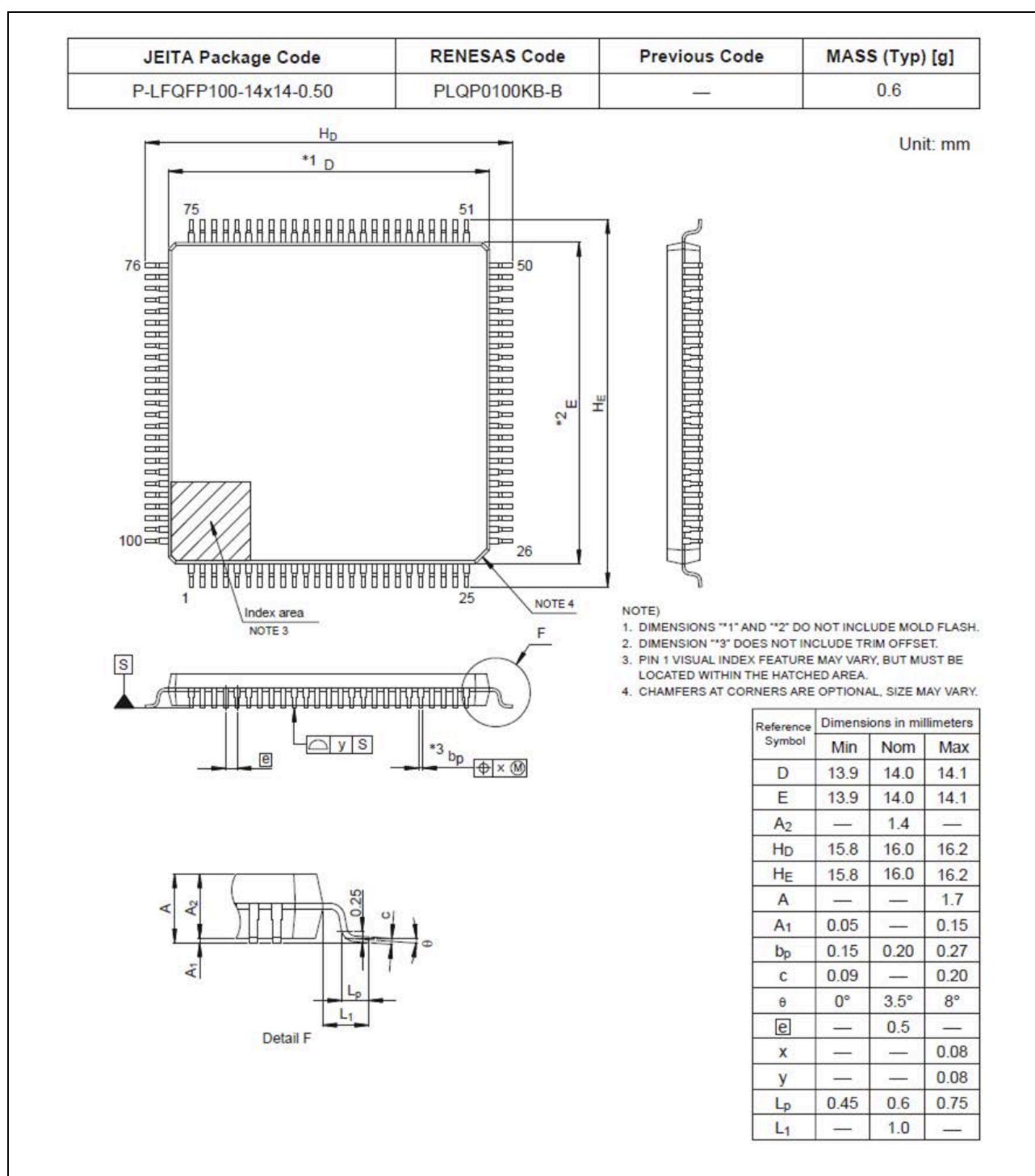


Figure D 100-Pin LFQFP (PLQP0100KB-B)

REVISION HISTORY	RX65N Group, RX651 Group Datasheet
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Rev.	Date	Description	
		Page	Summary
1.00	Aug 24, 2016	—	First edition, issued

General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

The following usage notes are applicable to all Microprocessing unit and Microcontroller unit products from Renesas. For detailed usage notes on the products covered by this document, refer to the relevant sections of the document as well as any technical updates that have been issued for the products.

1. Handling of Unused Pins

Handle unused pins in accordance with the directions given under Handling of Unused Pins in the manual.

¾ The input pins of CMOS products are generally in the high-impedance state. In operation with an unused pin in the open-circuit state, extra electromagnetic noise is induced in the vicinity of LSI, an associated shoot-through current flows internally, and malfunctions occur due to the false recognition of the pin state as an input signal become possible. Unused pins should be handled as described under Handling of Unused Pins in the manual.

2. Processing at Power-on

The state of the product is undefined at the moment when power is supplied.

¾ The states of internal circuits in the LSI are indeterminate and the states of register settings and pins are undefined at the moment when power is supplied.
In a finished product where the reset signal is applied to the external reset pin, the states of pins are not guaranteed from the moment when power is supplied until the reset process is completed. In a similar way, the states of pins in a product that is reset by an on-chip power-on reset function are not guaranteed from the moment when power is supplied until the power reaches the level at which resetting has been specified.

3. Prohibition of Access to Reserved Addresses

Access to reserved addresses is prohibited.

¾ The reserved addresses are provided for the possible future expansion of functions. Do not access these addresses; the correct operation of LSI is not guaranteed if they are accessed.

4. Clock Signals

After applying a reset, only release the reset line after the operating clock signal has become stable. When switching the clock signal during program execution, wait until the target clock signal has stabilized.

¾ When the clock signal is generated with an external resonator (or from an external oscillator) during a reset, ensure that the reset line is only released after full stabilization of the clock signal. Moreover, when switching to a clock signal produced with an external resonator (or by an external oscillator) while program execution is in progress, wait until the target clock signal is stable.

5. Differences between Products

Before changing from one product to another, i.e. to a product with a different part number, confirm that the change will not lead to problems.

¾ The characteristics of Microprocessing unit or Microcontroller unit products in the same group but having a different part number may differ in terms of the internal memory capacity, layout pattern, and other factors, which can affect the ranges of electrical characteristics, such as characteristic values, operating margins, immunity to noise, and amount of radiated noise. When changing to a product with a different part number, implement a system-evaluation test for the given product.

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